

SGA 6

Intersection Theory and Riemann–Roch

English translation

Contents

SGA 6, Front Matter, Exposé 0, Appendix, and Exposé I opening	2
Preface	3
Introduction	3
Table of Contents of SGA 6	4
1 Exposé 0 – Outline of a Programme for a Theory of Intersections on General Schemes	6
Exposé 0 – Outline of a Programme	6
1.1 Reminder on the Riemann–Roch formula	6
1.2 Removing the base field	7
1.3 The λ -filtration	8
1.4 General schemes	9
1.5 The sophisticated group $K^\bullet(X)$ and the cotangent complex	10
1.6 Analytic variant	12
1.7 The covariant invariant $K_\bullet(X)$	13
1.8 The determinant functor	14
Bibliography for Exposé 0	14
2 Appendix to Exposé 0 – Classes of Sheaves and the Riemann–Roch Theorem	16
Appendix – Classes of Sheaves and Riemann–Roch	16
Contents of the appendix	16
2.1 Chapter I. λ -rings: formal preliminaries	16
2.1.1 Definitions	16
2.1.2 Examples	18
2.1.3 The λ -ring defined by a graded ring	19
2.1.4 The operations $\rho^p(N, x)$	21
2.2 Chapter II. Coherent algebraic sheaves and Chern classes	23
2.2.1 Chow theory	23
2.2.2 Definition of Chern classes of coherent algebraic sheaves	25

2.2.3	Functorial generalities on $K(X)$	27
2.2.4	Riemann–Roch theorem in the report	28
3	Expose I – Generalities on Finiteness Conditions in Derived Categories	29
	Expose I – Generalities on Finiteness Conditions in Derived Categories	29
	Expose I, 0. Introduction	29
	Expose I, 1. Preliminary definitions	31
	Expose I, 1.1. Fibred categories	31
	Expose I, 1.4. Quasi-isomorphisms	31
	Expose I, 1.5. Finiteness conditions in derived categories	32
	SGA 6, Exposé I, Sections 2–8	34
	Corollary 1.4.5, proof continued	34
	2. Pseudo-coherent complexes	34
	3. Relation with the classical notion of coherence	44
	4. Perfect complexes	47
	5. Finite Tor-dimension and perfection	53
	6. Rank of a perfect complex	57
	7. Duality of perfect complexes	61
	8. Traces and cup-products	64
	Bibliography	66
	SGA 6, Exposé II and following sections	67
	1. General globalization criteria	67
	2. Application to certain categories of modules	68
	3. End of the comparison with quasi-coherent complexes	73

5. Applications: base change and semi-continuity theorems	74
1. Reminders and generalities on Grothendieck groups	79
2. The functors K_* and K'_* of a ringed topos	81
3. Complements on Grothendieck groups of schemes	84
1. Universal polynomials	88
2. Pre-lambda-rings	88
SGA 6, Exposé V.2.3.4 through Exposé VI	90
2. Pre-lambda-rings, continued	90
Grothendieck operations	90
3. The gamma-operations	92
4. Lambda-rings generated by generators and relations	95
5. The elements λ^p of N_*	98
6. Chern ring	102
7. Appendix: Adams operations	108
Exposé VI, 1. Projective bundles: locally free sheaves	111
Exposé VI, 2. Projective bundles: perfect complexes	115
Exposé VI, 3. Consequences of the structure theorem	118
Exposé VI, 4. Flag bundles	118
Exposé VI, 5. Applications and f^*	121
Exposé VI, 6. Filtration when X has an ample sheaf	123
SGA 6, Exposé VII	127

SGA 6, Exposé VIII, opening to 3.4	139
SGA 6, Exposé VIII.3.5 through Exposé XIII.2.11	144
SGA 6, Exposé XIII.3 through Exposé XIV	189

SGA 6, Front Matter, Exposé 0, Appendix, and Exposé I opening

Preface

The present edition of *Seminaire de Geometrie Algebrique 6* reproduces, up to minor changes, the first edition distributed by the IHES in the form of mimeographed exposés. The corrections concern mainly typographical errors, except in Exposés I and III, where L. Illusie made slight changes to a few statements which were incorrect in the original version. Footnotes have also been added, especially in Exposé XIV by A. Grothendieck, in order to point out recent progress on questions that were still open at the time of the seminar.

It should also be noted that Exposé XI, by D. Ferrand, on the theory of the determinant for perfect complexes, was not written up and will therefore not be published. To avoid errors in references, the numbering of the following exposés has nevertheless been left unchanged.

June 1971

Introduction

The purpose of this seminar is to develop a global theory of intersections and a Riemann–Roch formula for arbitrary schemes. A description of the programme of the seminar is given in Exposé 0. Already by the time of the well-known report of Borel–Serre in 1958, it was clear in principle how to prove a Riemann–Roch formula for general regular schemes, at least for a projective morphism. The extension to the general case is less evident. The programme in which it is placed – and which is partially carried out in this seminar – goes back to 1960. As in the theory of duality for coherent sheaves, one essential tool for formulating a satisfactory theory is Verdier’s theory of derived categories, and some familiarity with that theory is indispensable for understanding the seminar.

Apart from this theory, the seminar requires little beyond a general knowledge of the foundations of algebraic geometry as presented in EGA I, II, and III. Occasional references are made to EGA IV, mainly for facts concerning flat and smooth morphisms, and most of these also occur in the early exposés of SGA 1. In order to state some results with the generality required for applications, we sometimes use ringed sites and ringed topoi, for which we refer to SGA 4, Exposés I–IV. A reader who is not using the language of sites and topoi may replace them, at first reading, by ordinary topological spaces and their open sets; nevertheless the language of topoi is strongly recommended, since it gives an extremely convenient principle of unification.

The fundamental notion for the theory presented here is that of a perfect complex of modules, together with several variants developed in Exposés I, II, and III. These notions are also important in other parts of algebraic geometry, notably for formulas of Lefschetz–Verdier type in cohomology with discrete coefficients (SGA 5) or coherent coefficients. They should also play a role outside algebraic geometry, for example in the formulation of a complex analytic variant of the Riemann–Roch theorem treated here, or in suitable variants of the Atiyah–Singer index theorem. Some indications in this direction are given in Exposé II. At the time of the seminar, however, there was still no statement, even heuristic, that would simultaneously include both the algebraic

Riemann–Roch theorem and the index theorem. A new idea seems to be missing, just as a new idea is missing in algebraic geometry for proving Riemann–Roch beyond projective hypotheses; see Expose XIV, no. 2.

The local theory of intersections, as treated in Serre’s *Algebre locale. Multiplicites*, is not used in the seminar. We mention only that Serre’s course had an evident influence on the genesis of the theory in 1957. Nor do we use the theory of the Chow ring developed in the Chevalley Seminar of 1958. That theory is essentially tied to nonsingularity hypotheses, whereas the aim here is to develop an intersection theory on general schemes, and even on general locally ringed topoi. Expose XIV, nos. 4 and 8, comments on the relation between the present theory and Chow theory, and raises the question of a generalization of the latter. Thus the seminar presents a coherent and self-contained theory of intersections, but it is not exhaustive with respect to the different viewpoints that are useful, or even indispensable, in intersection theory. It should therefore be complemented by Serre’s and Chevalley’s treatments.

An appendix to Expose 0 reproduces Grothendieck’s 1957 report *Classes de faisceaux et theoreme de Riemann–Roch*, which served as the basis for the Borel–Serre seminar at Princeton in the same year and for their article. That report sketches two proofs of the Riemann–Roch theorem for nonsingular quasi-projective varieties. The first proof, for the time being valid only in characteristic zero, gives a slightly sharper result in the case of an immersion and had not yet appeared in the literature. The report can be read independently of the rest of the seminar and can serve, like Expose 0, as an introduction for readers not yet familiar with Borel–Serre. The proof just mentioned applies essentially unchanged to projective morphisms of complex analytic spaces and should apply in arbitrary characteristic once the problem of exterior-power operations in the derived category is solved. The lack of a study of that question is probably the most serious gap in the seminar from the point of view adopted here.

Two names appearing on the cover do not appear in the table of contents. J.-P. Jouanolou took an active part in the technical elaboration of the first part of the seminar, though he was prevented from participating in the oral lectures. In particular, he is responsible for the linear-independence assertion in the important statement VI 1.1 on the structure of the ring $K^\bullet$ of classes of locally free modules on a projective bundle. J.-P. Serre delivered two oral talks, logically independent of the rest of the seminar, and preferred to publish them separately under the title *Groupes de Grothendieck des schemas en groupes reductifs deployes*.

As in the other parts of the Seminaire de Geometrie Algebrique du Bois Marie, the sigla SGA 1 to SGA 6 refer to the different parts of the seminar. The following Roman numeral indicates the expose number, and the following Arabic numerals refer to the internal numbering of that expose. For references internal to the present seminar the siglum SGA 6 is omitted, so that a reference such as I 4.9 means statement 4.9 of Expose I. The siglum EGA refers to the *Elements de Geometrie Algebrique* of J. Dieudonne and A. Grothendieck.

Table of Contents of SGA 6

Expose	Title and author	Original page
0	<i>Outline of a Programme for a Theory of Intersections on General Schemes</i> , by A. Grothendieck	1
Appendix	<i>Classes of Sheaves and the Riemann–Roch Theorem</i> , by A. Grothendieck	20
I	<i>Generalities on Finiteness Conditions in Derived Categories</i> , by L. Illusie	56
II	<i>Existence of Global Resolutions</i> , by L. Illusie	160
III	<i>Relative Finiteness Conditions</i> , by L. Illusie	190
IV	<i>Grothendieck Groups of Ringed Topoi</i> , by L. Illusie	222
V	<i>Generalities on λ-Rings</i> , by P. Berthelot	297
VI	<i>The $K^\bullet$ of a Projective Bundle: Computations and Consequences</i> , by P. Berthelot	365
VII	<i>Regular Immersions and the Computation of the $K^\bullet$ of a Blown-Up Scheme</i> , by P. Berthelot	416
VIII	<i>The Riemann–Roch Theorem</i> , by P. Berthelot	466
IX	<i>Some Computations of $K_\bullet$-Groups</i> , by P. Berthelot	498
X	<i>Formalism of Intersections on Proper Algebraic Schemes</i> , by O. Jussila, with an appendix by A. Grothendieck on specialization in intersection theory	595
XI	<i>Not written</i>	–
XII	<i>A Relative Representability Theorem for the Picard Functor</i> , by M. Raynaud, written by S. Kleiman	595
XIII	<i>Finiteness Theorems for the Picard Functor</i> , by S. Kleiman	616
XIV	<i>Open Problems in Intersection Theory</i> , by A. Grothendieck	667

1 Expose 0 – Outline of a Programme for a Theory of Intersections on General Schemes

by A. Grothendieck

The present expose is introductory. Its reading is not logically necessary for the study of the seminar. It is addressed especially to readers who know the provisional form of the Riemann–Roch theorem as presented in the report of Borel–Serre, or in Grothendieck’s earlier report cited below as RRR and reproduced as the appendix to this expose.

1.1 Reminder on the Riemann–Roch formula

Let $f : X \rightarrow Y$ be a proper morphism of smooth quasi-projective schemes over a field k , and let F be a coherent sheaf on X . Denote by $\text{Cl}(F)$ the class of F in the group $K(X)$ of classes of coherent sheaves on X . The Riemann–Roch formula has the form

$$\text{Todd}(T_Y) \text{ch}_Y(f_!(\text{Cl}(F))) = f_*(\text{Todd}(T_X) \text{ch}_X(F)). \quad (1)$$

It holds in $A(Y) \otimes_{\mathbb{Z}} \mathbb{Q}$, where $A(Y)$ is the Chow ring of Y . The f_* on the right is the direct image of cycles, tensored with $\mathbb{Q}$,

$$f_* : A(X) \longrightarrow A(Y),$$

while the term $f_!(\text{Cl}(F))$ on the left is the Euler–Poincaré characteristic of F relative to f :

$$f_!(\text{Cl}(F)) = \sum_i (-1)^i \text{Cl}(R^i f_*(F)).$$

The symbols ch_X and ch_Y denote the Chern characters, and T_X and T_Y the tangent bundles. Since $\text{Todd}(-)$ and $\text{ch}(-)$ are universal polynomials with rational coefficients in the Chern classes, and the constant term of $\text{Todd}(-)$ is 1, the Todd class is invertible. After multiplying by $\text{Todd}(T_Y)^{-1}$, one obtains the more useful form

$$\text{ch}_Y(f_!(\text{Cl}(F))) = f_*(\text{Todd}(T_f) \text{ch}_X(F)), \quad (2)$$

where

$$T_f = T_X - f^*T_Y \in K(X). \quad (3)$$

Thus T_f plays the role of the virtual relative tangent bundle of X over Y . If f is smooth, then $T_f = T_{X/Y}$. If $f : X \rightarrow Y$ is an immersion, then

$$T_f = -\check{N}_{X/Y},$$

where $\check{N}_{X/Y}$ is the dual of the normal sheaf.

The main purpose of the seminar is to generalize (2) in two directions at once:

- (a) remove the hypothesis that a base field k exists;
- (b) replace regularity assumptions on X and Y by a local regularity condition on f .

Along the way one also studies a third problem: removing quasi-projectivity hypotheses, which in the absence of a base field are expressed by the existence of ample invertible modules on X and Y .

1.2 Removing the base field

We first examine (a), still retaining regularity and the existence of ample invertible modules on X and on Y . In this situation the definition of $K(X)$, $K(Y)$, and $f_! : K(X) \rightarrow K(Y)$ causes no new difficulty, because X and Y are regular. A natural way to give meaning to (2) would be to define Chow rings $A(X)$ and $A(Y)$, a direct image

$$f_* : A(X) \rightarrow A(Y),$$

Chern classes

$$c_i : K(X) \rightarrow A(X),$$

and the virtual relative tangent bundle $T_f \in K(X)$.

For the last of these a definition is essentially forced. The morphism f , using ample invertible modules on X and Y , can be factored as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y, \tag{4}$$

where i is a closed immersion and f' is smooth; one may take, for example, $X' = \mathbb{P}_Y^r$ for r sufficiently large. Set

$$T_f = i^* T_{X'/Y} - \check{N}_{X/X'}. \tag{5}$$

Here $\Omega_{X'/Y}^1$ is the locally free module of relative differentials, and $N_{X/X'}$ is the conormal module J/J^2 for the ideal J defining X in X' . Since X and X' are regular, this conormal module is locally free. Expose VIII proves that the element T_f defined by (5) is independent of the chosen factorization.

For Chow rings and Chern classes under these hypotheses, a theory analogous to the usual theory for smooth quasi-projective varieties should exist, but this work is not undertaken in the seminar. Instead one introduces another ring which plays the same role as the Chow ring and whose rationalization is isomorphic to $A(X) \otimes \mathbb{Q}$ when X is smooth quasi-projective over a field. This ring is the graded ring associated with a suitable filtration of $K(X)$.

One natural filtration, considered in RRR, is the topological filtration by codimension of support: $\text{Fil}^i K(X)$ is generated by the classes $\text{Cl}(F)$ of coherent sheaves whose support has codimension at least i . One must first know that this filtration is compatible with multiplication. In RRR this was done for X smooth over a field by using Chow's moving lemma. Assuming this, one gets a

graded ring $\mathrm{Gr}_{\mathrm{top}}^*(X)$. RRR defines a surjective ring homomorphism

$$\phi : A(X) \rightarrow \mathrm{Gr}_{\mathrm{top}}^*(X),$$

sending the class of a prime cycle Z to $\mathrm{Cl}_X(\mathcal{O}_Z)$; Expose XIV 4.2 shows that the kernel is a torsion group. Moreover, RRR explains how the images in $\mathrm{Gr}_{\mathrm{top}}^*(X)$ of the Chern classes $c_i(F)$ can be computed directly from $\mathrm{Cl}(F) \in K(X)$ and exterior-power operations in $K(X)$.

Thus for a noetherian regular scheme X with an ample invertible module, the ring $\mathrm{Gr}_{\mathrm{top}}^*(X) \otimes \mathbb{Q}$ is a convenient substitute for the hypothetical Chow ring. Its advantage is that many computations in RRR naturally take place in it. To fully justify this viewpoint, however, one would have to settle the compatibility of the topological filtration with multiplication, a problem closely tied to Chow's moving lemma – precisely the technical ingredient of Chow theory one hoped to avoid.

1.3 The λ -filtration

To bypass this problem, one equips $K(X)$ with a second filtration, finer than the topological filtration and describable in purely algebraic terms from the structure of $K(X)$ as an augmented λ -ring. The definition is suggested by the formula in RRR expressing the Chern classes of $x \in K(X)$ in $\mathrm{Gr}_{\mathrm{top}}^*(X) \otimes \mathbb{Q}$:

$$c_i(x) = \gamma^i(x - \varepsilon(x)) \pmod{\mathrm{Fil}_{\mathrm{top}}^{i+1} K(X)}. \quad (6)$$

Here γ^i is a certain integral combination of the operations λ^j , $j \leq i$, and ε is the augmentation. One proves that $\gamma^i(x - \varepsilon(x))$ has topological filtration at least i , so that (6) makes sense. The λ -filtration is then defined by ideals $\mathrm{Fil}_{\lambda}^i K(X)$ generated by finite sums of expressions

$$\gamma^{i_1}(x_1 - \varepsilon(x_1)) \cdots \gamma^{i_k}(x_k - \varepsilon(x_k)), \quad i_1 + \cdots + i_k \geq i.$$

The associated graded ring is denoted $\mathrm{Gr}^\bullet(X)$. It is this ring which replaces the Chow ring in the seminar. In the present regular ample situation the canonical map

$$\mathrm{Gr}^\bullet(X) \rightarrow \mathrm{Gr}_{\mathrm{top}}^*(X)$$

is an isomorphism modulo torsion. Thus, after tensoring with $\mathbb{Q}$, $\mathrm{Gr}^\bullet(X)$ plays the same role as $A(X) \otimes \mathbb{Q}$. The definition is also designed so that (6), with the λ -filtration in place of the topological filtration, gives a theory of Chern classes with values in $\mathrm{Gr}^\bullet(X)$. Expose V proves, using Expose VI, that $K(X)$ is a special λ -ring in the sense of RRR, and hence that these Chern classes have all the usual formal properties.

A substantial advantage of $\mathrm{Gr}^\bullet(X)$ over $A(X)$ or $\mathrm{Gr}_{\mathrm{top}}^*(X)$ is that it is defined for every scheme X , and more generally for every locally ringed topos. One takes $K(X)$ to be the group of classes of locally free modules on X ; it remains an augmented λ -ring and has the corresponding filtration. The price of this generality is that, unless one tensors with $\mathbb{Q}$, torsion phenomena make the properties of $\mathrm{Gr}^\bullet(X)$ less satisfactory than those of its competitors. For instance, for a proper morphism $f : X \rightarrow Y$ of noetherian regular schemes with ample invertible modules, it is not

generally true integrally that

$$f_* \operatorname{Fil}^i K(X) \subset \operatorname{Fil}^{i-d} K(Y),$$

where d is the relative dimension. Expose VII proves the corresponding assertion after tensoring with $\mathbb{Q}$, giving a degree $-d$ homomorphism

$$f_* : \operatorname{Gr}^\bullet(X) \otimes \mathbb{Q} \rightarrow \operatorname{Gr}^\bullet(Y) \otimes \mathbb{Q}$$

which plays the role of the direct image of cycles.

We therefore have all the structural elements needed to make sense of (2) without assuming a base field, when $f : X \rightarrow Y$ is a proper morphism of noetherian regular schemes admitting ample invertible modules.

1.4 General schemes

We next indicate how to give meaning to (2) after abandoning regularity of X and Y , replacing it by a local condition on f .

The first difficulty is that the observation underlying RRR – namely, that the group $K(X)$ may be defined either by coherent sheaves or by locally free sheaves – uses regularity in an essential way. For a general noetherian scheme one must distinguish two groups. We write

$$K_\bullet(X) \quad \text{and} \quad K^\bullet(X),$$

the first defined from coherent sheaves, covariant for proper morphisms, and the second from locally free sheaves, contravariant for arbitrary morphisms. The group $K^\bullet(X)$ is naturally a ring, and even an augmented λ -ring; $K_\bullet(X)$ is generally only a module over it. The map

$$x \longmapsto x \cdot \operatorname{Cl}(\mathcal{O}_X)$$

from $K^\bullet(X)$ to $K_\bullet(X)$ is not an isomorphism in general.

For the generalized Riemann–Roch formula the seminar works with $K^\bullet(X)$ and $K^\bullet(Y)$, not with $K_\bullet(X)$ and $K_\bullet(Y)$. These augmented λ -rings, with their natural filtrations, give graded rings $\operatorname{Gr}^\bullet(X)$ and $\operatorname{Gr}^\bullet(Y)$ and Chern-class maps

$$c_i : K^\bullet(X) \rightarrow \operatorname{Gr}^\bullet(X).$$

The problem is the direct image

$$f_* : K^\bullet(X) \rightarrow K^\bullet(Y), \tag{7}$$

which cannot be defined naively by

$$f_*(\operatorname{Cl}(F)) = \sum_i (-1)^i \operatorname{Cl}(R^i f_*(F)),$$

since even for a locally free F the sheaves $R^i f_*(F)$ need not be locally free, nor even of finite Tor-dimension.

Derived categories provide the solution. Although the individual sheaves $R^i f_*(F)$ may be bad, the derived direct image $Rf_*(F)$ often has good properties as a complex. If F has finite Tor-dimension over $\mathcal{O}_Y$, for example if f has finite Tor-dimension and F is locally free, then Expose III shows that $Rf_*(F)$ is perfect: its cohomology sheaves are coherent and it has globally finite Tor-dimension, or equivalently, on each affine open of Y , it is isomorphic in the derived category to a bounded complex of locally free modules of finite type.

Perfect complexes are as good as locally free modules from the viewpoint of homological algebra. They form a triangulated subcategory $\text{Parf}(Y)$ of $\mathcal{D}(Y)$. For any triangulated category C there is a Grothendieck group $K(C)$, generated by objects modulo the relations

$$\text{Cl}(K) - \text{Cl}(K') - \text{Cl}(K'')$$

attached to distinguished triangles

$$K' \rightarrow K \rightarrow K'' \rightarrow K'[1].$$

When Y has an ample invertible module, Expose II proves that every perfect complex on Y is globally isomorphic in $\mathcal{D}(Y)$ to a bounded complex of locally free modules of finite type. It follows, in Expose IV, that the canonical map from the naive $K^\bullet(Y)$ to $K(\text{Parf}(Y))$ is an isomorphism. Thus a perfect complex $L_\bullet$ on Y has a class

$$\text{Cl}(L_\bullet) \in K^\bullet(Y),$$

which is represented by

$$\sum_i (-1)^i \text{Cl}(L'_i)$$

for any bounded locally free complex $L'_\bullet$ quasi-isomorphic to it.

Consequently, if f is proper of finite Tor-dimension between noetherian schemes and Y has an ample invertible module, one defines (7) by

$$f_*(\text{Cl}(F)) = \text{Cl}(Rf_*(F)) \tag{8}$$

for locally free F on X . If X also has an ample invertible module, this is exactly the map on K -groups induced by the exact functor of triangulated categories

$$Rf_* : \text{Parf}(X) \rightarrow \text{Parf}(Y).$$

1.5 The sophisticated group $K^\bullet(X)$ and the cotangent complex

For an arbitrary scheme, or locally ringed topos, the right invariant replacing the ring $K(X)$ of RRR is not the naive group defined only from locally free modules, but rather the group

$K(\text{Parf}(X))$ of perfect complexes. We continue to denote it by $K^\bullet(X)$, reserving $K_{\text{naive}}^\bullet(X)$ for the group defined by locally free modules alone. This viewpoint is forced by the need to define direct images for proper morphisms of finite Tor-dimension: with perfect complexes, f_* is simply induced by Rf_* .

The new definition raises a homological-algebra problem: one must define a λ -ring structure on $K^\bullet(X)$, compatible with the ring structure coming from the derived tensor product. In other words one needs exterior-power operations in the derived category sending perfect complexes to perfect complexes. This question was not clarified in the seminar and is identified in the introduction as its most serious gap. Once it is solved, the construction of the filtration and of Chern classes proceeds as before.

To state the Riemann–Roch formula for a proper locally complete intersection morphism $f : X \rightarrow Y$, one must also define the virtual relative tangent class

$$T_f \in K^\bullet(X).$$

When f factors as in (4), by a closed immersion into a smooth Y -scheme, the formula (5) applies. It has a more derived form. The exterior differential gives a canonical morphism

$$N_{X/X'} \rightarrow \Omega_{X'/Y}^1 \otimes_{\mathcal{O}_{X'}} \mathcal{O}_X. \quad (9)$$

By placing the source in degree 1 and the target in degree 0, one obtains a two-term complex $L_{X/Y}$. Then

$$T_f = \text{Cl}(L_{X/Y}^\vee) \quad (10)$$

in $K^\bullet(X)$. For a locally complete intersection morphism, $L_{X/Y}$ is perfect, and Expose VIII proves that it is independent, up to canonical isomorphism in $\mathcal{D}(X)$, of the chosen factorization.

For an arbitrary morphism of schemes, and more generally for a morphism of commutatively ringed topoi, one can construct a canonical object $L_{X/Y}$ of $\mathcal{D}(X)$. Let $A = f^* \mathcal{O}_Y$ and let B be an A -algebra. Choose a map of sheaves of sets $T \rightarrow B$ generating B as an A -algebra, so that

$$A[T] \rightarrow B$$

is an epimorphism. If J is its kernel, consider the complex

$$J/J^2 \rightarrow \Omega_{A[T]/A}^1 \otimes_{A[T]} B. \quad (11)$$

It is not hard to see that its isomorphism class in $\mathcal{D}(X)$ is independent of the choice of T and $T \rightarrow B$. This is the relative cotangent complex. In the locally complete intersection case it is perfect and gives the class T_f by (10). This gives another justification for using the sophisticated $K^\bullet(X)$ of perfect complexes instead of the naive group.

Subject to the exterior-power question above, all structural ingredients are available for giving a meaning to (2) for a proper locally complete intersection morphism of arbitrary schemes. This does not mean that the formula is proved in that generality, even for a proper smooth three-dimensional

variety over an algebraically closed field of characteristic $p > 0$; see Expose XIV, no. 2.

1.6 Analytic variant

The form of Riemann–Roch just discussed also makes sense for a proper locally complete intersection morphism of complex analytic spaces, or of rigid analytic spaces in Tate’s sense. The proof in RRR applies only when the morphism is projective. As in the algebraic case, a new idea seems necessary for the general case.

For nonsingular complex analytic spaces there is another version. If X is a complex analytic space and X^{top} denotes the same space with the sheaf of continuous complex-valued functions, there is a canonical morphism of ringed spaces

$$\epsilon : X^{\text{top}} \rightarrow X$$

and hence a homomorphism

$$K^\bullet(X) \rightarrow K^\bullet(X^{\text{top}}). \quad (12)$$

If X^{top} is compact, Expose II shows that every perfect complex on X^{top} is globally represented by a bounded complex of complex vector bundles. Thus $K^\bullet(X^{\text{top}})$ is the usual topological K -group.

For a proper morphism $f : X \rightarrow Y$ of compact nonsingular complex analytic spaces, Atiyah–Hirzebruch attach a homomorphism

$$f_!^{\text{top}} : K^\bullet(X^{\text{top}}) \rightarrow K^\bullet(Y^{\text{top}}).$$

Using also the analytic direct image $f_!$, defined by (8) and Grauert’s finiteness theorem, one obtains a square

$$\begin{array}{ccc} K^\bullet(X) & \longrightarrow & K^\bullet(X^{\text{top}}) \\ \downarrow f_! & & \downarrow f_!^{\text{top}} \\ K^\bullet(Y) & \longrightarrow & K^\bullet(Y^{\text{top}}). \end{array}$$

The Riemann–Roch assertion in this form is the commutativity of the square. Unlike the Chow-theoretic formula, this assertion does not discard torsion and takes place in a discrete group. It is known for nonsingular projective varieties. It is plausibly a variant of the Atiyah–Singer index theorem and should have a common generalization with it.

Once this square is known to commute, the differentiable Riemann–Roch theorem of Atiyah–Hirzebruch gives a cohomological version, namely the commutativity of

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{\text{ch}} & H^{2*}(X, \mathbb{Q}) \\ \downarrow f_! & & \downarrow f_* \\ K^\bullet(Y) & \xrightarrow{\text{ch}} & H^{2*}(Y, \mathbb{Q}), \end{array}$$

where the horizontal maps come from ordinary topological Chern classes and the right vertical arrow is the Gysin homomorphism in rational cohomology.

1.7 The covariant invariant $K_\bullet(X)$

So far the discussion has focused on the contravariant invariant $K^\bullet(X)$. The covariant invariant $K_\bullet(X)$ plays a complementary role. If $K^\bullet(X)$ is regarded as analogous to integral cohomology, then $K_\bullet(X)$ is analogous to integral homology, and its $K^\bullet(X)$ -module structure corresponds to cap product. When X is regular, the natural map $K^\bullet(X) \rightarrow K_\bullet(X)$ being an isomorphism is analogous to Poincaré duality.

For a noetherian scheme X , $K_\bullet(X)$ is given an increasing filtration by setting $\text{Fil}_i K_\bullet(X)$ to be generated by classes $\text{Cl}_\bullet(F)$ of coherent sheaves with $\dim \text{supp } F \leq i$. Exposé X proves compatibility with the filtration of $K^\bullet(X)$:

$$\text{Fil}^i K^\bullet(X) \cdot \text{Fil}_j K_\bullet(X) \subset \text{Fil}_{j-i} K_\bullet(X). \quad (13)$$

The associated graded group $\text{Gr}_\bullet(X)$ is thereby a module over $\text{Gr}^\bullet(X)$. Its elements may be interpreted as classes of algebraic cycles, for an equivalence relation coarser than rational equivalence. For a proper morphism $f : X \rightarrow Y$ of noetherian schemes, the direct image

$$f_* : K_\bullet(X) \rightarrow K_\bullet(Y)$$

is compatible with the filtrations and gives a graded map

$$f_* : \text{Gr}_\bullet(X) \rightarrow \text{Gr}_\bullet(Y),$$

satisfying a projection formula. In the proposed theory, a manageable intersection theory on nonregular noetherian schemes should be the combined study of the contravariant invariants $K^\bullet(X), \text{Gr}^\bullet(X)$ and the covariant invariants $K_\bullet(X), \text{Gr}_\bullet(X)$. They have different functorial characters and complement each other. For example, the covariant groups behave better under certain nonproper fibrations through homotopy exact sequences; one has canonical isomorphisms

$$K_\bullet(X[t]) \simeq K_\bullet(X), \quad \text{Gr}_\bullet(X[t]) \simeq \text{Gr}_\bullet(X),$$

which generally fail for $K^\bullet$ when X is nonregular.

If X is proper over a field, the projection $\pi_X : X \rightarrow \text{point}$ defines the Euler–Poincaré characteristic

$$\chi = \pi_{X*} : K_\bullet(X) \rightarrow K_\bullet(\text{point}) = \mathbb{Z},$$

and on $\text{Gr}_0(X)$ it induces the degree of zero-cycles

$$\deg : \text{Gr}_0(X) \rightarrow \mathbb{Z}.$$

Multiplication followed by degree gives pairings

$$\text{Gr}^i(X) \times \text{Gr}_i(X) \rightarrow \mathbb{Z},$$

explaining in what sense $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_\bullet(X)$ are dual, as cohomology and homology are dual. The numerical relations in intersection theory over a field come from these maps and pairings. Expose X develops this point of view and also studies specialization.

1.8 The determinant functor

As in every intersection theory, one must locate the role of the Picard group. Under a mild restriction, automatically satisfied for instance when X is quasi-compact, Expose X proves a canonical isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Gr}^1(X),$$

which gives the exact role of the Picard group in the proposed theory. This motivates the more detailed study of the Picard group in Exposes XII and XIII, which are logically independent of the main text. Kleiman proves in Expose XIII finiteness theorems and results on numerical equivalence for the Picard group, using methods of Kleiman and Mumford that are substantially simpler than Grothendieck's original arguments. The one result not covered by those methods is EGA III, 7.7, whose proof and refinements, including representability theorems for the Picard functor, occupy Expose XII.

Finally, Expose XI was to study on an arbitrary locally ringed topos a determinant functor

$$\det : \mathrm{Parf}(X) \rightarrow \mathrm{Inv}(X),$$

from the category of perfect complexes, with only isomorphisms in the derived category as morphisms, to the category of invertible modules on X . It is essentially characterized by locality and by the formula

$$\det(L_\bullet) = \bigotimes_i \det(L_i)^{(-1)^i}$$

for a bounded complex of locally free modules. Although this expose is independent of the rest of the seminar except Expose I, it is an important part of the general “yoga” developed in Exposes I–XI.

Bibliography for Expose 0

- [1] M. Atiyah and F. Hirzebruch, *Vector bundles and homogeneous spaces*, Symposia in Pure Mathematics, vol. 3, Differential Geometry, pp. 7–38, 1961.
- [2] A. Borel and J.-P. Serre, *Le theoreme de Riemann–Roch*, Bulletin de la Societe Mathematique de France 86 (1958), 97–136.
- [3] H. Grauert, *Ein Theorem der analytischen Garbentheorie*, Publications Mathematiques de l’IHES 5 (1960), 5–64.
- [4] A. Grothendieck, *Classes de faisceaux et theoreme de Riemann–Roch*, mimeographed notes, Princeton, 1957; cited as RRR and reproduced as the appendix to Expose 0.

- [5] A. Grothendieck, *Technique de descente et theoremes d'existence en geometrie algebrique*, in *Fondements de la Geometrie Algebrique*, extracts from the Bourbaki Seminar 1957–1962.
- [6] W. Shih, *Seminaire de Topologie a l'IHES*, 1966/67.
- [7] M. Atiyah and F. Hirzebruch, *The Riemann–Roch theorem for analytic embeddings*, *Topology* 1 (1962), 151–166.

2 Appendix to Expose 0 – Classes of Sheaves and the Riemann–Roch Theorem

by A. Grothendieck

This appendix is the reproduction of the report cited in the introduction. Footnotes marked by special signs were added in October 1967.

Contents of the appendix

Chapter I: λ -Rings (formal preliminaries). Definitions; examples; the λ -ring defined by a graded ring; the operations $\rho^p(N, x)$.

Chapter II: Classes of coherent algebraic sheaves and Chern classes. Chow theory; definition of Chern classes of coherent algebraic sheaves; functorial generalities on $K(X)$; technical results; sheaf-theoretic definitions of Chern classes and applications to injective morphisms; the Riemann–Roch theorem.

2.1 Chapter I. λ -rings: formal preliminaries

2.1.1 Definitions

A λ -ring is a commutative ring K with unit, equipped with maps

$$\lambda^i : K \rightarrow K \quad (i \geq 0)$$

satisfying

$$\lambda^0(x) = 1, \quad \lambda^1(x) = x, \quad \lambda^n(x + y) = \sum_{i=0}^n \lambda^i(x) \lambda^{n-i}(y). \quad (14)$$

Equivalently, if

$$\lambda_t(x) = \sum_{n \geq 0} \lambda^n(x) t^n \in K[[t]],$$

then $x \mapsto \lambda_t(x)$ is a homomorphism from the additive group of K to the multiplicative group of formal power series of augmentation 1, lifting $x \mapsto 1 + xt$.

The ring $\mathbb{Z}$ has a unique λ -structure compatible with its ring structure and satisfying $\lambda_t(1) = 1 + t$; hence

$$\lambda_t(n) = (1 + t)^n, \quad \lambda^i(n) = \binom{n}{i}.$$

A λ -homomorphism is a ring homomorphism compatible with all the λ^i . An augmented λ -ring is a λ -ring equipped with a λ -homomorphism to $\mathbb{Z}$.

To define the special λ -rings, let k be a commutative ring with unit and put

$$\widetilde{K} = 1 + k[[t]]^+,$$

the group of formal series of augmentation 1. We put on this group the structure of a λ -ring for which addition is multiplication of formal series. The multiplication, denoted $f \circ g$, is characterized by universal integral polynomials in the coefficients and by the formula on linear factors

$$(1 + at) \circ (1 + bt) = 1 + abt. \quad (15)$$

The polynomials are obtained from elementary symmetric functions. Similarly, the operations $\lambda^i(f)$ are determined by universal polynomials and by the rule

$$\lambda^i(1 + at) = 1 \quad (i > 1),$$

where 1 is the zero element of the additive group $1 + k[[t]]^+$. With these operations $\widetilde{K}$ is a λ -ring, with zero element 1 and unit $1 + t$, and

$$(1 + at) \circ f = f(at). \quad (16)$$

A λ -ring K is called *special* if the additive homomorphism

$$K \rightarrow 1 + K[[t]]^+, \quad x \mapsto \lambda_t(x),$$

is a λ -homomorphism. Thus

$$\lambda_t(1) = 1 + t, \quad (17)$$

$$\lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y), \quad (18)$$

$$\lambda_t(\lambda^n(x)) = \lambda^n(\lambda_t(x)). \quad (19)$$

Equivalently, the operations satisfy universal polynomial identities

$$\lambda^i(1) = 0 \quad (i > 1),$$

$$\lambda^n(xy) = P_n(\lambda^1 x, \dots, \lambda^n x; \lambda^1 y, \dots, \lambda^n y),$$

$$\lambda^m(\lambda^n x) = P_{m,n}(\lambda^1 x, \dots, \lambda^{mn} x).$$

The P_n and $P_{m,n}$ are precisely the universal polynomials furnished by symmetric-function computations.

2.1.2 Examples

Let G be a group and k a commutative ring. The group $K(G)$ is the quotient of the free group generated by finite type k -linear representations of G by relations

$$(\rho \oplus \rho') - (\rho) - (\rho').$$

Tensor product and exterior powers define a λ -ring structure. One can also pass to a stronger quotient by imposing the relation

$$(\rho) - (\rho') - (\rho'')$$

whenever ρ is an extension of ρ' by ρ'' which is split as an extension of k -modules. This gives the reduced group of representation classes, denoted here $K_\bullet(G)$, and the λ -operations descend to it.

If k is algebraically closed of characteristic zero, then $K(G)$, and hence also $K_\bullet(G)$, is a special λ -ring. In positive characteristic this need not remain true for $K(G)$, but $K_\bullet(G)$ is still special under the hypotheses used in the report. Variants include algebraic groups and rational representations, topological groups or Lie groups with continuous representations, and analogous constructions for quadratic forms or Lie-algebra representations.

Let X be an algebraic variety over k . One may take the quotient of the free group generated by vector bundles on X by the direct-sum relations. Exterior powers and tensor products again give a λ -ring. If X is complete, Krull–Remak–Schmidt identifies this group with the free group generated by indecomposable vector bundles, so that its knowledge is essentially the classification of vector bundles together with their elementary tensor operations. One may also impose extension relations and obtain the group $K(\mathcal{E}(X))$ attached to the additive category of vector bundles. Under the same hypotheses as above this gives a special λ -ring.

A particularly important computation is the following. Let k be algebraically closed, and let G be a connected affine algebraic group which is globally reductive, meaning that its radical is a torus. Only rational linear representations are considered. In characteristic zero, complete reducibility shows that

$$K(G) = K_\bullet(G)$$

is the free group on irreducible rational representations. If T is a maximal torus, then

$$K(T) = K_\bullet(T) = \mathbb{Z}[\widehat{T}],$$

where $\widehat{T}$ is the character group. The Weyl group W acts on $\widehat{T}$, hence on $K(T)$, and restriction gives a homomorphism

$$K_\bullet(G) \rightarrow K_\bullet(T)^W. \tag{20}$$

Theorem. *The homomorphism (20) is an isomorphism.*

It follows that $K_\bullet(G)$ depends only on the type of G , i.e. on the action of W on a free abelian group. It is a special λ -ring. If $\rho_1, \dots, \rho_r$ are the fundamental representations of $G/\text{rad } G$ and

$\sigma_1, \dots, \sigma_s$ a basis of rational homomorphisms from G to k^* , then

$$K_\bullet(G) \subset \mathbb{Z}[\rho_1, \dots, \rho_r; \sigma_1, \sigma_1^{-1}, \dots, \sigma_s, \sigma_s^{-1}]$$

inside the rational function field generated by these symbols. In particular $K_\bullet(G)$ has no zero divisors. For $G = \prod_i \mathrm{GL}(n_i, k)$ this description becomes completely explicit in terms of the exterior powers of the standard representations.

Remarks 2.1. In the representation examples the augmentation is the degree, i.e. the dimension of the representation. In the vector-bundle examples the augmentation is the rank. Chevalley supplied the positive characteristic form of the theorem just used.

2.1.3 The λ -ring defined by a graded ring

Let A be a graded commutative ring with unit, for simplicity concentrated in nonnegative degrees with $A^0 = \mathbb{Z}$. Write A_+ for the positive-degree part and consider

$$\mathbf{A} = \mathbb{Z} \times (1 + A_+),$$

with elements $[n, x]$ and addition

$$[n, x] + [n', x'] = [n + n', xx'].$$

The zero element is $[0, 1]$. We put on $\mathbf{A}$ the structure of an augmented λ -ring with augmentation $[n, x] \mapsto n$.

The product is characterized by a universal formula

$$[m, x][n, y] = [mn, x^n y^m (x * y)], \quad (21)$$

where the components of $x * y \in 1 + A_+$ are integral universal polynomials in the homogeneous components of x and y . The operation $x * y$ is distributive with respect to multiplication in $1 + A_+$ and, on linear factors, is specified by

$$[1, 1 + x][1, 1 + y] = [1, 1 + x + y]. \quad (22)$$

The polynomials are the usual polynomials expressing Chern classes of a tensor product in terms of the Chern classes and ranks of the factors.

The filtration of $\mathbf{A}$ is defined by $\mathbf{A}^0 = \mathbf{A}$ and by letting $\mathbf{A}^n$ be the set of $[0, x]$ with $x^i = 0$ for $1 \leq i < n$. Then

$$[0, x] \in \mathbf{A}^m, [0, y] \in \mathbf{A}^n \implies [0, x][0, y] - [0, 1] \in \mathbf{A}^{m+n}.$$

The associated graded group is additively identified with A , but not directly as a ring. The map

$$\eta : A \rightarrow G(\mathbf{A}), \quad a \in A^i \mapsto \mathrm{cl}([1, 1 + a])$$

is a ring homomorphism.

The operations λ^i on $\mathbf{A}$ are determined by universal integral polynomials in the components of x , together with

$$\lambda^i[1, 1+x] = [0, 1]quad(i > 1).$$

These polynomials are the symmetric-function formulas for the Chern classes of exterior powers of a vector bundle. Thus $\mathbf{A}$ is a special augmented λ -ring and the filtration ideals $\mathbf{A}^i$ are stable under all λ^j .

For $x = [n, 1+x^1+\cdots+x^n]$ with $x^i = 0$ for $i > n$, one has $\lambda^i(x) = 0$ for $i > n$, $\lambda^n(x) = [1, 1+x^n]$, and

$$\lambda_t(x) = \sum_{i=0}^n [1, 1+x^i] t^i \lambda_{-t}(\bar{x}^{n-i}), \quad (23)$$

where the bar denotes the principal involution of the graded ring, changing signs according to degree. Formula (23) remains valid without the finite truncation assumption as soon as the augmentation is n .

For an arbitrary λ -ring K define

$$\gamma^n(x) = (-1)^n \sum_{i=0}^n (-1)^i \binom{n-1+i}{n-i} \lambda^i(x) = \lambda^n(x+n-1). \quad (24)$$

Then $\gamma^0(x) = 1$, $\gamma^1(x) = x$, and

$$\gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n = \lambda_{t/(1-t)}(x).$$

Conversely $\lambda_s(x) = \gamma_{s/(1+s)}(x)$. Therefore the γ^i define another λ -structure. For the ring $\mathbf{A}$ one obtains

$$\gamma^n([0, 1+x] - \varepsilon([0, 1+x])) = [0, 1+(n-1)!x^n + \cdots], \quad (25)$$

so the degree n component can be recovered, up to the factor $(n-1)!$, by reducing modulo $\mathbf{A}^{n+1}$.

The Chern homomorphism. For every graded ring A as above, define an augmented ring homomorphism

$$\text{ch} : \mathbf{A} \rightarrow A \otimes \mathbb{Q}. \quad (26)$$

It is characterized by additivity, by sending 1 to 1, by universal homogeneous rational polynomials in positive degrees, and by

$$\text{ch}([1, 1+x^1]) = \exp(x^1 t) = \sum_{n \geq 0} \frac{(x^1)^n}{n!} t^n. \quad (27)$$

Let η be the additive endomorphism of $A \otimes \mathbb{Q}$ sending a homogeneous element x^i of degree i to $(-1)^{i-1}(i-1)!x^i$. Then

$$\text{ch}\left([n, 1 + \sum_{i=1}^N x^i]\right) = n + \eta^{-1} \log\left(1 + \sum_{i=1}^N \eta(x^i)\right). \quad (28)$$

If A has only finitely many homogeneous components, this gives an isomorphism

$$\mathbf{A} \otimes \mathbb{Q} \simeq A \otimes \mathbb{Q}.$$

A formal power series $f \in \mathbb{Q}[[t]]$ defines by universal formulas an additive homomorphism $f_* : \mathbf{A} \otimes \mathbb{Q} \rightarrow \mathbf{A} \otimes \mathbb{Q}$. For

$$f(t) = \frac{t}{1 - e^{-t}}$$

one writes $f_* = \text{Todd}$ and extends Todd to all $x \in \mathbf{A}$. The following identity is central.

Proposition. *Let*

$$N = [q, 1 + N^1 + \cdots + N^q]$$

with $N^i = 0$ for $i > q$. If

$$\gamma_t(N - q) = \sum_{i \geq 0} \gamma^i(N - q)t^i, \quad \lambda_t(N) = \sum_{i=0}^q \lambda^i(N)t^i,$$

then

$$\text{ch}(\lambda_{-t}(N)) = (-1)^q \text{Todd}(N) \text{ch}(\gamma_t(N - q)). \quad (29)$$

In particular,

$$\text{ch}(\lambda_{-1}(N)) = (-1)^q \text{Todd}(N). \quad (30)$$

2.1.4 The operations $\rho^p(N, x)$

Let $\mathbf{A}$ now denote the ring of formal power series with integral coefficients in two infinite families of variables, written $\lambda^i N$ and $\lambda^j x$ for $i, j \geq 1$, with $\lambda^0 N = N$ and $\lambda^0 x = x$. Equip it with the unique completed special λ -ring structure for which the variables represent the exterior powers of N and x and the operations are continuous. Form the invertible series

$$\lambda_{-t}(N) = \sum_{i \geq 0} (-1)^i \lambda^i(N) t^i.$$

The element $\rho^p(N, x)$ is defined by

$$\rho^p(N, x) \lambda_{-t}(N) = \lambda_t(\rho^p(N, x) \lambda_{-1}(N)). \quad (31)$$

Theorem. *Let $\mathfrak{I}$ be the closed ideal generated by the $\lambda^i N$ with $i > q$. After reducing modulo $\mathfrak{I}$, and identifying the quotient with the ring of formal series in the remaining variables, the element*

$\rho^p(N, x)$ is a polynomial, homogeneous of weight p in the variables attached to x .

The proof reduces first to $x = 1$ and then to the representation ring of $G = \mathrm{GL}(N)$, where N is a vector space of dimension q over an algebraically closed field of characteristic zero. Since $K(G)$ identifies with a subring generated by the exterior powers of N and the inverse of the top exterior power, one obtains a polynomial identity by comparing in the common quotient field. The general case follows by checking the same identity in the representation ring of $\mathrm{GL}(N) \times \mathrm{GL}(F)$ and using that this ring has no zero divisors.

More explicitly, if N and F are finite-dimensional vector spaces, let n_0 be the kernel of

$$(a_1, \dots, a_q) \longmapsto a_1 + \dots + a_q.$$

The symmetric group acts on suitable exterior powers, and taking the alternating part gives a graded module for $\mathrm{GL}(N) \times \mathrm{GL}(F)$. Then

$$\rho^p(N, F) = \sum_i (-1)^i \mathrm{Cl}(\Lambda^i n_0^p \otimes \Lambda^p F). \quad (32)$$

This equality characterizes the universal polynomial $\rho^p(N, x)$ when $\dim F \geq p$. Grothendieck notes that the same assertion should hold over a non-algebraically-closed field of characteristic zero, and that the alternating-part construction is naturally a characteristic-zero operation.

For any special λ -ring K and any element N such that $\lambda^i N = 0$ for $i \gg 0$, the operations $x \mapsto \rho^p(N, x)$ are defined by universal integral polynomials. They satisfy $\rho^1(N, x) = x$ and (31). It is useful to introduce a new special λ -ring K_N , obtained by adjoining a unit to the additive group of K with multiplication determined by the ρ -operations and with

$$\lambda_t(x) = \sum_{p \geq 0} \rho^p(N, x) t^p, \quad \rho^0(N, x) = 1.$$

This construction turns K_N into a special λ -ring.

Guided by the preceding formulas, put

$$\gamma^n(N, x) = \sum_{i=0}^n \binom{n}{i} \rho^i(N, x). \quad (33)$$

Equivalently, it is $\rho^n(N, x_1)$, where x_1 denotes x regarded as an element of K_N .

Proposition. *Let A be a graded commutative ring with unit, let*

$$N = [q, 1 + N^1 + \dots + N^q]$$

and let

$$x = [n, 1 + \sum_{i \geq 1} x^i]$$

be an arbitrary element of $\mathbf{A}$. Then for every $j \geq 0$ one has

$$\gamma^j(N, x) \in \mathbf{A}^j,$$

and its degree j component has the form

$$(\gamma^j(N, x))^j = C_{q,j}(N^1, \dots, N^q; x^1, \dots, x^j) x^j, \quad (34)$$

where $C_{q,j}$ is a universal polynomial with integer coefficients characterized by

$$C_{q,j}(N^1, \dots, N^q; t, t^2, \dots, t^j) = \binom{n+q+j-1}{j} t^j. \quad (35)$$

The proof reduces to a polynomial ring and uses

$$\gamma^p(N, x) \lambda_{-1}(N) = \gamma^p(N \lambda_{-1}(N)),$$

together with the filtration estimates supplied by the preceding formulas.

2.2 Chapter II. Coherent algebraic sheaves and Chern classes

2.2.1 Chow theory

For simplicity assume in this paragraph that the ground field k is algebraically closed. Most, and perhaps all, of what follows remains valid without this restriction. All algebraic spaces and regular maps are taken over k .

An algebraic space is called quasi-projective if it is isomorphic to a locally closed subset of a projective space. Let X be a connected nonsingular quasi-projective variety of dimension n . The Chow ring $A(X)$ is the graded ring of cycles on X modulo rational equivalence, with $A^i(X)$ consisting of classes of cycles of dimension $n - i$. Its ring structure is based on Chow's moving theorem:

Theorem (Chow). *Every cycle on X is rationally equivalent to a nonsingular cycle. Given classes in $A^i(X)$ and $A^j(X)$, one can choose nonsingular representatives Y^i and Z^j which meet transversely everywhere.*

A cycle is called nonsingular when its irreducible components are nonsingular varieties. Two cycles meet transversely when every component of one meets every component of the other transversely, i.e. at intersection points the relevant tangent spaces are in general position.

If $f : X \rightarrow Y$ is a morphism between nonsingular quasi-projective varieties, the preceding theorem defines a graded ring homomorphism

$$f^* : A(Y) \rightarrow A(X). \quad (36)$$

Thus $A(X)$ is contravariant. If f is proper, it also defines an additive direct image

$$f_* : A(X) \rightarrow A(Y), \quad (37)$$

which preserves dimensions of cycles, hence shifts codimension by $\dim Y - \dim X$. The classical projection formula is

$$f_*(x f^* y) = f_*(x) y. \quad (38)$$

Chow theory also attaches Chern classes to vector bundles. For a vector bundle E on X , one has

$$c^i(E) \in A^i(X), \quad c^0(E) = 1,$$

and

$$c_t(E) = \sum_{i \geq 0} c^i(E) t^i.$$

The defining properties are the Whitney product formula for extensions

$$c(E) = c(E') c(E'') \quad (39)$$

whenever E is an extension of E' by E'' ; compatibility with the λ -ring structure,

$$c(E \otimes F) = c(E) \circ c(F), \quad c(\Lambda^i E) = \lambda^i(c(E)); \quad (40)$$

the normalization

$$c^1(L(D)) = \text{Cl}(D) \quad (41)$$

for the line bundle associated with a divisor D ; and functoriality

$$c^i(f^* E) = f^* c^i(E). \quad (42)$$

Equivalently, if $K(Y) \rightarrow K(X)$ is the λ -homomorphism induced by inverse image of vector bundles, then the diagram

$$\begin{array}{ccc} K(Y) & \longrightarrow & K(X) \\ \downarrow c_Y & & \downarrow c_X \\ A(Y) & \longrightarrow & A(X) \end{array}$$

commutes.

Theorem. *There exists a theory of Chern classes satisfying (39), (40), (41), and (42).*

We shall also use the following proposition.

Proposition. *Let Y be a closed subset of X and let $U = X \setminus Y$. If $x \in A(X)$ restricts to zero on U , then x has a representative which is a cycle supported on Y .*

Corollary. *Let $p = \dim X - \dim Y$. If $i < p$, then $A^i(X) \rightarrow A^i(U)$ is bijective. Every element of the kernel of $A^p(X) \rightarrow A^p(U)$ is a linear combination of the components of dimension $n - p$ of Y ; in particular it is proportional to $\text{Cl}(Y)$ when Y is irreducible.*

2.2.2 Definition of Chern classes of coherent algebraic sheaves

Let $\mathcal{C}$ be an abelian category and $\mathcal{C}_1$ a subclass of its objects. The group $K(\mathcal{C}_1)$ is the quotient of the free abelian group generated by isomorphism classes of objects of $\mathcal{C}_1$ by relations

$$(E) - (E') - (E'')$$

whenever E is an extension of E' by E'' and all three objects lie in $\mathcal{C}_1$. We write $\gamma_{\mathcal{C}_1}(E)$ for the class of E , omitting the subscript when no confusion is possible. This group has the expected universal property for additive functions on $\mathcal{C}_1$.

If $F : \mathcal{C} \rightarrow \mathcal{C}'$ is an additive functor with derived functors $R^q F$, and if $R^q F(A)$ belongs to a suitable subclass $\mathcal{C}'_1$ and vanishes for $q \gg 0$ for all $A \in \mathcal{C}_1$, then F defines a homomorphism

$$K(\mathcal{C}_1) \rightarrow K(\mathcal{C}'_1), \quad \gamma_{\mathcal{C}_1}(E) \mapsto \sum_{i \geq 0} (-1)^i \gamma_{\mathcal{C}'_1}(R^i F(E)). \quad (43)$$

The same applies to cohomological functors with only finitely many nonzero values. If F is exact, this reduces to

$$F(\gamma_{\mathcal{C}_1}(E)) = \gamma_{\mathcal{C}'_1}(F(E)).$$

It also gives the canonical homomorphism $K(\mathcal{C}'_1) \rightarrow K(\mathcal{C}_1)$ when $\mathcal{C}'_1 \subset \mathcal{C}_1$.

Theorem. *Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}'_1 \subset \mathcal{C}_1$ be two subclasses whose isomorphism classes form sets. Suppose:*

- (i) *In every exact sequence $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$, if A and A'' lie in $\mathcal{C}_1$ (respectively in $\mathcal{C}'_1$), then A' also lies in $\mathcal{C}_1$ (respectively in $\mathcal{C}'_1$).*
- (ii) *If $u : A \rightarrow B$ and $v : C \rightarrow B$ with $A, B, C \in \mathcal{C}_1$ and u surjective, then there exist $L \in \mathcal{C}'_1$ and maps $u' : L \rightarrow C$, $v' : L \rightarrow A$ such that u' is surjective and $vu' = uv'$.*
- (iii) *For every $A \in \mathcal{C}_1$ there is an integer $d(A)$ such that, for every left resolution L of A by objects of $\mathcal{C}'_1$, the $d(A)$ -cycles $Z_{d(A)}(L)$ lie in $\mathcal{C}'_1$.*

Then the canonical homomorphism $K(\mathcal{C}'_1) \rightarrow K(\mathcal{C}_1)$ is an isomorphism.

Proof. [Principle of proof] For $A \in \mathcal{C}_1$, choose a finite resolution L of A by objects of $\mathcal{C}'_1$ and put

$$f(A) = \sum_i (-1)^i \gamma_{\mathcal{C}'_1}(L^i).$$

The conditions allow one to compare any two such resolutions by a third resolution mapping surjectively to both. The difference of the corresponding Euler characteristics is the Euler characteristic of an acyclic complex in $\mathcal{C}'_1$, hence zero. Thus $f(A)$ is independent of the resolution. One checks similarly that f is additive, hence gives a homomorphism $K(\mathcal{C}_1) \rightarrow K(\mathcal{C}'_1)$ inverse to the canonical map.

For checking condition (ii), it is enough to know that $\mathcal{C}_1$ is contained in a subclass $\mathcal{C}_2$ satisfying: every object of $\mathcal{C}_2$ is a quotient of an object of $\mathcal{C}'_1$; and kernels of morphisms between objects of $\mathcal{C}_2$ again lie in $\mathcal{C}_2$.

Corollary. *Let X be a quasi-projective algebraic space. Let $\mathcal{F}(X)$ be the category of coherent algebraic sheaves on X , let $\xi(X)$ be the category of locally free coherent sheaves, and let $\mathcal{F}_0(X)$ be the coherent sheaves whose stalks have finite cohomological dimension. Then the canonical map*

$$K(\xi(X)) \rightarrow K(\mathcal{F}_0(X)) \quad (44)$$

is an isomorphism.

The conditions of the theorem are verified as follows. Conditions (i) and (iii) come from the definition of cohomological dimension and from the fact that a finite projective module over a noetherian local ring is free. For (ii) it suffices to use the following theorem of Serre.

Proposition (Serre). *Every coherent algebraic sheaf on a quasi-projective algebraic space is a quotient of a locally free coherent sheaf.*

For a closed subspace of projective space this is the usual theorem that $F(n)$ is generated by global sections for $n \gg 0$. The general quasi-projective case is reduced to it by extending coherent sheaves from an open subset to the ambient algebraic space. If F is coherent on an open $U \subset X$, one extends by Zorn's lemma to a maximal open, reduces to X affine, and constructs a quasi-coherent extension whose restriction to U is generated by finitely many sections; these sections generate a coherent subsheaf restricting to F .

If now X is quasi-projective and nonsingular, the syzygy theorem gives $\mathcal{F}_0(X) = \mathcal{F}(X)$. Hence

$$K(\xi(X)) \simeq K(\mathcal{F}(X)). \quad (45)$$

Equivalently, for any abelian group A , additive functions on vector bundles on X are in bijection with additive functions on coherent algebraic sheaves on X . If

$$0 \rightarrow \mathcal{O}_X(E_m) \rightarrow \cdots \rightarrow \mathcal{O}_X(E_0) \rightarrow F \rightarrow 0$$

is a finite locally free resolution of a coherent sheaf F , then the corresponding value is

$$g(F) = \sum_i (-1)^i f(E_i). \quad (46)$$

In particular the total Chern class function on vector bundles extends additively to coherent sheaves. Its components

$$F \mapsto c^i(F) \in A^i(X)$$

are called the Chern classes of the coherent sheaf F . They are computed from a finite locally free resolution by (46), interpreted multiplicatively for total Chern classes. The degree-zero component is the rank of F .

2.2.3 Functorial generalities on $K(X)$

For an algebraic space X write

$$K(X) = K(\mathcal{F}(X)).$$

If X is nonsingular and quasi-projective, we identify this group with $K(\xi(X))$ by (45); hence $K(X)$ inherits a λ -ring structure. It is important, however, to describe the operations directly in terms of coherent sheaves rather than by passing to vector bundles.

For a vector bundle E or a coherent sheaf F on X , write $\gamma(E)$ or $\gamma(F)$ for its class in $K(X)$. If Y is an irreducible subvariety of X , put

$$\gamma(Y) = \gamma(\mathcal{O}_Y),$$

where $\mathcal{O}_Y$ is regarded as a coherent sheaf on X ; extend linearly to cycles. There is a filtration on $K(X)$ by codimension of support: $K(X)^i$ is generated by classes of coherent sheaves F for which $\dim \text{supp } F \leq \dim X - i$. The associated graded group compares with the Chow group by sending a cycle to the class of its structure sheaf.

For nonsingular quasi-projective X and Y and a morphism $f : X \rightarrow Y$, inverse image of vector bundles gives a map

$$f^! : K(Y) \rightarrow K(X),$$

which is compatible with filtrations. On associated graded it matches the usual pullback of Chow classes:

$$\begin{array}{ccc} A(Y) & \xrightarrow{f^*} & A(X) \\ \downarrow & & \downarrow \\ G(K(Y)) & \xrightarrow{G(f^!)} & G(K(X)). \end{array}$$

The proof reduces, via the graph of f , to projections and immersions.

Proposition (Cellular decomposition). *Let P be an algebraic space with an increasing family of closed subsets*

$$P_0 \subset P_1 \subset \cdots \subset P_n = P$$

such that the connected components of $P^i - P^{i-1}$ are affine spaces. Then for every algebraic space X , the natural map

$$K(X) \otimes K(P) \rightarrow K(X \times P)$$

coming from tensor products of sheaves is surjective. Moreover the classes of the strata generate $K(P)$.

The proof is by induction on n , using the exact sequence for an open complement and a closed subset. Grothendieck notes that he does not know in this generality whether the map is always injective.

2.2.4 Riemann–Roch theorem in the report

Let $f : Y \rightarrow X$ be a proper morphism from a nonsingular quasi-projective variety Y to another such variety X , and let $y \in K(Y)$. The Riemann–Roch formula is

$$f_*(\text{ch}(y) \text{Todd}(Y)) = \text{ch}(f_! y) \text{Todd}(X) \quad \text{in } A(X) \otimes \mathbb{Q}. \quad (47)$$

The proof begins with three formal reductions. If $Z \xrightarrow{g} Y \xrightarrow{f} X$ are proper morphisms, then validity for f and g implies validity for fg ; under suitable surjectivity or injectivity hypotheses one can also infer one of the two from the other two. Since a proper morphism is factored as an immersion followed by a projection $X \times \mathbb{P}^r \rightarrow X$, one reduces to the cases of a projection and an immersion. The projection case is a standard computation, using the cellular decomposition of projective space.

For a closed immersion $i : Y \hookrightarrow X$, the formula becomes

$$\text{ch}(i_! y) = i_*(\text{ch}(y) \text{Todd}(E)^{-1}), \quad (48)$$

where E is the normal bundle of Y in X . If $y = i^* x$ for some $x \in K(X)$, the formula follows from

$$\text{ch}(\lambda_{-1}(E^*)) = c^p(E) \text{Todd}(E)^{-1}, \quad (49)$$

where $p = \text{codim}_X Y$. In particular, it holds for hypersurfaces in the appropriate form.

The general immersion case is reduced by blowing up X along Y . Let X' be the blow-up, Y' the inverse image of Y , and use the notation customary for the exceptional divisor and the normal bundle. One has identities of the form

$$\begin{aligned} i'_!(y) &= j_!(y \lambda_{-1}(F^*)), \\ \lambda_{-1}(F^*) &\equiv \xi^{p-1}(E) \pmod{(1 - \mathcal{L})}, \\ f_!(1) &= 1, \quad f_! f^! = 1, \\ g^* i_!(a) &= j_!(a c^p(E)) \pmod{\text{Ker } f_!}. \end{aligned}$$

The first follows from the Tor spectral sequence, the second from projective-space cohomology and Kunneth formulas, and the last from the compatibility of top Chern classes with the exceptional divisor. These formulas imply Riemann–Roch in the case where a certain filtration condition is satisfied; in particular when $\dim Y < \dim X - 1$.

Finally, one reduces to that inequality by replacing the immersion with a related one after product with a projective space and a suitable deformation. The report ends by noting that a remaining equality in the blow-up computation should follow directly from the same formula used above, and that at worst the argument is correct after passing to the associated graded, losing only possible torsion.

3 Expose I – Generalities on Finiteness Conditions in Derived Categories

by *L. Illusie*

0. Introduction

The purpose of Exposes I–IV is to develop, with the degree of generality required in this seminar, the formalism of finiteness conditions in the derived categories of ringed topoi.

As was said in Expose 0, the need to define, on arbitrary schemes, “Grothendieck groups” with good variance properties is what forces one to generalize and loosen the finiteness notions used up to now. A classical notion such as that of a coherent sheaf on a ringed space $(X, \mathcal{O}_X)$ loses much of its usefulness as soon as $\mathcal{O}_X$ itself is no longer coherent. One might try to replace it by finite presentation, but finite presentation has the defect that the kernel of an epimorphism of finitely presented modules need not be finitely presented.

If X is a ringed space whose local rings are noetherian, a coherent $\mathcal{O}_X$ -module F locally admits a finite left resolution by finite free modules; in other words, locally there is an exact sequence

$$0 \longrightarrow L_n \longrightarrow L_{n-1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

where the L_i are finite free. When $\mathcal{O}_X$ is no longer assumed coherent, we shall say that a $\mathcal{O}_X$ -module F is *pseudo-coherent* if it satisfies the analogous condition in all finite stages, i.e. if it is of finite n -presentation for every n . Pseudo-coherence then has, with respect to short exact sequences, the same stability property as coherence: if two terms of a short exact sequence of $\mathcal{O}_X$ -modules are pseudo-coherent, then so is the third.

This notion is extended without difficulty to complexes of sheaves. A complex F of $\mathcal{O}_X$ -modules is called *pseudo-coherent* if, locally, it can be approximated in all degrees by complexes whose terms are finite free in the relevant range. The precise formulation is given below in terms of n -pseudo-coherence.

Pseudo-coherent complexes have excellent stability properties, described in Sections 1 and 2 of this expose. First, pseudo-coherence is a local property. Secondly, if

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

is a morphism of ringed spaces, the derived inverse image Lf^* of a pseudo-coherent complex on Y is pseudo-coherent on X , provided that the complex has finite Tor-dimension relative to f , i.e. relative to the sheaf of rings $f^{-1}(\mathcal{O}_Y)$.

A coherent complex over a noetherian local ringed space also admits locally a finite left resolution by finite free modules. It is therefore natural, for an arbitrary sheaf of local rings $\mathcal{O}_X$, to single out the modules having this property. Such modules are called *perfect*. More generally, a complex

F of $\mathcal{O}_X$ -modules is called *perfect* if, locally, there is a quasi-isomorphism

$$L \longrightarrow F,$$

where L is a bounded complex whose components are finite free modules. Section 3 proves that the full subcategory $D(X)_{\text{parf}}$ of $D(X)$ formed by perfect complexes is, like the pseudo-coherent part, a triangulated subcategory, stable under inverse image and under the derived tensor product.

Global resolutions of perfect complexes are treated in Expose II. In particular, global resolutions exist in the following cases: X is the Zariski topos of an affine scheme; more generally, X is the Zariski topos of a quasi-compact scheme carrying an ample invertible sheaf, or even an ample family of invertible sheaves in the sense of Kleiman; and also when X is the topos of sheaves of sets on a compact topological space and $\mathcal{O}_X$ is the sheaf of continuous complex-valued functions. As an illustration, and to show the flexibility of the notions introduced, the appendix gives a purely sheaf-theoretic definition of the index of a family of elliptic operators.

Expose III studies the stability of finiteness conditions under derived direct image. To obtain useful results one is led to introduce the notion of a complex *pseudo-coherent relative* to a morphism $f : X \rightarrow Y$: this means pseudo-coherent relative to f and of finite Tor-dimension relative to f , again relative to the sheaf of rings $f^{-1}(\mathcal{O}_Y)$.

The central result of Expose III is the *finiteness theorem*. In a slightly more precise form, it says that if $f : X \rightarrow Y$ is a proper morphism of S -schemes locally of finite type, then the functor Rf_* carries complexes pseudo-coherent relative to S to complexes pseudo-coherent relative to S . In this generality the theorem remains a conjecture in the seminar, but it is proved in the two following important cases:

- a) S is locally noetherian;
- b) f is projective.

The extension to the general case seems to be of the same order of difficulty as the corresponding theorem in analytic geometry, namely Grauert's finiteness theorem.

Combining the finiteness theorem with an essentially formal identity called the *projection formula*, one obtains usable criteria for the stability of relative perfection under direct image. As a corollary one recovers Grauert's continuity and semicontinuity theorems, in the form of EGA III, 7.6.

Expose IV translates the results of Exposés I–III into the language of Grothendieck groups. For a ringed topos $(X, \mathcal{O}_X)$ one denotes by $K^*(X)$, respectively by $K_*(X)$, the Grothendieck group of the category of perfect complexes of finite Tor-dimension, respectively of pseudo-coherent complexes with bounded cohomology. These groups carry the functorialities needed in intersection theory.

1. Preliminary definitions

1.1. Fibred categories with additive, abelian, or triangulated fibres

Definition 1.1.1. Let S be a category, and let $\mathbf{C}$ be an S -fibred category, in the sense of SGA 1, VI, 6.1, whose fibres are additive categories, respectively abelian categories, respectively triangulated categories. We shall say that $\mathbf{C}$ is an additive, respectively abelian, respectively triangulated, S -category if, for every arrow $f : X \rightarrow Y$ of S , the inverse-image functor

$$f^* : \mathbf{C}_Y \longrightarrow \mathbf{C}_X$$

which is determined up to unique isomorphism, is additive, respectively exact, respectively triangulated.

Definition 1.1.2. Let $\mathbf{C}$ be a triangulated S -category, and let $\mathbf{C}'$ be a full sub- S -category of $\mathbf{C}$. We say that $\mathbf{C}'$ is a triangulated sub- S -category of $\mathbf{C}$ if, for every object X of S , the fibre $\mathbf{C}'_X$ is a triangulated subcategory of $\mathbf{C}_X$.

Definition 1.1.3. Let $\mathbf{C}$ be an additive S -category. One defines an additive, respectively triangulated, S -category

$$\mathbf{C}(\mathbf{C}), \quad \mathbf{K}(\mathbf{C}),$$

whose fibre at $X \in S$ is the category of complexes in $\mathbf{C}_X$, respectively the homotopy category of complexes in $\mathbf{C}_X$. If $\mathbf{C}$ is a flat abelian S -category, one also defines a triangulated S -category $\mathbf{D}(\mathbf{C})$ whose fibre at X is the derived category $\mathbf{D}(\mathbf{C}_X)$. For an arrow $f : X \rightarrow Y$ of S , the inverse-image functor on derived categories

$$f^* : \mathbf{D}(\mathbf{C}_Y) \longrightarrow \mathbf{D}(\mathbf{C}_X)$$

is obtained from the functor on homotopy categories by passing to the quotient.

1.4. Quasi-isomorphisms and finiteness conditions

Lemma (1.4.1). *Let S be a scheme, let $n \in \mathbb{Z}$, and let $u : E \rightarrow F$ be a morphism of perfect complexes on S . Suppose that $H^i(u) = 0$ for $i > n$ and that $H^n(u)$ is of finite presentation. Then there exists a commutative diagram of perfect complexes*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & & \downarrow u' \\ F & \xrightarrow{\cong} & F, \end{array}$$

such that $H^i(u') = 0$ for $i \geq n$ and v is an isomorphism in degrees $< n$.

Lemma (1.4.2). *Let $\mathbf{C}$ be an abelian category, let $u : E \rightarrow F$ be a morphism of complexes of $\mathbf{C}$, or a morphism in $\mathbf{D}(\mathbf{C})$, and let G be the cone of u . For a fixed integer n , the following conditions are equivalent:*

- (i) $H^i(G) = 0$ for $i \geq n$;
- (ii) $H^i(u)$ is an isomorphism for $i > n$ and an epimorphism for $i = n$.

The proof is immediate from the long exact cohomology sequence.

Definition 1.4.3. A morphism u satisfying the equivalent conditions of Lemma 1.4.2 is called an n -quasi-isomorphism, respectively an n -isomorphism, according as it is considered at the level of complexes, respectively in the derived category.

From Lemma 1.4.1 one obtains the following local improvement criterion.

Corollary (1.4.4). Let S , $\mathbf{C}$, and $\mathbf{C}_0$ be as in the preceding discussion. Let $u : E \rightarrow F$ be an $(n+1)$ -quasi-isomorphism of complexes in $\mathbf{C}_S$, with $n \in \mathbb{Z}$, and let G be its cone. Suppose that $H^n(G)$ is of finite $\mathbf{C}_0$ -type. Then the collection of objects X over S on which one can modify u as in 1.4.1 so as to obtain an n -quasi-isomorphism is a refinement of S .

In less formal terms, the finite-type condition on $H^n(G)$ permits one, locally, to gain one degree: one changes u in degree n by adding to E^n an object of $\mathbf{C}_0$, so that the resulting morphism becomes an n -quasi-isomorphism.

Corollary (1.4.5). Let $\mathbf{C}$ be an abelian category, let $n \in \mathbb{Z}$, and let $u : E \rightarrow F$ be an n -quasi-isomorphism of complexes of $\mathbf{C}$. There is a commutative diagram in the category of complexes

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & & \downarrow u' \\ F & \xrightarrow{\cong} & F, \end{array}$$

such that u' is an n -isomorphism and v is an $(n+1)$ -quasi-isomorphism.

1.5. Finiteness conditions in derived categories

Definition 1.5.0. We shall say that $\mathbf{C}_0$ is *locally stable under kernels of epimorphisms* if the following condition holds. For every object X of S and every epimorphism $u : E \rightarrow F$ in $\mathbf{C}_X$ with E and F objects of $\mathbf{C}_{0,X}$, the kernel of u is locally in $\mathbf{C}_{0,X}$; that is, the family of arrows $f : U \rightarrow X$ such that

$$f^*(\text{Ker } u) \in \mathbf{C}_{0,U}$$

is a refinement of X .

By decomposing a long exact sequence into short exact sequences, this condition is equivalent to the apparently stronger condition that, for every exact sequence

$$0 \longrightarrow E^0 \longrightarrow E^1 \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0$$

in $\mathbf{C}_X$, if E^i lies in $\mathbf{C}_{0,X}$ for $1 \leq i \leq n$, then E^0 is locally in $\mathbf{C}_{0,X}$.

We shall say that $\mathbf{C}_0$ is *locally quasi-stable under kernels of epimorphisms* if, for every exact sequence of the displayed form, E^0 is of finite $\mathbf{C}_0$ -type.

Definition 1.5.0 bis. We shall say that $\mathbf{C}_0$ is *locally quasi-liftable* in $\mathbf{C}$ if the following weaker condition holds. For every object X of S and every epimorphism $u : E \rightarrow F$ in $\mathbf{C}_X$, with F in $\mathbf{C}_{0,X}$ and E in $\mathbf{C}_X$, there exists locally an epimorphism $v : E' \rightarrow E$ with E' in $\mathbf{C}_{0,X}$ such that uv is an epimorphism in $\mathbf{C}_{0,X}$.

Theorem (1.5.1). *Assume that $\mathbf{C}_0$ is locally stable under kernels of epimorphisms, respectively locally quasi-stable under kernels of epimorphisms, respectively locally quasi-liftable in $\mathbf{C}$. Then the full subcategory $\mathbf{D}_{\mathbf{C}_0}(\mathbf{C})$ of $\mathbf{D}(\mathbf{C})$ formed by the complexes whose cohomology is locally in $\mathbf{C}_0$, respectively of finite $\mathbf{C}_0$ -type, respectively locally quasi-liftable, is a triangulated subcategory of $\mathbf{D}(\mathbf{C})$.*

SGA 6, Exposé I, Sections 2–8

Corollary 1.4.5, proof continued

In the situation of Corollary 1.4.5, u' is a quasi-isomorphism and v is a monomorphism inducing the identity in degrees $\geq n$, that is, such that

$$E'^i = E^i, \quad v^i = \text{Id}(E^i) \quad (i \geq n).$$

Proof. Applying (1.4.4) in the case where $\underline{S}$ is the punctual site and $\underline{C}_0 = \underline{C}$, one obtains a commutative triangle in $C(\underline{C})$

$$\begin{array}{ccc} E & \xrightarrow{v_{n-1}} & E'_{n-1} \\ u \downarrow & & \\ F & \xleftarrow{u'_{n-1}} & \end{array}$$

where u'_{n-1} is an $(n-1)$ -quasi-isomorphism and v_{n-1} is a monomorphism inducing the identity in degrees $\geq n$. Iterating this procedure, one obtains, for $i \leq n-1$, a sequence of commutative triangles in $C(\underline{C})$

$$\begin{array}{ccc} E'_i & \xrightarrow{v_{i-1}} & E'_{i-1} \\ u'_i \downarrow & & \\ F & \xleftarrow{u'_{i-1}} & \end{array}$$

where u'_i is an i -quasi-isomorphism and v_{i-1} is a monomorphism inducing the identity in degrees $\geq i$. The corollary follows by passing to the limit. More precisely, writing d'_i for the differential of E'_i , define E' by

$$E'^i = E^i, d'^i = d^i \quad \text{for } i \geq n,$$

and

$$E'^i = (E'_i)^i, \quad d'^i = (d'_i)^i \quad \text{for } i < n.$$

Then define v and u' by

$$v^j = (v_i)^j, \quad u'^j = (u'_i)^j \quad \text{for } i \leq j.$$

It is then immediate to check that E' , v , and u' so defined have the required properties.

2. Pseudo-coherent complexes

2.0

In this section $\underline{S}$ denotes a site, S an object of $\underline{S}$, $\underline{C}$ a flat abelian $\underline{S}$ -category in the sense of (1.1), and $\underline{C}_0$ a strictly full additive sub- $\underline{S}$ -category of $\underline{C}$. We assume that $\underline{C}_0$ is locally quasi-stable under kernels of epimorphisms (1.2), and locally quasi-liftable in $\underline{C}$ (1.2).

Definition (2.1). Let $S \in \text{Ob } \underline{S}$, let E be a complex of $\underline{C}_S$, and let $n \in \mathbb{Z}$. We shall say that E is

strictly $\underline{C}_0$ - n -pseudo-coherent (respectively $\underline{C}_0$ -pseudo-coherent, respectively $\underline{C}_0$ -perfect) if $E^i = 0$ for i sufficiently large and $E^i \in \text{Ob } \underline{C}_{0S}$ for $i \geq n$ (respectively for every i , respectively for every i and $E^i = 0$ for almost all i).

When no confusion is possible, we shall sometimes say simply “strictly n -pseudo-coherent” instead of “strictly $\underline{C}_0$ - n -pseudo-coherent”, and similarly for the other variants.

Lemma (2.1.1). *Let $E \in \text{Ob } C(\underline{C}_S)$ and let $n \in \mathbb{Z}$. If E is strictly n -pseudo-coherent and if $H^i(E) = 0$ for $i > n$, then $H^n(E)$ is of finite type.*

Proof. The sequence

$$0 \longrightarrow B^n(E) \longrightarrow Z^n(E) \longrightarrow H^n(E) \longrightarrow 0$$

is exact, and $E^i \in \text{Ob } \underline{C}_{0S}$ for $i \geq n$. Since $\underline{C}_{0S}$ is locally quasi-stable under kernels of epimorphisms, $Z^n(E)$ is of finite type; so is its quotient $H^n(E)$.

Proposition (2.2). *Let $F \in \text{Ob } D(\underline{C}_S)$ and let $n \in \mathbb{Z}$.*

a) *If F is strictly n -pseudo-coherent, relatively to $\underline{C}_0$, and if $u : E \rightarrow F$ is a quasi-isomorphism, then the family of objects X over S for which there exists a quasi-isomorphism*

$$v : F' \longrightarrow E_X$$

with F' strictly n -pseudo-coherent is a refinement of S .

b) *The following conditions are equivalent:*

(i) *the family of X over S for which there exists a quasi-isomorphism $u : E \rightarrow F_X$ in $C(\underline{C}_S)$, with E strictly n -pseudo-coherent, is a refinement of S ;*

(ii) *the same condition as (i), but with u an isomorphism in $D(\underline{C}_S)$;*

(iii) *the family of X over S for which there exists an n -quasi-isomorphism (1.4.3) $u : E \rightarrow F_X$ in $C(\underline{C}_S)$, with E strictly perfect, is a refinement;*

(iv) *the same condition as (iii), but with u an n -isomorphism in $D(\underline{C}_S)$.*

Proof. For a), choose, for an integer i large enough, an i -quasi-isomorphism $v_i : E_i \rightarrow E$ with E_i strictly n -pseudo-coherent; one may initially take $E_i = 0$. Suppose $i > n$. We show that, after refining S , there is an $(i-1)$ -quasi-isomorphism $v_{i-1} : E'_{i-1} \rightarrow E$, with E'_{i-1} strictly n -pseudo-coherent.

By (1.4.4), it is enough to show that $H^i(M^\bullet)$ is of $\underline{C}_0$ -finite type, where $M^\bullet$ is the cone of v_i . If $M'^\bullet$ denotes the cone of $u \circ v_i$, the fact that u is a quasi-isomorphism and (1.4.2) imply

$$H^j(M^\bullet) \xrightarrow{\sim} H^j(M'^\bullet) \quad \text{for all } j,$$

and $H^j(M'^\bullet) = 0$ for $j > i$. Moreover $M'^\bullet$ is strictly $(i-1)$ -pseudo-coherent. Hence, by (2.1.1), $H^i(M^\bullet) \simeq H^i(M'^\bullet)$ is of finite type. This proves the assertion. Iterating, and after refining S , one

obtains an n -quasi-isomorphism $v_0 : E_0 \rightarrow E$, where E_0 is strictly n -pseudo-coherent. By (1.4.5) one can then extend v_0 to a quasi-isomorphism $v : F' \rightarrow E$ without changing anything in degrees $> n$; in particular F' is strictly n -pseudo-coherent. This proves a).

For b), the equivalence of (i) and (ii) follows immediately from a) and from the definition of isomorphisms in $D(\underline{C}_S)$. If $E \in \text{Ob } D(\underline{C}_S)$ is strictly n -pseudo-coherent, there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

where E' is strictly perfect and $E''^i = 0$ for $i > n$; hence $E' \rightarrow E$ is an n -quasi-isomorphism, which proves (i) $\Rightarrow$ (iii). The implication (iii) $\Rightarrow$ (iv) is immediate.

It remains to prove (iv) $\Rightarrow$ (i). Let $u : E \rightarrow F$ be an n -isomorphism in $D(\underline{C}_S)$, with E strictly perfect. It is represented by a diagram in $C(\underline{C}_S)$

$$\begin{array}{ccc} E & & F \\ & \searrow & \uparrow \wr \\ & & E' \end{array}$$

where s is a quasi-isomorphism and u' is an n -quasi-isomorphism. By a), after refining S there exists a quasi-isomorphism $t : E'' \rightarrow E'$ with E'' strictly n -pseudo-coherent. Applying (1.4.5) to $u' \circ t : E'' \rightarrow F$, one extends it to a quasi-isomorphism $E''' \rightarrow F$ with E''' strictly n -pseudo-coherent. This proves (iv) $\Rightarrow$ (i), and completes the proof of (2.2).

Definition (2.3). *Under the hypotheses of (2.2), an object $F \in \text{Ob } D(\underline{C}_S)$ is called n - $\underline{C}_0$ -pseudo-coherent if it satisfies the equivalent conditions b) of (2.2). It is called $\underline{C}_0$ -pseudo-coherent if it is n - $\underline{C}_0$ -pseudo-coherent for every $n \in \mathbb{Z}$.*

When there is no risk of confusion, we shall simply say n -pseudo-coherent, pseudo-coherent, instead of n - $\underline{C}_0$ -pseudo-coherent, $\underline{C}_0$ -pseudo-coherent.

Remark. [Remark 2.3.1] Let $F \in \text{Ob } D(\underline{C}_S)$ be a pseudo-coherent complex. The family of X over S for which there exists a quasi-isomorphism $E \rightarrow F_X$ with E strictly pseudo-coherent (2.1) is not, in general, a refinement of S . Later, however, we shall see important cases in which this is true, namely in (2.7.a) and in Expose II.

The proof of (2.2 a)) gives the following result.

Proposition (2.4). *Let $n, i \in \mathbb{Z}$, and let $u : E \rightarrow F$ be an i -quasi-isomorphism in $C(\underline{C}_S)$, with F n -pseudo-coherent and E strictly n -pseudo-coherent. Then, after refining S , there exists a commutative triangle in $C(\underline{C}_S)$*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ & \searrow & \swarrow \wr \\ & & F \end{array}$$

where E' is strictly n -pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degrees $> i$.

Proposition (2.5). a) If $F \in \text{Ob } D(\underline{C}_S)$ is n -pseudo-coherent, respectively pseudo-coherent, then $F[m]$ is $(n - m)$ -pseudo-coherent, respectively pseudo-coherent.

b) Let

$$\begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ E & \longrightarrow & F \end{array}$$

be a distinguished triangle in $D(\underline{C}_S)$. If E and G are n -pseudo-coherent, respectively pseudo-coherent, then F is n -pseudo-coherent, respectively pseudo-coherent.

Proof. The assertion a) is immediate from the definitions. For b), it is enough to prove the non-parenthetical assertion. By rotation of the triangle, it is enough to show that if E is $(n + 1)$ -pseudo-coherent and F is n -pseudo-coherent, then G is n -pseudo-coherent. We may localize and suppose that E , respectively F , is strictly $(n + 1)$ -pseudo-coherent, respectively strictly n -pseudo-coherent. The morphism $E \rightarrow F$ in the derived category is represented by a pair of morphisms in $C(\underline{C}_S)$,

$$\begin{array}{ccc} & & E' \\ & \swarrow & \downarrow \\ E & & F, \end{array}$$

where one arrow is a quasi-isomorphism. By (2.2 a)), after localizing further, there is a quasi-isomorphism $t : E'' \rightarrow E'$ with E'' strictly $(n + 1)$ -pseudo-coherent. We are therefore reduced to the case where the morphism is represented by an actual morphism in $C(\underline{C}_S)$. The assertion then follows from the definitions and from the formula for the cone,

$$C(u)^\bullet = E[1] \oplus F^\bullet.$$

Scholium 2.6. By rotating triangles, pseudo-coherence properties imposed on two of the three objects in a triangle imply the corresponding property for the third. The following table summarizes the cases; the entries are pseudo-coherence degrees, and each row expresses an implication in a distinguished triangle $E \rightarrow F \rightarrow G \rightarrow E[1]$:

E	F	G
$n + 1$	n	n
n	n	$n - 1$.

We write $D(\underline{C})_{n\text{-coh}}$, respectively $D(\underline{C})_{\text{coh}}$, for the strictly full fibred subcategory of $D(\underline{C})$ whose fibre over $X \in \text{Ob } \underline{S}$ consists of the objects of $D(\underline{C}_X)$ that are n -pseudo-coherent, respectively pseudo-coherent, relative to $\underline{C}_0$. By (2.5), $D(\underline{C})_{\text{coh}}$ is a triangulated sub- $\underline{S}$ -category of $D(\underline{C})$.

Proposition (2.7). Suppose that $\underline{S}$ is the punctual site, so that $\underline{C} = C$ and $\underline{C}_0 = C_0$.

a) If $F \in \text{Ob } D(C)$ is pseudo-coherent, there exists a quasi-isomorphism $E \rightarrow F$ with E strictly pseudo-coherent.

b) Let $\mathcal{A}^{\text{ac}}(C_0)$ be the full triangulated subcategory of $K(C_0)$ formed by the acyclic complexes. The canonical functor

$$K(C_0)/\mathcal{A}^{\text{ac}}(C_0) \longrightarrow D(C)_{\text{coh}}$$

is fully faithful and has essential image $D(C)_{\text{coh}}$.

Proof. For a), let $F \in \text{Ob } D(C)_{\text{coh}}$. Suppose that an i -quasi-isomorphism $u_i : E_i \rightarrow F$ has been constructed in $C(C)$, with E_i strictly pseudo-coherent; this is possible for i large enough with $E_i = 0$. Let G_i be the cone of u_i . By (1.4.2) and (2.1.1), $H^i(G_i) \in \text{Ob } C_0$ is of finite C_0 -type. Hence, by (1.4.4), there exists a commutative triangle in $C(C)$

$$\begin{array}{ccc} E_i & \xrightarrow{v_i} & E'_i \\ u_i \downarrow & & \swarrow u'_i \\ F & & \end{array}$$

where E'_i is strictly pseudo-coherent, u'_i is an $(i-1)$ -quasi-isomorphism, and v_i is a monomorphism inducing the identity in degrees $> i$. Passing to the limit, as in the proof of (1.4.5), gives a quasi-isomorphism $u : E \rightarrow F$ with E strictly pseudo-coherent.

For b), one applies Verdier's criterion, chapter I, §2, 4.2 b)(i), together with a).

The proof of a) gives the following more precise result, generalizing (1.4.5).

Proposition (2.7.1). *Under the hypotheses of (2.7), let $u : E \rightarrow F$ be an n -quasi-isomorphism in $C(C)$, with F pseudo-coherent and E strictly pseudo-coherent. There exists a commutative triangle in $C(C)$*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ & \searrow & \swarrow u' \\ & & F \end{array}$$

where E' is strictly pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degrees $> n$.

Definition (2.8). Let $n \in \mathbb{N}$ and $F \in \text{Ob } \underline{C}_S$. We say that F is of n - $\underline{C}_0$ -finite presentation if the family of objects X over S on which there exists an exact sequence

$$E^{-n} \longrightarrow E^{-n+1} \longrightarrow \dots \longrightarrow E^0 \longrightarrow F_X \longrightarrow 0,$$

with $E^i \in \text{Ob } \underline{C}_{0X}$ for $-n \leq i \leq 0$, is a refinement of S . We say that F is of ∞ - $\underline{C}_0$ -finite presentation if F is of n - $\underline{C}_0$ -finite presentation for every $n \in \mathbb{N}$.

When no ambiguity is possible, we shall simply say n -finite presentation. We shall say “finite presentation” for “1-finite presentation”, and “finite type” for “0-finite presentation”; the latter notion coincides with the one introduced in (1.2).

Proposition (2.9). Let $n \in \mathbb{N}$ and $F \in \text{Ob } \underline{C}_S$. The following conditions are equivalent:

- (i) F is of n -finite presentation, respectively of ∞ -finite presentation;

(ii) F is $(-n)$ -pseudo-coherent, respectively pseudo-coherent.

When $\underline{S}$ is the punctual site, F is pseudo-coherent if and only if there exists an exact sequence

$$\cdots \longrightarrow E^{-n} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F \longrightarrow 0$$

with $E^i \in \text{Ob } C_0$ for every i .

Proof. For the equivalence it is enough to treat the non-parenthetical case. The implication (i) $\Rightarrow$ (ii) is immediate from the definitions. Conversely, (ii) $\Rightarrow$ (i) follows at once from (2.4): the morphism $0 \rightarrow F$ is a quasi-isomorphism, and one obtains, after localizing, a quasi-isomorphism $E \rightarrow F$ with E strictly n -pseudo-coherent and $E^i = 0$ for $i \leq -1$, that is, an n -finite presentation of F . The second assertion is proved in the same way, using (2.7.1).

Proposition (2.10). a) *Let $m, n \in \mathbb{Z}$ with $n \leq m$. If $F \in \text{Ob } D^-(\underline{C}_S)$ is n -pseudo-coherent and acyclic in degrees $\geq m$, then locally there is a quasi-isomorphism $E \rightarrow F$ with E strictly n -pseudo-coherent and $E^i = 0$ for $i > m$. If $\underline{S}$ is punctual, and F is pseudo-coherent and acyclic in degrees $\geq m$, then there is a quasi-isomorphism $E \rightarrow F$ with E strictly pseudo-coherent and $E^i = 0$ for $i > m$.*

b) *Let $F \in \text{Ob } D^-(\underline{C}_S)$ be acyclic in degrees $> n$. Then F is n -pseudo-coherent if and only if $H^n(F)$ is of finite type.*

Proof. For a), since F is acyclic in degrees $> m$, the morphism $0 \rightarrow F$ is an m -quasi-isomorphism. The assertion follows from (2.4) and (2.7.1).

For b), there is a distinguished triangle in $D(\underline{C}_S)$

$$\begin{array}{ccc} & & F' \\ & \nearrow^{+1} & \downarrow \\ F & \longrightarrow & H^n(F)[-n]. \end{array}$$

To see this, first truncate F by replacing F^n by $Z^n(F)$ and by putting $F^i = 0$ for $i > n$. The long exact cohomology sequence gives $H^i(F') = 0$ for $i \geq n - 1$, hence F' is $(n - 1)$ -pseudo-coherent. The conclusion follows from (2.5) and (2.6).

Proposition (2.11). *Let $n \in \mathbb{Z}$.*

a) *Let $F \in \text{Ob } D(\underline{C}_S)$. If, for every $i \geq n$, respectively for every i , the component F^i is of $(i - n)$ -finite presentation, respectively of ∞ -finite presentation, then F is n -pseudo-coherent, respectively pseudo-coherent.*

b) *Let $F \in \text{Ob } D^-(\underline{C}_S)$. If, for every $i \geq n$, respectively for every i , the cohomology object $H^i(F)$ is of $(i - n)$ -finite presentation, respectively of ∞ -finite presentation, then F is pseudo-coherent.*

Proof. It is enough to treat the non-parenthetical assertions. For a), reason by induction on an integer N such that $F^i = 0$ for $i > n + N$. If $N = 0$, the assertion follows immediately from (2.10

b). If $N \geq 1$, one has an exact sequence

$$0 \longrightarrow F''[-n-N] \longrightarrow F \longrightarrow F' \longrightarrow 0.$$

By induction, F' is n -pseudo-coherent; on the other hand $F''[-n-N]$ is n -pseudo-coherent by the hypothesis, (2.9), and (2.5 a). Hence F is n -pseudo-coherent by (2.5 b).

For b), reason by induction on N such that $H^i(F) = 0$ for $i > n + N$. The case $N = 0$ is (2.10 b). If $N \geq 1$, there is a distinguished triangle

$$\begin{array}{ccc} & & F' \\ & \nearrow & \downarrow \\ F & \longrightarrow & H^{n+N}(F)[-n-N]. \end{array}$$

The long exact sequence gives

$$H^i(F') = 0 \quad \text{for } i \geq n + 1,$$

and $H^i(F') \xrightarrow{\sim} H^i(F)$ for $i \leq n$. By the induction hypothesis, applied with $n + 1$ in place of n , F' is $(n + 1)$ -pseudo-coherent. Also $H^{n+N}(F)[-n-N]$ is n -pseudo-coherent by (2.9) and (2.5 a). The conclusion follows from (2.6).

Remark. [Remark 2.11.1] If F is pseudo-coherent, the components F^i , and even the cohomology objects $H^i(F)$, need not in general be pseudo-coherent, nor even of finite type. In the next section, however, in the sorites on coherence, we shall see important examples where pseudo-coherence of F implies pseudo-coherence of the $H^i(F)$.

Proposition (2.12). *Let $F', F'' \in \text{Ob } D(\underline{C}_S)$ and put $F = F' \oplus F''$. Then F is n -pseudo-coherent, respectively pseudo-coherent, if and only if F' and F'' are n -pseudo-coherent, respectively pseudo-coherent.*

Proof. It is enough to prove the non-parenthetical assertion. Sufficiency is obvious. For necessity, suppose by induction that F' and F'' have already been shown to be $(n + 1)$ -pseudo-coherent. After localizing, suppose even that F' and F'' are strictly $(n + 1)$ -pseudo-coherent and that $F = F' \oplus F''$ in $C(\underline{C}_S)$. Let F'_+ , respectively F'_- , be the complex obtained from F' by replacing F'^i by 0 for $i \geq n$, respectively for $i \leq n$; define F''_+ and F''_- similarly. There is an exact sequence

$$0 \longrightarrow F'_+ \oplus F''_+ \longrightarrow F \longrightarrow F'_- \oplus F''_- \longrightarrow 0.$$

Since $F'_+ \oplus F''_+$ is strictly perfect and F is n -pseudo-coherent, (2.6) implies that $F'_- \oplus F''_-$ is n -pseudo-coherent. By (2.10 b), this means that $H^n(F'_- \oplus F''_-)$ is of finite type. But

$$H^n(F'_- \oplus F''_-) = H^n(F'_-) \oplus H^n(F''_-),$$

so $H^n(F'_-)$ and $H^n(F''_-)$ are of finite type. Again by (2.10 b), F'_- and F''_- are n -pseudo-coherent. Applying (2.5) to

$$0 \longrightarrow F'_+ \longrightarrow F' \longrightarrow F'_- \longrightarrow 0$$

and similarly for F'' , one concludes that F' and F'' are n -pseudo-coherent.

Proposition (2.13). *Let $(\underline{S}', \underline{C}', \underline{C}'_0)$ be a triple satisfying the same hypotheses (2.0) as $(\underline{S}, \underline{C}, \underline{C}_0)$. Let $f : \underline{S}' \rightarrow \underline{S}$ be a morphism of sites, corresponding to an underlying functor $\psi : \underline{S} \rightarrow \underline{S}'$, and let*

$$u : D^*(\underline{C}) \longrightarrow D^*(\underline{C}') \quad (* = -, +, b)$$

be a Cartesian S -functor above f . Assume that u is exact on each fibre and that there exist integers a, b satisfying the following conditions for every $S \in \text{Ob } \underline{S}$:

(i) if $E \in \text{Ob } \underline{C}_0 S$, then $u(E)$ is a - $\underline{C}'_0$ -pseudo-coherent, respectively $\underline{C}'_0$ -pseudo-coherent;

(ii) if $F \in \text{Ob } D^(\underline{C}_S)$ is acyclic in degrees > 0 , then $u(F)$ is acyclic in degrees $\leq b$.*

Then, if $F \in \text{Ob } D^(\underline{C}_S)$ is m -pseudo-coherent relative to $\underline{C}_0$ and acyclic in degrees $> n$, the object $u(F) \in \text{Ob } D^*(\underline{C}'_{S'})$, where $S' = f^{-1}(S)$, is*

$$\sup(m + a, n + b)\text{-pseudo-coherent}$$

relative to $\underline{C}'_0$, respectively $(m + b)$ -pseudo-coherent, and is acyclic in degrees $> n + b$. In particular, in the parenthetical case, if F is pseudo-coherent relative to $\underline{C}_0$, then $u(F)$ is pseudo-coherent relative to $\underline{C}'_0$.

Proof. It is enough to treat the non-parenthetical case. First suppose that F is strictly perfect and that $F^i = 0$ for $i > n$. We prove that $u(F)$ is $(n + a)$ - $\underline{C}'_0$ -pseudo-coherent by induction on the number N of indices i for which $F^i \neq 0$. If $N = 1$, then $F = F^p[-p]$ for some $p \leq n$, and $u(F)$ is $(p + a)$ - $\underline{C}'_0$ -pseudo-coherent, hence a fortiori $(n + a)$ -pseudo-coherent. If $N > 1$, write F in a short exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

where F' and F'' are strictly perfect of length $< N$ and vanish in degrees $> n$. Applying (2.5 b) to the corresponding triangle gives the claim by induction.

Now let F be m - $\underline{C}_0$ -pseudo-coherent and acyclic in degrees $> n$. After localizing and using (2.10 a), we may suppose that F is strictly m -pseudo-coherent and that $F^i = 0$ for $i > n$. There is an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

with F'' zero in degrees $> m$, and F' strictly perfect and zero in degrees $> n$. In the corresponding distinguished triangle for $u(F)$, the object $u(F'')$ is acyclic in degrees $> m + b$, hence $(m + b)$ -pseudo-coherent, while $u(F')$ is $(n + a)$ -pseudo-coherent by the preceding paragraph. The conclusion follows again from (2.5 b).

Corollary (2.14). *Let $\underline{C}_1$ be a strictly full additive sub- $\underline{S}$ -category of $\underline{C}$ such that $\underline{C}_0 \subset \underline{C}_1 \subset \underline{C}$. Assume that every object of $\underline{C}_1$ is $\underline{C}_0$ -pseudo-coherent. Then:*

(i) $\underline{C}_1$ is locally quasi-liftable in $\underline{C}$ and locally quasi-stable under kernels of epimorphisms;

(ii) for $S \in \text{Ob } \underline{S}$ and $F \in \text{Ob } D(\underline{C}_S)$, F is n -pseudo-coherent, respectively pseudo-coherent, relative to $\underline{C}_1$ if and only if it is so relative to $\underline{C}_0$.

Proof. For (i), let $E \rightarrow F$ be an epimorphism in $\underline{C}_S$ with $F \in \text{Ob } \underline{C}_{1S}$. Since F is $\underline{C}_0$ -pseudo-coherent, hence in particular of $\underline{C}_0$ -finite type, after localizing there is an epimorphism $E' \rightarrow F$ with $E' \in \text{Ob } \underline{C}_{0S}$. Using quasi-liftability of $\underline{C}_0$ in $\underline{C}$, one may, after localizing again, suppose that this epimorphism factors through E ; this proves quasi-liftability of $\underline{C}_1$. Similarly, if $E \in \text{Ob } \underline{C}_{1S}$ admits a finite right resolution by objects of $\underline{C}_0$, then E is $\underline{C}_0$ -pseudo-coherent by (2.11 a), hence of $\underline{C}_0$ -finite type.

For (ii), it is enough to prove necessity. One may apply either (2.11 a) or (2.13), with $\underline{S}' = \underline{S}$, $\underline{C}' = \underline{C}$, $f = \text{Id}$, $u = \text{Id}$, and with $\underline{C}_0$ and $\underline{C}'_0$ replaced by $\underline{C}_1$ and $\underline{C}_0$ respectively.

Scholium 2.14.1. There is therefore a largest strictly full additive sub- $\underline{S}$ -category of $\underline{C}$ giving the same notions of n -pseudo-coherence and pseudo-coherence as $\underline{C}_0$, namely the subcategory formed by the $\underline{C}_0$ -pseudo-coherent objects. This category is locally stable under kernels of epimorphisms.

Example 2.15. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}, \underline{C}_0$ be the $\underline{S}$ -categories considered in (1.3.d). The fibred category $D(\underline{C})$ will often be denoted $D(\underline{S}, A)$, or $D(\underline{S})$, or even $D(S)$ if there is no ambiguity about A . If S is the final object of $\underline{S}$, an object of $D(S)$ will be called n -pseudo-coherent, respectively pseudo-coherent, if it is n - $\underline{C}_0$ -pseudo-coherent, respectively $\underline{C}_0$ -pseudo-coherent. The same convention applies to the local notions. By (2.14) the same notion is obtained if $\underline{C}_0$ is replaced by the $\underline{S}$ -category whose fibre over U is the category of A_U -modules locally direct summands of finite free modules. A direct summand of a finite free module is not, in general, locally free of finite type; nevertheless one has the following theorem.

Theorem (2.15.1). Assume that $\underline{S}$ satisfies one of the following two conditions:

- (i) $\underline{S}$ has enough points, and for every point s of $\underline{S}$, the finitely generated projective left A_s -modules are free. This last condition holds, for example, if $A_s/\text{Rad}(A_s)$ is a field, or if $\text{Rad}(A_s)$ is nilpotent.
- (ii) $(\underline{S}, A)$ is a locally ringed topos in the sense of Giraud: A is commutative and, for every $X \in \text{Ob } \underline{S}$ and every $f \in \Gamma(X, A)$, one has $X = X_f \amalg X_{1-f}$, where X_f , respectively X_{1-f} , denotes the largest subobject of X on which f , respectively $1 - f$, is invertible.

Then every direct summand of a finite free left A -module is locally free of finite type.

Proof. Assume first (i). If E and F are A -modules and E is of finite presentation, the canonical map

$$\text{Hom}(E, F)_s \longrightarrow \text{Hom}(E_s, F_s)$$

is an isomorphism for every point s of S . It follows that if E and F are both of finite presentation and $E_s \simeq F_s$ at a point s , then there exists an s -pointed object U of $\underline{S}$ such that $E|U \simeq F|U$. The conclusion is immediate.

Assume (ii). Let $n \in \mathbb{N}$ and let D^n be the functor on locally ringed topoi defined by

$$T \longmapsto \Gamma(T, \mathcal{O}_T)^n.$$

It is known that the functor $\underline{\text{End}}_{\mathcal{O}_T}(D^n)$ is represented by $X_n = \text{Spec } \mathbb{Z}[T_{ij}]$ with its endomorphism u of $\mathcal{O}_{X_n}^n$ defined by the matrix (T_{ij}) . Let $p : A^n \rightarrow A^n$ be an idempotent. We must show that $p(A^n)$ is locally free of finite type. The preceding representability gives a morphism $s : \underline{S} \rightarrow X_n$ of locally ringed topoi such that $s^*(u) = p$. Since the corresponding matrix is idempotent, s factors through the closed subscheme $M \subset X_n$ of idempotent matrices. Hence $p = s'^* i^*(u)$, and the claim follows because, by the case already proved, the image of $i^*(u)$ is locally free of finite type.

Here an s -pointed object means one through which the point s factors, i.e. an object U with s factoring through S/U .

Corollary (2.16). *Let $f : (\underline{S}', A') \rightarrow (\underline{S}, A)$ be a morphism of ringed topoi, defined by a functor $\psi : \underline{S} \rightarrow \underline{S}'$ and a morphism of sheaves of rings $\psi^*(A) \rightarrow A'$. Let $D(\underline{S}', A')$ denote the derived category of A' - A -bimodules, left over A' and right over A , and let*

$$\otimes_A : D(\underline{S}', A') \times D(\underline{S}, A) \longrightarrow D(\underline{S}', A')$$

be the derived functor of $(E', E) \mapsto E' \otimes_{\psi^ A} \psi^*(E)$.*

Let $X \in \text{Ob } \underline{S}$, $X' = \psi(X)$, $E' \in \text{Ob } D(\underline{S}', A')$, and $E \in \text{Ob } D(\underline{S}, A)$. If E' is m' -pseudo-coherent and acyclic in degrees $> n'$ as an object of $D(\underline{S}', A')$, and if E is m -pseudo-coherent and acyclic in degrees $> n$, then $E' \otimes_A E$ is $\sup(m + n', m' + n)$ -pseudo-coherent and acyclic in degrees $> n + n'$.

Proof. Replacing $\underline{S}$ by $\underline{S}/X$, one may suppose $X = S$. Apply (2.13) to the functor

$$u = E' \otimes_A - : D(\underline{S}, A) \longrightarrow D(\underline{S}', A').$$

If $U \in \text{Ob } \underline{S}$, $U' = \psi(U)$, and E is a complex of left A -modules which is acyclic in degrees $> n$, then $E' \otimes_A E$ is acyclic in degrees $> n + n'$. Moreover, for $E = (A_U)^\sharp$, one has $u(E) = (E'_{U'})^\sharp$, which is m' -pseudo-coherent by hypothesis and (2.12). Thus the hypotheses of (2.13) are satisfied with $a = m'$ and $b = n'$.

Corollary (2.16.1). *Under the hypotheses of (2.16):*

a) *The functor $Lf^* : D^-(\underline{S}, A) \rightarrow D^-(\underline{S}', A')$, which is the functor $A' \otimes_A -$, induces a functor*

$$Lf^* : D(\underline{S}, A)_{\text{coh}} \longrightarrow D(\underline{S}', A')_{\text{coh}}.$$

b) *If A is commutative, the functor $\otimes_A : D^-(\underline{S}) \times_{\underline{S}} D^-(\underline{S}) \rightarrow D^-(\underline{S})$ induces a functor*

$$\otimes_A : D(\underline{S})_{\text{coh}} \times_{\underline{S}} D(\underline{S})_{\text{coh}} \longrightarrow D(\underline{S})_{\text{coh}}.$$

Exercise 2.16.2. Let $\underline{S}$ be a topos and let $A \rightarrow A'$ be a morphism of rings of $\underline{S}$. Assume either:

- (i) $A \rightarrow A'$ is surjective with nilpotent kernel;
- (ii) the functor $A' \otimes_A -$ on left A -modules is exact and faithful.

Show that $E \in \text{Ob } D(\underline{S}, A)$ is n -pseudo-coherent, respectively pseudo-coherent, if and only if $A' \otimes_A E \in \text{Ob } D(\underline{S}, A')$ is n -pseudo-coherent, respectively pseudo-coherent.

Indication. First show that a left A -module L is of finite type if and only if $A' \otimes_A L$ is of finite type, and then reduce to this case by truncation, using (2.5) and (2.10 b).

3. Relation with the classical notion of coherence

3.0

The hypotheses and notation are those of (2.0), except that $\underline{C}_0$ is no longer assumed locally quasi-stable under kernels of epimorphisms.

Definition (3.1). Let $F \in \text{Ob } \underline{C}_S$. We say that F is precoherent relative to $\underline{C}_0$ if, for every object U over S and every morphism $u : E \rightarrow F|U$ with $E \in \text{Ob } \underline{C}_{0U}$, the kernel of u is of $\underline{C}_0$ -finite type (1.2). We say that F is coherent relative to $\underline{C}_0$ if F is precoherent and of finite type.

The notion of precoherence is introduced here only as a technical intermediary; the notion to retain is coherence.

Remark. [Remark 3.2] It is clear from the definitions that if F is precoherent, respectively of finite type, then every subobject, respectively every quotient, of F is precoherent, respectively of finite type.

Lemma (3.3). a) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H$$

be an exact sequence in $\underline{C}_S$. If G is of finite type and H is precoherent, then F is of finite type.

b) Let

$$F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be exact in $\underline{C}_S$. If F is of finite type and G is precoherent, then H is precoherent.

c) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be exact. If F and H are precoherent, respectively of finite type, then so is G .

Proof. For a), after localizing, choose an epimorphism $G' \rightarrow G$ with $G' \in \text{Ob } \underline{C}_{0S}$. This gives a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & F' & \longrightarrow & G' & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H. \end{array}$$

Since H is precoherent, F' is of finite type; since $F' \rightarrow F$ is an epimorphism, F is of finite type. For b), let U be over S and let $H' \rightarrow H|_U$ be a morphism with $H' \in \text{Ob } \underline{C}_{0U}$. After changing notation, suppose $U = S$. By quasi-liftability of $\underline{C}_0$ in $\underline{C}$, and after localizing, there is a commutative square

$$\begin{array}{ccc} G' & \longrightarrow & H' \\ \downarrow & & \downarrow \\ G & \longrightarrow & H, \end{array}$$

where the horizontal arrows are epimorphisms and $G' \in \text{Ob } \underline{C}_{0S}$. Since $\text{Ker}(f)$ is a quotient of $\text{Ker}(fp)$, it is enough to show that $\text{Ker}(fp)$ is of finite type; one may therefore suppose $H' = G'$. The maps $F \rightarrow G$ and $H' \rightarrow H$ define $g : F \oplus H' \rightarrow G$. If $K = \text{Ker}(G \rightarrow H)$, the snake lemma applied to the natural diagram with exact rows shows that $\text{Ker}(g) \rightarrow \text{Ker}(f)$ is an epimorphism. But $F \oplus H'$ is of finite type and G is precoherent, so $\text{Ker}(g)$ is of finite type by a). This proves b).

For c), first prove the assertion for precoherence. Up to notation, one must show that if $\varphi : G' \rightarrow G$ is a morphism with $G' \in \text{Ob } \underline{C}_{0S}$, then $\text{Ker}(\varphi)$ is of finite type. Let H' be the image of $G' \rightarrow H$. The snake lemma applied to

$$\begin{array}{ccccccc} 0 & \longrightarrow & F' & \longrightarrow & G' & \longrightarrow & H' \longrightarrow 0 \\ & & \downarrow & & \downarrow \varphi & \downarrow h & \\ 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H \longrightarrow 0 \end{array}$$

with h a monomorphism, identifies $\text{Ker}(f)$ with $\text{Ker}(\varphi)$. Since F' is of finite type by a), another application of a) gives the required finite type property. The finite-type assertion follows by choosing, after localizing, epimorphisms $F' \twoheadrightarrow F$ and $H' \twoheadrightarrow H$ with $F', H' \in \text{Ob } \underline{C}_{0S}$, and then by lifting further so that the map to H factors through G ; the resulting epimorphism $F' \oplus H' \rightarrow G$ proves that G is of finite type.

Corollary (3.4). *The strictly full subcategory of $\underline{C}_S$ formed by the coherent objects is stable under finite projective and finite inductive limits. In particular, the kernel, cokernel, and image of a morphism between two coherent objects are coherent. If*

$$E^0 \longrightarrow E^1 \longrightarrow E^2 \longrightarrow E^3 \longrightarrow E^4$$

is exact and if E^0, E^1, E^3, E^4 are coherent, then E^2 is coherent.

Corollary (3.5). *Assume that the objects of $\underline{C}_0$ are $\underline{C}_0$ -coherent. This implies in particular that $\underline{C}_0$ is locally quasi-stable under kernels of epimorphisms, so that the hypotheses of (2.0) hold.*

a) *For $F \in \text{Ob } \underline{C}_S$, the following conditions are equivalent:*

- (i) *F is of finite presentation;*
- (ii) *F is coherent;*
- (iii) *F is of ∞ -finite presentation (2.8).*

b) For $F \in \text{Ob } D(\underline{C}_S)$ and $n \in \mathbb{Z}$, one has

$$F \text{ is } n\text{-pseudo-coherent} \iff H^i(F) \text{ is coherent for } i \geq n+1,$$

and $H^n(F)$ is of finite type.

In particular, F is pseudo-coherent if and only if $H^n(F)$ is coherent for every n .

Proof. The assertion a), and the implication $\Leftarrow$ of b), follow directly from (3.4). The implication $\Rightarrow$ in b) follows immediately from a) and (2.11 b).

Examples 3.6. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}_0 \subset \underline{C}$ be the $\underline{S}$ -categories of (1.3.d), cf. also (2.15). The objects of $\underline{C}_0$ are coherent if and only if A is coherent, i.e. precoherent. This holds, for example, in the following cases:

- A is constant with value a left noetherian ring;
- $\underline{S}$ is the topos of sheaves of sets for the Zariski topology on a locally noetherian scheme S , and $A = \mathcal{O}_S$;
- $\underline{S}$ is the topos of sheaves of sets on a complex analytic space S , and $A = \mathcal{O}_S$.

By contrast, the sheaf of continuous complex-valued functions on a topological space is in general not coherent.

If $(\underline{S}, A)$ is defined by a ringed space, the notion of coherence considered here coincides with Serre's usual notion. In accordance with the newer conventions, the word “scheme” is used for what older texts called a “prescheme”, and “separated scheme” for what older texts called a “scheme”.

Proposition (3.7). a) Let $f : (\underline{S}', A') \rightarrow (\underline{S}, A)$ be a morphism of ringed topoi. Assume that f is flat on the right, i.e. that $f^* = A' \otimes_A -$ is exact. For $F \in \text{Ob } D(\underline{S}, A)$ and $G \in \text{Ob } D^-(\underline{S}, A)$ there is a canonical functorial morphism

$$f^* \text{RHom}(F, G) \longrightarrow \text{RHom}(f^*F, f^*G).$$

It is an isomorphism when F is pseudo-coherent.

b) Assume A commutative and coherent, and let $F \in \text{Ob } D(\underline{S})$ and $G \in \text{Ob } D^-(\underline{S})$. If $H^i(F)$ and $H^i(G)$ are coherent for every i , then so is $\underline{\text{Ext}}^i(F, G)$.

Proof. For a), to define the morphism one may suppose G has injective components, and then take the composite

$$f^* \underline{\text{Hom}}(F, G) \longrightarrow \underline{\text{Hom}}(f^*F, f^*G) \longrightarrow \text{RHom}(f^*F, f^*G).$$

We prove that it is an isomorphism for F pseudo-coherent. Let $T = f^* \text{RHom}(-, G)$ and $T' = \text{RHom}(f^*(-), f^*G)$, and choose $a \in \mathbb{Z}$ such that $H^i(G) = 0$ for $i < a$. The assertion is clear for

F strictly perfect. Also, if F is acyclic in degrees $> n$, then TF and $T'F$ are acyclic in degrees $< a - n$. Since the question is local, we may suppose that F is strictly n -pseudo-coherent, so that it fits into a distinguished triangle

$$\begin{array}{ccc} & & F' \\ & \nearrow & \downarrow \\ F'' & \longrightarrow & F, \end{array}$$

with F' strictly perfect and F'' acyclic in degrees $> n - 1$. The induced morphism of distinguished triangles

$$\begin{array}{ccccccc} TF'' & \longrightarrow & TF & \longrightarrow & TF' & \longrightarrow & \\ \downarrow & & \downarrow & & \downarrow & & \\ T'F'' & \longrightarrow & T'F & \longrightarrow & T'F' & \longrightarrow & \end{array}$$

has an isomorphism on the right, while the left terms are acyclic in degrees $< a - n$. Therefore $TF \rightarrow T'F$ is an isomorphism on cohomology in degrees $< a - n$. Since n can be taken arbitrarily negative, the morphism is an isomorphism.

For b), the proof is left to the reader; see, for example, Hartshorne, III, 15.3.5.

Remark. [Remark 3.7.1] A different compatibility of RHom with Lf^* will be given below in (7.1.2).

Corollary (3.7.2). *Let $(\underline{S}, A)$ be a ringed topos, let s be a point of $\underline{S}$, and let $E, F \in \mathrm{Ob} D(\underline{S}, A)_{\mathrm{coh}}$. If $E_s \simeq F_s$, then there exists an s -pointed object U of $\underline{S}$ such that $E|U \simeq F|U$.*

Proof. Let $u : E \rightarrow F$ and $v : F \rightarrow E$ be inverse isomorphisms on the fibre at s . By (2.2 a) and the compatibility of Hom with fibres for coherent objects, one has

$$\mathrm{Hom}(E, F)_s \simeq \mathrm{Hom}(E_s, F_s), \quad \mathrm{Hom}(F, E)_s \simeq \mathrm{Hom}(F_s, E_s).$$

Thus, on a suitable neighbourhood U of s , there exist sections u and v of the corresponding Hom sheaves. The endomorphisms vu of $E|U$ and uv of $F|U$ induce the identity at s ; after passing to a further neighbourhood V of s , they are the identity. Hence $E|V \simeq F|V$.

4. Perfect complexes

4.0

In this section $\underline{S}$ denotes a site, S an object of $\underline{S}$, $\underline{C}$ a flat abelian $\underline{S}$ -category, and $\underline{C}_0$ a strictly full additive sub- $\underline{S}$ -category of $\underline{C}$. Unless otherwise specified, $\underline{C}_0$ is assumed locally liftable in $\underline{C}$ and locally stable under kernels of epimorphisms.

Lemma (4.1). *Let $u : E \rightarrow F$ be a morphism in $C(\underline{C}_S)$. If E is strictly $\underline{C}_0$ -perfect and F is acyclic, then u is locally homotopic to zero.*

This lemma was communicated to the editor by D. Ferrand.

Proof. We must construct locally a homotopy k with $u = dk + kd$. Since E is strictly perfect, there exist $a \leq b$ with $E^i = 0$ for $i \notin [a, b]$; take $k^i = 0$ outside this range. Construct $k^i : E^i \rightarrow F^{i-1}$ by descending induction on i . Suppose k^i has been constructed so that

$$u^j = d^{j-1}k^j + k^{j+1}d^j \quad (j > i \geq a).$$

Then $u^i - d^{i-1}k^i$ factors through $Z^i(F)$. Since F is acyclic, $Z^i(F) = B^i(F)$. Because $\underline{C}_0$ is locally liftable in $\underline{C}$, after localizing there exists $k^{i-1} : E^{i-1} \rightarrow F^{i-2}$ such that

$$u^i = d^{i-1}k^i + k^{i+1}d^i.$$

The induction gives the required homotopy.

Corollary (4.2). *Let*

$$\begin{array}{ccc} P & \xrightarrow{p} & F \\ & \searrow & \uparrow \wr \\ & & E' \end{array}$$

be a diagram in $C(\underline{C}_S)$, where P is strictly $\underline{C}_0$ -perfect and the vertical arrow is a quasi-isomorphism. The family of X over S for which there exists a morphism $u : P_X \rightarrow E_X$ in $C(\underline{C}_X)$ making the evident square commute in $K(\underline{C}_X)$ is a refinement of S .

Proof. Let G be the cone of the quasi-isomorphism $E' \rightarrow F$. The exact sequence

$$\cdots \rightarrow \mathrm{Hom}_{K(\underline{C}_S)}(P, E') \rightarrow \mathrm{Hom}_{K(\underline{C}_S)}(P, F) \rightarrow \mathrm{Hom}_{K(\underline{C}_S)}(P, G) \rightarrow \cdots$$

shows that the obstruction to lifting p lies in $\mathrm{Hom}_K(P, G)$. Since G is acyclic, this obstruction is locally zero by (4.1). Thus, locally, the required lift exists.

Corollary (4.3). *Let $f : E \rightarrow F$ be a quasi-isomorphism in $C(\underline{C}_S)$ with F strictly perfect. The family of X over S for which there exists a quasi-isomorphism $g : F_X \rightarrow E_X$ such that $f_X g$ is homotopic to the identity is a refinement of S .*

Proof. This is the special case of (4.2) obtained by taking $P = F$ and $p = \mathrm{Id}_F$.

Scholium 4.4. Let $F \in \mathrm{Ob} K(\underline{C}_S)$. Using, for instance, a cleavage of $\underline{C}$, one can define a category (Qis/F) fibred over $\underline{S}/_S$ whose fibre at X over S is the opposite of the category of quasi-isomorphisms with target F_X . Then (4.3) says that, if F is strictly perfect, the morphism Id_F is locally cofinal in (Qis/F) .

Notation 4.5. Let $\underline{M}$ be an $\underline{S}$ -fibred category. For $X \in \mathrm{Ob} \underline{S}$ and $E, F \in \mathrm{Ob} \underline{M}_X$, write $\underline{\mathrm{Hom}}_X(E, F)$ for the sheaf on $\underline{S}/_X$ associated with the presheaf

$$U \mapsto \mathrm{Hom}_{\underline{M}_U}(E|U, F|U).$$

Even when $\underline{C}$ is a stack, the corresponding presheaf for $\underline{M} = K(\underline{C})$ or $D(\underline{C})$ is not in general a sheaf.

Corollary (4.6). *Let $E, F \in \text{Ob } C(\underline{C}_S)$, with E strictly perfect. The canonical map*

$$\underline{\text{Hom}}_{K(\underline{C}_S)}(E, F) \longrightarrow \underline{\text{Hom}}_{D(\underline{C}_S)}(E, F)$$

is an isomorphism.

Proof. This follows immediately from (4.4) and from the formula

$$\text{Hom}_{D(\underline{C}_X)}(E_X, F_X) = \varinjlim_{X' \rightarrow X \in (\text{Qis}/E_X)^\circ} \text{Hom}_{K(\underline{C}_{X'})}(E_{X'}, F_{X'})$$

for X over S .

Definition (4.7). *Let $F \in \text{Ob } D(\underline{C}_S)$ and let $a, b \in \mathbb{Z}$ with $a \leq b$. We say that F has $\underline{C}_0$ -perfect amplitude contained in $[a, b]$, and write*

$$\underline{C}_0\text{-parf-amp}(F) \subset [a, b],$$

if the family of X over S for which there exists a quasi-isomorphism $F' \rightarrow F$ in $C(\underline{C}_X)$ with F' strictly $\underline{C}_0$ -perfect and $F'^i = 0$ for $i \notin [a, b]$, is a refinement of S . By (4.3), this is equivalent to the condition obtained by replacing “quasi-isomorphism in $C(\underline{C}_X)$ ” by “isomorphism in $D(\underline{C}_X)$ ”. We say that F has finite $\underline{C}_0$ -perfect amplitude if this holds for some a, b , and that F is $\underline{C}_0$ -perfect if it is locally of finite $\underline{C}_0$ -perfect amplitude.

When no confusion is possible, the subscript $\underline{C}_0$ will be omitted.

Example 4.8. Let $(\underline{S}, A)$ be a ringed topos. Let $\underline{C}$ be the $\underline{S}$ -category whose fibre over U is the category of left A_U -modules, and let $\underline{C}_0$ be the sub- $\underline{S}$ -category of left A_U -modules which are direct summands of finite free modules. By (1.3.1), the pair $(\underline{C}, \underline{C}_0)$ satisfies the hypotheses of (4.0). An object F of $D(\underline{S}, A)$ is said to have perfect amplitude contained in $[a, b]$, to have finite perfect amplitude, or to be perfect, if this holds relative to $\underline{C}_0$. When direct summands of finite free modules are locally free of finite type, to say that F is perfect means that F is locally isomorphic in the derived category to a bounded complex whose components are finite free modules.

Notation 4.9. We write

$$D(\underline{C})_{\underline{C}_0\text{-parf}} \quad \text{or} \quad \text{Parf}(\underline{C}, \underline{C}_0)$$

for the strictly full sub- $\underline{S}$ -category of $D(\underline{C})$ formed by the $\underline{C}_0$ -perfect objects. In the situation of (4.8) we also write $D(\underline{S})_{A\text{-parf}}$, $\text{Parf}(\underline{S}, A)$, or simply $\text{Parf}(S)$ when A is clear.

Proposition (4.10). *The category $\text{Parf}(\underline{C}, \underline{C}_0)$ is a triangulated sub- $\underline{S}$ -category of $D(\underline{C})$. The canonical functor*

$$K^b(\underline{C}_0) \longrightarrow \text{Parf}(\underline{C}, \underline{C}_0)$$

and similarly the induced functor on categories of distinguished triangles, induce $\underline{S}$ -equivalences on the associated stacks. Explicitly:

- (i) for every $X \in \text{Ob } \underline{S}$ and $E, F \in \text{Ob } K^b(\underline{C}_{0X})$, the canonical map of Hom sets to $\text{Parf}(\underline{C}, \underline{C}_0)$ is an isomorphism, and the same holds for triangles;
- (ii) every object of $\text{Parf}(\underline{C}, \underline{C}_0)$, and similarly every triangle in it, is locally isomorphic in $D(\underline{C})$ to an object of $K^b(\underline{C}_0)$, respectively to a triangle in $K^b(\underline{C}_0)$.

Proof. The non-triangular form of (i) is a special case of (4.6), and the non-triangular form of (ii) is true by definition. The category $\text{Parf}(\underline{C}, \underline{C}_0)$ is clearly stable under translation. If

$$\begin{array}{ccc} & & G \\ & \nearrow^{+1} & \downarrow \\ E & \xrightarrow{u} & F \end{array}$$

is a distinguished triangle with E and F perfect, then locally E and F may be assumed to lie in $K^b(\underline{C}_0)$. After further localizing, the morphism u may be represented by a morphism of $K^b(\underline{C}_0)$; then G is isomorphic to the cone of u , hence is perfect. This proves that $\text{Parf}(\underline{C}, \underline{C}_0)$ is triangulated and also proves the triangular version of (ii).

For the triangular version of (i), let $E = (E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_1)$ and $F = (F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow F_1)$ be distinguished triangles in $K(\underline{C}_{0X})$. The injectivity of the map of Hom sets follows from the non-triangular assertion. If $(\alpha_i) : E \rightarrow F$ is a morphism of triangles in $D(\underline{C})$, then locally each α_i is represented by a morphism in $K^b(\underline{C}_0)$. The differences measuring commutativity of the squares vanish in $D(\underline{C})$ and hence, locally, in $K(\underline{C}_0)$ by the non-triangular assertion. Thus the morphism of triangles is locally represented in $K(\underline{C}_0)$. This proves surjectivity.

Remark. [Remark 4.10.1] In (4.10) one may replace $\text{Parf}(\underline{C}, \underline{C}_0)$ by $D^*(\underline{C})_{\underline{C}_0\text{-parf}} = D^*(\underline{C}) \cap \text{Parf}(\underline{C}, \underline{C}_0)$ for $*$ = $-$ or b .

Complement 4.11. The triangulated structure of $\text{Parf}(\underline{C}, \underline{C}_0)$ can be sharpened in terms of perfect amplitude. If E has perfect amplitude in $[a, b]$, then $E[n]$ has perfect amplitude in $[a - n, b - n]$. If

$$\begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ E & \longrightarrow & F \end{array}$$

is a distinguished triangle, the following table, analogous to (2.6), records the possible implications; the entries are perfect amplitudes:

E	F	G
$[a, b + 1]$	$[a, b]$	$[a, b]$
$[a, b]$	$[a, b]$	$[a - 1, b - 1]$.

Corollary (4.12). *Assume that $\underline{S}$ is the punctual site. Then $F \in \text{Ob } D(\underline{C})$ has perfect amplitude relative to $\underline{C}_0$ contained in $[a, b]$ if and only if there exists a quasi-isomorphism $F' \rightarrow F$ with F' strictly perfect and $F'^i = 0$ for $i \notin [a, b]$. The canonical functors*

$$K^b(\underline{C}_0) \longrightarrow \text{Parf}(\underline{C}, \underline{C}_0), \quad \text{Tr}(K^b(\underline{C}_0)) \longrightarrow \text{Tr}(\text{Parf}(\underline{C}, \underline{C}_0))$$

are equivalences of categories.

Remark. [Remark 4.12.1] When $\underline{S}$ is punctual, the objects of $\underline{C}_0$ are projective. If $\underline{C}_0$ contains all projective objects of $\underline{C}$, one says “projective amplitude” instead of “ $\underline{C}_0$ -perfect amplitude”.

Remark. [Remark 4.12.2] Let C be an abelian category, viewed as fibred over the punctual site, and let C_0 be a strictly full additive subcategory. Suppose that C_0 is stable under kernels of epimorphisms and quasi-liftable in C . An acyclic object of $K(C_0)$ is in general not zero, so the functor $K^b(C_0) \rightarrow D(C)$ cannot be faithful. However, if $K^{\text{ac}}(C_0)$ denotes the thick subcategory of $K^b(C_0)$ formed by the complexes which are acyclic in C , the canonical functor

$$K^b(C_0)/K^{\text{ac}}(C_0) \longrightarrow D(C)$$

is fully faithful. Indeed, one already knows from (2.7) that the functor to $D^b(C)$ is fully faithful; applying Verdier’s criterion shows that the quotient functor has the required full faithfulness.

Lemma (4.13). *Let $F \in \text{Ob } D(\underline{C}_S)$ and let $a, b, c \in \mathbb{Z}$ with $a \leq b$ and $a \leq c$. If F has perfect amplitude contained in $[a, c]$ and if $H^i(F) = 0$ for $i \geq b$, then F has perfect amplitude contained in $[a, d]$, where $d = \inf(b, c)$.*

Proof. We may suppose $b \leq c$ and that F is strictly perfect with $F^i = 0$ outside $[a, c]$. Then F is isomorphic to its truncation F' defined by $F'^i = F^i$ for $i \leq b$, $F'^i = 0$ for $i > b$, and $F'^b = Z^b(F^\bullet)$. The latter object has a finite right resolution by objects of $\underline{C}_0$, hence is locally an object of $\underline{C}_0$. This gives the claim.

Corollary (4.14). *Let $F \in \text{Ob } \underline{C}_S$ and let $n \in \mathbb{N}$. The following conditions are equivalent:*

- (i) $\underline{C}_0\text{-parf-amp}(F) \subset [-n, 0]$;
- (ii) locally, F admits a left resolution of length n by objects of $\underline{C}_{0S}$; that is, the family of X over S for which there is an exact sequence

$$0 \longrightarrow L^{-n} \longrightarrow L^{-n+1} \longrightarrow \dots \longrightarrow L^0 \longrightarrow F_X \longrightarrow 0$$

with all $L^i \in \text{Ob } \underline{C}_{0X}$ is a refinement of S .

Proof. This is an immediate consequence of (4.7) and (4.13).

Definition (4.14.1). We say that F has perfect dimension $\leq n$, and write $\text{parf.dim}(F) \leq n$, if F satisfies the equivalent conditions of (4.14). The perfect dimension of F is the smallest such n , if it exists; otherwise it is infinite.

When S is the punctual site and $\underline{C}_0$ is the full subcategory of projective objects of $\underline{C}$, perfect dimension is just the usual projective dimension.

Proposition (4.15). Let $a, b, n \in \mathbb{Z}$ with $a \leq b$ and $n \geq 0$, and let $F \in \text{Ob } D(\underline{C}_S)$ be such that $F^i = 0$, respectively $H^i(F) = 0$, for $i \notin [a, b]$. If, for every $i \in [a, b]$, the object F^i , respectively $H^i(F)$, has perfect dimension $\leq n + (i - a)$, then F has perfect amplitude contained in $[a - n, b]$. Consequently, if all F^i , respectively all $H^i(F)$, are perfect, then F is perfect.

Proof. Proceed by induction on b , with a and n fixed, using truncations and (4.11), as in the proof of (2.11).

Lemma (4.16). Let $a, b \in \mathbb{Z}$ with $a \leq b$, and let $F \in \text{Ob } D(\underline{C}_S)$ be such that $F^i = 0$ for $i \notin [a, b]$. Assume that $\underline{C}_0\text{-parf-amp}(F) \subset [a, b]$ and that $F^i \in \text{Ob } \underline{C}_S$ for $i > a$. Then F^a is locally an object of $\underline{C}_{0S}$.

Proof. Truncating F gives an exact sequence

$$0 \longrightarrow F' \longrightarrow F^\bullet \longrightarrow F^a[-a] \longrightarrow 0$$

with $\underline{C}_0\text{-parf-amp}(F') \subset [a + 1, b + 1]$. From (4.11) and (4.13) it follows that $\underline{C}_0\text{-parf-dim}(F') = 0$, i.e. F' is locally an object of $\underline{C}_{0S}$, and the assertion follows.

Proposition (4.17). Let $F', F'' \in \text{Ob } D(\underline{C}_S)$ and let $a, b \in \mathbb{Z}$, $a \leq b$. Consider:

- (i) $\underline{C}_0\text{-parf-amp}(F') \subset [a, b]$ and $\underline{C}_0\text{-parf-amp}(F'') \subset [a, b]$;
- (ii) $\underline{C}_0\text{-parf-amp}(F' \oplus F'') \subset [a, b]$.

Then (ii) implies (i). If $\underline{C}_0$ is locally stable under direct summands, then the two conditions are equivalent.

Proof. The implication (ii) $\Rightarrow$ (i) is a special case of (4.11). Conversely, put $F = F' \oplus F''$. Since F is pseudo-coherent, F' and F'' are pseudo-coherent by (2.12). Also $H^i(F) = 0$ for $i \notin [a, b]$ implies the same for F' and F'' . By (2.10 a), after localizing one may suppose $F'^i, F''^i \in \text{Ob } \underline{C}_S$ for $i \geq a + 1$ and zero for $i > b$, and after truncating also zero for $i < a$. Since F has perfect amplitude in $[a, b]$, (4.16) shows that F^a is locally in $\underline{C}_{0S}$. But $F^a = F'^a \oplus F''^a$, hence F'^a and F''^a are locally in $\underline{C}_{0S}$ if $\underline{C}_0$ is locally stable under direct summands.

Proposition (4.18). Let $(\underline{S}', \underline{C}', \underline{C}'_0)$ satisfy the same hypotheses (4.0) as $(\underline{S}, \underline{C}, \underline{C}_0)$. Let $f : \underline{S}' \rightarrow \underline{S}$ be a morphism of sites, defined by $\psi : \underline{S} \rightarrow \underline{S}'$, and let

$$u : D^*(\underline{C}) \longrightarrow D^*(\underline{C}') \quad (* = -, +, b)$$

be an S -functor above f . If u sends the objects of $\underline{C}_0$ to $\underline{C}'_0$ -perfect complexes, then u defines a functor

$$D^*(\underline{C})_{\underline{C}_0\text{-parf}} \longrightarrow D^*(\underline{C}')_{\underline{C}'_0\text{-parf}}.$$

Proof. Let $F \in \text{Ob } D^*(\underline{C})_{\underline{C}_0\text{-parf}}$. The question is local, so one may suppose F strictly perfect. The assertion follows by devissage, using the fact that $D^*(\underline{C}')_{\underline{C}'_0\text{-parf}}$ is a triangulated subcategory of $D^*(\underline{C}')$.

Remark. [Remark 4.18.1] If there exist $m, n \in \mathbb{Z}$, $m \leq n$, such that $F \in \text{Ob } \underline{C}_{0S}$ implies

$$\underline{C}'_0\text{-parf-amp}(u(F)) \subset [m, n],$$

then

$$\underline{C}_0\text{-parf-amp}(F) \subset [a, b] \implies \underline{C}'_0\text{-parf-amp}(u(F)) \subset [a + m, b + n].$$

The proof is an exercise, using (4.11).

Corollary (4.19). *Under the hypotheses of (2.16), if $E' \in \text{Ob } D(\underline{S}', A')$ is perfect as an object of $D(\underline{S}', A')$ and if $E \in \text{Ob } D(\underline{S}, A)$ is perfect, then $E' \otimes_A E$ is perfect.*

Proof. One may suppose that $E' \in \text{Ob } D(\underline{S}', A')$, where S' is the final object of $\underline{S}'$, and apply (4.18) to $u = E' \otimes_A - : D(\underline{S}, A) \rightarrow D(\underline{S}', A')$.

Corollary (4.19.1). *In particular, if $f : (\underline{S}', A') \rightarrow (\underline{S}, A)$ is a morphism of ringed topoi, the derived inverse image Lf^* sends perfect complexes to perfect complexes. If A is commutative, the derived tensor product of two perfect complexes is perfect.*

5. Finite Tor-dimension and perfection

5.0

The reader will have wondered what is missing from a pseudo-coherent complex in order for it to be perfect. Is it enough, for example, that it have bounded cohomology? The answer is no. Let $\underline{C}$ be the category of modules over a commutative noetherian ring A , and let $\underline{C}_0$ be the full subcategory of finite projective A -modules. Any finite A -module, viewed as a complex concentrated in degree zero, is pseudo-coherent and has bounded cohomology, but it is perfect only if it has finite projective dimension, equivalently finite Tor-dimension. This example suggests that the difference between pseudo-coherence and perfection is a cohomological-dimension condition. We now examine this in the category of modules on a ringed topos. The preliminary facts on Tor-dimension are recalled briefly; see Hartshorne, III, 4, and SGA 4, XVII, 4.9.

Proposition (5.1). *Let $(\underline{S}, A)$ be a ringed topos, let $X \in \text{Ob } \underline{S}$, let $F \in \text{Ob } D^-(X)$, and let $m, n \in \mathbb{Z}$ with $m \leq n$. The following conditions are equivalent:*

(i) F is isomorphic in $D^-(X)$ to a complex F' such that each F'^i is flat and $F'^i = 0$ for $i \notin [m, n]$;

(ii) for every right A_X -module E , one has

$$H^i(E \otimes_{A_X} F) = 0 \quad \text{for } i \notin [m, n].$$

Proof. The implication (i) $\Rightarrow$ (ii) is trivial. The converse follows, by truncation, from the following lemma.

Lemma (5.1.1). *Let $F \in \text{Ob } D^-(X)$. Assume that F satisfies condition (ii) of (5.1), that $F^i = 0$ for $i \notin [m, n]$, and that F^i is flat for $i \neq m$. Then F^m is flat.*

Proof. There is a distinguished triangle

$$\begin{array}{ccc} & & F^m[-m] \\ & \nearrow & \downarrow \\ F' & \longrightarrow & F, \end{array}$$

where F' has flat components and is zero outside $[m+1, n]$. Apply $E \otimes_{A_X} -$ for an arbitrary right A_X -module E , and write the associated exact cohomology sequence.

Definition (5.2). *An object F of $D^-(X)$ satisfying the equivalent conditions of (5.1) is said to have flat amplitude contained in $[m, n]$, and we write*

$$\text{tor-amp}(F) \subset [m, n].$$

We say that F has finite flat amplitude, or finite Tor-dimension, if this holds for some $m \leq n$.

5.3. For $X \in \text{Ob } \underline{S}$, denote by $D^-(X)_{\text{torfd}}$ the full subcategory of $D^-(X)$ formed by the objects of finite Tor-dimension. By criterion (ii) of (5.1), it is a triangulated subcategory of $D^-(X)$. As X varies, the categories $D^-(X)_{\text{torfd}}$ form a triangulated sub- $\underline{S}$ -category of $D(\underline{S})$, denoted $D(\underline{S})_{\text{torfd}}$. The tensor product induces an exact $\underline{S}$ -functor

$$\otimes_A : D(\underline{S})_{\text{torfd}} \times_{\underline{S}} D(\underline{S}) \longrightarrow D(\underline{S}).$$

For a left A -module F , to have flat amplitude contained in $[-n, 0]$ is equivalent to having Tor-dimension $\leq n$ in the ordinary sense, i.e. to admitting a left resolution of length n by flat A -modules.

5.4. Let $X \in \text{Ob } \underline{S}$ and $m \leq n$. An object $F \in \text{Ob } D^-(X)$ has flat amplitude contained in $[m, n]$ if and only if there is a covering family $(X_i \rightarrow X)$ such that $F|_{X_i}$ has flat amplitude contained in $[m, n]$ for all i . In particular, if $\underline{S}/_X$ is of finite type in the sense of SGA 4, VI, 1.7, and if F is locally of finite Tor-dimension, then F is of finite Tor-dimension.

Proposition (5.5). *Under the hypotheses of (2.16), if E' has flat amplitude contained in $[m', n']$ as an object of $D(\underline{S}', A')$ and E has flat amplitude contained in $[m, n]$, then $E' \otimes_A E$ has flat amplitude contained in the corresponding interval obtained by adding the bounds.*

Proof. This is immediate from the form (5.1)(i) of finite Tor-dimension and from the fact that if E is a flat left A -module, then ψ^*E is a flat left ψ^*A -module.

Corollary (5.5.1). a) *The functor $Lf^* : D(\underline{S}, A) \rightarrow D(\underline{S}', A')$ induces*

$$Lf^* : D(\underline{S}, A)_{\text{torfd}} \longrightarrow D(\underline{S}', A')_{\text{torfd}}.$$

b) *If A is commutative, the tensor product induces*

$$\otimes_A : D(\underline{S})_{\text{torfd}} \times_{\underline{S}} D(\underline{S}) \longrightarrow D(\underline{S})_{\text{torfd}}.$$

Remark. [Remark 5.5.2] Let $E \in \text{Ob } D^-(\underline{S}, A)$, with S the final object of $\underline{S}$, and let $m \leq n$. If E has flat amplitude contained in $[m, n]$, then so does every fibre E_s at a point s of $\underline{S}$. The converse holds if $\underline{S}$ has enough points, using criterion (5.1)(ii) and the fact that tensor product commutes with passage to fibres.

Exercises 5.6. a) Assume A commutative and $(\underline{S}', A')$ faithfully flat over $(\underline{S}, A)$. Then $E \in \text{Ob } D^-(\underline{S}, A)$ has flat amplitude contained in $[m, n]$ if and only if f^*E does.

b) For $F \in \text{Ob } D^+(\underline{S}, A)$, show the equivalence of:

- (i) F is isomorphic in $D^+(\underline{S}, A)$ to a complex F' whose components are injective and which is zero outside $[m, n]$;
- (ii) for every left A -module E , one has $\underline{\text{Ext}}^i(E, F) = 0$ for $i \notin [m, n]$.

Such an F is said to have injective amplitude contained in $[m, n]$. The full subcategory of $D^+(\underline{S}, A)$ formed by the objects of finite injective amplitude is triangulated and is denoted $D^+(\underline{S}, A)_{\text{injdf}}$.

c) If A is commutative, show that

$$\text{RHom} : D(\underline{S}, A)_{\text{torfd}}^\circ \times D(\underline{S}, A)_{\text{injdf}} \longrightarrow D(\underline{S}, A)_{\text{parf}}$$

is defined.

We can now answer the question raised in (5.0) and state the main result of this section.

Theorem (5.7). *Let $(\underline{S}, A)$ be a ringed topos, let $E \in \text{Ob } D(\underline{S}, A)$, and let $m, n \in \mathbb{Z}$ with $m \leq n$. The following conditions are equivalent:*

- (i) $\text{parf. amp}(E) \subset [m, n]$;

(ii) E is $(m - 1)$ -pseudo-coherent and $\text{tor-amp}(E) \subset [m, n]$.

Since “perfect” plainly implies “pseudo-coherent”, one obtains:

Corollary (5.7.1). *An object E is perfect if and only if it is pseudo-coherent and locally of finite Tor-dimension.*

For the proof we need a few lemmas.

Lemma (5.7.2). *Let B be a ring of $\underline{S}$. If E is a left A -module, G a right B -module, and F an A - B -bimodule, there is a canonical functorial homomorphism*

$$G \otimes_B \underline{\text{Hom}}_A(E, F) \longrightarrow \underline{\text{Hom}}_A(E, G \otimes_B F).$$

It is an isomorphism when E is a direct summand of a finite free A -module.

Proof. The morphism is the evident evaluation map. The assertion is clear for $E = A$ and is stable under finite sums and direct summands.

Lemma (5.7.3). *Let $E \in \text{Ob } D^-(\underline{S}, A)$ have finite Tor-dimension, and suppose E is $(m - 1)$ -pseudo-coherent. Then locally, after replacing E by a quasi-isomorphic complex, one may arrange that $E^i = 0$ for $i > n$ and that the terms in the relevant range are finite locally free, provided the Tor-amplitude is contained in $[m, n]$.*

Proof. This is obtained by applying the approximation process of Section 2 and then using (5.1.1) to force flatness in the bottom degree. The pseudo-coherent hypothesis supplies finite-type generators; the Tor-amplitude hypothesis eliminates the last obstruction.

Proof. [Proof of Theorem 5.7] If E has perfect amplitude contained in $[m, n]$, then it is locally represented by a complex of finite locally free modules in degrees $[m, n]$. It is therefore $(m - 1)$ -pseudo-coherent, and its Tor-amplitude is contained in $[m, n]$ by (5.1).

Conversely, assume (ii). By pseudo-coherence, locally E is represented, up to the required order, by a bounded-above complex of finite locally free modules. Since the Tor-amplitude is contained in $[m, n]$, the truncation argument of (5.1.1) and the lifting results of Section 2 show that this representation may be chosen with no non-zero terms outside $[m, n]$. Thus E has perfect amplitude contained in $[m, n]$.

Proposition (5.8). *Let $E \in \text{Ob } D(\underline{S}, A)$ be perfect. Then perfect amplitude and Tor-amplitude agree: if $\text{parf. amp}(E) \subset [a, b]$, then $\text{tor-amp}(E) \subset [a, b]$, and conversely finite Tor-amplitude together with pseudo-coherence gives the corresponding perfect amplitude by (5.7).*

Corollary (5.8.4). *Perfect complexes are locally of finite Tor-dimension. They are stable under Lf^* for morphisms of ringed topoi and, when the ring is commutative, under derived tensor product.*

Corollary (5.9). *Let $(\underline{S}, A)$ be a ringed topos. Suppose that every coherent A -module has locally finite projective dimension bounded in the required sense. Then every pseudo-coherent object of $D(\underline{S}, A)$ with bounded cohomology is perfect.*

Examples 5.10. Corollary (5.9) applies in particular when $(\underline{S}, A)$ is defined by a ringed space whose local rings are regular, for example a regular scheme, or a smooth complex analytic space over $\mathbb{C}$.

6. Rank of a perfect complex

6.0

Let $(X, \mathcal{O}_X)$ be a ringed space in local rings. To each locally free sheaf of finite type L on an open set U one attaches a locally constant integer-valued function $\text{rank}_U(L)$, called the rank of L . The family of such functions is characterized by compatibility with restriction, additivity in short exact sequences of finite locally free modules, and the condition $\text{rank}(\mathcal{O}_X) = 1$. We now extend the notion of rank to perfect complexes.

Definition (6.1). *Let C be an additive category, respectively a triangulated category, and let F be an abelian group. A function $f : \text{Ob}(C) \rightarrow F$ is called additive if, whenever $L \simeq L' \oplus L''$, respectively whenever there is a distinguished triangle*

$$L' \longrightarrow L \longrightarrow L'' \longrightarrow L'[1],$$

one has

$$f(L) = f(L') + f(L'').$$

The additive functions from C to F form an abelian group, denoted $\text{Add}(C, F)$.

6.2. If C is triangulated, any additive function on C is additive on the underlying additive category. In particular $f(0) = 0$, $f(L) = f(M)$ when $L \simeq M$, and

$$f(L[n]) = (-1)^n f(L)$$

for all L and all $n \in \mathbb{Z}$.

6.3. The functor $\text{Add}(C, -)$ is representable by an abelian group denoted $k(C)$, endowed with a universal additive function $\text{Ob}(C) \rightarrow k(C)$. If C is additive, $k(C)$ is the group completion of the monoid of isomorphism classes of objects. If C is triangulated, $k(C)$ is the quotient of the free abelian group on $\text{Ob}(C)$ by the relations $L = L' + L''$ for each distinguished triangle $L' \rightarrow L \rightarrow L'' \rightarrow L'[1]$.

We do not use the usual notation $K^0(C)$ here, because for an additive category C the notation $K^0(C)$ already denotes the homotopy category of complexes in C in the conventions being used.

The group $k(C)$ is functorial in C for additive, respectively exact, functors, and isomorphic functors induce the same map on $k(C)$.

Lemma (6.4). *Let C be an additive category. The canonical functor $C \rightarrow K^b(C)$ induces a functorial isomorphism*

$$k(C) \xrightarrow{\sim} k(K^b(C)).$$

Proof. This follows immediately from the formula

$$f(L) = \sum_i (-1)^i f(L^i) \quad (6.4.1)$$

for $L \in \text{Ob } K(C)$ and for any additive function f on $K(C)$.

Definition (6.5). *Let $\underline{S}$ be a category, $\underline{C}$ an additive, respectively triangulated, $\underline{S}$ -category, and F an abelian presheaf on $\underline{S}$. A family of functions*

$$f_X : \text{Ob}(\underline{C}_X) \longrightarrow F(X), \quad X \in \text{Ob } \underline{S},$$

is called an $\underline{S}$ -additive function from $\underline{C}$ to F if:

- (i) *each f_X is additive in the sense of (6.1);*
- (ii) *for every arrow $u : X \rightarrow Y$ of $\underline{S}$, the diagram*

$$\begin{array}{ccc} \text{Ob } \underline{C}_Y & \xrightarrow{f_Y} & F(Y) \\ \downarrow & & \downarrow F(u) \\ \text{Ob } \underline{C}_X & \xrightarrow{f_X} & F(X) \end{array}$$

commutes.

The $\underline{S}$ -additive functions from $\underline{C}$ to F form an abelian group denoted $\text{Add}_{\underline{S}}(\underline{C}, F)$.

Remarks 6.6. Let $\underline{C}^{\text{iso}}$ be the fibred $\underline{S}$ -category whose fibre over X has the same objects as $\underline{C}_X$ and whose morphisms are only the isomorphisms, and let $\underline{F}$ be the fibred category with discrete fibre $F(X)$ over X . An $\underline{S}$ -additive function is a Cartesian $\underline{S}$ -functor $\underline{C}^{\text{iso}} \rightarrow \underline{F}$ whose restrictions to the fibres are additive.

If $\underline{C}$ is triangulated, additivity can be expressed by the exact sequence

$$0 \longrightarrow \text{Add}_{\underline{S}}(\underline{C}, F) \longrightarrow \text{Cart}_{\underline{S}}(\underline{C}^{\text{iso}}, \underline{F}) \xrightarrow{\Delta} \text{Cart}_{\underline{S}}(\text{Tr}(\underline{C})^{\text{iso}}, \underline{F}), \quad (6.6.1)$$

where $\Delta(f)(T) = f(L) - f(L'') - f(L')$ for a distinguished triangle $T = (L' \rightarrow L \rightarrow L'' \rightarrow L'[1])$.

For a morphism $u : X \rightarrow Y$ in $\underline{S}$, inverse-image functors, all isomorphic to one another, define a unique homomorphism $k(\underline{C}_Y) \rightarrow k(\underline{C}_X)$. Thus $X \mapsto k(\underline{C}_X)$ is an abelian presheaf on $\underline{S}$, denoted $k(\underline{C})$, and

$$\text{Add}_{\underline{S}}(\underline{C}, F) = \text{Hom}(k(\underline{C}), F).$$

Proposition (6.7). *Let $\underline{S}$ be a site, $\underline{C}$ a flat abelian $\underline{S}$ -category, and $\underline{C}_0$ a strictly full additive sub- $\underline{S}$ -category satisfying the hypotheses of (4.0). The morphism of presheaves*

$$k(\underline{C}_0) \longrightarrow k(\text{Parf}(\underline{C}, \underline{C}_0))$$

induced by the inclusion $\underline{C}_0 \subset \text{Parf}(\underline{C}, \underline{C}_0)$ becomes an isomorphism after sheafification. Equivalently, for every abelian sheaf $\underline{F}$ on $\underline{S}$, the restriction homomorphism

$$\text{Add}_{\underline{S}}(\text{Parf}(\underline{C}, \underline{C}_0), \underline{F}) \longrightarrow \text{Add}_{\underline{S}}(\underline{C}_0, \underline{F})$$

is an isomorphism.

Proof. The restriction factors as

$$\text{Add}_{\underline{S}}(\text{Parf}(\underline{C}, \underline{C}_0), \underline{F}) \xrightarrow{a} \text{Add}_{\underline{S}}(K^b(\underline{C}_0), \underline{F}) \xrightarrow{b} \text{Add}_{\underline{S}}(\underline{C}_0, \underline{F}).$$

By (6.4), b is an isomorphism. It remains to see that a is an isomorphism. Consider the commutative diagram whose rows are the exact sequences (6.6.1) and whose vertical arrows are restrictions from perfect complexes to bounded complexes in $\underline{C}_0$. Since $\underline{F}$ is a sheaf, $\underline{F}$ is a stack; by (4.10), the two right vertical arrows are isomorphisms. Hence so is a .

Remark. [Remark 6.7.1] In (6.7) one may replace $\text{Parf}(\underline{C}, \underline{C}_0)$ by $D^*(\underline{C})_{\underline{C}_0\text{-parf}}$ for $*$ = $-$ or b .

6.8. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}$ and $\underline{C}_0$ be the $\underline{S}$ -categories considered in (4.8). There are homomorphisms of abelian presheaves

$$k(\underline{C}_0) \xrightarrow{\alpha} k(D^-(A\text{-parf})) \xrightarrow{\beta} k(\text{Parf}(A\text{-parf})),$$

which induce isomorphisms after sheafification by (6.7). The homomorphism α is compatible with inverse image: for a morphism of ringed topoi $u : (\underline{S}', A') \rightarrow (\underline{S}, A)$, the square obtained from Lu^* commutes. If A is commutative, the derived tensor product gives ring-presheaf structures and α is a homomorphism of ring presheaves.

6.9. Assume now that $(\underline{S}, A)$ is locally ringed. Then, by (2.15.1), $\underline{C}_{0X}$ is the category of locally free A_X -modules of finite type. Let $\underline{C}_1$ be the $\underline{S}$ -category whose fibre over X is the category of finite free A_X -modules. An object of $D(\underline{S})$ is perfect if and only if it is perfect relative to $\underline{C}_1$. The homomorphisms

$$k(\underline{C}_1) \longrightarrow k(\underline{C}_0) \longrightarrow k(D^-(\underline{S})_{\text{parf}}) \longrightarrow k(\text{Parf}(\underline{S}))$$

induce isomorphisms after sheafification. We shall show that these associated sheaves are canonically isomorphic to the constant sheaf $\mathbb{Z}_{\underline{S}}$.

Lemma (6.10). *Let $(\underline{S}, A)$ be a locally ringed topos, let U be an object different from the empty object, and let $p, q \in \mathbb{N}$. If $A_U^p \simeq A_U^q$, then $p = q$.*

Proof. By the same kind of argument as in the proof of (2.15.1), one reduces to the locally ringed topoi defined by a prescheme. There the assertion is trivial, by evaluating at a point of U .

Definition (6.11). *The rank, denoted rank , is the $\underline{S}$ -additive function from $\underline{C}_1$ to $\mathbb{Z}_{\underline{S}}$ defined by*

$$\text{rank}(L) = p \quad \text{if } L \simeq A_X^p, \quad X \neq \emptyset,$$

and by $\text{rank}(L) = 0$ for objects over the empty object.

By (6.7), the rank function extends uniquely to an $\underline{S}$ -additive function from $\text{Parf}(\underline{S})$ to $\mathbb{Z}_{\underline{S}}$, still denoted rank.

Proposition (6.12). a) *For $L, M \in \text{Ob } D^-(X)_{\text{parf}}$, one has*

$$\text{rank}(LLM) = \text{rank}(L) \text{rank}(M).$$

b) *If $u : (\underline{S}', A') \rightarrow (\underline{S}, A)$ is a morphism of locally ringed topoi and $M \in \text{Ob } D^-(\underline{S})_{\text{parf}}$, then*

$$\text{rank}(Lu^*M) = u^*(\text{rank}(M)).$$

Equivalently, rank defines a homomorphism of ring presheaves

$$k(D^-(\underline{S})_{\text{parf}}) \longrightarrow \mathbb{Z}_{\underline{S}}$$

compatible with inverse images.

Proof. It is enough to check these statements after passing to the associated sheaves. By (6.9), one may replace $k(D^-(\underline{S})_{\text{parf}})$ by $k(\underline{C}_1)$, where the assertions reduce to finite free modules and are immediate.

6.14. Let $(\underline{S}, A)$ be locally ringed, let $X \in \text{Ob } \underline{S}$, and let $n \in H^0(X, \mathbb{Z}_{\underline{S}})$. Define an object A_X^n of $\text{Parf}(X)$ as follows. If n is constant with value an integer $n \geq 0$, set $A_X^n = A_X^{\oplus n}$; if $n < 0$, set $A_X^n = A_X^{\oplus |n|}[1]$. In general, decompose X into disjoint pieces on which n is constant and glue the corresponding objects. This gives a homomorphism of ring presheaves

$$i : \mathbb{Z}_{\underline{S}} \longrightarrow k(\text{Parf}(\underline{S})), \quad i_X(n) = \text{cl}(A_X^n).$$

Proposition (6.14). *Under the hypotheses above,*

$$\text{rank} \circ i = \text{Id}_{\mathbb{Z}_{\underline{S}}}.$$

Thus $k(\text{Parf}(\underline{S}))$, viewed as a presheaf of $\mathbb{Z}_{\underline{S}}$ -algebras via i , is augmented by rank. Moreover,

$$\text{rank} : k(\text{Parf}(\underline{S})) \longrightarrow \mathbb{Z}_{\underline{S}}$$

becomes an isomorphism after sheafification.

Proof. The first assertion follows from the relations $\text{cl}(L[1]) = -\text{cl}(L)$ and $\text{cl}(L^n) = n \text{cl}(L)$. For the second, i factors through $k(\underline{C}_0)$, and since $\text{rank} \circ i$ is the identity, it is enough, by (6.9), to prove that $i \circ \text{rank} : k(\underline{C}_0) \rightarrow k(\underline{C}_0)$ induces an isomorphism after sheafification. This says that for $L \in \text{Ob } \underline{C}_0$, the object $i \circ \text{rank}(L)$ is locally isomorphic to L , which is immediate.

Corollary (6.15). *Under the hypotheses of (6.13), for every abelian sheaf $\underline{F}$ on $\underline{S}$, the group of $\underline{S}$ -additive functions from $\text{Parf}(\underline{S})$ to $\underline{F}$ is canonically identified with $H^0(\underline{S}, \underline{F})$, where $\underline{S}$ denotes the final object of $\underline{S}$. For $\underline{F} = \mathbb{Z}_{\underline{S}}$, the rank function corresponds to the unit section of $\mathbb{Z}_{\underline{S}}$.*

7. Duality of perfect complexes

7.0

In this section $(\underline{S}, A)$ denotes a ringed topos in commutative rings, and S is the final object of $\underline{S}$. The aim is to generalize to perfect complexes the usual duality notions for locally free sheaves of finite type.

Proposition (7.1). *Let $E \in \text{Ob } D(\underline{S})_{\text{parf}}$ and $F \in \text{Ob } D^+(\underline{S})$.*

a) *If F has locally bounded cohomology, so does $\text{RHom}(E, F)$. If F is pseudo-coherent, respectively locally of finite Tor-dimension, respectively perfect, then the same is true of $\text{RHom}(E, F)$.*

7.1.1. Assume moreover that E has finite Tor-dimension, as holds by (5.8.4 b). Let a, b, m, n be integers such that

$$\text{tor-amp}(E) \subset [a, b]$$

(equivalently, by (5.8), $\text{parf-amp}(E) \subset [a, b]$), and $H^i(F) = 0$ for $i \notin [m, n]$, respectively $\text{parf-amp}(F) \subset [m, n]$. Then

$$H^i(\text{RHom}(E, F)) = 0 \quad \text{for } i \notin [m - b, n - a],$$

i.e. $\text{tor-amp}(\text{RHom}(E, F)) \subset [m - b, n - a]$.

Proof. The assertion a) is local; we may suppose E strictly perfect and then argue by devissage on E , reducing to $E = A$. For b), one may suppose E strictly perfect with $E^i = 0$ for $i \notin [c, d]$, and choose a representative of F in the indicated degrees, or with flat components in the indicated interval. Then the assertion follows from the next lemma.

Lemma (7.1.1). *Let $E \in \text{Ob } D(\underline{S})$ and $F \in \text{Ob } D^+(\underline{S})$. If E is strictly pseudo-coherent, with components direct summands of finite free modules, the canonical map in $D^+(\underline{S})$*

$$\text{Hom}^\bullet(E, F) \longrightarrow \text{RHom}(E, F)$$

is an isomorphism.

Proof. The spectral sequence

$$E_2^{pq} = H^p(\operatorname{Ext}^q(E^\bullet, F^\bullet)) \Rightarrow \operatorname{Ext}^{p+q}(E, F)$$

degenerates because $\operatorname{Ext}^q(E^i, F^j) = 0$ for $q \neq 0$. This gives the required isomorphisms between the cohomology of $\operatorname{Hom}^\bullet(E, F)$ and of $\operatorname{RHom}(E, F)$.

Proposition (7.1.2). *Let $f : (\underline{S}', A') \rightarrow (\underline{S}, A)$ be a morphism of ringed topoi. Let $E \in \operatorname{Ob} D^-(\underline{S})$ and $F \in \operatorname{Ob} D^+(\underline{S})$ be such that $\operatorname{RHom}(E, F) \in \operatorname{Ob} D^+(\underline{S})$ and $Lf^*(F) \in \operatorname{Ob} D^+(\underline{S}')$. There is a canonical functorial morphism*

$$Lf^* \operatorname{RHom}(E, F) \longrightarrow \operatorname{RHom}(Lf^* E, Lf^* F).$$

It is an isomorphism if E is perfect.

Proof. By the adjunction between Lf^* and Rf_* , it is enough to define a map

$$\operatorname{RHom}(E, F) \longrightarrow Rf_* \operatorname{RHom}(Lf^* E, Lf^* F).$$

Use the adjunction morphism $F \rightarrow Rf_* Lf^* F$ and the duality isomorphism

$$Rf_* \operatorname{RHom}(Lf^* E, Lf^* F) \simeq \operatorname{RHom}(E, Rf_* Lf^* F).$$

If E is perfect, the assertion is local and follows by devissage from the trivial case $E = A$.

Proposition (7.2). *For $E \in \operatorname{Ob} D(\underline{S})$ and $F \in \operatorname{Ob} D^+(\underline{S})$ there is a canonical functorial morphism*

$$E \longrightarrow \operatorname{RHom}(\operatorname{RHom}(E, F), F).$$

It is an isomorphism if $E \in \operatorname{Ob} D(\underline{S})_{\text{parf}}$ and $F = A$.

Proof. Resolve F injectively and use the canonical morphism in the homotopy category

$$E \longrightarrow \operatorname{Hom}^\bullet(\operatorname{Hom}^\bullet(E, F), F).$$

For E perfect and $F = A$, the assertion is local and follows by devissage from $E = A$.

7.3. For $E \in \operatorname{Ob} D(\underline{S})$ put

$$E^\vee = \operatorname{RHom}(E, A),$$

and call this the *dual* of E . By (7.1) and (7.2), if E is perfect, respectively perfect and of finite Tor-dimension, then $E^\vee$ has the same property, and the canonical morphism $E \rightarrow E^{\vee\vee}$ is an isomorphism. In other words, $\operatorname{RHom}(-, A)$ is an anti-equivalence of $D(\underline{S})_{\text{parf}}$, respectively of the finite Tor-dimension part, with itself.

Proposition (Cartan isomorphism 7.4). *For $E \in \text{Ob } D(\underline{S})$, $F \in \text{Ob } D^-(\underline{S})$, and $G \in \text{Ob } D^+(\underline{S})$, there are canonical functorial isomorphisms*

$$\text{RHom}(E \mathbb{L} F, G) \simeq \text{RHom}(E, \text{RHom}(F, G)),$$

and the corresponding non-underlined Hom-isomorphisms obtained by taking global sections.

Proof. This is standard: resolve G injectively and F by flat modules.

Lemma (7.5). *For $E \in \text{Ob } D(\underline{S})$ and $F \in \text{Ob } D^+(\underline{S})$, there is a canonical functorial homomorphism*

$$E^\vee \mathbb{L} \text{RHom}(E, F) \longrightarrow F.$$

Proposition (7.6). *Let $E \in \text{Ob } D(\underline{S})$ and let $F, G \in \text{Ob } D^+(\underline{S})$. Under either of the following hypotheses:*

- (i) G has finite Tor-dimension;*
- (ii) E is perfect, F has finite Tor-dimension, and $G \in \text{Ob } D(\underline{S})$,*

there is a canonical functorial homomorphism

$$\text{RHom}(E, F) \mathbb{L} G \longrightarrow \text{RHom}(E, F \mathbb{L} G).$$

It is an isomorphism when E is perfect.

Proof. Under (i), define the map by resolving F injectively and G flatly. Under (ii), by (7.4) it is equivalent to define a map

$$E \mathbb{L} \text{RHom}(E, F) \mathbb{L} G \longrightarrow F \mathbb{L} G,$$

which is obtained by tensoring the map of (7.5) with G . When E is perfect, reduce locally by devissage to $E = A$.

Corollary (7.7). *For $E \in \text{Ob } D(\underline{S})_{\text{parf}}$ and $F \in \text{Ob } D^+(\underline{S})$, there is a functorial isomorphism*

$$E^\vee \mathbb{L} F \xrightarrow{\sim} \text{RHom}(E, F).$$

Proof. Apply (7.6) with F and G replaced by A and F , and use (7.1) to know that $E^\vee$ is perfect.

Corollary (7.8). *For $E, F \in \text{Ob } D(\underline{S})_{\text{parf}}$, there is a natural morphism*

$$\text{RHom}(E, F) \mathbb{L} \text{RHom}(F, E) \longrightarrow A$$

which is a perfect duality: modulo (7.4), it identifies each of the two complexes with the dual of the other.

Proof. The natural pairings $E^\vee \mathbb{L}E \rightarrow A$ and $F^\vee \mathbb{L}F \rightarrow A$ give, by tensor product, a pairing

$$E^\vee \mathbb{L}F \mathbb{L}F^\vee \mathbb{L}E \longrightarrow A.$$

Using (7.7) this is the asserted pairing on RHom complexes. Perfectness of the duality follows from the canonical isomorphisms

$$\mathrm{RHom}(F^\vee \mathbb{L}E, A) \simeq \mathrm{RHom}(F^\vee, E^\vee) \simeq F^{\vee\vee} \mathbb{L}E^\vee \simeq F \mathbb{L}E^\vee,$$

with the usual compatibility check; alternatively, reduce locally to $E = F = A$.

Exercise 7.9. For $E, F \in \mathrm{Ob} D(\underline{S})_{\mathrm{parf}}$, show that there is a canonical isomorphism

$$(E \mathbb{L}F)^\vee \simeq E^\vee \mathbb{L}F^\vee.$$

8. Traces and cup-products

8.0

In this section we generalize to perfect complexes the trace of an endomorphism of a finitely generated projective module. Applications are given in SGA 5, Expose III. The hypotheses and notation are those of (7.0).

8.1. Let $E \in \mathrm{Ob} D(\underline{S})_{\mathrm{parf}}$. There is a natural morphism

$$\mathrm{tr} : \mathrm{RHom}(E, E) \longrightarrow A,$$

obtained by composing the inverse of the isomorphism (7.7),

$$\mathrm{RHom}(E, E) \simeq E^\vee \mathbb{L}E,$$

with the canonical map $E^\vee \mathbb{L}E \rightarrow A$ of (7.5). For $U \in \mathrm{Ob} \underline{S}$, applying $H^0(U, -)$ gives an $H^0(U, A)$ -module homomorphism

$$\mathrm{Tr}_{E|U} : \mathrm{Hom}_{D(\underline{S})}(E|U, E|U) \longrightarrow H^0(U, A),$$

called the trace homomorphism. The finite Tor-dimension hypothesis on E is in fact unnecessary for defining this trace; see (8.5).

8.1.1. If $E = A$, then $\mathrm{Tr}_{A|U} : \mathrm{Hom}(A|U, A|U) \rightarrow H^0(U, A)$ is the usual isomorphism. If E and F are direct summands of finite free modules, and if an endomorphism of $(E \oplus F)|U$ is represented by a block matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$\mathrm{Tr}_{E \oplus F|U}(u) = \mathrm{Tr}_{E|U}(a) + \mathrm{Tr}_{F|U}(d).$$

8.1.2. Let E be a strictly perfect complex, let $U \in \mathrm{Ob} \underline{S}$, and let $u : E|U \rightarrow E|U$ be a morphism of complexes with components $u^i : E^i|U \rightarrow E^i|U$. Then

$$\mathrm{Tr}_{E|U}(u) = \sum_i (-1)^i \mathrm{Tr}_{E^i|U}(u^i).$$

Proof. The question is local. One may suppose that u is represented by a section of $H^0(E^\vee \otimes E)$, and even by a sum of elementary tensors $\sum_i x_i \otimes x'_i$, where x_i and x'_i are sections of E^i and of $E^{\vee i}$. Writing explicitly the map $E^\vee \otimes E \rightarrow A$, one obtains

$$\mathrm{Tr}_{E|U}(u) = \sum_i (-1)^i \langle x_i, x'_i \rangle = \sum_i (-1)^i \mathrm{Tr}_{E^i|U}(u^i).$$

8.2. Let $E, F \in \mathrm{Ob} D(\underline{S})_{\mathrm{parf}}$. By (7.8) there is a canonical homomorphism

$$\cup : \mathrm{RHom}(E, F) \mathbb{L} \mathrm{RHom}(F, E) \longrightarrow A.$$

For $U \in \mathrm{Ob} \underline{S}$, and for

$$u \in H^0(U, \mathrm{RHom}(E, F)), \quad v \in H^0(U, \mathrm{RHom}(F, E)),$$

the image under $H^0(\cup)$ defines an $H^0(U, A)$ -bilinear pairing

$$\mathrm{Hom}_{D(\underline{S})}(E|U, F|U) \otimes_{H^0(U, A)} \mathrm{Hom}_{D(\underline{S})}(F|U, E|U) \longrightarrow H^0(U, A),$$

called the cup-product and denoted $\langle u, v \rangle$.

Proposition (8.3). *In the situation of (8.2), for $u : E|U \rightarrow F|U$ and $v : F|U \rightarrow E|U$ one has*

$$\mathrm{Tr}_{E|U}(vu) = \mathrm{Tr}_{F|U}(uv) = \langle u, v \rangle.$$

In particular, for $E = F$,

$$\mathrm{Tr}_{E|U}(u) = \langle u, \mathrm{Id}_E \rangle = \langle \mathrm{Id}_E, u \rangle.$$

Proof. It is enough, for notation, to take $U = S$. The maps u and v may be viewed as

$$u : A \rightarrow E^\vee \mathbb{L} F, \quad v : A \rightarrow F^\vee \mathbb{L} E.$$

The composites giving vu and uv are obtained by inserting the corresponding evaluation morphisms. The result follows from the commutativity of the standard diagrams comparing the two possible ways of evaluating

$$E^\vee \mathbb{L} F \mathbb{L} F^\vee \mathbb{L} E$$

and the analogous expression with E and F interchanged. This verification is purely formal.

Proposition (8.4). *Let E, F, U be as in (8.2). Let $s : E|U \rightarrow F|U$ and let $s' : E|U \rightarrow F|U$ be an isomorphism. Then*

$$\mathrm{Tr}_{F|U}(s's^{-1}) = \mathrm{Tr}_{E|U}(s^{-1}s').$$

Proof. This is immediate from (8.3).

8.5. Let $\mathcal{E}(\underline{C}_0)$ be the $\underline{S}$ -fibred category whose fibre over U has objects the pairs (L, u) with $L \in \mathrm{Ob} \underline{C}_0$ and $u \in \mathrm{End}_{\underline{C}_0}(L)$, and whose morphisms $(L, u) \rightarrow (M, v)$ are the morphisms $f : L \rightarrow M$ satisfying $fu = vf$. By (8.4), the trace may be viewed as an $\underline{S}$ -functor from $\mathcal{E}(\underline{C}_0)^{\mathrm{iso}}$ to A , where the superscript means that only isomorphisms are kept and A is regarded as the fibred category with discrete fibres. Since every perfect complex is locally of finite Tor-dimension, the forgetful $\underline{S}$ -functor

$$\mathcal{E}(\underline{C}_0)^{\mathrm{iso}} \longrightarrow \mathcal{E}(D(\underline{S})_{\mathrm{parf}})^{\mathrm{iso}}$$

induces an isomorphism on the associated stacks. Hence the trace homomorphism extends to perfect complexes without assuming finite Tor-dimension.

Bibliography

- [A] Hartshorne, R. *Residues and Duality*, Lecture Notes in Mathematics, no. 20, Springer, 1966.
- [B] Verdier, J.-L. *Catégories dérivées*, I.H.E.S., 1963.
- [C] Giraud, J. *Cohomologie non abélienne*, to appear with North-Holland.
- [D] *Cohomologie étale des schémas*, by M. Artin, A. Grothendieck, and J.-L. Verdier, to appear with North-Holland.
- [E] *Cohomologie l -adique et fonctions L* , by A. Grothendieck, to appear with North-Holland.
- [F] Godement, R. *Topologie algébrique et théorie des faisceaux*, Hermann, 1958.
- [G] Cartan, H. and Eilenberg, S. *Homological Algebra*, Princeton, 1956.

SGA 6, Exposé II and following sections

1. General globalization criteria

1.0

In this section we fix a site $\underline{S}$, a flat abelian $\underline{S}$ -category $\underline{C}$ in the sense of I, 1.1, and a strictly full additive sub- $\underline{S}$ -category $\underline{C}_0$. We suppose that $\underline{C}_0$ is locally quasi-liftable in $\underline{C}$ and locally quasi-stable under kernels of epimorphisms.

Proposition (1.1). *Let $S \in \text{Ob } \underline{S}$, and assume:*

- (i) $\underline{C}_{0S}$ is quasi-liftable in $\underline{C}_S$;
- (ii) every object of $\underline{C}_S$ of $\underline{C}$ -finite type, respectively of finite $\underline{C}_0$ -presentation, is of $\underline{C}_{0S}$ -finite type.

Then:

- a) $\underline{C}_{0S}$ is quasi-stable under kernels of epimorphisms.
- b) If $E \in \text{Ob } \mathcal{D}(\underline{C}_S)$ is n - $\underline{C}_0$ -pseudo-coherent, then E is n - $\underline{C}_{0S}$ -pseudo-coherent, respectively $(n+1)$ - $\underline{C}_{0S}$ -pseudo-coherent. Equivalently,

$$\mathcal{D}(\underline{C}_S)_{n-\underline{C}_0\text{-coh}} \subset \mathcal{D}(\underline{C}_S)_{n-\underline{C}_{0S}\text{-coh}}$$

with the corresponding $(n+1)$ -variant.

- c) An object $E \in \mathcal{D}(\underline{C}_S)$ is pseudo-coherent relative to $\underline{C}_0$ if and only if it is pseudo-coherent relative to $\underline{C}_{0S}$; thus

$$\mathcal{D}(\underline{C}_S)_{\underline{C}_0\text{-coh}} = \mathcal{D}(\underline{C}_S)_{\underline{C}_{0S}\text{-coh}}.$$

Moreover the canonical functor

$$K(\underline{C}_{0S})/K_{\underline{C}_S}^{\underline{C}_{0S}-\emptyset} \longrightarrow \mathcal{D}(\underline{C}_S)$$

is fully faithful and has essential image $\mathcal{D}(\underline{C}_S)_{\underline{C}_{0S}\text{-coh}}$.

If $\underline{C}_{0S}$ is liftable in $\underline{C}_S$, then $K_{\underline{C}_S}^{\underline{C}_{0S}-\emptyset}$ is reduced to zero and

$$K(\underline{C}_{0S}) \longrightarrow \mathcal{D}(\underline{C}_S)_{\underline{C}_{0S}\text{-coh}}$$

is fully faithful with the same essential image.

Proof. For a), let

$$0 \longrightarrow L^0 \longrightarrow L^1 \longrightarrow L^2 \longrightarrow \dots$$

be an exact sequence in $\underline{C}_{0S}$ with all L^i in $\underline{C}_{0S}$. One has to prove that L is of $\underline{C}_{0S}$ -finite type. By I, 2.2 it is $\underline{C}_S$ -pseudo-coherent, hence by I, 2.9 of finite $\underline{C}_S$ -presentation, and the assertion follows from (ii).

For b), one argues by descending induction on i for $i \geq n$ (respectively $i \geq n + 1$). After replacing E by a strictly pseudo-coherent representative and taking the usual cone, one reduces to a complex with $E^i = 0$ above the critical degree. By I, 2.10, the relevant cohomology object is of $\underline{C}_0$ -finite type, respectively of finite $\underline{C}_0$ -presentation; by I, 1.4 it is then of $\underline{C}_{0S}$ -finite type. Applying I, 2.10 again gives the required pseudo-coherence. Part c) follows from b) and I, 2.7.

Proposition (1.2). *Assume that $\underline{C}_0$ is locally liftable in $\underline{C}$ and locally stable under kernels of epimorphisms, and keep the hypotheses of 1.1. Let $\underline{C}_{0,\text{loc}}$ be the strictly full subcategory of $\underline{C}$ consisting of objects locally lying in $\underline{C}_0$. Then:*

a) $\underline{C}_{0,\text{loc}}$ is stable under kernels of epimorphisms and quasi-liftable in $\underline{C}$.

b) For $E \in \text{Ob } \mathcal{D}(\underline{C}_*)$ and an interval $[m, n] \subset \mathbb{Z}$, E has $\underline{C}_0$ -perfect amplitude contained in $[m, n]$ if and only if it is isomorphic in $\mathcal{D}(\underline{C}_*)$ to a complex F with $F^i = 0$ outside $[m, n]$, with $F^i \in \underline{C}_{0,\text{loc}}$ for $i > m$, and with $F^m \in \underline{C}_{0,\text{loc}}$.

c) The canonical functor

$$K^b(\underline{C}_{0,\text{loc}})/K^{b,\emptyset}(\underline{C}_{0,\text{loc}}) \longrightarrow \mathcal{D}(\underline{C}_*)$$

is fully faithful and has essential image the full subcategory of complexes of finite $\underline{C}_0$ -perfect amplitude.

d) If $\underline{C}_0$ is liftable, one may replace in b) “is isomorphic to F in $\mathcal{D}(\underline{C}_*)$ ” by “there exists a quasi-isomorphism $F \rightarrow E$ ”, and $K^{b,\emptyset}(\underline{C}_{0,\text{loc}})$ is zero.

Proof. Stability of $\underline{C}_{0,\text{loc}}$ under kernels of epimorphisms is clear. Since objects of $\underline{C}_{0S}$ are $\underline{C}_S$ -pseudo-coherent, hence $\underline{C}$ -pseudo-coherent by 1.1 c), quasi-liftability follows from I, 2.14. For b), only necessity has to be proved. From 1.1 c) one already knows that E is $\underline{C}_{0,\text{loc}}$ -pseudo-coherent. Since E is acyclic above n , one may, after replacing it by an isomorphic complex, suppose it strictly $(m + 1)$ - $\underline{C}_{0,\text{loc}}$ -pseudo-coherent and zero above n . Since E has perfect amplitude in $[m, n]$, I, 4.16 implies that E^m is locally an object of $\underline{C}_0$, hence of $\underline{C}_{0,\text{loc}}$. Part c) follows from a), b), and I, 4.12.2; part d) is immediate from I, 4.6.

2. Application to certain categories of modules

2.0. Notation

Let $(\underline{S}, \underline{A})$ be a ringed topos, where $\underline{A}$ is a sheaf of rings, not necessarily commutative. We write $\underline{C}$ for the $\underline{S}$ -category of left $\underline{A}$ -modules and $\underline{C}_0$ for the sub- $\underline{S}$ -category of finite free left $\underline{A}$ -modules. Then $\underline{C}$ is flat and $\underline{C}_0$ is locally liftable and locally stable under kernels.

2.1. Lifting lemmas

Lemma (2.1.1). *Let A be a ring and let C be the category of left A -modules. Let C_0 be the full additive subcategory of finite free modules. Then:*

- a) C_0 is liftable in C .
- b) C_0 is stable under kernels of epimorphisms if and only if every finite left A -module is finitely presented.

The proof is the elementary lifting argument for free finite modules; b) is the usual kernel criterion for finite presentation.

Lemma (2.1.2). *Let $(\underline{S}, \underline{A})$ be a ringed topos, $\underline{C}$ the $\underline{S}$ -category of left $\underline{A}$ -modules, and $\underline{C}_0$ the subcategory of finite free modules. Let $S \in \text{Ob } \underline{S}$, and let $\underline{C}'_S$ be a strictly full additive subcategory of $\underline{C}_S$ containing $\underline{C}_{0S}$ and stable under kernels, cokernels, and extensions. Suppose that*

- (i) $\text{Ext}^1_{\underline{C}_S}(G, F) = 0$ for all $F \in \underline{C}'_S$ and all $G \in \underline{C}_{0S}$;
- (ii) every extension of an object $F \in \underline{C}'_S$ by an object $G \in \underline{C}_S$ such that $\text{Hom}(G, \underline{A})$ lies in $\underline{C}_{0S}$ and G is locally a direct factor of a finite free module is split.

Then $\underline{C}'_S$ is liftable in $\underline{C}_S$.

Lemma (2.1.3). *Let A and B be abelian categories and let $\theta : A \rightarrow B$ be an exact functor.*

- a) *For $M \in A$, the following conditions are equivalent:*

- (i) *for every $L \in A$, the canonical maps*

$$\text{Ext}_A^n(L, M) \longrightarrow \text{Ext}_B^n(\theta L, \theta M)$$

are isomorphisms for all n ;

- (ii) *for every $L \in A$, the canonical map*

$$\text{Hom}_A(L, M) \longrightarrow \text{Hom}_B(\theta L, \theta M)$$

is an isomorphism, and for every $n \geq 1$ and every $x \in \text{Ext}_A^n(L, M)$ there is an epimorphism $L' \twoheadrightarrow L$ which kills x .

- b) *The following are equivalent:*

- (i) *for all $L, M \in A$ and all n , the canonical map on Ext^n is an isomorphism;*

- (ii) *the canonical functor*

$$\mathcal{D}^*(A) \longrightarrow \mathcal{D}^*(B)$$

is fully faithful and has essential image the full subcategory of complexes F such that every $H^i(F)$ lies in the essential image of θ .

Proof. For a), the implication (i) implies (ii) because a class in $\mathrm{Ext}_A^n(L, M)$ is represented, in Yoneda's interpretation, by an exact sequence

$$0 \rightarrow M \rightarrow E_n \rightarrow \cdots \rightarrow E_1 \rightarrow L \rightarrow 0,$$

and the epimorphism $E_1 \rightarrow L$ kills the class. Conversely, set

$$F^n(L) = \mathrm{Ext}_A^n(L, M), \quad G^n(L) = \mathrm{Ext}_B^n(\theta L, \theta M).$$

Effaceability gives both $F^n(L)$ and $G^n(L)$, for $n \geq 1$, as filtered colimits of the cokernels associated with short exact sequences

$$0 \rightarrow L'' \rightarrow L' \rightarrow L \rightarrow 0.$$

Since $F^0 \rightarrow G^0$ is an isomorphism, induction on n proves that $F^n \rightarrow G^n$ is an isomorphism for all n . Part b) follows by the standard devissage in the derived category. The dual formulation is obtained by reversing arrows; if the categories have enough injectives, the same conditions are equivalent to the corresponding assertion for RHom on $\mathcal{D}^- \times \mathcal{D}^+$.

2.2. Noetherian schemes and divisorial schemes

Definition (2.2.1). *For a scheme S , let $\mathrm{Qcoh}(S)$ and $\mathrm{Coh}(S)$ denote the categories of quasi-coherent and coherent $\mathcal{O}_S$ -modules. For $E \in \mathcal{D}(\mathrm{Qcoh}(S))$, pseudo-coherence, perfect amplitude, and perfection are understood relative to the fibred category on the Zariski site of S given by $U \mapsto \mathrm{Loclib}(U)$. We write $\mathcal{D}(\mathrm{Qcoh}(S))_{\mathrm{pcoh}}$ and $\mathcal{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$ for the corresponding full triangulated subcategories.*

Proposition (2.2.2). *Let S be a noetherian scheme. The canonical functor*

$$\mathcal{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathcal{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has essential image $\mathcal{D}(\mathrm{Qcoh}(S))_{\mathrm{pcoh}}$.

Proof. Apply 1.1 on the Zariski site of S , with $\underline{C}(U) = \mathrm{Qcoh}(U)$ and $\underline{C}_0(U) = \mathrm{Coh}(U)$. A coherent object is pseudo-coherent, so pseudo-coherence in $\mathcal{D}(\mathrm{Qcoh}(U))$ is the same as pseudo-coherence relative to $\underline{C}_0$. The two hypotheses of 1.1 are the ordinary facts that a quotient of a coherent module is coherent and that every quasi-coherent module on a noetherian scheme is the filtered direct limit of its coherent submodules.

Corollary (2.2.2.1). *For S noetherian, respectively noetherian of finite dimension, the canonical functor*

$$\mathcal{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathcal{D}^-(S) \quad \text{respectively} \quad \mathcal{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathcal{D}^b(S)$$

is fully faithful and has essential image $\mathcal{D}^-(S)_{\mathrm{pcoh}}$, respectively $\mathcal{D}^b(S)_{\mathrm{pcoh}}$.

Proposition (2.2.3). *Let S be quasi-compact and quasi-separated, and let $(\mathcal{L}_i)_{i \in I}$ be a family of*

invertible sheaves. For an $\mathcal{O}_S$ -module F and $(i, n) \in I \times \mathbb{Z}$, let $F_{(i,n)}$ be the submodule of $F \otimes \mathcal{L}_i^{\otimes n}$ generated by its global sections. The following conditions are equivalent:

- (i) the opens S_f , where $f \in \Gamma(S, \mathcal{L}_i^{\otimes n})$, $i \in I$, $n \in \mathbb{N}$, form a basis for the topology of S ;
- (ii) those among the S_f which are affine cover S ;
- (iii) every quasi-coherent finite type $\mathcal{O}_S$ -module F admits an epimorphism

$$\bigoplus_{(i,n)} \mathcal{L}_i^{\otimes -n} \longrightarrow F;$$

- (iv) for every finite type quasi-coherent F and finite type quasi-coherent submodule $F' \subset F$, there are $(i, n) \in I \times \mathbb{N}^+$ and $u : \mathcal{L}_i^{\otimes -n} \rightarrow F$ with $\text{Im}(u) \not\subset F'$.

Proof. This is a paraphrase of EGA II, 4.5.2 and 4.5.5. The implication from the existence of affine basic opens to the epimorphism is obtained by extending finitely many local generators after multiplying them by sufficiently high powers of the chosen sections. Conversely, applying the generator condition to ideals defining complements of open neighbourhoods gives the required basic opens.

Definition (2.2.4). Such a family $(\mathcal{L}_i)$ is called *ample*. A scheme is called *divisorial* if it admits an ample family of invertible modules.

Lemma (2.2.6). Let X be locally noetherian and let $U \subset X$ be an open such that, at every maximal point x of $X - U$, one has $\dim \mathcal{O}_{X,x} \leq 1$. Then U contains the maximal points of X .

Proof. Put $Y = X - U$ and localize at a maximal point x of Y . The pullback Y' is the closed point of $X' = \text{Spec } \mathcal{O}_{X,x}$, and U' is affine. If the dimension of X' were at least two, local cohomology would give $H_{Y'}^1(X', \mathcal{O}_{X'}) \neq 0$, while affineness of U' forces the relevant higher local cohomology to vanish, a contradiction.

Proposition (2.2.7). Every noetherian separated locally factorial scheme is divisorial. In particular every noetherian regular separated scheme is divisorial.

Proof. After reducing to the integral case, choose a finite affine open cover (S_i) . By the lemma, $S - S_i$ is purely of codimension one, hence the support of an effective divisor. Thus there are invertible sheaves $\mathcal{L}_i$ and sections f_i such that $S_{f_i} = S_i$. The family $(\mathcal{L}_i)$ is ample.

The separatedness hypothesis is essential: the affine plane with the origin doubled has only trivial invertible sheaves, and global functions cannot separate the two origins.

Lemma (2.2.8). Let S be divisorial, let $(\mathcal{L}_i)$ be an ample family, and let $\underline{L}(S)$ be the strictly full additive subcategory of $\text{Qcoh}(S)$ generated by the $\mathcal{L}_i^{\otimes -n}$, $(i, n) \in I \times \mathbb{N}^+$. For $E \in \mathcal{D}(\text{Qcoh}(S))$:

- a) $\underline{L}(S)$ and $\text{Loclib}(S)$ are quasi-liftable in $\text{Qcoh}(S)$ and quasi-stable under kernels of epimorphisms.

- b) E is n -pseudo-coherent, respectively pseudo-coherent, if and only if it is so relative to $\underline{L}(S)$ or to $\text{Loclib}(S)$. The canonical functors

$$K^-(\underline{L}(S))/K^{-,\emptyset}(\underline{L}(S)) \rightarrow \mathcal{D}(\text{Qcoh}(S))$$

and

$$K^-(\text{Loclib}(S))/K^{-,\emptyset}(\text{Loclib}(S)) \rightarrow \mathcal{D}(\text{Qcoh}(S))$$

are fully faithful and have essential image $\mathcal{D}^-(\text{Qcoh}(S))_{\text{pcoh}}$.

- c) E has perfect amplitude contained in $[m, n]$ if and only if it is isomorphic in $\mathcal{D}(\text{Qcoh}(S))$ to a complex F with $F^i = 0$ for $i \notin [m, n]$, with $F^i \in \underline{L}(S)$ for $i > m$, and with F^m locally free of finite type. Moreover

$$K^b(\text{Loclib}(S))/K^{b,\emptyset}(\text{Loclib}(S)) \rightarrow \mathcal{D}(\text{Qcoh}(S))$$

is fully faithful with essential image $\mathcal{D}(\text{Qcoh}(S))_{\text{parf}}$.

Proof. Given an epimorphism $F \rightarrow G$ with G of finite type, take a finite type quasi-coherent submodule $F' \subset F$ still mapping epimorphically to G , then cover F' by a sum of negative powers of the ample invertible sheaves. This proves quasi-liftability. The remaining assertions follow from 1.1 applied on the Zariski site, using that objects of $\underline{L}$ and of Loclib are pseudo-coherent and using the ampleness criterion above.

Proposition (2.2.9). *Assume in addition that S is separated or noetherian. Then:*

- a) *The canonical functors*

$$K^{-,b}(\underline{L}(S))/K^{-,b,\emptyset}(\underline{L}(S)) \rightarrow \mathcal{D}^-(S), \quad K^{-,b}(\text{Loclib}(S))/K^{-,b,\emptyset}(\text{Loclib}(S)) \rightarrow \mathcal{D}^-(S)$$

are fully faithful with essential image $\mathcal{D}^-(S)_{\text{pcoh}}$. If S has finite cohomological dimension, the analogous bounded functors are equivalences.

- b) *The canonical functor*

$$K^{-,b}(\text{Loclib}(S))/K^{-,b,\emptyset}(\text{Loclib}(S)) \rightarrow \mathcal{D}(S)$$

is fully faithful and has essential image $\mathcal{D}(S)_{\text{parf}}$; the same explicit description of perfect amplitude as in 2.2.8 holds in $\mathcal{D}(S)$.

This is an immediate consequence of 2.2.8 together with the comparison between $\mathcal{D}(\text{Qcoh}(S))$ and $\mathcal{D}(S)$ in the cases considered.

Remark (2.2.10). *If $S = \text{Spec } A$ is affine, the single sheaf $\mathcal{O}_S$ is ample and $\underline{L}(S) = \text{Lib}(S)$. Thus $R\Gamma$ identifies the pseudo-coherent and perfect pieces of $\mathcal{D}(S)$ with the corresponding derived categories of A -modules defined by finite free and finite projective modules. In particular, a pseudo-coherent complex with bounded cohomology is locally isomorphic in $\mathcal{D}(S)$ to a strictly pseudo-coherent complex.*

3. End of the comparison with quasi-coherent complexes

The coherator, being of finite cohomological dimension, admits a right derived functor

$$\underline{R}Q : \mathcal{D}(S) \longrightarrow \mathcal{D}(\mathrm{Qcoh}(S)).$$

For $E \in \mathcal{D}(\mathrm{Qcoh}(S))$ there is a functorial canonical isomorphism

$$E \xrightarrow{\sim} \underline{R}Q(E),$$

because the terms are Q -acyclic and satisfy $E^i \xrightarrow{\sim} Q(E^i)$. For $E \in \mathcal{D}(S)$ there is a canonical morphism

$$\underline{R}Q(E) \longrightarrow E.$$

These morphisms make $\underline{R}Q$ a right adjoint to the canonical functor

$$\phi : \mathcal{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathcal{D}(S).$$

Since the first morphism is an isomorphism, ϕ is fully faithful. The spectral sequence

$$R^pQ(H^q(E)) \Rightarrow R^*Q(E)$$

shows that the second morphism is an isomorphism for $E \in \mathcal{D}(S)_{\mathrm{Qcoh}}$. Hence the essential image of ϕ is $\mathcal{D}(S)_{\mathrm{Qcoh}}$.

SGA 6 – Expose III, continued

Applications: Base Change and Semi-continuity Theorems

5. Applications: base change and semi-continuity theorems

Lemma (5.1). *Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos, let $\mathbf{Mod}(X)$ be the category of $\mathcal{O}_X$ -modules, and let $E \in \mathcal{D}^-(X)$ and $p \in \mathbb{Z}$. Consider:*

- a) $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ is right exact on $\mathbf{Mod}(X)$;
- b) the canonical map $H^p(E) \otimes M \rightarrow H^p(E \otimes^{\mathbf{L}} M)$ is an isomorphism for all M ;
- c) $M \mapsto H^{p+1}(E \otimes^{\mathbf{L}} M)$ is left exact;
- d) the canonical map

$$H^{p+1}(E \otimes^{\mathbf{L}} M) \rightarrow \mathrm{Hom}(H^{-p-1}(E^\vee), M)$$

is an isomorphism for all M .

Then $a \Leftrightarrow b \Leftrightarrow c \Leftarrow d$. If E is pseudo-coherent, all four conditions are equivalent.

The map in d) is obtained from the evaluation $E^\vee \otimes^{\mathbf{L}} E \rightarrow \mathcal{O}_X$, hence

$$E \otimes^{\mathbf{L}} M \rightarrow \mathrm{RHom}(E^\vee, M),$$

followed on cohomology by

$$\mathrm{Ext}^{p+1}(E^\vee, M) \rightarrow \mathrm{Hom}(H^{-p-1}(E^\vee), M).$$

Proof. The implications $b \Rightarrow a \Leftrightarrow c \Leftarrow d$ are formal. To prove $a \Rightarrow b$, reduce by a presentation of M to the case $M = \mathcal{O}_{U,X}$. Both sides then identify with extension by zero from U , and exactness of $i_!$ gives the result. If E is pseudo-coherent, it suffices for d to test injective M . For such M the last displayed map on Ext is an isomorphism, and the assertion follows from the fact that

$$E \otimes^{\mathbf{L}} M \rightarrow \mathrm{RHom}(E^\vee, M)$$

is an isomorphism for pseudo-coherent E and injective M , after localization, truncation, and devissage to the case $E = \mathcal{O}_X$.

Remark (5.2). *These conditions are local on the topos. If X has enough points, they may be checked stalkwise. For a complex represented by flat terms, the criterion recovers the usual EGA III criterion: right exactness is equivalent to flatness of a suitable cokernel. In particular, for a*

perfect complex the condition for E at degree p is dual to the corresponding condition for $E^\vee$ at degree $-p-1$.

For a quasi-separated scheme X and a complex with quasi-coherent locally bounded cohomology, the same assertions hold after restricting to quasi-coherent modules: the quasi-coherent versions of the four conditions are equivalent in the same way, and for pseudo-coherent E all are equivalent.

Proposition (5.3). *Let $f : X \rightarrow Y$ be a morphism of quasi-compact and quasi-separated schemes, let $E \in \mathcal{D}^b(X)$ have quasi-coherent cohomology and finite flat amplitude relative to f , and let $p \in \mathbb{Z}$. Consider:*

a) $M \mapsto R^p f_*(E \otimes_Y^{\mathbf{L}} M)$ is right exact on $\mathrm{Qcoh}(Y)$;

b) the canonical map

$$R^p f_*(E) \otimes M \rightarrow R^p f_*(E \otimes_Y^{\mathbf{L}} M)$$

is an isomorphism for all quasi-coherent M ;

c) $M \mapsto R^{p+1} f_*(E \otimes_Y^{\mathbf{L}} M)$ is left exact;

d) the canonical map

$$R^{p+1} f_*(E \otimes_Y^{\mathbf{L}} M) \rightarrow \mathrm{Hom}(H^{-p-1}(Rf_*(E)^\vee), M)$$

is an isomorphism for all quasi-coherent M ;

e) for every base change $Y' \rightarrow Y$, the canonical map

$$R^p f_*(E) \otimes_Y \mathcal{O}_{Y'} \rightarrow H^p(Rf_*(E) \otimes_Y^{\mathbf{L}} \mathcal{O}_{Y'})$$

is an isomorphism.

Then $a \Leftrightarrow b \Leftrightarrow c \Leftrightarrow e \Leftarrow d$. If f is proper, E is pseudo-coherent relative to f , and the finiteness theorem applies to f , these conditions are all equivalent.

Proof. The implications involving a), b), c), and d) follow from the projection formula and Lemma 5.1. It remains to compare b) and e). This follows from the following base-change form: if X is quasi-separated and $E \in \mathcal{D}^-(X)$ has locally bounded quasi-coherent cohomology, then the condition of Lemma 5.1 a) is equivalent to requiring

$$H^p(E) \otimes_X \mathcal{O}_{X'} \rightarrow H^p(E \otimes_X^{\mathbf{L}} \mathcal{O}_{X'})$$

to be an isomorphism for every $X' \rightarrow X$. The proof is affine: write $X = \mathrm{Spec} A$, $X' = \mathrm{Spec} A'$, and $E = F^\sim$, and reduce to

$$H^p(F) \otimes_A A' \rightarrow H^p(F \otimes_A^{\mathbf{L}} A').$$

Conversely, for a quasi-coherent module N take $A' = D_A(N)$.

If E is pseudo-coherent, the same argument applied to $E^\vee$ gives the compatibility of

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}((E \otimes_X^\mathbf{L} \mathcal{O}_{X'})^\vee),$$

which completes the proof.

Remark (5.3.2). *The proposition applies in particular when E is a bounded complex of quasi-coherent flat $\mathcal{O}_X$ -modules of finite presentation relative to f , since such a complex has finite perfect amplitude relative to f .*

Definition (5.4). *Let X be a topos, $\mathrm{Pt}(X)$ the set of its points, M an ordered set, and $f : \mathrm{Pt}(X) \rightarrow M$. We say that f is upper semi-continuous at x (respectively lower semi-continuous, respectively locally constant) if there exists an x -pointed object U of X such that, for every point y of U , one has $f(y) \leq f(x)$ (respectively $f(y) \geq f(x)$, respectively $f(y) = f(x)$). The global notions mean the corresponding pointwise conditions.*

If $s \in H^0(X, M)$ is a section of the constant sheaf, then $x \mapsto x^*s$ is locally constant. In a locally ringed topos, a perfect complex has a locally constant rank. If P is a finite type module, then

$$x \longmapsto \mathrm{rank}_{k(x)}(P_x \otimes_{\mathcal{O}_{X,x}} k(x))$$

is upper semi-continuous, by Nakayama's lemma.

Lemma (5.5). *Let $(X, \mathcal{O}_X)$ be a locally ringed topos, let $E \in \mathcal{D}^-(X)$, and let $p \in \mathbb{Z}$.*

(i) *If E is perfect, then the vector spaces $H^i(E_x \otimes_{\mathcal{O}_{X,x}}^\mathbf{L} k(x))$ are finite-dimensional and zero except for finitely many i , and*

$$x \mapsto \sum_i (-1)^i \mathrm{rank}_{k(x)} H^i(E_x \otimes_{\mathcal{O}_{X,x}}^\mathbf{L} k(x))$$

is locally constant.

(ii) *If E is $(p-1)$ -pseudo-coherent, then $H^p(E_x \otimes_{\mathcal{O}_{X,x}}^\mathbf{L} k(x))$ is finite-dimensional and its dimension is upper semi-continuous. If moreover $M \mapsto H^p(E \otimes_{\mathcal{O}_X}^\mathbf{L} M)$ is exact, then $H^p(E)$ is locally free of finite type and the same function is locally constant.*

(iii) *If E is p -pseudo-coherent, respectively $(p-1)$ -pseudo-coherent, and if the stalk functor $M \mapsto H^p(E_x \otimes_{\mathcal{O}_{X,x}}^\mathbf{L} M)$ is right exact, respectively left exact, then this exactness holds on a suitable x -pointed neighbourhood.*

Proof. For (i), use the compatibility of the rank of a perfect complex with inverse images. For (ii), after truncating and shifting, reduce to a strictly perfect complex concentrated in degrees $[-1, 1]$. The Euler characteristic identity reduces upper semi-continuity of H^0 to that of the neighbouring

cohomology sheaves and to local constancy of rank. Exactness gives flatness, hence finite local freeness, of the relevant cokernels, and the exact sequence

$$0 \rightarrow H^p(E) \rightarrow Z'^p(E) \rightarrow E^{p+1} \rightarrow Z'^{p+1}(E) \rightarrow 0$$

then shows that $H^p(E)$ is locally free of finite type. Part (iii) is the same openness argument applied to the flatness of Z'^p or Z'^{p+1} at the chosen point.

Corollary (5.6). *Let X be a scheme, $p \in \mathbb{Z}$, and $E \in \mathcal{D}^-(X)$ a $(p-1)$ -pseudo-coherent complex. Then:*

(i) $H^p(E \otimes_X^{\mathbf{L}} k(x))$ is finite-dimensional over $k(x)$ and

$$x \mapsto \text{rank}_{k(x)} H^p(E \otimes_X^{\mathbf{L}} k(x))$$

is constructible and upper semi-continuous on X .

(ii) If X is quasi-separated and E has locally bounded quasi-coherent cohomology, exactness of $M \mapsto H^p(E \otimes_X^{\mathbf{L}} M)$ on quasi-coherent modules implies that the preceding function is locally constant and that $H^p(E)$ is locally free of finite type. Conversely, if X is locally noetherian and reduced and the function is locally constant, the functor is exact on quasi-coherent modules.

Proof. On each irreducible component the generic point satisfies the exactness condition, hence exactness holds on a non-empty open neighbourhood and the rank is locally constant there. This gives constructibility; the remaining assertions follow from Lemma 5.5 and from the classical EGA III criterion after localization and truncation.

Proposition (5.7). *Let $f : X \rightarrow Y$ be a proper morphism of quasi-compact schemes, let $E \in \mathcal{D}^b(f)_{\text{parf}}$, and let $p \in \mathbb{Z}$. Assuming the finiteness theorem applies to f – for instance if f is projective or Y is noetherian – one has:*

(i) the $H^i(Rf_* E \otimes_Y^{\mathbf{L}} k(y))$ are finite-dimensional over $k(y)$ and zero except for finitely many i , and their alternating rank is locally constant on Y ;

(ii) $y \mapsto \text{rank}_{k(y)} H^p(Rf_* E \otimes_Y^{\mathbf{L}} k(y))$ is constructible and upper semi-continuous;

(iii) if Y is quasi-separated and $M \mapsto R^p f_*(E \otimes_Y^{\mathbf{L}} M)$ is exact on quasi-coherent modules, then $R^p f_*(E)$ is locally free of finite type and the function in (ii) is locally constant. Conversely, if Y is noetherian and reduced and the function in (ii) is locally constant, the same functor is exact.

Proof. By the finiteness theorem, $Rf_* E$ is perfect. The first two assertions are therefore 5.5 (i) and 5.6 (i), and the third follows from 5.6 (ii) together with the projection formula

$$Rf_* E \otimes_Y^{\mathbf{L}} M \xrightarrow{\sim} Rf_*(E \otimes_Y^{\mathbf{L}} M).$$

Remark (5.8). *For a fibre inclusion $i_y : X_y \rightarrow X$, under the usual flatness or flat-complex hypotheses, Kunnet's formula gives*

$$H^i(Rf_*E \otimes_Y^{\mathbf{L}} k(y)) \xrightarrow{\sim} H^i(X_y, E_y),$$

*where $E_y = \mathrm{Li}_y^*E$ or i_y^*E in the flat case. An analogous analytic theory exists for complex analytic spaces; its essential inputs are Grauert's finiteness theorem for proper maps and the analytic projection formula.*

Bibliography

Grauert, H. *Ein Theorem der Analytischen Garbentheorie*, Publ. IHES, no. 5.

Hartshorne, R. *Residues and Duality*, Lecture Notes in Mathematics, no. 20, Springer, 1966.

Godement, R. *Theorie des faisceaux*, Hermann, 1958.

Grothendieck Groups of Ringed Topoi

L. Illusie

1. Reminders and generalities on Grothendieck groups

1.1

Let C be a triangulated category. An application f from $\text{Ob } C$ to an abelian group G is called additive if

$$f(E) = f(E') + f(E'')$$

for every distinguished triangle $E' \rightarrow E \rightarrow E'' \rightarrow E'[1]$. Additive maps form an abelian group functorial in C , represented by an abelian group $k(C)$ and a universal additive map

$$\text{cl}_C : \text{Ob } C \rightarrow k(C).$$

The group $k(C)$ is functorial in exact functors, and equivalent exact functors induce the same homomorphism.

Lemma (1.2). *For $L \in C$ and $n \in \mathbb{Z}$, one has*

$$\text{cl}(L[n]) = (-1)^n \text{cl}(L).$$

If $L \simeq L' \oplus L''$, then $\text{cl}(L) = \text{cl}(L') + \text{cl}(L'')$. If for some n the coproduct $\coprod_{i \geq 0} L[2ni]$ is representable, then $\text{cl}(L) = 0$.

Proof. The first assertions follow immediately from the definitions. For the last, use

$$\coprod_{i \geq 0} L[2ni] \simeq L \oplus \left(\coprod_{i \geq 0} L[2ni] \right)[2n].$$

Lemma (1.3). *If A is a thick subcategory of a triangulated category C , then the inclusion and quotient functors give an exact sequence*

$$k(A) \longrightarrow k(C) \longrightarrow k(C/A) \longrightarrow 0.$$

Lemma (1.4). *Let A be an additive, respectively abelian, category, and let f be additive on $\text{Ob } K^b(A)$, respectively $\text{Ob } \mathcal{D}^b(A)$. Then*

$$f(E) = \sum_i (-1)^i f(E^i), \quad \text{respectively} \quad f(E) = \sum_i (-1)^i f(H^i(E)).$$

1.5

Let C be abelian and let A be a full subcategory stable under finite sums. A map $f : \text{Ob } A \rightarrow G$ is additive if

$$\sum_i (-1)^i f(E^i) = 0$$

for every bounded acyclic complex E of C whose components lie in A . When A is stable under kernels of epimorphisms this is equivalent to additivity on short exact sequences with terms in A . If $K^{b,\emptyset}(A)$ denotes the thick subcategory of acyclic bounded complexes, then

$$\text{cl}_A : \text{Ob } A \rightarrow k(K^b(A)/K^{b,\emptyset}(A))$$

is universal for such additive maps. We write this group as $k(A)$, or $k(A, C)$ if the ambient abelian category must be specified. In particular $k(\mathcal{D}^b(C)) = k(C)$.

Lemma (1.6). *Let C be abelian and let A be a thick abelian subcategory. Let $\mathcal{D}_A(C)$ be the full subcategory of $\mathcal{D}(C)$ consisting of complexes whose cohomology objects lie in A . Then $\mathcal{D}_A(C)$ is triangulated, and the canonical functor $\mathcal{D}^b(A) \rightarrow \mathcal{D}_A(C)$ induces an isomorphism*

$$k(A) \xrightarrow{\sim} k(\mathcal{D}_A^b(C)).$$

Proof. Triangulatedness follows from the long exact cohomology sequence. The inverse map sends E to

$$\sum_i (-1)^i \text{cl}_A(H^i(E)),$$

which is additive, and the assertion follows from Lemma 1.4.

Definition (1.7). *For a triangulated, respectively abelian, category C , a subset $C' \subset \text{Ob } C$ is exact if, in every distinguished triangle, respectively short exact sequence, whenever two of the three terms lie in C' , so does the third. It is dense if $\text{Ob } C$ is the smallest exact subset containing C' .*

Lemma (1.8). *If $C' \subset \text{Ob } C$ is dense, then the classes $\text{cl}_C(E)$ for $E \in C'$ generate $k(C)$.*

Proof. Let $C_0 = C'$ and define C_i inductively by closing under the third term of a triangle or short exact sequence whose other two terms lie in C_{i-1} . The subgroups generated by the classes of the C_i are all equal, and the filtration is exhaustive.

Lemma (1.9). *Let C be abelian and A a full subcategory stable under finite sums, subobjects, and quotients, and suppose $\text{Ob } A$ is dense in $\text{Ob } C$. Then the homomorphism $k(A, C) \rightarrow k(C)$ defined by $E \mapsto \text{cl}_C(E)$ is an isomorphism.*

2. The functors K . and K' of a ringed topos

2.1

Let (X, A) be a ringed topos. We use the following notation: $\mathcal{D}^b({}_AX)_{\text{coh}}$ is the category of pseudo-coherent complexes with bounded cohomology of left A -modules; $\mathcal{D}({}_AX)_{\text{parf}}$ is the category of complexes of finite perfect amplitude; $\text{Loclib}({}_AX)$ is the category of left A -modules locally direct factors of finite free modules; and $\text{Coh}({}_AX)$ is the category of coherent left A -modules.

Definition (2.2). *Unless otherwise stated,*

$$K({}_AX) = k(\mathcal{D}^b({}_AX)_{\text{coh}}), \quad K'({}_AX) = k(\mathcal{D}({}_AX)_{\text{parf}}),$$

and

$$K({}_AX)_{\text{naive}} = k(\text{Coh}({}_AX)), \quad K'({}_AX)_{\text{naive}} = k(\text{Loclib}({}_AX)).$$

Thus K . is the Grothendieck group of pseudo-coherent complexes and K' the Grothendieck group of perfect complexes; the naive versions are computed directly from coherent, respectively locally free, modules.

If the final object of X is quasi-compact, the local bounded variants coincide with these groups. If A is coherent as a left A -module, the canonical map

$$K({}_AX)_{\text{naive}} \longrightarrow K({}_AX)$$

is an isomorphism, because coherent modules form a thick abelian subcategory and pseudo-coherent bounded complexes have coherent cohomology.

There is a natural homomorphism

$$K'({}_AX) \longrightarrow K({}_AX)$$

induced by inclusion of perfect complexes among pseudo-coherent bounded complexes. It is not generally injective or surjective. If the final object is quasi-compact, X has enough points, and all stalk modules have finite Tor-dimension, then

$$\mathcal{D}({}_AX)_{\text{parf}} = \mathcal{D}^b({}_AX)_{\text{coh}},$$

so this map is an isomorphism; for example this holds for compact spaces ringed by regular local rings.

2.6. Pairings

Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi. For bimodules left over A' and right over $f^{-1}A$, tensor product induces

$$\mathrm{Loclib}_{A'}(A'X'_A) \times \mathrm{Loclib}(AX) \circ \mathrm{Loclib}(A'X')$$

and hence

$$K'_{\mathrm{naive}}(A'X'_A)_{A'} \otimes K'_{\mathrm{naive}}(AX) \circ K'_{\mathrm{naive}}(A'X').$$

Derived tensor product similarly gives

$$\mathbf{L} \otimes_A : \mathcal{D}(A'X'_A)_{A' - \mathrm{parf}} \times \mathcal{D}(AX)_{\mathrm{parf}} \circ \mathcal{D}(A'X')_{\mathrm{parf}}$$

and therefore

$$K'(A'X'_A)_{A'} \otimes K'(AX) \circ K'(A'X').$$

The naive and derived pairings are compatible with the canonical maps from naive to derived groups.

2.7. Contravariance and ring structure

For a morphism of ringed topoi $f : (X', A') \rightarrow (X, A)$, the functor Lf^* sends perfect complexes to perfect complexes and gives

$$f^* : K'(AX) \circ K'(A'X')$$

with an analogous naive map. These maps satisfy $(gf)^* = f^*g^*$.

If $(X, \mathcal{O}_X)$ is commutatively ringed, the tensor product makes $K'(X)$ and $K'(X)_{\mathrm{naive}}$ into commutative unital rings, with unit $\mathrm{cl}(\mathcal{O}_X)$, and the canonical naive-to-derived map is a unital ring homomorphism. Pullback maps are ring homomorphisms.

2.8. Augmentation by rank

If $(X, \mathcal{O}_X)$ is locally ringed, rank defines an augmentation

$$\mathrm{rank} : K'(X) \rightarrow H^0(X, \mathbb{Z}),$$

compatible with the section $i : H^0(X, \mathbb{Z}) \rightarrow K'(X)_{\mathrm{naive}}$ defined by locally free modules of prescribed rank. These structures are preserved by inverse image. Later lambda-ring structures refine this augmented algebra structure.

2.9. Comparison of K' and the naive group

The canonical map

$$K'(X)_{\mathrm{naive}} \rightarrow K'(X)$$

need not be an isomorphism in general. The global resolution criteria of Expose II show that it is an isomorphism in particular when:

- (i) X is the Zariski topos of a separated or noetherian divisorial scheme;
- (ii) X is the topos of a compact space and the sheaf of rings is soft;
- (iii) X is a Stein topos with quasi-compact final object and $A = \mathcal{O}_X$.

2.10. The pairing between K' and K .

For a commutatively ringed topos,

$$\mathbf{L} \otimes : \mathcal{D}(X)_{\text{parf}} \times \mathcal{D}^b(X)_{\text{coh}} \rightarrow \mathcal{D}^b(X)_{\text{coh}}$$

defines

$$K'(X) \otimes K.(X) \rightarrow K.(X),$$

so that $K.(X)$ is a $K'(X)$ -module. If $f : X \rightarrow Y$ is a morphism, then $K.(X)$ is also a $K'(Y)$ -module via pullback.

2.11. Covariance on K . and the projection formula

Let $f : X \rightarrow Y$ be proper and pseudo-coherent between quasi-compact schemes. Assuming the finiteness conjecture for f – for example if Y is noetherian or f is projective – the functor Rf_* induces

$$f_* : K.(X) \rightarrow K.(Y).$$

It is $K'(Y)$ -linear: for $y \in K'(Y)$ and $x \in K.(X)$,

$$f_*(yx) = yf_*(x),$$

which follows from the derived projection formula

$$Rf_*(E) \otimes^{\mathbf{L}} M \xrightarrow{\sim} Rf_*(E \otimes_Y^{\mathbf{L}} M).$$

For a composable proper morphism $g : Y \rightarrow Z$ satisfying the same hypotheses, $g_*f_* = (gf)_*$.

The same statements hold for proper morphisms of finite-dimensional analytic spaces by Grauert's finiteness theorem and the analytic projection formula.

2.12. Pullback on K . and pushforward on K'

If $f : X \rightarrow Y$ has finite Tor-dimension, then pullback defines

$$f^* : K.(Y) \rightarrow K.(X), \quad (gf)^* = f^*g^*.$$

If, in addition, one is in the proper finite-type situation above, then Rf_* sends perfect complexes to perfect complexes and gives

$$f_* : K'(X) \rightarrow K'(Y).$$

This is not usually a ring homomorphism, but it satisfies the projection formula

$$f_*(f^*(y)x) = yf_*(x)$$

for $y \in K(Y)$, respectively $K'(Y)$, and $x \in K'(X)$. For separated divisorial schemes this construction yields the previously non-obvious homomorphism

$$f_* : K'(X)_{\text{naive}} \rightarrow K'(Y)_{\text{naive}},$$

which is one of the main motivations for the finiteness and resolution formalism of the preceding exposes.

3. Complements on Grothendieck groups of schemes

Proposition (3.1.0). *Let*

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{h} & Y' \end{array}$$

be a cartesian square of schemes, with Y quasi-compact and f quasi-compact and quasi-separated. Let $E \in \mathcal{D}^b(X)$ have quasi-coherent cohomology. Suppose X and Y' are Tor-independent over Y , and that either h has finite Tor-dimension or E has finite flat amplitude relative to f . Then the base-change morphism

$$\mathrm{L}h^* Rf_* E \longrightarrow Rf'_* \mathrm{L}g^* E$$

is defined and is an isomorphism.

Proof. The map is defined using the trivial duality formula for h ,

$$\mathrm{Hom}(\mathrm{L}h^* L, M) \xrightarrow{\sim} \mathrm{Hom}(L, Rh_* M).$$

It is enough to define $Rf_* E \rightarrow Rf_* Rg_* \mathrm{L}g^* E$, and this is the image under Rf_* of the adjunction map $E \rightarrow Rg_* \mathrm{L}g^* E$. To prove that the map is an isomorphism, localize first on Y and Y' , then use the Leray spectral sequence for an affine open cover of X to reduce to the affine case. The assertion then becomes the ring identity

$$B \otimes_A^{\mathbf{L}} A' \xrightarrow{\sim} B'$$

together with Tor-independence, and hence

$$E \otimes_A^{\mathbf{L}} A' \xrightarrow{\sim} E \otimes_B^{\mathbf{L}} B'.$$

Proposition (3.1.1). *In the cartesian square above, assume Y and Y' quasi-compact, X and Y' Tor-independent over Y , f proper and pseudo-coherent, and the finiteness theorem valid for f and f' . If f is perfect, respectively if h has finite Tor-dimension, then for $x \in K'(X)$, respectively $x \in K.(X)$,*

$$h^* f_*(x) = f'_* g^*(x).$$

3.2. Passage to the limit

Let $(X_i, f_{ij})_{i \in I}$ be a filtered projective system of quasi-compact and quasi-separated schemes with affine transition morphisms. Put $X = \varprojlim X_i$ and let $f_i : X \rightarrow X_i$ be the canonical morphisms.

Proposition (3.2.1). *The pullback maps induce an isomorphism*

$$\varinjlim K'(X_i)_{\text{naive}} \xrightarrow{\sim} K'(X)_{\text{naive}}.$$

Indeed, a locally free module of finite type on X descends to some X_i , and its class in the filtered colimit is independent of the descent after increasing i .

Proposition (3.2.2). *The canonical map*

$$\varinjlim K'(X_i) \longrightarrow K'(X)$$

is an isomorphism.

This follows from Deligne's localization technique, which identifies the category of perfect complexes on X with the filtered colimit of the corresponding categories on the X_i .

Proposition (3.2.3). *If all X_i and X are noetherian and the transition morphisms are flat, then*

$$\varinjlim K.(X_i) \xrightarrow{\sim} K.(X).$$

The module structures over K'_{naive} are compatible with these limit isomorphisms.

3.3. Relative Grothendieck groups

Let S' be an S -scheme, let X' be an S' -scheme, and let $h : X' \rightarrow X$ be an S -morphism. We assume that inverse image on X' of a vector bundle on X is again a vector bundle on X' , so that there is a canonical homomorphism

$$h^* : K^\bullet(X) \rightarrow K^\bullet(X').$$

Suppose first that $h : X' \rightarrow X$ is a closed immersion. One considers the usual exact localization diagrams and defines

$$K_{S'/S}^\bullet(X) = \ker(K_{S'}^\bullet(X') \rightarrow K^\bullet(X' \times_S (S - S'))).$$

Equivalently, this is the kernel of the natural map from $K_{S'}^\bullet(X')$ to $K^\bullet(X')$ induced by the inclusion. Pullback by h then gives a homomorphism

$$K_S^\bullet(X) \longrightarrow K_{S'/S}^\bullet(X)$$

and a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \rightarrow & K_S^\bullet(X) & \rightarrow & K^\bullet(X) & \rightarrow & K^\bullet(X \times_S (S - S')) & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & K_{S'/S}^\bullet(X) & \rightarrow & K_{S'}^\bullet(X') & \rightarrow & K^\bullet(X' \times_S (S - S')) & \rightarrow & 0. \end{array}$$

If S' is empty this recovers $K_S^\bullet(X)$, while for $X = S$ and $X' = S'$ one gets $K_{S'/S}^\bullet(S) = 0$.

Closed immersions $S'' \rightarrow S' \rightarrow S$ give transition maps

$$K_{S'/S}^\bullet(X) \rightarrow K_{S''/S}^\bullet(X),$$

so that closed immersions over S define a functor to graded abelian groups. If S' is a closed subscheme of S , $X' = X \times_S S'$, and X is regular, then

$$K_{S'/S}^\bullet(X) \longrightarrow K_S^\bullet(X)$$

is an isomorphism. The inverse is obtained from the commutative square relating $K_S^\bullet(X)$, $K_{S'/S}^\bullet(X)$, $K^\bullet(X)$, and $K_{S'}^\bullet(X')$.

For a regular scheme X and a closed subscheme $S \subset X$ we write

$$K_S^\bullet(X) = K^\bullet(X \text{ on } S).$$

If $S' \subset S$ is closed, there is a commutative diagram of exact sequences

$$\begin{array}{ccccccccc} 0 & \rightarrow & K_{S'}^\bullet(X) & \rightarrow & K_S^\bullet(X) & \rightarrow & K_{S-S'}^\bullet(X - S') & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & K^\bullet(S') & \rightarrow & K^\bullet(S) & \rightarrow & K^\bullet(S - S') & \rightarrow & 0. \end{array}$$

The support of an element of $K^\bullet(X)$ is the smallest closed subset Z such that the element comes from $K_Z^\bullet(X)$; the group $K^\bullet(X)$ is the union of the subgroups $K_Z^\bullet(X)$ as Z runs through closed subsets of X .

Bibliography

A. Borel and J.-P. Serre, *Le theoreme de Riemann-Roch*, Bull. Soc. Math. France 86 (1958), 97–136.

R. Bott, *Lectures on $K(X)$* , Benjamin, 1969.

A. Grothendieck, *La theorie des classes de Chern*, Bull. Soc. Math. France 86 (1958), 137–154.

A. Grothendieck, *Classes de faisceaux et theoreme de Riemann-Roch*, mimeographed notes, Princeton, 1957.

M. Karoubi, *Espaces classifiants en K -theorie*, Trans. Amer. Math. Soc. 147 (1970), 75–115.

Yu. Manin, *Lectures on the K -functor in algebraic geometry*, Russian Math. Surveys 24 (1969), 1–89.

SGA 6 – Expose V

Generalities on λ -rings

P. Berthelot

This paper contains reminders and clarifications on Grothendieck's notion of a λ -ring. It also corrects and supplements the exposition, adding material needed later in the seminar.

1. Universal polynomials

Let $\mathcal{C}_0$ be the category of commutative rings with unit. We shall use polynomial laws with integral coefficients in finite or infinite families of indeterminates. For indeterminates $X_1, \dots, X_n$ put

$$\sigma_i^{(n)}(X_1, \dots, X_n) = \sum_{1 \leq k_1 < \dots < k_i \leq n} X_{k_1} \cdots X_{k_i},$$

the elementary symmetric functions.

For two families X and Y , the universal polynomials P_n are defined by

$$\prod_{i,j} (1 + X_i Y_j T) = \sum_{n \geq 0} P_n(\sigma_1(X), \dots, \sigma_n(X); \sigma_1(Y), \dots, \sigma_n(Y)) T^n.$$

Each P_n is isobaric of weight n with respect to both families of elementary symmetric functions.

If $Y_i = X_i^s$, then

$$\sigma_i(Y_1, \dots, Y_n) = Q_{i,s}(\sigma_1(X), \dots, \sigma_i(X)),$$

where $Q_{i,s}$ is a universal isobaric polynomial of weight is . Similarly, if the variables Z are the products $X_i Y_j$, then

$$\sigma_k(Z) = R_{n,m,k}(\sigma_1(X), \dots, \sigma_n(X); \sigma_1(Y), \dots, \sigma_m(Y)),$$

with $R_{n,m,k}$ universal and isobaric of weight k . We shall also use universal polynomials $C_{n,m,k}$ governing elementary symmetric functions of sums of variables.

An augmented-ring functor is a functor F from $\mathcal{C}_0$ to commutative unital rings equipped with a natural augmentation $F \rightarrow \text{id}$. We write $F(A)_+$ for the kernel of $F(A) \rightarrow A$, so that $F(A) = A \oplus F(A)_+$. The category of F -algebras consists of rings A equipped with an augmentation $F(A) \rightarrow A$, with morphisms compatible with these structures.

2. Pre- λ -rings

Let K be a commutative ring with unit. A pre- λ -ring structure on K is a homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]_+$$

where the target is made into a multiplicative group. Traditionally one writes

$$\lambda_t(x) = \sum_{i \geq 0} \lambda^i(x) t^i,$$

with $\lambda^0(x) = 1$ and $\lambda^1(x) = x$.

For a scheme X , the Grothendieck group of vector bundles $K^\bullet(X)$ has such a structure, where λ^i is induced by exterior powers. More generally the Grothendieck group of a suitable exact category has the same structure.

The universal polynomials define on $1 + K[[t]]_+$ a multiplication

$$(1 + \sum a_i t^i) \circ (1 + \sum b_j t^j) = 1 + \sum_{n \geq 1} P_n(a_1, \dots, a_n; b_1, \dots, b_n) t^n,$$

with unit $1 + t$, and analogous formulas define the operations λ^i . With this notation

$$\lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y),$$

so that λ_t is multiplicative for this universal product.

One also defines universal polynomials L_n by

$$\lambda_t(1 + \sum x_i t^i) = 1 + \sum_{n \geq 1} L_n(x_1, \dots, x_n) t^n.$$

Finally the Adams operations are introduced by

$$\psi_t(x) = -t \frac{d}{dt} \log \lambda_{-t}(x) = \sum_{n \geq 1} \psi^n(x) t^n.$$

They are additive endomorphisms satisfying

$$\psi^1(x) = x, \quad \psi^m(x + y) = \psi^m(x) + \psi^m(y), \quad \psi^m(xy) = \psi^m(x) \psi^m(y).$$

If x has dimension 1, meaning $\lambda^i(x) = 0$ for $i \geq 2$, then $\psi^m(x) = x^m$.

SGA 6, Exposé V.2.3.4 through Exposé VI

2. Pre- λ -rings, continued

2.3.4

For every $x \in K$ and every integer $n \geq 1$ one has

$$(-1)^{n-1} n \lambda^n(x) = \sum_{i=1}^n (-1)^{i-1} \lambda^{n-i}(x) \psi^i(x). \quad (50)$$

This formula makes it possible to express the operations λ^n in terms of the operations ψ^n , and conversely.

2.3.5

Put

$$\lambda_{-1/t}(x)(-t)^n = \sum_{k=0}^n (-1)^k \lambda^k(x) t^{n-k}. \quad (51)$$

Then

$$\lambda^n(\lambda^m(x)) = P_{n,m}(\lambda^1(x), \dots, \lambda^{nm}(x)), \quad (52)$$

where $P_{n,m}$ is a universal isobaric polynomial of weight nm .

2.3.6

Let K be a pre- λ -ring. We define a filtration $\text{Fil}^\bullet K$ of K by putting

$$\text{Fil}^0 K = K, \quad \text{Fil}^1 K = K, \quad \text{Fil}^n K = \text{the subgroup generated by the } x \text{ such that } \lambda^i(x) = 0 \text{ for } i < n. \quad (53)$$

One checks that

$$\text{Fil}^p K \cdot \text{Fil}^q K \subset \text{Fil}^{p+q} K.$$

Grothendieck operations

2.3.7

Let K be a pre- λ -ring. The *Grothendieck operations* γ^i are defined by

$$\gamma_t(x) = \lambda_{t/(1-t)}(x) = \sum_{n \geq 0} \gamma^n(x) t^n. \quad (54)$$

Equivalently,

$$\gamma^n(x) = \lambda^n(x + n - 1) = \sum_{i=0}^n (-1)^i \binom{n-1}{i} \lambda^{n-i}(x). \quad (55)$$

2.3.8

The following formulas hold:

$$\gamma^0(x) = 1, \quad \gamma^1(x) = x, \quad \gamma^n(x + y) = \sum_{i+j=n} \gamma^i(x) \gamma^j(y). \quad (56)$$

The operations γ^i define a new pre- λ -ring structure, called the *opposite* structure.

2.3.9

One has

$$\gamma^n(xy) = P'_n(\gamma^1(x), \dots, \gamma^n(x); \gamma^1(y), \dots, \gamma^n(y)), \quad (57)$$

where P'_n is a universal polynomial.

2.4

A λ -ring is a pre- λ -ring K such that the homomorphism

$$\lambda_t : K \longrightarrow 1 + K[[t]]_+$$

is a λ -homomorphism, that is, is compatible with the pre- λ -ring structures. This means that

$$\lambda^n(\lambda^m(x)) = P_{n,m}(\lambda^1(x), \dots, \lambda^{nm}(x)) \quad (58)$$

for all x and all n, m , where $P_{n,m}$ is the universal polynomial of 2.3.5.

2.4.1

The following conditions are equivalent:

- (i) K is a λ -ring.
- (ii) The Adams operations satisfy $\psi^n \circ \psi^m = \psi^{nm}$ for all n, m .
- (iii) For every finite-dimensional element x , the elements $\lambda^n(x)$ satisfy the universal relations.

2.4.2

If K is a λ -ring, the Adams operations ψ^n are ring endomorphisms and satisfy

$$\psi^n \circ \psi^m = \psi^{nm}. \quad (59)$$

If K is also augmented, the operations ψ^n are compatible with the augmentation.

2.5

Let K be a pre- λ -ring. Denote by $\mathfrak{W}(K)$ the multiplicative group $1 + K[[t]]_+$. We have seen that $\mathfrak{W}(K)$ carries a pre- λ -ring structure. If K is a λ -ring, then $\mathfrak{W}(K)$ is again a λ -ring.

2.5.1

The element $1 + t$ is the unit for the product $\circ$. One has

$$\lambda^n(1 + t) = 1 \quad (n \geq 2), \quad \psi^n(1 + t) = 1 + t^n. \quad (60)$$

2.6

Let K be a ring. We denote by $\mathfrak{W}_\circ(K)$ the functor $\mathfrak{W}$ equipped with the product $\circ$ defined in 2.3 and with its operations λ^i . We denote by $\mathfrak{W}_\gamma(K)$ the same functor equipped with the product and operations arising from the operations γ^i .

2.6.1

There is a commutative diagram

$$\begin{array}{ccc} \mathfrak{W}_\circ(K) & \xrightarrow{\lambda_t} & \mathfrak{W}_\circ(\mathfrak{W}_\circ(K)) \\ \downarrow & & \downarrow \\ \mathfrak{W}_\gamma(K) & \xrightarrow{\gamma_t} & \mathfrak{W}_\gamma(\mathfrak{W}_\gamma(K)). \end{array}$$

3. The γ -operations

3.1

Let K be a ring and let $\mathfrak{W}(K) = 1 + K[[t]]_+$. The opposite pre- λ -ring structure on K defines operations γ^i , related to the operations λ^i by

$$\gamma_t(x) = \lambda_{t/(1-t)}(x). \quad (61)$$

3.2

For every $x \in K$ and every integer $n \geq 0$,

$$\gamma^n(x) = \lambda^n(x + n - 1). \quad (62)$$

This follows from the definition and the binomial formula for exterior powers.

3.3

Let K be a pre- λ -ring. We say that K is a γ -ring if the operations γ^i define a λ -ring structure on K , that is, if

$$\gamma_t : K \longrightarrow \mathfrak{W}_\gamma(K)$$

is a λ -homomorphism.

3.4

The following conditions are equivalent:

- (i) K is a γ -ring.
- (ii) The operations γ^i satisfy the universal relations.
- (iii) The operations ψ^n satisfy $\psi^n \circ \psi^m = \psi^{nm}$.

3.5

It is immediate that a λ -ring is a γ -ring and conversely; indeed the conditions expressed in terms of the ψ^n are identical. Moreover,

$$\gamma^n(xy) = \sum_{i+j=n} P'_{i,j}(\gamma^1(x), \dots; \gamma^1(y), \dots), \quad (63)$$

where the $P'_{i,j}$ are the universal polynomials of 2.3.9.

3.6

Let K be a ring. The functor $\mathfrak{W}_\gamma(K)$ is a γ -ring, just as $\mathfrak{W}_\circ(K)$ is a λ -ring. There is a canonical isomorphism

$$\mathfrak{W}_\circ(K) \simeq \mathfrak{W}_\gamma(K), \quad (64)$$

given by the change of variable $t \mapsto t/(1 - t)$.

3.7

Let K be a λ -ring and let A be a K - λ -algebra. The canonical γ -filtration of A is defined by

$$\text{Fil}_\gamma^n A = \text{extthe ideal generated by the products } \gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k) \quad (65)$$

with $\sum i_j \geq n$ and $\varepsilon(x_j) = 0$. It is decreasing:

$$A = \text{Fil}_\gamma^0 A \supset \text{Fil}_\gamma^1 A \supset \cdots \supset \text{Fil}_\gamma^n A \supset \cdots,$$

and

$$\text{Fil}_\gamma^0 A = A, \quad \text{Fil}_\gamma^1 A = \text{Ker}(\varepsilon), \quad \text{Fil}^p A \cdot \text{Fil}^q A \subset \text{Fil}^{p+q} A. \quad (66)$$

3.8

Let K be a binomial ring and let A be a K - λ -algebra. Then

$$\gamma_t(x) = (1/(1-t))^x = \exp(-x \log(1-t)). \quad (67)$$

If $a \in K$, multiplication by $\gamma_t(a)$ for the γ -law is characterized by its value on $1 + Xt$:

$$\gamma_t(a)(1 + Xt) = (1 + Xt)^a = \exp(a \log(1 + Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \quad (68)$$

In particular the n -th universal polynomial defining multiplication by $\gamma_t(a)$ for the γ -law is the polynomial $R_{n,a}$ of 2.9.

3.9–3.10

Let A be an augmented K - λ -algebra, with augmentation $\varepsilon : A \rightarrow K$. The *canonical filtration* of A is the ring filtration for which $\text{Fil}^n A$ is the ideal generated by products

$$\gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k), \quad \sum i_j \geq n, \quad \varepsilon(x_j) = 0. \quad (69)$$

Thus

$$\text{Fil}^0 A = A, \quad \text{Fil}^1 A = \text{Ker } \varepsilon. \quad (70)$$

This filtration is functorial in A : every λ -homomorphism of augmented K - λ -algebras is compatible with the filtrations. Unless expressly stated otherwise, this is the filtration considered on an augmented λ -ring.

3.11

Let A be a λ -ring. Denote by $\text{Fil}_\lambda^n A$ the filtration defined using the operations λ^i , equivalently the canonical γ -filtration of the opposite structure. One has

$$\text{Fil}_\lambda^n A = \text{Fil}_\gamma^n A. \quad (71)$$

Indeed the two filtrations are defined by the same conditions, in view of the equivalence between λ -rings and γ -rings.

3.11.1–3.11.2

Let A be a λ -ring and let $x_1, \dots, x_n$ be elements of A with $\varepsilon(x_i) = 0$. Then

$$\gamma^n(x_1 + \dots + x_n) \equiv x_1 \cdots x_n \pmod{\text{Fil}^{n+1} A}. \quad (72)$$

If $x_1, \dots, x_n$ are elements of filtration at least 1, then

$$\gamma^n(x_1 + \dots + x_n) \equiv \sum_{i=1}^n x_i \pmod{\text{Fil}^2 A}. \quad (73)$$

4. λ -rings generated by generators and relations

4.1

We work throughout over $\mathcal{C}_0$. Let K be a λ -ring. Consider a family of indeterminates $(X_i)_{i \in I}$ and put

$$A = K[(X_i), (\lambda^n X_i)_{i \in I, n \geq 2}].$$

We wish to define on A a λ -ring structure extending that of K . To give a homomorphism $\lambda_t : A \rightarrow \mathfrak{W}(A)$, it is enough to give the images of the X_i and of the $\lambda^n X_i$; these determine a ring homomorphism $A \rightarrow \mathfrak{W}(A)_o$. We put

$$\lambda_t(X_i) = 1 + X_i t + \sum_{n \geq 2} \lambda^n X_i t^n, \quad \lambda_t(\lambda^n X_i) = \lambda^n(\lambda_t(X_i)). \quad (74)$$

Since the relevant diagram is a diagram of ring homomorphisms, it is enough to check commutativity on the X_i and $\lambda^n X_i$; this follows from the definitions and from the fact that λ_u and $\mathfrak{W}^u(\lambda_t)$ are λ -homomorphisms.

4.2

The λ -ring A is characterized by the following universal property.

Proposition 3.1. *For every K - λ -algebra B and every family $(b_i)_{i \in I}$ of elements of B , there exists a unique K - λ -homomorphism $f : A \rightarrow B$ such that $f(X_i) = b_i$ for all $i \in I$.*

For f to be a λ -homomorphism it is necessary that $f(\lambda^j X_i) = \lambda^j(b_i)$ for all i, j , and this determines f uniquely. The resulting homomorphism is a λ -homomorphism because the equality $\lambda_t \circ f = \mathfrak{W}(f) \circ \lambda_t$ need only be checked on generators of A over K .

4.3–4.4

One may make the same construction using the γ 's instead of the λ 's. Let

$$A' = K[(X_i), (\gamma^n X_i)_{i \in I, n \geq 2}].$$

The structure is defined by

$$\gamma_t(X_i) = 1 + X_i t + \sum_{n \geq 2} \gamma^n X_i t^n, \quad \gamma_t(\gamma^n X_i) = \gamma^n(\gamma_t(X_i)). \quad (75)$$

The same universal property as in 4.2 holds, hence A' is canonically isomorphic, as a λ -ring, to A . The ring A is called the *free λ -ring* generated over K by the family $(X_i)_{i \in I}$.

Let A be a λ -ring and J an ideal of A . We say that J is a λ -ideal if for every $x \in J$ and every $n \geq 1$ one has $\lambda^n(x) \in J$. The kernel of a λ -homomorphism is a λ -ideal. Conversely, if J is a λ -ideal, then A/J inherits a λ -ring structure.

4.5–4.7

An intersection of λ -ideals is a λ -ideal. Hence there is a smallest λ -ideal containing any given family $(x_i)_{i \in I}$; by the relations of 2.4.2 it is the ordinary ideal generated by the $\lambda^n(x_i)$, $n \geq 1$.

Let $(P_{i'})_{i' \in I'}$ be a family of elements in the free λ -ring $A = K[(X_i), (\lambda^n X_i)]$. If J is the λ -ideal generated by the $P_{i'}$, then A/J is called the *λ -ring generated* over K by the generators $(X_i)_{i \in I}$ subject to the relations $(P_{i'})_{i' \in I'}$. The same construction can be made with relations among the X_i and the $\gamma^n X_i$, since an ideal is a λ -ideal if and only if it is stable under all operations γ^n .

If B is a K - λ -algebra and (b_i) is a family of elements of B , the relations $P_{i'}$ can be evaluated at $b_i, \lambda^n(b_i)$. If all these evaluations vanish, then there is a unique K - λ -homomorphism $A/J \rightarrow B$ sending X_i to b_i .

4.8–4.9

Let $n \geq 0$. In the free λ -ring on one indeterminate X over K , the ideal generated by the $\lambda^k X$, $k > n$, is a λ -ideal. Indeed, if $A = K[X, \lambda^k X]_{k \geq 2}$ and $B = A/J$, one has to show that the composite $A \xrightarrow{\lambda_t} \mathfrak{W}(A) \rightarrow \mathfrak{W}(B)$ vanishes on J . This reduces to the following elementary fact: for any ring C and any series $\varphi \in \mathfrak{W}(C)$ whose coefficients vanish in degrees $> n$, one has $\lambda^k(\varphi) = 1$

for $k > n$. It is enough to prove this for

$$C = K[X_1, \dots, X_n], \quad \varphi = 1 + X_1 t + \dots + X_n t^n.$$

After adjoining roots by the elementary symmetric functions, one may write

$$1 + X_1 t + \dots + X_n t^n = \prod_{i=1}^n (1 + T_i t),$$

and therefore

$$\lambda_u(1 + X_1 t + \dots + X_n t^n) = \prod_{i=1}^n \lambda_u(1 + T_i t) = \prod_{i=1}^n (1 + \{1 + T_i t\}u),$$

which has degree n in u .

Corollary 3.2. *The λ -ring generated over K by one generator X subject to the relations $\lambda^k(X) = 0$ for $k > n$ is isomorphic to $K[X, \dots, \lambda^n X]$.*

The same assertions hold with λ replaced by γ . One may also replace X by a family (X_i) subject to relations $\lambda^k(X_i) = 0$ for $k > n_i$.

4.10–4.15

Let K be a binomial ring and A an augmented K - λ -algebra. Let

$$B = A[X, \gamma^k X]_{k \geq 2}$$

be the free λ -ring on X over A , augmented by sending X to 0. We grade B by the weight in the variables $\gamma^k X$, giving $\gamma^k X$ weight k , and denote by M_k the set of monomials of weight k .

Lemma 3.3. *Let $a \in A$. The coefficient of a monomial of weight n in*

$$\gamma^k(aX) = P'_k(\gamma^i(a), \gamma^j X)$$

has filtration at least $k - n$ in A .

The polynomial P'_k has no isobaric component of weight $< k$, so the lemma is clear when a has augmentation zero. In general one writes

$$\gamma_t(aX) = \gamma_t((a - \varepsilon(a))X) \gamma_t(\varepsilon(a)X),$$

and uses

$$\gamma_t(\varepsilon(a)X) = (\gamma_t X)^{\varepsilon(a)} = \sum_{n=0}^{\infty} R_{n, \varepsilon(a)}(\gamma^j X) t^n,$$

with $R_{n, a}$ isobaric of weight n .

Proposition 3.4. *One has*

$$\mathrm{Fil}^k B = \mathrm{Fil}^k A + \mathrm{Fil}^{k-1} A \cdot M_1 + \cdots + \mathrm{Fil}^0 A \cdot M_k + B \cdot M_{k+1}. \quad (76)$$

The inclusion of the right hand side in the left is clear. Conversely, since the right hand side defines a ring filtration, it suffices to treat the generators $\gamma^k(b)$ with $\varepsilon(b) = 0$. Writing $b = \sum a_i m_i$ with m_i a monomial in the $\gamma^j X$, additivity reduces the assertion to $\gamma^k(am_i)$. Using the lemma and the A - λ -homomorphism sending X to m_i , one reduces further to $\gamma^k(m_i)$. Finally the absence of components of weight $< k$ in the polynomials P'_k and $Q'_{j,k}$ gives the result.

The decomposition in (76) is a direct sum decomposition of A -modules: if

$$b = \sum_{j \leq k} a_{i,j} m_{i,j} + b'$$

with $m_{i,j} \in M_j$ and $b' \in B \cdot M_{k+1}$, then $b \in \mathrm{Fil}^k B$ implies $a_{i,j} \in \mathrm{Fil}^{k-j} A$. The assertion also remains valid after quotienting by any λ -ideal contained in the augmentation ideal, and with any family of indeterminates instead of the single X .

Corollary 3.5.

$$\mathrm{Gr}^k B \simeq \mathrm{Gr}^k A \oplus \mathrm{Gr}^{k-1} A \cdot M_1 \oplus \cdots \oplus \mathrm{Gr}^0 A \cdot M_k.$$

If B' is the λ -ring generated over K by X with relations $\gamma^k(X) = 0$ for $k > n$, so $B' = K[X, \dots, \gamma^n X]$, then its λ -ring filtration is the same as the weight filtration in the variables $\gamma^j X$.

5. The elements $\lambda^p(N, x)$

This section will only be used in Expose VII, section 4, for the study of regular immersions; it may be omitted on a first reading.

5.1–5.3

Let $d \geq 1$. Let A be the $\mathbb{Z}$ - λ -algebra generated by N and x , subject to the relations $\lambda^i(N) = 0$ for $i > d$. Thus

$$A = \mathbb{Z}[N, \dots, \lambda^d N; x, \dots, \lambda^j x, \dots].$$

Put

$$\lambda_{-1}(N) = \sum_{i=0}^d (-1)^i \lambda^i(N).$$

Lemma 3.6. *For every $p \geq 1$, $\lambda^p(x\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$.*

Since $\lambda^p(x\lambda_{-1}(N))$ is isobaric of weight p in the quantities $\lambda^j(\lambda_{-1}(N))$, it suffices to prove the lemma for $\lambda^p(\lambda_{-1}(N))$. Let B be the λ -ring generated over $\mathbb{Z}$ by $N_1, \dots, N_d$ with $\lambda^i(N_j) = 0$

for $i \geq 2$. Sending N to $\sum_i N_i$ defines a λ -homomorphism $A \rightarrow B$ which identifies A with the symmetric polynomials in the N_i . Then

$$\lambda_{-1}(N) = \prod_{i=1}^d (1 - N_i).$$

If $N_i = 1$ for some i , then $\lambda_t(\lambda_{-1}(N))$ becomes $\lambda_t(0) = 1$, so all $\lambda^p(\lambda_{-1}(N))$ vanish. Hence each is divisible by every $1 - N_i$, and therefore by their product.

Since A is integral, we define $\lambda^p(N, x)$ unambiguously by

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)), \quad p \geq 1. \quad (77)$$

Thus $\lambda^p(N, x)$ is a polynomial in the $\lambda^i(x)$ and $\lambda^j(N)$, isobaric of weight p in the $\lambda^i(x)$. If a λ -ring B contains an element N'' with $\lambda^k(N'') = 0$ for $k > d$, then $\lambda^p(N'', x'')$ is defined by substitution; it does not depend on the chosen bound d .

5.4

In the λ -ring generated by N, x, y subject to $\lambda^k(N) = 0$ for $k > d$, formula (77) gives

$$\begin{aligned} \lambda^p(N, x+y)\lambda_{-1}(N) &= \lambda^p((x+y)\lambda_{-1}(N)) \\ &= \sum_{i+j=p} \lambda^i(x\lambda_{-1}(N))\lambda^j(y\lambda_{-1}(N)). \end{aligned}$$

Since the ring is integral, this yields the identity

$$\lambda^p(N, x+y) = \sum_{\substack{i+j=p \\ i,j \neq 0}} \lambda^i(N, x)\lambda^j(N, y)\lambda_{-1}(N) + \lambda^p(N, x) + \lambda^p(N, y). \quad (78)$$

If N' is another element with $\lambda^k(N') = 0$ for $k > d'$, then $\lambda^k(N + N') = 0$ for $k > d + d'$, and additivity gives

$$\lambda^p(N + N', x)\lambda_{-1}(N) = \lambda^p(N', x\lambda_{-1}(N)). \quad (79)$$

5.5

Let A be any λ -ring and let $N \in A$ satisfy $\lambda^k(N) = 0$ for $k > d$. We associate to A a new λ -ring A_N . First define on A a new multiplication by

$$x \cdot_N y = xy\lambda_{-1}(N). \quad (80)$$

This ring generally has no unit, so we adjoin a unit and consider $\mathbb{Z} \times A$, with

$$(n, x) + (n', x') = (n + n', x + x')$$

and

$$(n, x)(n', x') = (nn', nx' + n'x + x \cdot_N x'). \quad (81)$$

The ring A embeds by $x \mapsto (0, x)$. Define a pre- λ -ring structure by

$$\lambda_t(n, x) = (1 + t)^n \sum_{p=0}^{\infty} \lambda^p(N, x) t^p. \quad (82)$$

It is a λ -ring: by substitution it suffices to treat the universal case in which A is generated over $\mathbb{Z}$ by N, X, Y with $\lambda^k(N) = 0$ for $k > d$. There is then a homomorphism $A_N \rightarrow A$ sending $(1, 0)$ to 1 and $(0, x)$ to $x\lambda_{-1}(N)$; by (80) and (77) it is a λ -homomorphism, and it is injective in the universal integral case. Since A is a λ -ring, so is A_N .

In A_N one can form the γ -operations. For $x \in A$,

$$\gamma^p(0, x) = \sum_{i=0}^{p-1} \binom{p-1}{i} \lambda^i(N, x),$$

which lies in A ; equivalently it is characterized by

$$\gamma^p(N, x)\lambda_{-1}(N) = \gamma^p(x\lambda_{-1}(N)). \quad (83)$$

5.6

Let A be the λ -ring generated over $\mathbb{Z}$ by N, L, x subject to $\lambda^k(N) = 0$ for $k > d$ and $\lambda^k(L) = 0$ for $k > 1$. We compute $\lambda^p(N + L, x)$ in terms of the $\lambda^q(N, x)$. By definition,

$$\lambda^p(N + L, x)\lambda_{-1}(N)\lambda_{-1}(L) = \lambda^p(x\lambda_{-1}(N)\lambda_{-1}(L)),$$

hence

$$\begin{aligned} 1 + \lambda_{-1}(N)\lambda_{-1}(L) \sum_{k \geq 1} \lambda^k(N + L, x)t^k &= \lambda_t(x\lambda_{-1}(N)) \circ \lambda_t(1 - L) \\ &= \frac{\lambda_t(x\lambda_{-1}(N))}{\lambda_{Lt}(x\lambda_{-1}(N))} \\ &= 1 + \frac{\sum_{k \geq 1} \lambda^k(N, x)\lambda_{-1}(N)(1 - L^k)t^k}{1 + \sum_{k \geq 1} \lambda^k(N, x)\lambda_{-1}(N)L^k t^k}. \end{aligned}$$

5.7

Let K be a binomial ring, A an augmented K - λ -algebra, and $N \in A$ such that $\lambda^k(N) = 0$ for $k > d$. Then for every $x \in A$ and every $p \geq 1$,

$$\gamma^p(N, x) \in \text{Fil}^{p-d} A. \quad (84)$$

Indeed, let $A_{\circ}$ be the K - λ -algebra generated by $N'_{\circ}, X'_{\circ}$ with $\lambda^k(N'_{\circ}) = 0$ for $k > d$, and augmented by $\varepsilon(N'_{\circ}) = \varepsilon(X'_{\circ}) = 0$. This is the polynomial ring over K in the variables $\gamma^i(N'_{\circ})$, $1 \leq i \leq d$, and $\gamma^j(X'_{\circ})$, $j \geq 1$; its λ -filtration is the weight filtration. Put

$$N_{\circ} = N'_{\circ} + d, \quad X_{\circ} = X'_{\circ} + \varepsilon(x).$$

For $N_{\circ}, X_{\circ}$ the assertion is equivalent to

$$\gamma^p(N_{\circ}, X_{\circ})\gamma^d(N'_{\circ}) \in \text{Fil}^p A_{\circ}.$$

Since $\gamma^d(N'_{\circ}) = (-1)^d \lambda_{-1}(N_{\circ})$, this becomes

$$(-1)^d \gamma^p(X_{\circ})\gamma^d(N'_{\circ}) \in \text{Fil}^p A_{\circ},$$

which holds because $\gamma^d(N'_{\circ})$ has zero augmentation. A K - λ -homomorphism $A_{\circ} \rightarrow A$ compatible with augmentations sends $N'_{\circ}$ to $N - d$ and $X'_{\circ}$ to $x - \varepsilon(x)$; compatibility with filtrations gives the desired result.

End of Expose V

SGA 6 – Expose V, continued

The Chern ring and Adams operations

6. Chern ring

Throughout this section the base ring is a fixed binomial ring K . We shall use the category $\mathcal{G}$ of 1.1, and therefore return to the notation of section 1: G denotes the functor defined on $\mathcal{G}$, and G_i its restriction to $\mathcal{G}_0$.

6.1

In 3.5 and 3.6 we defined morphisms of functors

$$G_i \times G_i \xrightarrow{*} G_i, \quad G \circ i \xrightarrow{\gamma'} G \circ i \circ G_S, \quad G_i \xrightarrow{a} G_i,$$

the last being multiplication, for the law $\circ$, by $\gamma^t(a)$ with $a \in K$. By 1.9 these yield morphisms

$$G \times G \xrightarrow{*} G_g, \quad G \xrightarrow{\gamma'} G \circ i \circ G_S, \quad G \xrightarrow{a} G_g.$$

The isobaricity of the polynomials $P'_n, Q'_{i,j}, R_{n,a}$ implies that the images of the first and third functors lie in G , and that the image of the second lies in the subgroup of formal series in u whose non-zero degree coefficients lie in G .

Composing the first and third morphisms with the canonical morphism $G \rightarrow 1 + G^+$ gives two morphisms of functors on $\mathcal{G}$:

$$(1 + G^+) \times (1 + G^+) \xrightarrow{*} 1 + G^+, \quad 1 + G^+ \xrightarrow{a} 1 + G^+.$$

They define on $1 + G^+$ a multiplication and operations of K . The multiplication, denoted $*$, is determined by its value on $(1 + X, 1 + Y)$:

$$(1 + X) * (1 + Y) = \frac{1 + X + Y}{(1 + X)(1 + Y)}. \quad (85)$$

The n -th universal polynomial for this law is the isobaric component of weight n of P'_n , denoted P''_n . The operations of K are again written exponentially; the universal polynomials defining them are the lowest-degree isobaric components of the $R_{n,a}$, and since the $R_{n,a}$ are themselves isobaric, these components are the $R_{n,a}$.

For $A \in \text{Obj}(\mathcal{G})$, the ring $1 + \widehat{A}^+$ defined by these laws generally has no unit. We extend it by adjoining the scalars of K , obtaining the ring $K \times (1 + \widehat{A}^+)$ with

$$(a, \varphi) + (a', \varphi') = (a + a', \varphi\varphi')$$

and

$$(a, \varphi)(a', \varphi') = (aa', \varphi^{a'} \varphi'^a (\varphi * \varphi')). \quad (86)$$

The unit is $(1, 1)$, and $1 + \hat{A}^+$ is the ideal $(0) \times (1 + \hat{A}^+)$. It is naturally a K -algebra, with

$$a \cdot (a', \varphi) = (a, 1)(a', \varphi) = (aa', \varphi^a). \quad (87)$$

The preceding construction of γ' and the canonical morphism $G \rightarrow 1 + G^+$ give a morphism

$$G \xrightarrow{\gamma'} G \circ i \circ (K \times (1 + G^+)),$$

and therefore a morphism of functors

$$1 + G^+ \longrightarrow G \circ i \circ (K \times (1 + G^+)).$$

The universal polynomials defining it are the coefficients of $\gamma'^i(1 + X_1 + \cdots + X_n + \cdots)$; the j -th coefficient is the isobaric component of weight j of $Q'_{i,j}$, denoted $Q''_{i,j}$.

Extending this morphism to $K \times (1 + G^+)$ using the λ -structure of K , one obtains a pre- λ -ring structure whose corresponding γ -operations satisfy

$$\gamma_t(a, \varphi) = \gamma_t(a) \left(1 + \sum_{p \geq 1} \gamma'^p(\varphi) t^p \right). \quad (88)$$

We write these operations simply as γ . The verification that this is a λ -ring structure reduces to the commutativity of the diagram of 3.7.2; it is known on K and follows on $1 + G^+$ from the corresponding diagram for G by passing to the quotient.

Definition 3.7. The functor defined above on graded K -algebras, with values in augmented K - λ -algebras, is called the *Chern ring*. Its value on A is denoted $\text{Ch}(A)$, or $\text{Ch}_K(A)$ if the base must be specified.

From (85), (86), (88), and 3.6.2, one obtains

$$(1, 1 + X)(1, 1 + Y) = (1, 1 + X + Y), \quad \lambda^i(1, 1 + X) = 0 \quad (i \geq 2). \quad (89)$$

These are the relations used to define the Chern ring in Grothendieck's treatment. Conversely, they determine (85) and (88).

This construction is motivated as follows. Suppose that A is the associated graded ring of the $K^\bullet$ of a commutatively locally ringed topos, for the filtration of 3.10, and that this $K^\bullet$ is a λ -ring. We shall define Chern classes of elements of an augmented λ -ring with values in its associated graded ring. To a locally free sheaf of finite type one then associates an element of $\text{Ch}(A)$: its scalar component is the rank, and its $1 + \hat{A}^+$ component is the sum of its Chern classes. The formulas (89) then become the usual formulas for tensor products of invertible sheaves and exterior powers of an invertible sheaf.

6.3. The Chern character

For every graded K -algebra A , denote by $A_{\mathbb{Q}}$ the graded K -algebra $A \otimes_{\mathbb{Z}} \mathbb{Q}$. Define a functorial homomorphism

$$1 + \hat{A}^+ \xrightarrow{\text{ch}} \hat{A}_{\mathbb{Q}}^+$$

by its value on $1 + X$:

$$\text{ch}(1 + X) = \exp(X) - 1. \quad (90)$$

This homomorphism is compatible with the multiplication of 6.1. Indeed, it is enough to check this on $1 + X, 1 + Y$, and one computes

$$\begin{aligned} \text{ch}((1 + X) * (1 + Y)) &= \text{ch}\left(\frac{1 + X + Y}{(1 + X)(1 + Y)}\right) \\ &= \text{ch}(1 + X + Y) - \text{ch}(1 + X) - \text{ch}(1 + Y) \\ &= \exp(X + Y) - \exp(X) - \exp(Y) + 1 \\ &= \text{ch}(1 + X) \text{ch}(1 + Y). \end{aligned}$$

Let η be the functorial endomorphism of $\hat{A}_{\mathbb{Q}}^+$ defined by

$$\eta\left(\sum_{k \geq 1} a_k t^k\right) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{(k-1)!} a_k t^k.$$

Then, for $\varphi \in 1 + \hat{A}^+$,

$$\text{ch}(\varphi) = \eta(\text{Log } \varphi). \quad (91)$$

It follows that ch commutes with the operations of K , since this may again be checked on $1 + X$:

$$\text{ch}((1 + X)^a) = \eta(a \text{Log}(1 + X)) = a\eta(\text{Log}(1 + X)) = a \text{ch}(1 + X).$$

Thus ch extends to a homomorphism of K -algebras

$$\text{ch} : K \times (1 + \hat{A}^+) \longrightarrow K \oplus \hat{A}_{\mathbb{Q}}^+, \quad \text{ch}(a, \varphi) = a + \eta(\text{Log } \varphi). \quad (92)$$

This homomorphism is called the *Chern character*. If A has characteristic zero, the Chern character is an isomorphism, with inverse

$$a + \psi_1 \longmapsto (a, \exp(\eta^{-1}\psi_1)).$$

Proposition 3.8. *Let A be a graded K -algebra. Give $\text{Ch}(A)$ the degree filtration*

$$(\text{Ch } A)_0 = \text{Ch } A, \quad (\text{Ch } A)_n = 0 \times (1 + \hat{A}^{\geq n}) \quad (n \geq 1), \quad (93)$$

where $\hat{A}^{\geq n} = \prod_{i \geq n} A^i$. Then:

(i) If $\varphi \in (\text{Ch } A)_n$, $n \geq 1$, and $\varphi = 1 + a_n + \cdots$, then

$$\text{ch}(\varphi) = \frac{(-1)^{n-1}}{(n-1)!} a_n + \cdots. \quad (94)$$

(ii) If $\varphi = 1 + a_m + \cdots \in (\text{Ch } A)_m$ and $\psi = 1 + b_n + \cdots \in (\text{Ch } A)_n$, then

$$\varphi * \psi = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!} a_m b_n + \cdots. \quad (95)$$

In particular the degree filtration is a ring filtration.

The first congruence follows immediately from (91). For the second, by substitution one reduces to a torsion-free universal polynomial ring, embeds it in its tensor product with $\mathbb{Q}$, and uses the injectivity of ch . If $\varphi * \psi = 1 + Z_p + \cdots$ is the first non-zero term, multiplicativity of ch and (94) identify Z_p .

6.6. Comparison of filtrations

The λ -ring $\text{Ch } A$ is augmented over K by $(a, \varphi) \mapsto a$. Hence it has its canonical filtration as an augmented λ -ring. We compare it with the degree filtration.

Proposition 3.9. *Let $x \in \text{Ch } A$ have the form $x = (n, 1 + \cdots + a_n)$. Then:*

- (i) $\lambda^i(x) = 0$ for $i > n$;
- (ii) $\lambda^n(x) = (1, 1 + a_1)$;
- (iii) $\sum_{i=0}^n (-1)^i \lambda^i(x) = (0, 1 - (n-1)!a_n + \cdots)$, where the omitted terms have degree $> n$.

It is enough, by functoriality, to prove this for $A = K[X_1, \dots, X_n]$ filtered by weight, with X_i of weight i , and $x = (n, 1 + X_1 + \cdots + X_n)$. Embedding by symmetric functions into $K[T_1, \dots, T_n]$ gives

$$(n, 1 + X_1 + \cdots + X_n) = \sum_{i=1}^n (1, 1 + T_i).$$

Then (89) gives the assertions on $\lambda^k(x)$, and the product formula (95) gives the final congruence for $\lambda_{-1}(x)$.

Corollary 3.10. *The relation 6.6.1(iii) holds for every element x of augmentation n .*

For this one uses that each $(\text{Ch } A)_n$ is a λ -ideal. If $x = (0, \varphi) \in (\text{Ch } A)_n$, then the polynomials defining the coefficients of $\lambda^i(x)$ depend only on the first coefficients of φ ; hence all coefficients of degree $< n$ vanish. Thus $\lambda^i(x) \in (\text{Ch } A)_n$.

Consequently, for $x = (n, 1 + a_1 + \cdots + a_n + \cdots)$,

$$\lambda_{-1}(x) = (0, 1 - (n-1)!a_n + \cdots).$$

Equivalently, if ε is the augmentation of $\text{Ch } A$, then

$$\lambda^n(x - \varepsilon(x)) = (0, 1 + (-1)^{n-1}(n-1)!a_n + \cdots). \quad (96)$$

It follows that the canonical λ -ring filtration satisfies

$$\text{Fil}^n \text{Ch } A \subset (\text{Ch } A)_n. \quad (97)$$

If A is a finite graded K -algebra of characteristic zero, the two filtrations are in fact identical. Indeed, if $A^{\geq p} = 0$ and $x = (0, 1 + a_n + \cdots + a_p) \in (\text{Ch } A)_n$, put

$$x' = \left(0, 1 + \frac{(-1)^{n-1}}{(n-1)!}a_n + a_{n+1} + \cdots + a_p\right).$$

Then $\lambda^n(x') = (0, 1 + a_n + \cdots)$, and one proceeds by induction on n .

6.7. Chern classes in an augmented λ -ring

Let A be an augmented λ -ring over K , with augmentation ε . Since A has its canonical filtration, one may form the Chern ring of $\text{Gr } A$. For $x \in A$, the element $\gamma_t(x - \varepsilon(x))$ lies in $G(A)$; its image in

$$G(A)/G'(A) = 1 + \widehat{\text{Gr } A}^+$$

is denoted $c(x)$ and is called the *total Chern class* of x . Its i -th coefficient is denoted $c^i(x)$:

$$c^i(x) = \text{cl}_{\text{Gr}^i A}(\lambda^i(x - \varepsilon(x))), \quad c(x) = 1 + c^1(x) + \cdots. \quad (98)$$

Put

$$\tilde{c}(x) = (\varepsilon(x), c(x)), \quad (99)$$

called the *completed Chern class* of x .

Proposition 3.11. *The completed Chern class*

$$\tilde{c}: A \longrightarrow \text{Ch}(\text{Gr } A)$$

is a homomorphism of augmented K - λ -algebras.

Additivity is clear. For multiplicativity, write

$$\begin{aligned} \gamma_t(xy - \varepsilon(xy)) &= \gamma_t(\varepsilon(y)(x - \varepsilon(x)))\gamma_t(\varepsilon(x)(y - \varepsilon(y))) \\ &\quad \cdot \gamma_t((x - \varepsilon(x))(y - \varepsilon(y))) \\ &= (\gamma_t(x - \varepsilon(x)))^{\varepsilon(y)}(\gamma_t(y - \varepsilon(y)))^{\varepsilon(x)} \\ &\quad \cdot (\gamma_t(x - \varepsilon(x))\bar{\circ}\gamma_t(y - \varepsilon(y))), \end{aligned}$$

and pass to the associated graded, using (86). Compatibility with scalars and augmentations is

immediate. Compatibility with the λ -operations follows by checking the diagram

$$\begin{array}{ccc} A & \xrightarrow{\gamma^t} & G_g(A) \\ \downarrow \tilde{c} & & \downarrow G_g(\tilde{c}) \\ \text{Ch}(\text{Gr } A) & \xrightarrow{\gamma^t} & G_g(\text{Ch}(\text{Gr } A)) \end{array}$$

on K and on elements of augmentation zero.

Proposition 3.12. *Let A be an augmented K - λ -algebra and let $z \in \text{Fil}^m A$. Then*

$$\gamma^m(z) \equiv (-1)^{m-1}(m-1)! z \pmod{\text{Fil}^{m+1} A}. \quad (100)$$

Equivalently,

$$c^m(z) = \text{cl}_{\text{Gr}^m A}((-1)^{m-1}(m-1)! z).$$

One writes z as a sum of products $a\gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k)$ with $\varepsilon(x_i) = 0$ and $i_1 + \cdots + i_k \geq m$. Since $\tilde{c}$ is a λ -homomorphism and (96) controls the first non-zero term of the Chern class, formula (95) gives the displayed congruence.

Proposition 3.13. *Let K be a binomial ring, R an augmented K - λ -algebra, $N \in R$ such that $\lambda^k(N) = 0$ for $k > d$, and $x \in \text{Fil}^i R$. Then*

$$\gamma^{d+i}(N, x) - (-1)^{d+i-1}(d+i-1)! x \in \text{Fil}^{i+1} R. \quad (101)$$

Choose a presentation of x by products of γ -operations, and pass to the universal K - λ -algebra R_0 generated by N'_0 and the auxiliary variables, with $\gamma^k(N'_0) = 0$ for $k > d$ and zero augmentation. The filtration of R_0 is the weight filtration. After replacing N'_0 by $N_0 + d$, the desired assertion becomes

$$\gamma^{d+i}(N_0, x_0)\gamma^d(N'_0) - (-1)^{d+i-1}(d+i-1)!x_0\gamma^d(N'_0) \in \text{Fil}^{d+i+1} R_0.$$

Since $\lambda_{-1}(N_0) = (-1)^d\gamma^d(N'_0)$, this is exactly Proposition 6.9 applied to $z = (-1)^d x_0 \gamma^d(N'_0)$. Functoriality then gives the general case.

Proposition 3.14. *Assume that K has characteristic zero. Let $n \geq 0$. Then the functor Ch is an equivalence from the category of graded K -algebras A with $A_0 = K$ and $A_k = 0$ for $k > n$ to the category of augmented K - λ -algebras Λ with $\text{Fil}^{n+1} \Lambda = 0$. A quasi-inverse is $\Lambda \mapsto \text{Gr } \Lambda$.*

If A is finite graded, the two filtrations on $\text{Ch } A$ coincide. Define

$$u : A \longrightarrow \text{Gr}_{\deg}(\text{Ch } A)$$

by the identity on A_0 and, for $i > 0$, by

$$u(a_i) = \text{cl}_{\text{Gr}^i}(0, 1 + (-1)^{i-1}(i-1)!a_i).$$

Formula (95) shows that u is a homomorphism of graded rings, compatible with the K -operations.

Its inverse sends a class in $\text{Gr}^i(\text{Ch } A)$ to $(-1)^{i-1}a_i/(i-1)!$, where a_i is the i -th coefficient of any lift.

For an augmented λ -algebra Λ , the isomorphism

$$\Lambda \longrightarrow \text{Ch}(\text{Gr } \Lambda)$$

is given by the completed Chern class $\tilde{c}$. It is enough to check this on associated graded rings, and then on generators of the canonical filtration, i.e. the classes $\gamma^i(x)$ with $\varepsilon(x) = 0$. Formula (96) shows that the resulting composite is the identity.

Corollary 3.15. *If K has characteristic zero and Λ is a K - λ -algebra with discrete filtration, then*

$$\tilde{c} : \Lambda \longrightarrow \text{Ch}(\text{Gr } \Lambda)$$

is an isomorphism.

Definition 3.16. Let A be a graded K -algebra and Λ an augmented K - λ -algebra. A *Chern theory for Λ with values in A* is a λ -homomorphism

$$\Lambda \longrightarrow \text{Ch}(A)$$

compatible with augmentations.

The functor sending A to the set of such Chern theories is representable. When K has characteristic zero and Λ has finite filtration, the representing graded algebra is $\text{Gr } \Lambda$ by Proposition 6.11.

7. Appendix: Adams operations ψ^k

We follow here a presentation due to J.-P. Serre.

7.1

Let K be a pre- λ -ring. For $x \in K$ define elements $\psi^k(x)$ by

$$\frac{d}{dt} \lambda_t(x) / \lambda_t(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \psi^k(x) t^{k-1}. \quad (102)$$

The maps $\psi^k : K \rightarrow K$ are additive, since logarithmic derivation turns products into sums. They are isobaric polynomials of weight k in $\lambda^1, \dots, \lambda^k$; for example

$$\psi^1(x) = x, \quad \psi^2(x) = x^2 - 2\lambda^2(x).$$

7.2–7.3

Conversely, if K is a ring of characteristic zero and additive maps $\psi^k : K \rightarrow K$ are given with $\psi^1 = \text{id}_K$, then

$$\lambda_t(x) = \exp \left(\sum_{k=1}^{\infty} (-1)^{k-1} \psi^k(x) t^k / k \right) \quad (103)$$

defines a pre- λ -structure on K . Thus in characteristic zero giving a pre- λ -structure is equivalent to giving the corresponding operations ψ^k . It is also clear from (102) that the ψ^k commute with λ -homomorphisms; in characteristic zero a homomorphism of pre- λ -rings is a λ -homomorphism if and only if it commutes with all ψ^k .

7.4

For $K = \mathbb{Z}$ with its λ -ring structure, all ψ^k are the identity. Indeed

$$\frac{d}{dt} \lambda_t(n) / \lambda_t(n) = n(1+t)^{n-1} / (1+t)^n = n \sum_{k \geq 1} (-1)^{k-1} t^{k-1}.$$

On the category $\mathcal{G}_0$ of $\mathbb{Z}$ -algebras, the operations ψ^k define additive endomorphisms of the λ -ring functor G . They are determined by

$$\psi^k(1 + Xt) = 1 + X^k t. \quad (104)$$

Proposition 3.17. *(i) If K is a λ -ring, the operations ψ^k satisfy*

$$\psi^k(1) = 1, \quad \psi^k(xy) = \psi^k(x)\psi^k(y), \quad \psi^k \circ \psi^{k'} = \psi^{kk'}. \quad (105)$$

(ii) Conversely, if the ψ^k satisfy these identities and K has characteristic zero, then K is a λ -ring.

For the first assertion, $\lambda_t(1) = 1+t$ gives $\psi^k(1) = 1$. To prove multiplicativity, consider the diagram expressing the product $K \times K \rightarrow K$ and the product $\circ$ on $G(K)$; after applying logarithmic derivative and extracting the $k-1$ coefficient, it is enough to check the universal pair $(1+Xt, 1+Yt)$. Then

$$\psi^k((1+Xt) \circ (1+Yt)) = \psi^k(1+XYt) = (XY)^k,$$

which equals $X^k Y^k$.

For the composition law it is enough to check the universal element again:

$$\psi_u^{k'}(\lambda_u(1+Xt)) = \psi_u^{k'}(1+(1+Xt)u) = (1+Xt)^{\circ k'} = 1+X^{k'}t,$$

and applying ψ^k gives $X^{kk'}$. If K has characteristic zero, logarithmic derivation is an isomorphism from $G(K)$ to $K[[t]]$; hence the relations in (105) force λ_t to be a ring homomorphism and then a λ -homomorphism. The relation $\lambda_t(1) = 1+t$ follows by integration from $\psi^k(1) = 1$.

Bibliography for Expose V

- [1] A. Grothendieck, *La theorie des classes de Chern*, Bulletin de la Societe Mathematique de France 86 (1958), 137–154.
- [2] F. Hirzebruch, *Neue topologische Methoden in der algebraischen Geometrie*, Ergebnisse der Mathematik, Springer, 1956.

SGA 6 – Expose VI

$K^\bullet$ of a Projective Bundle: Calculations and Consequences

Pierre Berthelot

1. Computation of K^* of a projective bundle: locally free sheaves of finite type

In this section $K^*(X)$ denotes the Grothendieck group of locally free sheaves of finite type on a scheme X . The aim is the following theorem.

Theorem 3.18 (1.1). *Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$ -module of rank $r + 1$, and let $X = \mathbb{P}(E)$ be the associated projective bundle. The structural morphism $f : X \rightarrow S$ gives $K^*(X)$ the structure of a $K^*(S)$ -algebra. Let $\xi \in K^*(X)$ be the class of $\mathcal{O}_X(1)$. Then $K^*(X)$ is a free $K^*(S)$ -module with basis*

$$1, \xi, \dots, \xi^r.$$

1.2. Plan of the proof

The proof has three parts.

1°) The powers of ξ generate $K^*(X)$ as a $K^*(S)$ -module. This is proved by showing:

- (a) If $\mathcal{F}$ is locally free of finite type on X , there is a graded $\mathrm{Sym}(E)$ -module $\mathcal{M}$ of finite presentation, flat over $\mathcal{O}_S$, such that $\mathcal{F} = \mathrm{Proj}_0(\mathcal{M})$.
- (b) Such an $\mathcal{M}$ has a finite graded resolution by graded $\mathrm{Sym}(E)$ -modules $\mathcal{L}_i$ locally free of finite type:

$$0 \rightarrow \mathcal{L}_{r+1} \rightarrow \dots \rightarrow \mathcal{L}_0 \rightarrow \mathcal{M} \rightarrow 0.$$

- (c) Each $\mathcal{L}_i$ has a finite filtration whose quotients are of the form $\mathcal{N}_j \otimes_{\mathcal{O}_S} \mathrm{Sym}(E)(j)$ with $\mathcal{N}_j$ locally free of finite type on S .

2°) The elements $1, \xi, \dots, \xi^{r+1}$ are linearly dependent over $K^*(S)$.

3°) The elements $1, \xi, \dots, \xi^r$ are linearly independent over $K^*(S)$.

1.3. Graded modules

Let S be a scheme, $\mathcal{B}$ a graded $\mathcal{O}_S$ -algebra, and $\mathcal{M}$ a graded $\mathcal{B}$ -module. For a graded $\mathcal{B}$ -module $\mathcal{N}$, write $\mathcal{N}(n)$ for the graded module with

$$(\mathcal{N}(n))_k = \mathcal{N}_{n+k}.$$

A graded module is locally free, respectively locally free of finite type, if locally on S it is a direct sum, respectively a finite direct sum, of modules $\mathcal{B}(n)$. It is of finite presentation if locally it has a presentation by finite free graded modules.

For every n there is a canonical homomorphism of graded $\mathcal{B}$ -modules

$$\mathcal{M}_n \otimes_{\mathcal{O}_S} \mathcal{B}(-n) \longrightarrow \mathcal{M}, \quad (106)$$

sending $x \otimes a$ to $a \cdot x$. A graded module free of finite type decomposes as

$$\mathcal{M} \simeq \bigoplus_{n=n_0}^{n_1} (\mathcal{B}(-n))^{k_n}. \quad (107)$$

The integers k_n in such a decomposition are independent of the chosen decomposition. This is proved by induction, comparing the lowest non-zero homogeneous components and then passing to the quotient.

Proposition 3.19. *Let S be quasi-compact, E a locally free $\mathcal{O}_S$ -module of finite type, $X = \mathbb{P}(E)$, and $\mathcal{F}$ an $\mathcal{O}_X$ -module of finite presentation, flat over S . There exists an integer n_0 such that for all $n \geq n_0$:*

- (i) $R^i f_*(\mathcal{F}(n)) = 0$ for $i \geq 1$;
- (ii) $f_*(\mathcal{F}(n))$ is locally free of finite type on S ;
- (iii) $\mathcal{M} = \bigoplus_{n \geq n_0} f_*(\mathcal{F}(n))$ is a graded $\mathrm{Sym}(E)$ -module of finite presentation.

The assertion is local on S , hence the quasi-compactness allows reduction to S affine. In the noetherian case, (i) follows from Serre's theorem, (ii) from EGA III, 7.9.10, and (iii) from the comparison between $\mathcal{F}$ and a quasi-coherent graded module defining it. For the general affine case, write S as a filtered projective limit of noetherian affine schemes and descend E and $\mathcal{F}$ to some stage. The necessary flatness and finite-presentation hypotheses descend by EGA IV, and the noetherian case applies after base change.

Corollary 3.20. *Under the hypotheses of Proposition VI.1.3, there is a graded $\mathrm{Sym}(E)$ -module $\mathcal{M}$ of finite presentation, flat over $\mathcal{O}_S$, such that $\mathcal{F} = \mathrm{Proj}_0(\mathcal{M})$.*

Lemma 3.21. *Let A be a ring, B an A -algebra of finite presentation, and M a B -module of finite presentation. Assume that B and M are flat over A , and that for every residue field k of A , the module $M \otimes_A k$ has projective dimension $\leq n$ over $B \otimes_A k$. Then M has a resolution of length n by projective B -modules of finite type:*

$$0 \rightarrow L_n \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0.$$

This follows immediately from EGA IV, 12.3.2.

Proposition 3.22. *Let S be quasi-compact, E locally free of rank $r + 1$, $\mathcal{B} = \text{Sym}(E)$ with its usual grading, and $\mathcal{M}$ a graded $\mathcal{B}$ -module of finite presentation, flat over S . Then there is a graded resolution*

$$0 \rightarrow \mathcal{L}_{r+1} \rightarrow \cdots \rightarrow \mathcal{L}_0 \rightarrow \mathcal{M} \rightarrow 0,$$

with the $\mathcal{L}_i$ graded $\mathcal{B}$ -modules locally free of finite type.

First, any such $\mathcal{M}$ admits a graded surjection $\mathcal{L} \rightarrow \mathcal{M}$ with $\mathcal{L}$ locally free of finite type: choose N such that $\mathcal{M}$ is generated in degrees $\leq N$ and put

$$\mathcal{L} = \bigoplus_{i \leq N} \mathcal{M}_i \otimes_{\mathcal{O}_S} \mathcal{B}(-i).$$

The map is the one of (106). Iterating this construction on kernels gives a global graded resolution. After $r + 1$ steps, the kernel is locally projective over $\mathcal{B}$ on affine opens by Lemma VI.1.5, hence locally free as a graded $\mathcal{B}$ -module; truncating gives the asserted resolution.

Proposition 3.23. *Let S be quasi-compact, E locally free of finite type, $\mathcal{B} = \text{Sym}(E)$, and $\mathcal{L}$ a graded $\mathcal{B}$ -module locally free of finite type. Then $\mathcal{L}$ has a decreasing finite filtration by graded submodules $\mathcal{L}^{(m)}$ such that*

$$(i) \quad \mathcal{L}^{(m)} = \mathcal{L} \text{ for } m \leq m_0;$$

$$(ii) \quad \mathcal{L}^{(m)} = 0 \text{ for } m \geq m_1;$$

$$(iii) \quad \text{gr}^m \mathcal{L} \text{ is isomorphic, as a graded } \mathcal{B}\text{-module, to } \mathcal{N}_m \otimes_{\mathcal{O}_S} \mathcal{B}(m) \text{ with } \mathcal{N}_m \text{ locally free of finite type on } S.$$

Since S is quasi-compact, the degrees of local homogeneous bases lie between fixed integers n_0 and n_1 . Induction on $n_1 - n_0$ proves the assertion. The first step is the canonical map $\mathcal{M}_{n_0} \otimes \mathcal{B}(-n_0) \rightarrow \mathcal{M}$, which locally identifies the summand of lowest homogeneous degree; its cokernel has local generators in higher degrees, and induction applies.

1.8–1.9. Generation and the relation

Let $\mathcal{F}$ be locally free of finite type on X . By Corollary VI.1.4 and Proposition VI.1.6, resolve the corresponding graded module by $\mathcal{L}_i$'s. Applying Proj_0 gives an exact sequence and hence in $K^*(X)$

$$\text{cl}(\mathcal{F}) = \sum_i (-1)^i \text{cl}(\text{Proj}_0 \mathcal{L}_i).$$

Using the filtration of Proposition VI.1.7, each term is a sum of classes

$$\text{cl}(f^* \mathcal{N}_{ij}(j)) = f^*(\text{cl} \mathcal{N}_{ij}) \xi^j.$$

Thus $K^*(X)$ is generated as a $K^*(S)$ -module by the powers of ξ .

To get a dependence relation, use the canonical surjection $f^*E \rightarrow \mathcal{O}_X(1)$ with locally free kernel $\mathcal{F}$ of rank r . Since $\lambda^{r+1}(f^*E - \xi) = 0$, and $\lambda_t(-\xi) = 1/(1 + \xi t)$, one obtains

$$\sum_{i=0}^{r+1} (-1)^{r+1-i} \xi^{r+1-i} \lambda^i(E) = 0. \quad (108)$$

1.10–1.11. Independence

The proof of independence is due to J.-P. Jouanolou.

Lemma 3.24. *With the hypotheses of Theorem VI.1.1, let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves of finite type on X with equal classes in $K^*(X)$. Then there is n_0 such that for all $n \geq n_0$:*

(i) $f_*(\mathcal{F}(n))$ and $f_*(\mathcal{G}(n))$ are locally free of finite type on S ;

(ii) their classes in $K^*(S)$ are equal.

This follows by writing the equality of classes in $K^*(X)$ as a finite integral combination of relations coming from short exact sequences, twisting by $\mathcal{O}_X(n)$, and applying f_* . For n sufficiently large all higher direct images vanish and all direct images are locally free, so the relations descend to $K^*(S)$.

Suppose

$$f^*(x_0) + f^*(x_1)\xi + \cdots + f^*(x_r)\xi^r = 0. \quad (109)$$

Write each x_j as a difference of classes of locally free sheaves, and construct locally free sheaves $\mathcal{F}, \mathcal{F}'$ on X such that (109) says $\text{cl}(\mathcal{F}) = \text{cl}(\mathcal{F}')$. By the lemma, for large n ,

$$\sum_{j=0}^r x_j \sigma_{n+j} = 0, \quad (110)$$

where σ_p is the class of $\text{Sym}^p(E)$ in $K^*(S)$.

The Koszul complex

$$0 \rightarrow \lambda^{r+1}(E) \otimes \text{Sym}(E) \rightarrow \lambda^r(E) \otimes \text{Sym}(E) \rightarrow \cdots \rightarrow E \otimes \text{Sym}(E) \rightarrow \text{Sym}(E) \rightarrow 0$$

is a graded resolution of $\mathcal{O}_S$. Its homogeneous components give

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) \sigma_{p-i} = 0 \quad (p \geq 1), \quad (111)$$

with $\sigma_k = 0$ for $k < 0$. From (110) for all large n , relation (111) allows descending induction on n , and gives (110) for all $n \geq -r$. Taking $n = -r$ gives $x_r = 0$, and continuing downward gives $x_j = 0$ for all j . This proves independence and Theorem VI.1.1.

Remark 3.25. Let $\sigma_i(E)$ denote the class of $\text{Sym}^i(E)$. If $\sigma_t(E) = 1 + \sum_{i \geq 1} \sigma_i(E)t^i$, relation (111)

is equivalent to

$$\lambda_t(E)\sigma_{-t}(E) = 1. \quad (112)$$

This shows the additivity of σ_t and extends it to a homomorphism $K^*(S) \rightarrow 1 + K^*(S)[[t]]^+$.

Remark 3.26. Any sequence of $r + 1$ consecutive powers of ξ is a basis of $K^*(X)$ over $K^*(S)$. Put $x = 1 - \xi^{-1}$, the class of a hyperplane section. Then $1, x, \dots, x^r$ is also a basis, and the dependence relation is

$$\sum_{k=0}^{r+1} \lambda^{r+1-k}(\check{E} - r - 1)x^k = 0. \quad (113)$$

For $E = \mathcal{O}_S^{r+1}$, so $X = \mathbb{P}_S^r$, this reduces to

$$x^{r+1} = 0. \quad (114)$$

Remark 3.27. Using relative schemes in the sense of Hakim, Theorem VI.1.1 also makes sense when S is a locally ringed topos. If the final object of the topos is quasi-compact, the proof transports without difficulty.

2. Computation of K^* of a projective bundle: perfect complexes

This section will not be used later in the seminar. Here $K^*(X)$ denotes the Grothendieck group of perfect complexes on X .

Theorem 3.28 (2.1). *Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$ -module of rank $r + 1$, and let $X = \mathbb{P}(E)$. Let ξ be the class in $K^*(X)$ of the complex consisting of $\mathcal{O}_X(1)$ in degree zero. Then $K^*(X)$ is a free $K^*(S)$ -module with basis*

$$1, \xi, \dots, \xi^r.$$

Let $\mathcal{B} = \text{Sym}(E)$. Let $\mathcal{A}$ be the category of $\mathcal{O}_X$ -modules, $\mathcal{G}$ the category of graded $\mathcal{B}$ -modules, $\mathcal{G}_0$ its full subcategory of graded $\mathcal{B}$ -modules locally free of finite type, and $\mathcal{G}'$ the category of $\mathcal{O}_S$ -modules. For a complex L^* on X , write $L^*(n) = L^* \otimes \mathcal{O}_X(n)$.

2.3. The functor $\underline{\text{Proj}}_0$

We need to apply Proj_0 to arbitrary graded $\mathcal{B}$ -modules, not only quasi-coherent ones. If $Y = \text{Spec}(\mathcal{A})$ and $f : Y \rightarrow S$, the isomorphism $\mathcal{A} \simeq f_*\mathcal{O}_Y$ gives a morphism of ringed spaces

$$\tilde{f} : (Y, \mathcal{O}_Y) \longrightarrow (S, \mathcal{A})$$

and an exact inverse-image functor

$$\tilde{f}^*(\mathcal{M}) = f^{-1}\mathcal{M} \otimes_{f^{-1}\mathcal{A}} \mathcal{O}_Y.$$

For a quasi-coherent $\mathcal{A}$ -module this is the usual associated module.

Now let $\mathcal{B}$ be graded and quasi-coherent, $\mathcal{M}$ a graded $\mathcal{B}$ -module, and $X = \text{Proj}(\mathcal{B})$. For an affine open $U \subset S$ and a homogeneous section s of positive degree, put

$$\mathcal{M}_{(s),U} = ((\mathcal{M}|_U) \otimes_{\mathcal{B}|_U} (\mathcal{B}|_U)_s)_0.$$

On $D_+(U, s)$ define

$$\underline{\text{Proj}}_0(\mathcal{M})_{U,s} = \tilde{f}_{U,s}^*(\mathcal{M}_{(s),U}).$$

The usual localization isomorphisms

$$\mathcal{M}_{(st),U} \simeq (\mathcal{M}_{(s),U})_{t^d/s^e}$$

show that these modules glue. This gives an exact functor $\underline{\text{Proj}}_0$ on graded modules and hence a functor

$$\underline{\text{Proj}}_0 : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{A}),$$

compatible with tensor products.

2.4–2.6. Acyclic approximation

For an $\mathcal{O}_X$ -module $\mathcal{L}$ and an integer k , set

$$F_k(\mathcal{L}) = \bigoplus_{n \geq k} f_*(\mathcal{L}(n)). \quad (115)$$

The functor F_k is left exact, hence has a derived functor

$$\mathbb{R}F_k : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{G}). \quad (116)$$

If $\mathcal{L}$ is locally free of finite type, then for k sufficiently large it is F_k -acyclic, by the vanishing theorem used in 1.3.

There is a functorial morphism

$$\beta : \underline{\text{Proj}}_0(F_k(\mathcal{F})) \longrightarrow \mathcal{F} \quad (117)$$

constructed on each standard open $D_+(U, s)$ by sending a fraction z/s^n to $z\alpha_d(s^n)^{-1}$. It agrees with the usual EGA morphism when $\mathcal{F}$ is quasi-coherent. Passing to derived categories gives

$$\underline{\text{Proj}}_0 \circ \mathbb{R}F_k \longrightarrow \text{id}_{D^+(\mathcal{A})}. \quad (118)$$

Proposition 3.29. *Let S be quasi-compact, E locally free of finite type, $X = \mathbb{P}(E)$, and L^* a perfect complex on X . For all sufficiently large k :*

(i) $\mathbb{R}F_k(L^*)$ is $\mathcal{G}_0$ -perfect;

(ii) the canonical morphism

$$\underline{\mathrm{Proj}}_0 \mathbb{R}F_k(L^*) \longrightarrow L^*$$

is an isomorphism in $D^+(\mathcal{A})$.

The assertion is local on S , so assume S affine. Since $\mathcal{O}_X(1)$ is ample, one may replace L^* by a strictly perfect complex. For large k all its finitely many terms are F_k -acyclic and the modules $F_k(L^i)$ are $\mathcal{G}_0$ -perfect, giving (i); the morphism is then an isomorphism term by term, giving (ii).

2.7–2.8. Decomposition and proof

For each n , the functor taking the n -th homogeneous component gives

$$\mathrm{Comp}_n : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{G}').$$

The canonical map (106) gives a morphism

$$\mathrm{Comp}_n \otimes_{\mathcal{O}_S} \mathcal{B}(-n) \longrightarrow \mathrm{id}_{D^+(\mathcal{G})}. \quad (119)$$

If M^* is $\mathcal{G}_0$ -perfect, one constructs a finite sequence of distinguished triangles

$$N_i^* \rightarrow P_{i-1}^* \rightarrow P_i^* \rightarrow N_i^*[1]$$

with $P_0^* = M^*$, final P_k^* acyclic, and

$$N_i^* = K_i^* \otimes_{\mathcal{O}_S} \mathcal{B}(d_i)$$

for perfect complexes K_i^* on S . The construction starts with the smallest degree d_0 appearing locally in a bounded locally free representative, uses (119), and iterates. Finiteness follows from the finite filtration of Proposition VI.1.7.

Now let L^* be perfect on X . For large k , put $M^* = \mathbb{R}F_k(L^*)$. Applying the preceding decomposition and then $\underline{\mathrm{Proj}}_0$, one gets in $K^*(X)$

$$\mathrm{cl}(L^*) = \sum_i \mathrm{cl}(f^* K_i^*) \xi^{d_i} = \sum_i f^*(\mathrm{cl} K_i^*) \xi^{d_i}.$$

Thus the powers of ξ generate $K^*(X)$ over $K^*(S)$. The dependence relation follows from (108), viewed in the perfect-complex K -group. For independence one uses the $K^*(S)$ -linear map $f_* : K^*(X) \rightarrow K^*(S)$. Since

$$f_*(1) = 1, \quad f_*(\xi^n) = 0 \quad (-r \leq n < 0),$$

multiplying a relation by appropriate powers of ξ^{-1} and applying f_* successively gives all coefficients zero. This proves Theorem VI.2.1.

3. Consequences of the structure theorem

Proposition 3.30. *Let S be quasi-compact, let $E_1, \dots, E_n$ be locally free $\mathcal{O}_S$ -modules of finite type, and let X be the fiber product over S of the corresponding flag schemes. Then the canonical homomorphism*

$$K^*(S) \longrightarrow K^*(X)$$

is injective, whether K^ is defined using locally free sheaves of finite type or perfect complexes.*

By constructing flag schemes by successive projective bundles, this reduces to the projective-bundle case, where it follows immediately from Theorem VI.1.1 or VI.2.1.

Theorem 3.31. *For every quasi-compact scheme S , the pre- λ -ring $K^*(S)$ defined from locally free sheaves of finite type is a λ -ring.*

It is enough to verify the universal λ -relations on classes of locally free sheaves. After decomposing S into finitely many disjoint opens one may assume constant ranks. Pulling back to $X = \mathbb{P}(E) \times_S \mathbb{P}(F)$ is injective on K^* , and on X the relevant classes split into sums of invertible sheaves. Thus it suffices to check the relations for classes ξ, η of invertible sheaves, where they become

$$1 + \xi\eta t = (1 + \xi t) \circ (1 + \eta t), \quad \lambda_u(1 + \xi t) = 1 + \{1 + \xi t\}u,$$

which are precisely the defining relations of the universal λ -ring structure of $1 + K^*(S)[[t]]_+$.

More generally, if $(T, \mathcal{A})$ is a commutatively ringed topos and K is the Grothendieck group of locally free $\mathcal{A}$ -modules of bounded rank on T , then the canonical pre- λ -structure on K is a λ -ring. The proof is the same after reducing constant-rank locally free modules to representations of the group schemes $\mathrm{GL}(n)$ and using the fact that the representation ring $R_{\mathbb{Z}}(G)$ is a λ -ring for finite products of linear group schemes.

4. Computation of $K^\bullet$ of a flag bundle

4.1

Let S be a scheme and let $\mathcal{E}$ be a locally free $\mathcal{O}_S$ -module of rank n . Let $\pi = (p_1, \dots, p_k)$ be a sequence of integers ≥ 1 with $\sum_i p_i = n$. The *flag functor of type π of $\mathcal{E}$* assigns to an S -scheme $X \rightarrow S$ the set of chains of quotients of $f^*\mathcal{E}$

$$(\mathcal{L}_{k-1}, \dots, \mathcal{L}_1),$$

where $\mathcal{L}_i$ is a quotient of $\mathcal{L}_{i+1}$, locally free of rank $p_1 + \dots + p_i$. This functor is represented by an S -scheme $\underline{D}_\pi(\mathcal{E})$ equipped with the universal quotient flag. For $\pi = (1, n-1)$ this is $\mathbb{P}(\mathcal{E})$; for $\pi = (r, n-r)$ it is the Grassmannian; if all $p_i = 1$, it is the usual flag bundle $\underline{D}(\mathcal{E})$.

Lemma 3.32 (4.2). *Let $\pi = (p_1, \dots, p_i, \dots, p_k)$ and split $p_i = p' + p''$. The flag bundle of type*

$$\pi' = (p_1, \dots, p_{i-1}, p', p'', p_{i+1}, \dots, p_k)$$

is obtained from $\underline{D}_\pi(\mathcal{E})$ by taking the flag bundle of type (p', p'') of the kernel of the relevant universal quotient. This identification is canonical.

The proof is immediate from the functorial definition of flag schemes.

Lemma 3.33 (4.3). *Let A be a commutative ring, and let $t_1, \dots, t_m \in A$. Consider the algebra generated by variables $\xi_i^{(j)}$ subject to the elementary symmetric relations*

$$\sigma_r(\xi_1^{(j)}, \dots, \xi_m^{(j)}) = e_r(t_1, \dots, t_m).$$

For a partition $p_1 + \dots + p_k = n$, the corresponding block monomials

$$\prod_{j=1}^k \prod_{\alpha=1}^{p_j} (\xi_{p_1+\dots+p_{j-1}+\alpha}^{(j)})^{a_{j,\alpha}}, \quad 0 \leq a_{j,\alpha} \leq n - \alpha,$$

form a basis over the subalgebra generated by the block elementary symmetric functions, and the relations among those symmetric functions are generated by the isobaric relations from

$$\prod_{j=1}^k \prod_{\alpha=1}^{p_j} (1 + \eta_j^{(\alpha)} T) = \prod_{i=1}^m (1 + t_i T).$$

Corollary 3.34 (4.4). *The same block monomials form a basis over A .*

The lemma and corollary are borrowed from Chevalley's seminar.

4.5–4.8

Let S be quasi-compact, let $\mathcal{E}$ be locally free of rank n , and put $X = \underline{D}_\pi(\mathcal{E})$. Let $\mathcal{L}_i$ be the universal quotients and $\mathcal{F}_i = \text{Ker}(\mathcal{L}_i \rightarrow \mathcal{L}_{i-1})$. Denote by F_i the class of $\mathcal{F}_i$.

Theorem 3.35 (4.6). *With this notation, the $K^\bullet(S)$ -algebra $K^\bullet(X)$ is generated by the Chern roots $\xi_i^{(\alpha)}$ of the F_i . The ideal of relations among them is generated by the relations deduced from*

$$\prod_{i=1}^k \lambda_t(F_i) = \lambda_t(E). \tag{120}$$

Equivalently, $K^\bullet(X)$ is the $K^\bullet(S)$ - λ -algebra generated by elements $F_i^{(\alpha)}$ subject to

$$(i) \quad \lambda^r(F_i^{(\alpha)}) = 0 \text{ for } r > p_i;$$

$$(ii) \quad \prod_{i,\alpha} (1 + F_i^{(\alpha)} t) = \lambda_t(E).$$

First consider the usual complete flag bundle. Induction on the rank, using the projective bundle theorem at each step, gives a $K^\bullet(S)$ -basis

$$\xi_1^{a_1} \xi_2^{a_2} \cdots \xi_n^{a_n}, \quad 0 \leq a_i \leq n - i.$$

The relation $\prod_i (1 + \xi_i t) = \lambda_t(E)$ holds by additivity. The abstract algebra defined by this relation has the same basis by Lemma 4.3, hence is isomorphic to $K^\bullet(X)$.

For a general flag type π , pass from $\underline{D}_\pi(\mathcal{E})$ to the complete flag bundle by successively splitting the $\mathcal{F}_i$. The complete flag K -ring is a free module both over the subalgebra generated by the $\xi_i^{(\alpha)}$ and over $K^\bullet(\underline{D}_\pi(\mathcal{E}))$, with the same basis. Therefore these subrings coincide. The relations follow from Lemma 4.3.

Corollary 3.36 (4.8). *With the hypotheses above, put*

$$c_i^{(j)} = \gamma^i(F_j - p_j), \quad 1 \leq j \leq k, \quad 1 \leq i \leq p_j.$$

Then $K^\bullet(X)$ is generated over $K^\bullet(S)$ by the $c_i^{(j)}$, and the ideal of relations is generated by those deduced from

$$\gamma_t(E - n) = \prod_{j=1}^k \gamma_t(F_j - p_j). \quad (121)$$

This is just Theorem VI.4.6 rewritten in terms of the γ -operations.

4.9–4.10. Infinite Grassmannians

Let $\underline{G}_n^{r,n}$ be the Grassmannian of type $(n, r - n)$ of $\mathcal{O}_S^r$, with $r \geq n$. As r varies these form an inductive system, denoted $\underline{G}_n^{\infty,n}$, and

$$K^\bullet(\underline{G}_n^{\infty,n}) = \varinjlim_r K^\bullet(\underline{G}_n^{r,n}).$$

The universal quotients give compatible classes c_i in this limit.

Proposition 3.37 (4.10). *For S quasi-compact there is an isomorphism*

$$K^\bullet(\underline{G}_n^{\infty,n}) \simeq K^\bullet(S)[[c_1, \dots, c_n]]/I,$$

where I is the closed ideal generated by the relations arising from the universal γ -identities for the tautological quotient and its complement.

For finite r the presentation of Theorem VI.4.6 gives

$$K^\bullet(\underline{G}_n^{r,n}) = K^\bullet(S)[c_1, \dots, c_n]/I_r,$$

with I_r homogeneous and supported in weights $> r - n$. Passing to the limit amounts to completing for the topology defined by the I_r . By passing to the splitting algebra and using the nilpotence

of the corresponding $\xi_j - 1$, this topology is equivalent to the weight topology, which proves the assertion.

5. Applications to projective bundles; study of f_*

Let S be quasi-compact, E locally free of rank $r + 1$, and $X = \mathbb{P}(E)$ with structural morphism $f : X \rightarrow S$. On X there is the canonical exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow f^*E \rightarrow \mathcal{O}_X(1) \rightarrow 0.$$

Let E, F, ξ denote the corresponding classes in $K^\bullet$. Put $x = \xi - 1$.

5.2

For a locally free sheaf $\mathcal{G}$ on X , the higher direct images $R^i f_* \mathcal{G}$ need not be locally free of finite type on S , so the direct image $f_* : K^\bullet(X) \rightarrow K^\bullet(S)$ is not defined by alternating sums of higher direct images in this setting. Theorem VI.1.1 provides a substitute. Every $y \in K^\bullet(X)$ has a unique expansion

$$y = a_0 + a_1 \xi + \cdots + a_r \xi^r, \quad a_i \in K^\bullet(S), \quad (122)$$

and we put

$$f_*(y) = a_0. \quad (123)$$

This is additive and satisfies the projection formula. Moreover

$$f_*(1) = 1, \quad f_*(\xi^n) = 0 \quad (-r \leq n < 0), \quad f_*(\xi^n) = s_n \quad (n \geq 0), \quad (124)$$

where s_n is the n -th complete symmetric function. The last relations follow by induction from the dependence relation and the Koszul relation (111).

Proposition 3.38 (5.3). *For every $k \geq 0$,*

$$\mathrm{Fil}^k K^\bullet(X) = \mathrm{Fil}^k K^\bullet(S) + \mathrm{Fil}^{k-1} K^\bullet(S)x + \cdots + \mathrm{Fil}^{k-r} K^\bullet(S)x^r, \quad (125)$$

as a direct sum of $K^\bullet(S)$ -modules, where the filtration is the canonical filtration of augmented λ -algebras.

Here x has zero augmentation and

$$\gamma_t(x) = \lambda_{t/(1-t)}(1 - \xi^{-1}) = \frac{1}{1 - xt},$$

so $\gamma^i(x) = x^i$. The result follows from the filtration computation for free λ -algebras in V, 4.12 and 4.13, together with the relation (113). Directness follows from the independence of $1, \dots, x^r$.

Corollary 3.39 (5.4). *The filtration induced on $K^\bullet(S)$ by $K^\bullet(X)$ is its original filtration.*

Corollary 3.40 (5.5). *If X is a product over S of flag bundles of locally free sheaves, then*

$$\mathrm{Gr} K^\bullet(S) \longrightarrow \mathrm{Gr} K^\bullet(X)$$

is injective.

Corollary 3.41 (5.6). *With the projective-bundle notation above, $\mathrm{Gr} K^\bullet(X)$ is a free $\mathrm{Gr} K^\bullet(S)$ -module with basis $1, x, \dots, x^r$, and*

$$\mathrm{Gr}^k K^\bullet(X) = \mathrm{Gr}^k K^\bullet(S) \oplus \mathrm{Gr}^{k-1} K^\bullet(S)x \oplus \dots \oplus \mathrm{Gr}^{k-r} K^\bullet(S)x^r.$$

The elements $1, x, \dots, x^{r+1}$ satisfy

$$\sum_{i=0}^{r+1} (-1)^i c_i(E) x^{r+1-i} = 0, \quad (126)$$

where $c_i(E)$ is the i -th Chern class of E .

Corollary 3.42 (5.7). *For a flag bundle as in 4.5, the $\mathrm{Gr} K^\bullet(S)$ -algebra $\mathrm{Gr} K^\bullet(X)$ is generated by the classes $\bar{\xi}_i^{(\alpha)}$ of $\xi_i^{(\alpha)} - 1$, with relations generated by*

$$c_t(E) = \prod_{i,\alpha} c_t(\xi_i^{(\alpha)}),$$

where c_t denotes the total Chern class.

Corollary 3.43 (5.8). *For every k ,*

$$f_*(\mathrm{Fil}^k K^\bullet(X)) \subset \mathrm{Fil}^k K^\bullet(S).$$

This is immediate from Proposition VI.5.3 and the projection formula.

Proposition 3.44 (5.9). *With the notation of 5.1, for every $y \in K^\bullet(X)$ one has*

$$f_*(\lambda_{-1}(\mathcal{F})y) = 0 \quad \text{and} \quad f_*(\gamma^*(\mathcal{F})y) = 0.$$

In particular,

$$f_*(\lambda_{-1}(\mathcal{F})) = 1.$$

Define coefficientwise

$$\xi_t : 1 + K^\bullet(X)[[t]]^+ \rightarrow 1 + K^\bullet(S)[[t]]^+, \quad \xi_t \left(\sum a_i t^i \right) = \sum f_*(a_i) t^i.$$

Although not generally a group homomorphism, it satisfies

$$\xi_t(m f^* y) = \xi_t(m) y.$$

Using $E = F + \xi$ and $\lambda_t(E) = \lambda_t(F)(1 + \xi t)$ gives the asserted identities.

Proposition 3.45 (5.10). *Let $z \in K^\bullet(X)$. The condition $z(1 - \xi) = 0$ is equivalent to the existence of $y \in K^\bullet(S)$ such that*

$$(i) \quad z = y\lambda_{-1}(\mathcal{F});$$

$$(ii) \quad y\lambda_{-1}(E) = 0.$$

The sufficiency follows from

$$\lambda_{-1}(E) = \lambda_{-1}(F)\lambda_{-1}(\xi) = \lambda_{-1}(F)(1 - \xi).$$

Conversely write $z = \sum_{i=0}^r a_i \xi^i$. Combining the dependence relation for the ξ^i with $z(1 - \xi) = 0$ and using the independence of $1, \dots, \xi^r$ expresses z as a multiple of $\lambda_{-1}(F)$, and the top relation gives $y\lambda_{-1}(E) = 0$.

Proposition 3.46 (5.11). *Let $z \in \text{Gr } K^\bullet(X)$. The condition $zx = 0$ is equivalent to the existence of $y \in \text{Gr } K^\bullet(S)$ such that*

$$(i) \quad z = y c^{\text{top}}(F);$$

$$(ii) \quad y c^{\text{top}}(E) = 0.$$

The proof is the same as that of 5.10, after passing to associated graded rings and using 5.6.

6. Study of the filtration of $K^\bullet(X)$ when X has an ample sheaf

Let X be a scheme possessing an ample sheaf $\mathcal{L}$. Then the two definitions of $K^\bullet(X)$, by locally free sheaves of finite type or by perfect complexes, coincide. Let

$$\varepsilon : K^\bullet(X) \rightarrow H^0(X, \mathbb{Z})$$

be the augmentation.

Proposition 3.47 (6.1). *If X is quasi-compact and has an ample sheaf, then $\text{Fil}^1 K^\bullet(X)$ is a nilideal.*

It suffices to show that $E - \varepsilon(E)$ is nilpotent for the class E of a locally free sheaf. By decomposing X into finitely many disjoint opens and then passing to the flag bundle of E , one reduces to the case where E is the class of an invertible sheaf. Let L be the class of an ample invertible sheaf. For n large, $E^\vee(n)$ is generated by global sections, giving a surjection $(E(-n))^m \rightarrow \mathcal{O}_X$ with locally free kernel. Thus

$$mEL^{-n} = 1 + F.$$

The corresponding γ -relation shows that $1 - EL^{-n}$, hence $L^n - E$, is nilpotent. Since $E - 1 = (L^n - 1) - (L^n - E)$ and $L^n - 1$ is also nilpotent for large n , the result follows.

Proposition 3.48 (6.2). *Let R be an augmented sub- λ -algebra of $K^\bullet(X)$ generated by the classes of finitely many sheaves. Then the λ -ring filtration of R is discrete and $\text{Fil}^1 R$ is nilpotent. Moreover R is a finite module over its image in $H^0(X, \mathbb{Z})$.*

After reducing to constant ranks, R is generated by finitely many elements $x_i = E_i - \varepsilon(E_i)$ and their γ -operations. Proposition 6.1 makes all these elements nilpotent. Hence the augmentation ideal of R is nilpotent. Since only finitely many non-zero monomials in the x_i and $\gamma^j(x_i)$ remain, the filtration is discrete and R is finite over the augmentation ring.

Corollary 3.49 (6.3). *Let A be a commutative ring. An element y of $K^\bullet(X) \otimes_{\mathbb{Z}} A$ is invertible if and only if its augmentation is invertible. In particular, the class of any nowhere-zero locally free sheaf of finite type becomes invertible in $K^\bullet(X) \otimes_{\mathbb{Z}} A$.*

The augmentation ideal is nil, so invertibility is detected after quotienting by it.

Corollary 3.50 (6.4). *Let X be quasi-compact with an ample sheaf, and let $f : X' \rightarrow X$ be proper and perfect. Assume either f is projective or X is noetherian, so that the finiteness theorem applies. If there is $x' \in K^\bullet(X' \otimes A)$ such that the augmentation of $f_*(x')$ is invertible, then*

$$f^* : K^\bullet(X \otimes A) \rightarrow K^\bullet(X' \otimes A)$$

is injective.

Indeed, by the projection formula,

$$f_*(f^*y \cdot x') = y \cdot f_*(x'),$$

and $f_*(x')$ is invertible by 6.3.

6.5–6.6. The codimension filtration

Assume now that X is noetherian of dimension d and has an ample sheaf. Define a filtration $K^\bullet(X)_i$ as follows: $x \in K^\bullet(X)_i$ if and only if for every closed subscheme $Y \subset X$ there is a perfect complex $M^\bullet(Y)$ with

- (a) $\text{cl}(M^\bullet) = x$;
- (b) $\text{codim}(\text{Supp } H^j(M^\bullet) \cap Y, Y) \geq i + j$, with $H^j(M^\bullet) = 0$ for $j \gg 0$.

These $K^\bullet(X)_i$ are additive subgroups.

Proposition 3.51 (6.6). *The filtration of 6.5 is a ring filtration, and $K^\bullet(X)_{d+1} = 0$.*

If x and x' are represented by perfect complexes $M^\bullet$ and $M'^\bullet$ satisfying the codimension estimates, then xx' is represented by $M^\bullet \otimes M'^\bullet$. The support of $H^n(M^\bullet \otimes M'^\bullet)$ is contained in the union of the intersections of supports with total degree n , and the codimension estimate follows from EGA 0_{IV}, 14.2.2.3. Finally $K^\bullet(X)_{d+1} = 0$ because a support in codimension $> d$ is empty.

Lemma 3.52 (6.7). *Let $X = \operatorname{Spec} A$ be affine noetherian, let $x_1, \dots, x_n$ be finitely many points of X , and let $\mathcal{E}$ be invertible. There is a section $s \in \Gamma(X, \mathcal{E})$ which is a basis of $\mathcal{E}_{x_i}$ for all i .*

The proof uses closed specializations x'_i with distinct maximal ideals $\mathfrak{m}_i$, the isomorphisms $M/\mathfrak{m}_i M \simeq A/\mathfrak{m}_i$ for $M = \Gamma(X, \mathcal{E})$, and Nakayama's lemma.

Lemma 3.53 (6.8). *Let X be noetherian with an ample sheaf $\mathcal{L}$, let $\mathcal{E}$ be invertible, and let $x_1, \dots, x_n$ be points of X . There is an integer N such that, for every $k \geq 1$, $\mathcal{E}(kN)$ has a global section which is a basis at every x_i .*

Choose a power of the ample sheaf and a section whose non-vanishing locus is affine and contains all x_i , apply Lemma 6.7 there, and extend after multiplying by a further power.

Theorem 3.54 (6.9). *Let X be a noetherian scheme of dimension d with an ample sheaf. Then*

$$\operatorname{Fil}^{d+1} K^\bullet(X) = 0,$$

and therefore $(\operatorname{Fil}^1 K^\bullet(X))^{d+1} = 0$.

First, if $\mathcal{E}$ and $\mathcal{E}'$ are invertible, with classes E, E' , then $E - E' \in K^\bullet(X)_1$. For a closed $Y \subset X$, choose one point in each irreducible component. Lemma 6.8 gives morphisms from suitable negative powers of the ample sheaf to $\mathcal{E}$ and $\mathcal{E}'$ which are isomorphisms at those points. The two-term complex built from these morphisms has class $E - E'$ and is acyclic at the chosen generic points of Y , hence its cohomology supports have codimension at least 1 in Y .

Now take $x \in \operatorname{Fil}^{d+1} K^\bullet(X)$. It suffices to treat products

$$\gamma^{i_1}(E_1 - F_1) \cdots \gamma^{i_k}(E_k - F_k), \quad \sum i_j \geq d+1,$$

with E_i, F_i represented by locally free sheaves of the same constant rank. Pull back to the product of the flag bundles of these sheaves. The pullback is injective on $K^\bullet$ and splits each class into sums of invertible classes. For invertible classes ξ, η ,

$$\gamma_t(\xi - \eta) = \frac{\gamma_t(\xi - 1)}{\gamma_t(\eta - 1)} = \frac{1 + (\xi - 1)t}{1 + (\eta - 1)t},$$

so $\gamma^i(\xi - \eta) = (-1)^{i-1}(\xi - \eta)(\eta - 1)^{i-1} \in K^\bullet(X')_i$. Thus the pullback of x belongs to $K^\bullet(X')_{d+1}$. Multiplying by a basis element of $K^\bullet(X')$ over $K^\bullet(X)$ of relative codimension r gives an element of $K^\bullet(X')_{d+1+r} = 0$, since $\dim X' = d + r$. By the basis property and injectivity, $x = 0$.

Problem 3.55 (6.10). Under the hypotheses of Theorem VI.6.9, is it true that for every k one has

$$K^\bullet(X)_k = \operatorname{Fil}^k K^\bullet(X)?$$

Bibliography for Expose VI

- [1] H. Bass, *K-Theory and stable algebra*, Publications Mathematiques de l'IHES, no. 22.

- [2] C. Chevalley, *Seminaire 1958: Anneaux de Chow et applications*.
- [3] M. Hakim, *Topos localement anneles et schemas relatifs*, thesis in preparation.
- [4] R. Hartshorne, *Residues and duality*, Lecture Notes in Mathematics 20, Springer-Verlag.
- [5] J.-P. Serre, *Le groupe de Grothendieck des schemas en groupes reductifs*, Publications Mathematiques de l'IHES, no. 34.

SGA 6, Exposé VII

1. Generalities on regular immersions

The classical notion of regular immersion from EGA IV, 16.9.2, is not adequate in the non-noetherian case. We shall use a weaker notion, which nevertheless has the required properties and which agrees with the usual definition in the noetherian case.

Definition 1.1. Let A be a commutative ring, let E be a finite projective A -module, and let $u : E \rightarrow A$ be an A -linear homomorphism. We say that u is *regular* if the Koszul complex $K_\bullet(u)$ defined by u is acyclic in positive degrees; equivalently, it is a resolution of $B = A/u(E)$.

Recall that the Koszul complex attached to u is the exterior algebra $\wedge(E)$ as a graded A -module, with boundary the interior multiplication by u , considered as an element of $\check{E}$. This is the antiderivation of degree -1 of $\wedge(E)$ which is equal to u on E .

Lemma 1.2. Let $u : E \rightarrow A$ be regular. Let A' be an A -algebra, let $u' : E \otimes_A A' \rightarrow A'$ be obtained from u by base change, and suppose that u' is regular. Put $J = u(E)$ and $J' = JA' = u'(E \otimes_A A')$. Then, for every integer $n \geq 0$, one has

$$\begin{aligned} J^n/J^{n+1} \otimes_A A' &\xrightarrow{\sim} J'^n/J'^{n+1}, \\ \mathrm{Tor}_i^A(J^n/J^{n+1}, A') &= 0 \quad (i \geq 1), \\ J^n \otimes_A A' &\xrightarrow{\sim} J'^n, \\ \mathrm{Tor}_i^A(J^n, A') &= 0 \quad (i \geq 1). \end{aligned}$$

Proof. First suppose that the quotients J^n/J^{n+1} are projective over $B = A/J$. Since J is of finite type, they are locally free of finite type over B . The assertions are local on $\mathrm{Spec} A$, so we may assume that A is local and that $u(E)$ is contained in the maximal ideal. The quotients J^n/J^{n+1} are then free over B , and the vanishing of their higher Tors after tensoring with A' is reduced to the vanishing of $\mathrm{Tor}_i^A(B, A')$. These groups are the homology groups of $K_\bullet(u) \otimes_A A' = K_\bullet(u')$, and vanish in positive degrees by the regularity of u' .

It follows by induction on n , using

$$0 \rightarrow J^n/J^{n+1} \rightarrow A/J^{n+1} \rightarrow A/J^n \rightarrow 0,$$

that $\mathrm{Tor}_i^A(A/J^n, A') = 0$ for $i \geq 1$. The identifications of J^n/J^{n+1} after base change then give the identifications of J^n by using

$$0 \rightarrow J^n \rightarrow A \rightarrow A/J^n \rightarrow 0,$$

and the remaining Tor-vanishing follows from

$$0 \rightarrow J^{n+1} \rightarrow J^n \rightarrow J^n/J^{n+1} \rightarrow 0.$$

It remains to know that J^n/J^{n+1} is projective over B . This is local; assume E is free with basis mapping to $f_1, \dots, f_d$. Let $A_0 = \mathbb{Z}[T_1, \dots, T_d]$ and map T_i to f_i . The corresponding universal homomorphism is regular, and the preceding part applies to the base change $A_0 \rightarrow A$. Since J_0^n/J_0^{n+1} is free of finite type in the universal case, $J^n/J^{n+1} \simeq J_0^n/J_0^{n+1} \otimes_{A_0} A$ is locally free of finite type. This proves the lemma.

Proposition 1.3. With the hypotheses of 1.2, the ideal $J = u(E)$ is quasi-regular in the sense of EGA IV, 16.9.7. Thus J is of finite type, J/J^2 is a finite projective A/J -module, and the canonical homomorphism

$$\mathrm{Sym}_{A/J}(J/J^2) \longrightarrow \mathrm{gr}_J(A) = \bigoplus_{n \geq 0} J^n/J^{n+1}$$

is an isomorphism.

Proof. We have just proved that J/J^2 is finite projective. Locally, using the same universal polynomial algebra as in the proof of 1.2, the displayed morphism is obtained by base change from the corresponding morphism for the ideal generated by the variables. In that case it is an isomorphism. Hence it is an isomorphism in general.

Definition 1.4. Let $(X, \mathcal{A})$ be a commutatively locally ringed site, let $\underline{E}$ be an $\mathcal{A}$ -module locally free of finite type, and let $u : \underline{E} \rightarrow \mathcal{A}$ be an $\mathcal{A}$ -linear homomorphism. We say that u is regular if its Koszul complex is acyclic in positive degrees. An ideal $\mathcal{J}$ of $\mathcal{A}$ is called regular if it is the image of some surjective regular homomorphism $\underline{E} \rightarrow \mathcal{J}$.

Definition 1.5. Let $i : Y \rightarrow X$ be an immersion of schemes. Let U be an open subset of X containing $i(Y)$ such that i is a closed immersion into U . We say that i is *regular* if the ideal of $\mathcal{O}_U$ defining $i(Y)$ is regular; this condition does not depend on U . The immersion is regular at $y \in Y$ if this is true after restricting to some neighbourhood of $i(y)$.

This definition replaces the one of EGA IV, 16.9.2. In the noetherian case the two definitions are equivalent, by EGA IV, 19.5.1. Any immersion which is locally defined by a regular sequence is regular, and every regular immersion is quasi-regular.

1.4.1–1.4.3. Let $\mathcal{J}$ be a finite type ideal of $\mathcal{O}_X$, and let $x \in \mathrm{Supp}(\mathcal{O}_X/\mathcal{J})$. A family of sections f_i of $\mathcal{J}$ near x is a minimal system of generators near x when its images form a basis of $\mathcal{J} \otimes k(x)$. If $\mathcal{J}/\mathcal{J}^2$ is locally free, a basis of $\mathcal{J}/\mathcal{J}^2$ near x can be lifted to a minimal system of generators of $\mathcal{J}$ near x , by Nakayama's lemma. Thus regularity may be checked on any minimal system of local generators: the corresponding Koszul complex must be acyclic near the point.

Proposition 1.5. A regular immersion remains regular after flat base change. Conversely, regularity may be checked locally for the quasi-compact faithfully flat topology.

Proof. Flat base change carries a Koszul resolution to the base-changed Koszul complex and preserves acyclicity. Conversely, if after a faithfully flat quasi-compact base change the relevant

Koszul complex is acyclic, then its positive homology sheaves vanish after a faithful flat pullback and therefore vanish already before base change.

Let $i : Y \rightarrow X$ be a regular closed immersion defined by $\mathcal{J}$. The *conormal sheaf* of Y in X is

$$\underline{N}_{X/Y} = \mathcal{J}/\mathcal{J}^2.$$

It is locally free of finite type. Its rank at a point is the codimension of the immersion at that point. We write it as $\text{codim}(i)$ when it is constant.

Proposition 1.6. Let $i : Y \rightarrow X$ and $j : Z \rightarrow Y$ be two immersions. Then there is a natural exact sequence

$$j^*(\underline{N}_{X/Y}) \rightarrow \underline{N}_{X/Z} \rightarrow \underline{N}_{Y/Z} \rightarrow 0.$$

Proof. Let $\mathcal{I}$ be the ideal of Y in X , $\mathcal{J}$ the ideal of Z in X , and $\mathcal{K}$ the ideal of Z in Y . From

$$0 \rightarrow \mathcal{I} \rightarrow \mathcal{J} \rightarrow \mathcal{K} \rightarrow 0$$

and after tensoring first by $\mathcal{O}_Y$ and then by $\mathcal{O}_Z$, one obtains

$$j^*(\mathcal{I}/\mathcal{I}^2) \rightarrow \mathcal{J}/\mathcal{J}^2 \rightarrow \mathcal{K}/\mathcal{K}^2 \rightarrow 0,$$

which is the asserted sequence.

Proposition 1.7. Let $i : Y \rightarrow X$ and $j : Z \rightarrow Y$ be immersions.

i) If i and j are regular, then $i \circ j$ is regular, and

$$0 \rightarrow j^*(\underline{N}_{X/Y}) \rightarrow \underline{N}_{X/Z} \rightarrow \underline{N}_{Y/Z} \rightarrow 0$$

is exact. In particular $\text{codim}(i \circ j) = \text{codim}(i) + \text{codim}(j)$.

ii) If $i \circ j$ and i are regular, and the preceding conormal sequence is exact and locally split, then j is regular.

iii) If X is noetherian, and if $i \circ j$ and j are regular, then i is regular at the points of Z .

Proof. The assertion is local, so write $X = \text{Spec } A$, $Y = \text{Spec } B$, $Z = \text{Spec } C$, with $B = A/I$ and $C = A/J = B/K$. Choose minimal systems of generators $f_1, \dots, f_n$ for I and $\bar{g}_1, \dots, \bar{g}_m$ for K , lifted to elements g_j of A . The situation is obtained from the standard diagram

$$\mathbb{Z}[T_1, \dots, T_{n+m}] \rightarrow \mathbb{Z}[T_1, \dots, T_m] \rightarrow \mathbb{Z}$$

by base change. The regularity conditions are expressed by the vanishing of the corresponding Tor

groups. The transitivity spectral sequence

$$E_{pq}^2 = \mathrm{Tor}_p^{B_0}(C_0, \mathrm{Tor}_q^{A_0}(B_0, A)) \implies \mathrm{Tor}_{p+q}^{A_0}(C_0, A)$$

then gives (i) and (ii). The rank computation for the conormal sheaves gives exactness on the left. Part (iii) is the noetherian statement of EGA IV, 19.1.5, using the equivalence of definitions in the noetherian case.

Proposition 1.8. Let X be a scheme, let $\underline{E}$ be a locally free $\mathcal{O}_X$ -module of finite type, and let $u : \underline{E} \rightarrow \mathcal{O}_X$ be regular with image $\mathcal{J}$. Let Y be the closed subscheme defined by $\mathcal{J}$, let X' be the blow-up of X along Y , let $P = \mathbb{P}(\underline{E})$ with structural map $p : P \rightarrow X$, and consider on P the exact sequence

$$0 \rightarrow \underline{E} \rightarrow p^* \underline{E} \xrightarrow{v} \mathcal{O}_P(1) \rightarrow 0.$$

Then:

- i) The formation of X' commutes with every base change $T \rightarrow X$ for which the pullback of u is regular.
- ii) The closed immersion $j : X' \rightarrow P$ defined by the surjection $S(\underline{E}) \rightarrow \bigoplus_{n \geq 0} \mathcal{J}^n$ is regular. Its ideal is the image of the regular morphism v ; if $\mathrm{codim}(Y, X)$ is constant, then $\mathrm{codim}(j) = \mathrm{codim}(i) - 1$.
- iii) The graded $S(\underline{E})$ -module $\bigoplus_{n \geq 0} \mathcal{J}^n$ is perfect as a graded module: locally it has a finite graded resolution by graded free modules of finite type.

Proof. The first assertion follows from $X' = \mathrm{Proj}(\bigoplus \mathcal{J}^n)$ and the base-change formulas of 1.2. For (ii), the question is local. In the standard case $A = \mathbb{Z}[X_1, \dots, X_d]$, with $u(e_i) = X_i$, one works on the standard open $D^+(X_1)$ of the projective bundle. The kernel of $p^* \underline{E} \rightarrow \mathcal{O}_P(1)$ is generated by $e_i - (X_i/X_1)e_1$, $i > 1$, and the functions defining the image of v form a regular sequence; they generate exactly the ideal of the blow-up in the projective bundle. The general case is obtained by base change from this standard case, and acyclicity follows from the Tor-vanishing in 1.2. For (iii), construct in the standard noetherian case a finite graded free resolution, and tensor it with the given ring. The homology in positive degree is computed by the Tor groups $\mathrm{Tor}_i(A, J_0^n)$, which vanish by 1.2; hence the tensor complex is the desired resolution.

Proposition 1.9. Let $i : Y \rightarrow X$ be a regular closed immersion, let X' be the blow-up of X along Y , and let $f : X' \rightarrow X$ be the structural morphism. Then i and f are perfect morphisms.

Proof. Locally the structure sheaf of Y has a finite locally free resolution given by a Koszul complex, hence i is perfect. By 1.8, f factors as a regular immersion into a projective bundle followed by the smooth structural morphism of the projective bundle. Both factors are perfect.

Proposition 1.10. If X and Y are smooth schemes over S , then every immersion $X \rightarrow Y$ is regular. In particular every section of a smooth morphism is a regular immersion.

This follows at once from EGA IV, 17.12.1, because in the noetherian smooth case the EGA notion of regular immersion agrees with the present one.

2. Computations for regular immersions

2.1–2.3. Let A be a ring and let B, C be A -algebras. The graded module

$$\bigoplus_{i \geq 0} \mathrm{Tor}_i^A(B, C)$$

has a natural structure of an associative graded algebra over $B \otimes_A C$. Indeed, take a projective resolution $P_\bullet$ of B . The product is obtained by composing the canonical homology product

$$H_i(P_\bullet \otimes_A C) \otimes_A H_j(P_\bullet \otimes_A C) \rightarrow H_{i+j}((P_\bullet \otimes_A C) \otimes_A (P_\bullet \otimes_A C))$$

with a lifting $P_\bullet \otimes_A P_\bullet \rightarrow P_\bullet$ of the multiplication $B \otimes_A B \rightarrow B$, and then with the multiplication of C . The result is independent of choices and is functorial in B and C .

If the resolution $P_\bullet$ is itself a differential graded algebra, then this product is computed directly by the product of $P_\bullet$. In particular, for a quotient by a regular ideal one may use the Koszul resolution; hence the resulting Tor-algebra is alternating. These constructions globalize: for two X -schemes X' and X'' , the sheaf

$$\bigoplus_{i \geq 0} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''})$$

is a graded associative $\mathcal{O}_{X' \times_X X''}$ -algebra.

Lemma 2.4. Let X' and X'' be two X -schemes, and let $\mathcal{M}$ be a flat $\mathcal{O}_{X'}$ -module. Then there is a canonical isomorphism

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}) \otimes_{\mathcal{O}_{X'}} \mathcal{M} \xrightarrow{\sim} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_{X''}).$$

Proof. This is affine-local. If $X = \mathrm{Spec} A$, $X' = \mathrm{Spec} A'$, $X'' = \mathrm{Spec} A''$, and M is a flat A' -module, a projective A -resolution of A'' gives

$$M \otimes_{A'} (A' \otimes_A P_\bullet) \simeq M \otimes_A P_\bullet,$$

and passage to homology gives the assertion.

Proposition 2.5. Let Y and Z be closed subschemes of X , and put $T = Y \times_X Z$. Assume that locally near every point of T the schemes Y , Z , and T are defined by Koszul resolutions associated to systems of sections $f_1, \dots, f_n$, $g_1, \dots, g_m$, and their union. Then

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z)$$

for all $i \geq 0$, and

$$\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \mathrm{Ker}(i^* \underline{N}_{X/Y} \oplus j^* \underline{N}_{X/Z} \rightarrow \underline{N}_{X/T}),$$

where $i : T \rightarrow Y$ and $j : T \rightarrow Z$ are the induced immersions. In particular, if $Y \rightarrow X$ is a regular closed immersion, then

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z), \quad \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq i^* \underline{N}_{X/Y}.$$

Proof. The Tor-algebra is alternating by the Koszul calculation of 2.2, hence there is a unique algebra homomorphism from the exterior algebra on degree one. Locally, the tensor product of the two Koszul complexes is the Koszul complex for the union of the two systems. The spectral sequence of the associated bicomplex degenerates, and the homology is exactly the exterior algebra on the degree-one homology. The description of degree one follows from the exact sequence

$$0 \rightarrow \mathcal{I} \cap \mathcal{J} \rightarrow \mathcal{I} \oplus \mathcal{J} \rightarrow \mathcal{I} + \mathcal{J} \rightarrow 0$$

and the Tor exact sequence; it identifies the degree-one Tor with the kernel of the difference map between the two conormal sheaves and the conormal sheaf of T .

2.6. Suppose that X has an ample sheaf. If $i : Y \rightarrow X$ is a regular closed immersion, then i is perfect and the additive morphism

$$i_* : K'(Y) \rightarrow K'(X)$$

is defined by sending a perfect complex on Y to its direct image on X .

Proposition 2.7. Let X have an ample sheaf, let $i : Y \rightarrow X$ be a regular closed immersion, and let $\underline{N}$ be the conormal sheaf of Y in X . Then, for every $x \in K_\bullet(Y)$,

$$i^! i_*(x) = x \cdot \lambda_{-1}(\underline{N}), \quad \lambda_{-1}(\underline{N}) = \sum_{i \geq 0} (-1)^i \lambda^i(\underline{N}).$$

Proof. By additivity it is enough to take x to be the class of a locally free $\mathcal{O}_Y$ -module $\mathcal{M}$ of finite type. The homology sheaves of $\mathbf{L}i^* \mathbf{R}i_*(\mathcal{M})$ are

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_Y) \simeq \mathcal{M} \otimes_{\mathcal{O}_Y} \bigwedge^i \underline{N}$$

by 2.4 and 2.5. Their alternating sum in $K'(Y)$ is precisely $x \lambda_{-1}(\underline{N})$.

Corollary 2.8. Under the hypotheses of 2.7, the self-intersection formula is

$$i_*(y \cdot x \cdot \lambda_{-1}(\underline{N})) = i_*(y) \cdot i_*(x)$$

for $x, y \in K_\bullet(Y)$. This follows from 2.7 and the projection formula.

3. Computation of K' of a blow-up

Let $i : Y \rightarrow X$ be a regular closed immersion. Let X' be the blow-up of X along Y , put $Y' = Y \times_X X'$, and write $f : X' \rightarrow X$ and $g : Y' \rightarrow Y$ for the structural morphisms. The immersion $j : Y' \rightarrow X'$ is a regular closed immersion of codimension one, defined by an invertible ideal. If $\mathcal{J}$ is the ideal of Y in X , then $\mathcal{J}\mathcal{O}_{X'} = \mathcal{O}_{X'}(1)$.

Since

$$Y' = \text{Proj}\left(\bigoplus_{n \geq 0} \mathcal{J}^n / \mathcal{J}^{n+1}\right) = \text{Proj}(\text{Sym}(\underline{N})) = \mathbb{P}(\underline{N}),$$

where $\underline{N} = \mathcal{J}/\mathcal{J}^2$, there is on Y' an exact sequence

$$0 \rightarrow \underline{F} \rightarrow g^* \underline{N} \rightarrow \mathcal{O}_{Y'}(1) \rightarrow 0.$$

The sheaf $\mathcal{O}_{Y'}(1)$ is also the conormal sheaf of j .

Lemma 3.2. With these notations,

$$\text{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \text{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y), \quad \text{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \underline{F}.$$

Proof. Work locally where $\mathcal{J}$ is defined by a regular sequence $f_1, \dots, f_d$. On the standard open of the blow-up on which $f_1^{(1)}$ is invertible, change the basis of the free module defining the Koszul complex by setting $e'_1 = e_1$ and $e'_i = e_i - (f_i/f_1)e_1$ for $i > 1$. The differential sends e'_1 to f_1 and the other e'_i to zero. Since f_1 is a non-zero-divisor on that open, the homology is the exterior algebra on the image of the module spanned by $e'_2, \dots, e'_d$, which is $\underline{F}$. These local identifications are compatible with the algebra structure and glue.

3.3. If X has an ample sheaf, then so do X' , Y , and Y' . The immersions i and j , the morphism g (a projective bundle), and the morphism f are perfect. Hence we have additive maps

$$i_* : K'(Y) \rightarrow K'(X), \quad j_* : K'(Y') \rightarrow K'(X'), \quad f_* : K'(X') \rightarrow K'(X), \quad g_* : K'(Y') \rightarrow K'(Y).$$

Proposition 3.4. For every $x \in K'(Y)$,

$$f^* i_*(x) = j_*(g^*(x) \lambda_{-1}(\underline{F})).$$

Equivalently, after applying f_* to the two sides, this is the usual compatibility of the Gysin construction with the exceptional divisor.

Proof. It is enough by additivity to take x to be a locally free sheaf $\mathcal{M}$ on Y . The homology of

$\mathbf{L}f^*\mathbb{R}i_*(\mathcal{M})$ is $\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{M}, \mathcal{O}_{X'})$, and 2.4 together with Lemma 3.2 identifies this with $g^*\mathcal{M} \otimes \bigwedge^i \underline{F}$. Taking the alternating class gives the displayed formula.

Lemma 3.5. For the blow-up $f : X' \rightarrow X$,

- i) $R^i f_* \mathcal{O}_{X'}(n) = 0$ for $i \geq 1$ and $n \geq 0$;
- ii) $f_* \mathcal{O}_{X'}(n) = \mathcal{J}^n$ for $n \geq 0$.

Proof. The assertion is local on X . For $n \gg 0$ the vanishing follows from the standard acyclicity theorem for projective perfect morphisms. The exact sequences

$$0 \rightarrow \mathcal{O}_{X'}(n+1) \rightarrow \mathcal{O}_{X'}(n) \rightarrow \mathcal{O}_{Y'}(n) \rightarrow 0$$

and the known cohomology of $\mathcal{O}(n)$ on the projective bundle $Y' = \mathbb{P}(\underline{N})$ permit descending induction to all $n \geq 0$. The identification of the direct image with $\mathcal{J}^n$ is first obtained for large n from a finite graded presentation of $\bigoplus \mathcal{J}^n$, then descended by the same exact sequences.

Proposition 3.6. With the above notation,

$$f_* f^* = \mathrm{Id}_{K'(X)}.$$

Proof. The projection formula gives $f_* f^*(x) = x \cdot f_*(1)$. By Lemma 3.5, $\mathbb{R}f_* \mathcal{O}_{X'}$ has no higher cohomology and degree-zero cohomology $\mathcal{O}_X$, so $f_*(1) = 1$.

Theorem 3.7. Suppose X is quasi-compact and has an ample sheaf. Then the sequence

$$0 \longrightarrow K'(Y) \xrightarrow{u} K'(Y') \times K'(X) \xrightarrow{v} K'(X')$$

is exact, where

$$u(y) = (g^*(y)\lambda_{-1}(\underline{F}), -i_*(y)), \quad v(y', x) = j_*(y') + f^*(x).$$

Moreover u admits the left inverse $u'(y', x) = g_*(y')$. If, in addition, X is noetherian and Y is regular, then v is surjective, and hence

$$0 \rightarrow K'(Y) \rightarrow K'(Y') \times K'(X) \rightarrow K'(X') \rightarrow 0$$

is exact.

Proof. The identity $g_*\lambda_{-1}(\underline{F}) = 1$, together with the projection formula, gives the left inverse of u . The equality $v \circ u = 0$ is exactly Proposition 3.4. Conversely, if $j_*(y') + f^*(x) = 0$, applying f_* and using 3.6 gives $x = -i_*g_*(y')$. Put $y = g_*(y')$. It remains to prove $y' = g^*(y)\lambda_{-1}(\underline{F})$. The difference $z = y' - g^*(y)\lambda_{-1}(\underline{F})$ has $g_*(z) = 0$ and $j_*(z) = 0$. The self-intersection formula gives

$j^*j_*(z) = z(1 - L)$, where L is the conormal line of j . The projective-bundle computation implies that the simultaneous kernel of j_* and g_* is zero; hence $z = 0$.

If X is noetherian and Y regular, then Y' and X' are regular along the exceptional divisor, and the usual devissage and localization argument gives the surjectivity of v .

Remark 3.8. In the noetherian regular case, Theorem 3.7 determines the abelian group $K'(X')$ from the λ -rings $K'(X)$ and $K'(Y)$, the maps i_* and i^* , and the class of the conormal sheaf N in $K'(Y)$. The projective-bundle formula computes $K'(Y')$ from $K'(Y)$ and N . The next results describe the multiplicative and rational λ -structures from the same data.

Proposition 3.9. Let X be noetherian and Y regular. In the notation of 3.7, the exterior-power operations on the class represented by $f^*(x) = j_*(y') + x'$ are computed by the same universal formulas as in the projective-bundle case: if $y = g_*(y')$, then

$$f^*(\lambda^p(x)) = j_*\left(\sum_{i=0}^{p-1} (-1)^i \lambda^{p-1-i}(N) g^*(\lambda^i(y))\right) + \lambda^p(x')$$

whenever the displayed decomposition is defined. The proof reduces to the case where i is a section of a projective bundle, by using the blow-up square and functoriality of the operations λ^p .

4. Self-intersection formulas for the operations λ^p and γ^p

Proposition 4.1. Let $i : Y \rightarrow X$ be a regular immersion of codimension d , and suppose the ideal of Y is generated by a regular sequence of d sections. If N is the conormal class and $y \in K'(Y)$, then for every $p \geq 1$,

$$i^*i_*(\lambda^p(Ny)) = \lambda^p(N)\lambda^p(y).$$

The proof reduces to the zero section of a projective space bundle.

Propositions 4.2–4.3. The preceding formula holds for every regular immersion. One embeds the situation into successive projective bundles so that the composite immersion is given by a regular sequence, proves the formula for the composite, and descends step by step.

Consequently, for every regular immersion $i : Y \rightarrow X$ and every $y \in K'(Y)$,

$$i_*(\lambda^p(Ny)) = \sum_{k=0}^p (-1)^k \lambda^{p-k}(N) \lambda^k(i_*(y)) \quad \text{modulo torsion.}$$

Theorem 4.3. Let $i : Y \rightarrow X$ be a regular immersion with conormal class N . For every $y \in K'(Y)$ and every $p \geq 1$,

$$i_*(\gamma^p(Ny)) = \gamma^p(i_*(y)) \quad \text{modulo torsion.}$$

Proof. Apply the previous exterior-power formula to the generating functions defining the Adams and Grothendieck operations. If

$$u(t) = 1 + \sum_{k \geq 1} i_*(\lambda^k(Ny))t^k, \quad v(t) = \sum_{k \geq 0} \lambda^k(i_*(y))t^k,$$

then the logarithmic-derivative relation implies $d(u(t)/v(t))/dt \equiv 0$ modulo torsion. Since the constant term is 1, integration gives $u(t) \equiv v(t)$, which is equivalent to the stated congruences for the γ^p .

Corollary 4.4. The same congruence follows directly for the operations γ^p , since they are integral linear combinations of the exterior-power operations in the universal formulas.

Remark 4.5. The proof suggests that the congruence should in fact be an equality, not merely an equality modulo torsion. In characteristic zero this was proved in Riemann-Roch references by a geometric characterization of the classes $\lambda^p(Nx)$. Such a method should extend to all characteristics once exterior powers are available for perfect complexes.

4.6. Application to the multiplicative structure

Keep the notation of 3.1 and 3.3, and suppose X noetherian and Y regular. Theorem 3.7 describes $K'(X')$ as an abelian group. Its multiplication is determined by the rings $K'(X)$ and $K'(Y)$, the maps i_* and i^* , and the conormal class N .

More explicitly, for $y'_1, y'_2 \in K'(Y')$ and $x_1, x_2 \in K'(X)$,

$$\begin{aligned} (j_*(y'_1) + f^*(x_1))(j_*(y'_2) + f^*(x_2)) \\ = j_*(y'_1 g^*(g_*(y'_2))\lambda_{-1}(\underline{F})) + j_*(y'_1 f^* i^*(x_2)) + j_*(y'_2 f^* i^*(x_1)) + f^*(x_1 x_2). \end{aligned}$$

This is just the projection formula, the self-intersection formula for j , and Theorem 3.7.

4.7. Rational λ -ring structure of $K'(X')$

After tensoring with $\mathbb{Q}$, the λ -ring structure of $K'(X')$ is determined by the same data. Recall that over a $\mathbb{Q}$ -algebra the datum of a binomial-ring structure is equivalent to a λ -ring structure. If $y' \in K'(Y')$ and $y = g_*(y')$, then

$$j^* j_*(y') = y' g^*(N)$$

and the Adams operations satisfy

$$\psi^k(j_*(y')) = j_*(g^*(\psi^k(y))\lambda_{-1}(\underline{F})^{k-1}).$$

Together with the formulas of 3.9 and the known λ -structures on $K'(X)_{\mathbb{Q}}$ and $K'(Y)_{\mathbb{Q}}$, these identities determine the operations λ^p on $K'(X')_{\mathbb{Q}}$.

5. Complements and applications

5.1. Complete intersections

Proposition 5.1. Let X be a regular scheme with an ample sheaf, and let Y be a closed subscheme which is a complete intersection of order r . Then the ideal of Y is locally generated by an $\mathcal{O}_X$ -regular sequence of r sections.

Theorem 5.2. Let X be a regular scheme with an ample sheaf, let Z be an irreducible closed subscheme of X , and let z be its generic point. Suppose the maximal ideal of $\mathcal{O}_{X,z}$ can be generated by an $\mathcal{O}_{X,z}$ -regular sequence. Then there is an open neighbourhood U of z such that $Z \cap U \rightarrow U$ is a regular immersion.

Proof. The ideal of Z at its generic point is the maximal ideal of the regular local ring $\mathcal{O}_{X,z}$, and by hypothesis it has a regular system of generators. After shrinking X near z , these generators extend to sections defining a regular ideal. Hence $Z \cap U \rightarrow U$ is regular, of the expected codimension.

5.3. Filtrations and Gysin morphisms

For the filtration by codimension of support, a regular immersion $i : Y \rightarrow X$ of codimension d satisfies

$$i_*(\mathrm{Fil}^k K'(Y)) \subset \mathrm{Fil}^{k+d} K'(X).$$

For the blow-up $f : X' \rightarrow X$ along Y ,

$$f^*(\mathrm{Fil}^k K'(X)) \subset \mathrm{Fil}^k K'(X').$$

The exact sequence of Theorem 3.7 is compatible with these filtrations in the noetherian regular case.

5.6. Surjectivity after tensoring with $\mathbb{Q}$

Suppose X is noetherian and Y regular. The surjectivity in Theorem 3.7 remains true after tensoring with $\mathbb{Q}$. If $x' \in \mathrm{Fil}^k K'(X')_{\mathbb{Q}}$, one may choose $(y', x) \in K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}$ with

$$x' = v(y', x), \quad x \in \mathrm{Fil}^k K'(X)_{\mathbb{Q}}.$$

Filtering the product by

$$\mathrm{Fil}^k(K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}) = \mathrm{Fil}^{k-1} K'(Y')_{\mathbb{Q}} \times \mathrm{Fil}^k K'(X)_{\mathbb{Q}},$$

one obtains a surjection

$$\mathrm{Fil}^k(K'(Y')_{\mathbb{Q}} \times K'(X)_{\mathbb{Q}}) \longrightarrow \mathrm{Fil}^k K'(X')_{\mathbb{Q}}.$$

Indeed, if $x' = \gamma^k(x'_1)$ with augmentation zero, lift x'_1 to a pair (y'_1, x_1) with x_1 of filtration at least one; the formula for the γ -operation in the product ring gives a lift of x' in the required filtered degree.

SGA 6, Exposé VIII, opening to 3.4

1. Complete-intersection morphisms

Definition 1.1. Let $f : X \rightarrow Y$ be a morphism of schemes. We say that f is a *complete-intersection morphism* if, for every point $x \in X$, there exist an open neighbourhood U of x and a smooth Y -scheme V such that the restriction $f|_U$ factors as

$$U \xrightarrow{i} V \rightarrow Y,$$

where i is a regular immersion in the sense of Exposé VII. After replacing V by an open subscheme, one may suppose that i is a regular closed immersion.

In this definition one may replace V by an affine space $\mathbb{A}_Y^n$. Indeed, locally V embeds as a closed subscheme of an affine space over an affine open of Y ; this immersion is regular because V is smooth, and the composite of regular immersions is regular.

Proposition 1.2. Let $f : X \rightarrow Y$ be a morphism. Suppose that two factorizations of f through smooth Y -schemes are related by a smooth morphism between the intermediate schemes. Then the immersion in one factorization is regular if and only if the immersion in the other factorization is regular.

Proof. If $i : X \rightarrow V$ is regular and $V \rightarrow V'$ is smooth, form $X' = V \times_{V'} X$. The induced section $X \rightarrow X'$ is a regular immersion, being a section of a smooth morphism, and the base change of i is regular. Hence the composite immersion is regular. Conversely, when the smooth morphism is étale the comparison is immediate because the induced section is open; the general smooth case is reduced étale-locally to this situation.

Proposition 1.3. Suppose $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and $f : X \rightarrow Y$ factors through a regular closed immersion $j : X \rightarrow \mathbb{A}_Y^n$. Then one may find an analogous factorization through $\mathbb{A}_Y^{n+1}$ in which a section of the projection is a regular immersion and the remaining immersion is regular.

Proof. Choose lifts in A of the coordinate functions defining the original factorization. They define a section of the larger affine space. Since a section of a smooth morphism is a regular immersion, and since the conormal sequence for the two immersions is exact and locally split, the remaining immersion is regular by Exposé VII, 1.7.

Proposition 1.4. The composite of two complete-intersection morphisms is a complete-intersection morphism.

Proof. Factor each morphism locally as a regular immersion followed by a smooth morphism. After restricting to suitable opens, the composite factors through the fiber product of the two smooth intermediates; the new intermediate is smooth and the induced immersion is regular by Exposé VII, 1.7.

Proposition 1.5. Complete-intersection morphisms are stable under arbitrary base change.

Proof. This follows from the definition, from the stability of smooth morphisms under base change, and from the flat-base-change stability of regular immersions.

Proposition 1.6. A morphism which locally factors as a regular immersion followed by a smooth morphism is perfect. In particular every complete-intersection morphism is perfect.

Proof. A regular immersion is perfect by Expose VII, 1.9, and a smooth morphism is perfect. The composite of perfect morphisms is perfect.

Proposition 1.7. If a complete-intersection morphism is flat and of finite presentation, then locally it is a regular immersion followed by a smooth morphism of fixed pure relative dimension.

Proposition 1.8. Let $f : X \rightarrow Y$ be a complete-intersection morphism, and let

$$U \xrightarrow{i} V \xrightarrow{g} Y$$

be a local factorization near x , with i regular and g smooth. If $\mathcal{J}$ is the ideal of i , the integer

$$d_x = \text{rank}(\Omega_{V/Y}^1)_{i(x)} - \text{rank}(\mathcal{J}/\mathcal{J}^2)_x$$

does not depend on the chosen factorization.

Proof. Given two factorizations, compare them through the fiber product of the two smooth intermediate schemes over Y . The conormal sheaf for the induced immersion is obtained by adding the pullback of the relative differentials of the smooth comparison morphism. Hence the displayed difference of ranks is unchanged.

Definition 1.9. The integer d_x is called the *virtual relative dimension* of f at x . If it is independent of x , for instance when X is connected, it is called the virtual relative dimension of f and is denoted $d(f)$ or $d_{X/Y}$.

Proposition 1.10. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are complete-intersection morphisms, then

$$d_{g \circ f} = d_f + d_g.$$

Proposition 1.11. In a cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow f \\ Y' & \longrightarrow & Y, \end{array}$$

if f is a complete-intersection morphism, then so is its base change f' , and $d_{f'} = d_f$.

2. Projective morphisms and complete-intersection morphisms

Proposition 2.1. A projective smooth morphism is a complete-intersection morphism, and its virtual relative dimension is its relative dimension.

Proposition 2.2. A projective morphism which is a regular closed immersion is a complete-intersection morphism of virtual relative dimension $-\operatorname{codim}(X, Y)$.

Theorem 2.3. Let $f : X \rightarrow Y$ be projective. Suppose f factors as

$$X \xrightarrow{i} \mathbb{P}(E) \xrightarrow{p} Y,$$

where E is a vector bundle on Y , p is the canonical projection, and i is a regular immersion. Then f is a complete-intersection morphism and

$$d_f = \operatorname{rank}(E) - 1 - \operatorname{codim}(X, \mathbb{P}(E)).$$

Proof. The projection p is smooth of relative dimension $\operatorname{rank}(E) - 1$, and the immersion i has virtual relative dimension equal to the negative of its codimension. The formula follows from the additivity of virtual dimensions.

Corollary 2.4. If $f : X \rightarrow Y$ is projective and a complete-intersection morphism, then after choosing a sufficiently ample vector bundle on Y , f may be factored as a regular immersion into a projective bundle followed by the projection.

Proposition 2.5. Let $f : X \rightarrow Y$ be projective and a complete-intersection morphism.

- i) If f is finite, then $d_f \leq 0$.
- ii) If f is surjective, then $d_f \geq 0$.
- iii) If f is finite and surjective, then $d_f = 0$.

Remark 2.6. If Y is connected and $f : X \rightarrow Y$ is projective and a complete-intersection morphism, then d_f is constant on Y ; for a dominant morphism it is $\dim X - \dim Y$.

3. Complete-intersection morphisms and the Chern character

Proposition 3.1. Let $f : X \rightarrow Y$ be a projective complete-intersection morphism of virtual relative dimension d . Suppose Y is quasi-compact and has an ample sheaf. Then, for every integer k ,

$$f_*(\operatorname{Fil}^k K'(X)_{\mathbb{Q}}) \subset \operatorname{Fil}^{k-d} K'(Y)_{\mathbb{Q}}.$$

Proof. Factor f as $X \xrightarrow{i} P \xrightarrow{p} Y$, with p smooth of relative dimension n and i a regular immersion of codimension $n - d$. The Gysin map for i raises the support-codimension filtration by $n - d$, while the smooth projective projection p lowers it by n . The result follows.

Proposition 3.2. The preceding inclusion defines graded homomorphisms

$$f_{\text{gr}} : \text{Gr}^k K'(X)_{\mathbb{Q}} \longrightarrow \text{Gr}^{k-d} K'(Y)_{\mathbb{Q}}.$$

3.3. Reminder on the Chern character and the Todd class

For a graded K -algebra A , with K a binomial ring, Expose V defined

$$\text{Ch}(A) = K \times (1 + \hat{A}^+).$$

For every augmented K - λ -algebra Λ there is a K - λ -homomorphism, the completed Chern class,

$$\tilde{c} = (\varepsilon, c) : \Lambda \longrightarrow \text{Ch}(\text{Gr}_{\Lambda}).$$

There is also a functorial ring homomorphism, the Chern character,

$$\text{ch} : \text{Ch}(A) \longrightarrow K \otimes_{\mathbb{Z}} \mathbb{Q},$$

and a multiplicative endomorphism, the Todd character,

$$\text{Todd} : 1 + \hat{A}^+ \longrightarrow 1 + \hat{A}^+.$$

Both are multiplicative in the expected sense: $\text{ch}(xy) = \text{ch}(x)\text{ch}(y)$ and $\text{Todd}(x \oplus y) = \text{Todd}(x)\text{Todd}(y)$.

3.4. The Todd character of a morphism

Definition 3.4. Let $f : X \rightarrow Y$ be a complete-intersection morphism of virtual relative dimension d . Assume Y is quasi-compact and has an ample sheaf. The *Todd character* of f , denoted $\tau(f)$ or $\tau_{X/Y}$, is the element of $\text{Gr} K'(X)_{\mathbb{Q}}$ defined by

$$\tau(f) = \text{Todd}(N_f)^{-1},$$

where N_f is the virtual conormal class of f in $K'(X)$.

Proposition 3.4. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are complete-intersection morphisms, then

$$\tau(g \circ f) = \tau(f) \cdot f^* \tau(g).$$

Proof. The virtual conormal class satisfies $N_{g \circ f} = N_f \oplus f^* N_g$. The multiplicativity of the Todd

character gives

$$\mathrm{Todd}(N_{g \circ f}) = \mathrm{Todd}(N_f) f^* \mathrm{Todd}(N_g),$$

and taking inverses gives the formula.

SGA 6, Exposé VIII.3.5 through Exposé XIII.2.11

Exposé VIII – The Riemann–Roch theorem, continuation

3.5. The “Grothendieckian” Riemann–Roch theorem

Theorem 3.5. Let $f : X \rightarrow Y$ be a projective complete-intersection morphism, of virtual relative dimension d . Assume that Y is quasi-compact and admits an ample sheaf. Then, for every $x \in K'(X)$, one has

$$f_*(c(x) \tau(f)) = c(f_*(x)).$$

Equivalently, the following square is commutative:

$$\begin{array}{ccc} K'(X) & \xrightarrow{c(\cdot)\tau(f)} & \mathrm{Gr} K'(X)_{\mathbb{Q}} \\ f_* \downarrow & & \downarrow f_{\mathrm{gr}} \\ K'(Y) & \xrightarrow{c} & \mathrm{Gr} K'(Y)_{\mathbb{Q}}. \end{array}$$

Sketch of proof. The proof is by reduction to the case of a projective bundle. Choose a factorization

$$X \xrightarrow{i} \mathbb{P}(E) \xrightarrow{p} Y$$

with i a regular immersion and p the projection. The formula is first checked for p by the explicit computation of $K'(\mathbb{P}(E))$ as a free $K'(Y)$ -module with the usual powers of the tautological class as basis. For the immersion i the formula is the self-intersection formula, expressed by the inverse Todd character of the conormal bundle. The general assertion follows by the projection formula and by the multiplicativity

$$\tau(p \circ i) = \tau(i) i^* \tau(p).$$

Corollary 3.6. If $f : X \rightarrow Y$ is projective and smooth of relative dimension d , then

$$f_*(c(x) \mathrm{Todd}(T_f)) = c(f_*(x)),$$

where T_f is the relative tangent bundle. Indeed, for a smooth morphism the virtual conormal bundle is $-T_f$, hence $\tau(f) = \mathrm{Todd}(T_f)$.

3.7. Compatibility formulae

Proposition 3.7. Consider a cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow f' & & \downarrow f \\ Y' & \longrightarrow & Y, \end{array}$$

with f projective and a complete-intersection morphism, and with $Y' \rightarrow Y$ flat. Then, for every $x \in K'(X)$,

$$g^* f_*(x) = f'_* g'^*(x).$$

This is the usual flat base-change formula for proper direct image in K' -theory.

Proposition 3.8. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are projective complete-intersection morphisms, then

$$(g \circ f)_*(x) = g_* f_*(x) \quad (x \in K'(X)).$$

This is the transitivity of the Gysin morphisms.

4. The Todd class

Definition 4.1. For a regular scheme X admitting an ample sheaf, define

$$\text{Todd}(X) = \text{Todd}(T_X) \in \text{Gr } K'(X)_{\mathbb{Q}},$$

where T_X is the tangent bundle of X . In the general formulation one treats this class through the preceding functorial Todd character.

Proposition 4.2. If $f : X \rightarrow Y$ is smooth, then

$$\text{Todd}(X) = \text{Todd}(T_f) f^* \text{Todd}(Y).$$

Indeed $T_X = T_f \oplus f^* T_Y$.

Proposition 4.3. If $i : Y \rightarrow X$ is a regular closed immersion, then

$$\text{Todd}(Y) = \text{Todd}(i) i^* \text{Todd}(X).$$

This is the same multiplicativity, now applied to the virtual tangent bundle of the immersion.

Theorem 4.4. Let X and Y be regular schemes admitting ample sheaves, and let $f : X \rightarrow Y$ be a projective complete-intersection morphism. Then, for every $x \in K'(X)$,

$$f_*(c(x) \text{Todd}(X)) = c(f_*(x)) \text{Todd}(Y).$$

Proof. Apply Theorem 3.5 and use

$$\tau(f) = \frac{\text{Todd}(X)}{f^* \text{Todd}(Y)}.$$

This relation is precisely the expression, in $K'(X)$, of the equality of virtual tangent classes

$$T_X = T_f + f^*T_Y.$$

The projection formula then transports the factor $f^* \text{Todd}(Y)$ from the source to $\text{Todd}(Y)$ on the target.

Corollary 4.5. For a projective smooth morphism $f : X \rightarrow Y$ between regular schemes with ample sheaves,

$$f_*(c(x) \text{Todd}(T_f)) = c(f_*(x)).$$

This is Corollary 3.6 written in terms of the Todd class of the relative tangent bundle.

5. The Riemann–Roch theorem

5.1. Statement

Theorem 5.1. Let $f : X \rightarrow Y$ be a projective complete-intersection morphism of virtual relative dimension d . Assume that X and Y are quasi-compact and admit ample sheaves. Let $\mathcal{E}$ be a vector bundle on X , and let e be its class in $K'(X)$. Then

$$f_*(c(\mathcal{E}) \text{Todd}(T_f)) = c(f_!(\mathcal{E})) \quad \text{in } \text{Gr } K'(Y)_{\mathbb{Q}}.$$

Here $f_!$ denotes the alternating class of the higher direct images when this notation is available.

Proof. Factor f as $X \xrightarrow{i} \mathbb{P}(E) \xrightarrow{p} Y$, where i is a regular immersion. Then

$$f_!(\mathcal{E}) = p_! i_!(\mathcal{E}).$$

For the immersion, Riemann–Roch gives

$$c(i_!\mathcal{E}) = i_*(c(\mathcal{E}) \text{Todd}(N_i)^{-1}),$$

and for the projective bundle projection

$$c(p_!\mathcal{F}) = p_*(c(\mathcal{F}) \text{Todd}(T_p)).$$

Combining these two identities and using the projection formula gives

$$c(f_!\mathcal{E}) = f_*(c(\mathcal{E}) \text{Todd}(N_i)^{-1} i^* \text{Todd}(T_p)).$$

The virtual tangent bundle of f is the sum of the virtual tangent bundle of i and the pullback of T_p . Since $N_i = -T_i$ in K -theory, the displayed product is $\text{Todd}(T_f)$. This proves the formula.

Corollary 5.1. If $Y = \operatorname{Spec}(k)$ for a field k , then

$$\sum_i (-1)^i \dim_k H^i(X, \mathcal{E}) = \int_X c(\mathcal{E}) \operatorname{Todd}(T_X).$$

Thus the theorem recovers the usual Euler characteristic formula.

5.2. Projective bundles and the Hirzebruch formula

Let E be a vector bundle of rank $r + 1$ on a scheme Y , let $P = \mathbb{P}(E)$, and let $f : P \rightarrow Y$ be the projection. Denote by $\mathcal{O}_P(1)$ the tautological invertible sheaf and put $\xi = c_1(\mathcal{O}_P(1))$.

Proposition 5.2. There is an exact sequence on P

$$0 \longrightarrow \Omega_{P/Y}^1(1) \longrightarrow f^*E \longrightarrow \mathcal{O}_P(1) \longrightarrow 0.$$

Equivalently, after twisting,

$$0 \longrightarrow \Omega_{P/Y}^1 \longrightarrow f^*E(-1) \longrightarrow \mathcal{O}_P \longrightarrow 0.$$

Proof. Let V be the vector bundle associated with E , and let Z be the complement of the zero section. The morphism $Z \rightarrow P$ is the principal $\mathbb{G}_m$ -bundle associated with $\mathcal{O}_P(1)$. If

$$B = \bigoplus_{n \in \mathbb{Z}} \mathcal{O}_P(n),$$

then Z is affine over P with coordinate algebra B . The exact sequence of relative differentials for $Z \rightarrow P \rightarrow Y$ is a graded exact sequence of B -modules

$$0 \longrightarrow \Omega_{P/Y}^1 \otimes B \longrightarrow \Omega_{Z/Y}^1 \longrightarrow \Omega_{Z/P}^1 \longrightarrow 0.$$

The module $\Omega_{Z/Y}^1$ is obtained from f^*E , placed in degree 1, and $\Omega_{Z/P}^1$ is generated by the differential in the $\mathbb{G}_m$ -direction. Taking homogeneous degree 0 gives the asserted Euler sequence.

Proposition 5.3. For every integer $n \geq 0$,

$$f_*(c(\mathcal{O}_P(1)^n) \operatorname{Todd}(T_f)) = 1.$$

The Todd class of T_f is computed from the Euler sequence above.

Proof. In $K(P)$ one has

$$[T_f] = [f^*E^\vee \otimes \mathcal{O}_P(1)] - [\mathcal{O}_P].$$

The projective bundle formula writes $K'(P)$ as a free $K'(Y)$ -module on the powers of $\mathcal{O}_P(1)$, and the pushforward of these powers is the alternating class of the symmetric powers of $E^\vee$.

Substitution into the characteristic series defining the Todd class gives the claimed identity; it is the standard Hirzebruch computation for projective space, performed relative to Y .

Theorem 5.4. For $f : \mathbb{P}(E) \rightarrow Y$ and every $x \in K'(\mathbb{P}(E))$,

$$f_*(c(x) \text{ Todd}(T_f)) = c(f_*(x)).$$

By additivity it is enough to check this on the standard basis $1, \mathcal{O}(1), \dots, \mathcal{O}(r)$ of $K'(\mathbb{P}(E))$ over $K'(Y)$; Proposition 5.3 supplies exactly this check.

5.5. The general Riemann–Roch theorem

Theorem 5.5. Let $f : X \rightarrow Y$ be a projective complete-intersection morphism between regular schemes admitting ample sheaves. Then, for every $x \in K'(X)$,

$$f_*(c(x) \text{ Todd}(X)) = c(f_*(x)) \text{ Todd}(Y) \quad \text{in } \text{Gr } K'(Y)_{\mathbb{Q}}.$$

Proof. Choose a factorization $X \xrightarrow{i} P \xrightarrow{p} Y$ with $P = \mathbb{P}(E)$ and i a regular immersion. Apply the projective-bundle case to p and the self-intersection Riemann–Roch formula to i :

$$c(i_*x) = i_*(c(x) \text{ Todd}(N_i)^{-1}).$$

Then

$$c(f_*x) = p_*(i_*(c(x) \text{ Todd}(N_i)^{-1}) \text{ Todd}(T_p)) = f_*(c(x) \text{ Todd}(N_i)^{-1} i^* \text{ Todd}(T_p)).$$

The last Todd factor equals $\text{Todd}(T_f)$, and the passage from $\text{Todd}(T_f)$ to $\text{Todd}(X)/f^* \text{ Todd}(Y)$ gives the displayed formula.

Corollary 5.6. If $f : X \rightarrow Y$ is proper and smooth between regular schemes with ample sheaves, then, for every vector bundle $\mathcal{E}$ on X ,

$$f_*(c(\mathcal{E}) \text{ Todd}(T_f)) = c(f_!\mathcal{E}).$$

5.7. Examples and applications

For an invertible sheaf $\mathcal{L}$ one has

$$c(\mathcal{L}) = 1 + c_1(\mathcal{L}), \quad \text{Todd}(\mathcal{L}) = \frac{c_1(\mathcal{L})}{1 - e^{-c_1(\mathcal{L})}}.$$

If D is an effective Cartier divisor and $\mathcal{L} = \mathcal{O}(D)$, then

$$c(\mathcal{O}(D)) = 1 + [D].$$

For a regular proper surface X over a field and an invertible sheaf $\mathcal{L}$, Riemann–Roch gives

$$\chi(\mathcal{L}) = \chi(\mathcal{O}_X) + \frac{1}{2} \mathcal{L} \cdot (\mathcal{L} - K_X),$$

which is the classical surface formula. For a regular proper curve of genus g , and an invertible sheaf of degree d , the formula becomes

$$\chi(\mathcal{L}) = d + 1 - g.$$

Remark 5.8. The form above is broader than the classical theorem: it applies through K' -theory to singular schemes and to complete-intersection morphisms which need not be smooth.

5.9. Compatibilities and variants

Proposition 5.9. The Riemann–Roch theorem is compatible with flat base change. For a cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

with g flat, and for every $x \in K'(X)$,

$$g^* f_*(c(x) \text{Todd}(f)) = f'_*(c(g'^* x) \text{Todd}(f')).$$

Proposition 5.10. The theorem is compatible with open immersions: if $j : U \rightarrow X$ is open, then

$$j^*(c(x) \text{Todd}(X)) = c(j^* x) \text{Todd}(U).$$

Theorem 5.11. For a proper complete-intersection morphism $f : X \rightarrow Y$ between regular schemes admitting ample sheaves, the square

$$\begin{array}{ccc} K'(X) & \xrightarrow{c(\cdot) \text{Todd}(X)} & \text{Gr } K'(X)_{\mathbb{Q}} \\ f_* \downarrow & & \downarrow f_{\text{gr}} \\ K'(Y) & \xrightarrow{c(\cdot) \text{Todd}(Y)} & \text{Gr } K'(Y)_{\mathbb{Q}} \end{array}$$

is commutative. This is Theorem 5.5 rewritten using

$$\text{Todd}(f) = \text{Todd}(X)/f^* \text{Todd}(Y).$$

5.12. Conclusion of Expose VIII

The Riemann–Roch theorem in the form just stated is the culmination of the construction developed in the preceding exposes. It connects K' -theory with the graded rings attached to the

Chern character, and it inserts precisely the Todd correction accounting for the geometry of the morphism. The proof rests on the Gysin morphisms for proper complete-intersection maps, on the self-intersection formula for regular immersions, on the construction of the Chern and Todd characters in the lambda-ring formalism, and on the explicit verification for projective bundles.

Expose IX – Some computations of K -groups

Some computations of K -groups

P. Berthelot

This expose corresponds to no oral lecture and will not be used later in the seminar. It contains a number of computations of the groups $K_\bullet$ for various fibrations: vector bundles, projective bundles, torsors under a linear group, and related objects. Throughout, the base scheme is noetherian, and $K_\bullet(X)$ denotes the Grothendieck group of coherent sheaves on a noetherian scheme X . Recall that, for X noetherian, the natural homomorphism from the K -group of coherent $\mathcal{O}_X$ -modules to the K -group of pseudo-coherent complexes on X is an isomorphism.

The results are analogues of the corresponding computations for the Chow ring, as in Grothendieck's lectures in the Chevalley seminar. One may also ask to what extent statements of the same type hold for the graded group associated with $K_\bullet$ for the filtration by dimension of support. Even for the homotopy exact sequence below, which is one of the most frequently used tools in the computations, this is not known in full generality.

1. Vector bundles

Proposition 1.1 (homotopy exact sequence). Let X be a noetherian scheme, let X' be a closed subscheme of X , let $U = X - X'$ be the open complement, and let $i : X' \rightarrow X$ and $j : U \rightarrow X$ be the immersions. Then

$$K_\bullet(X') \xrightarrow{i_*} K_\bullet(X) \xrightarrow{j^*} K_\bullet(U) \longrightarrow 0$$

is exact.

Proof. For a coherent $\mathcal{O}_{X'}$ -module F' , the sheaf i_*F' is supported on X' , hence its restriction to U is zero. Thus $j^*i_* = 0$. We obtain a homomorphism

$$\rho : K_\bullet(X) / \text{Im } K_\bullet(X') \longrightarrow K_\bullet(U).$$

We construct its inverse. Let G be a coherent $\mathcal{O}_U$ -module. Since X is noetherian, there exists a coherent $\mathcal{O}_X$ -module F restricting to G . If F' is another such extension, then

$$\text{cl}(F) - \text{cl}(F') \in \text{Im } K_\bullet(X').$$

Indeed, embed G diagonally in $j^*(F \oplus F')$ and extend it to a coherent subsheaf $F'' \subset F \oplus F'$.

Comparing F'' with F and F' produces kernels and cokernels supported on X' . Since a coherent sheaf supported on X' is killed by a sufficiently high power of the ideal of X' , it may be devissed into coherent $\mathcal{O}_{X'}$ -modules. Hence its class lies in the image of $K_\bullet(X')$. The class of any extension of G is therefore well defined modulo this image, and this construction is additive in short exact sequences. It gives the inverse of ρ .

Proposition 1.2. Let $f : X \rightarrow S$ be a flat morphism of noetherian schemes. Let $(\xi_i)_{i \in I}$ be a family of elements of $K^*(X)$ and consider

$$\varphi : K_\bullet(S)^{(I)} \longrightarrow K_\bullet(X), \quad (x_i) \longmapsto \sum_i \xi_i f^*(x_i).$$

Assume that for every point $s \in S$ the images of the ξ_i in $K_\bullet(X_s)$ generate that group. Then φ is surjective.

Proof. Let S' be a closed subscheme of S and U its complement. Write $X_{S'}$ and X_U for the restrictions of X . The homomorphisms $\varphi_{S'}$ and φ_U fit into a commutative diagram with exact rows

$$\begin{array}{ccccccc} K_\bullet(S')^{(I)} & \rightarrow & K_\bullet(S)^{(I)} & \rightarrow & K_\bullet(U)^{(I)} & \rightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ K_\bullet(X_{S'}) & \rightarrow & K_\bullet(X) & \rightarrow & K_\bullet(X_U) & \rightarrow & 0. \end{array}$$

The commutativity of the left square is the compatibility of proper direct image with flat inverse image; the right square follows from functoriality of inverse image. If $\varphi_{S'}$ and φ_U are surjective, so is φ .

Noetherian induction on closed subschemes of S reduces the proof to the case where $S = \text{Spec}(A)$ is integral and affine. Let K be the fraction field and let s be the generic point. For $y \in K_\bullet(X)$, the class y_s in $K_\bullet(X_s)$ is, by hypothesis, a finite sum $\sum_i \xi_i n_i$. Since $K = \varinjlim A_f$ and $X_s = \varprojlim X \times_S \text{Spec}(A_f)$, the same equality holds over some non-empty open $U = \text{Spec}(A_f)$. By induction on the closed complement, the remaining difference lies in the image of φ . Thus φ is surjective.

Corollary 1.3. If $f : X \rightarrow S$ is flat and, for every $s \in S$, the map $K^*(X) \rightarrow K_\bullet(X_s)$ is surjective, then

$$K_\bullet(S) \otimes_{K^*(S)} K^*(X) \longrightarrow K_\bullet(X)$$

is surjective.

Corollary 1.4. If $f : X \rightarrow S$ is flat and the canonical map $\mathbb{Z} \rightarrow K_\bullet(X_s)$ is an isomorphism for every $s \in S$, then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is surjective. If, moreover, f has a section g of finite Tor-dimension, then f^* is an isomorphism, with inverse g^* .

Remark 1.5. For such a section g to have finite Tor-dimension it is necessary and sufficient that the fibres of X be regular at the points of $g(S)$; equivalently, f is regular at those points, and smooth there if f is flat of finite presentation.

Corollary 1.6. Let E be a locally free $\mathcal{O}_S$ -module of rank n , let $X = \mathbb{V}(E)$ be the corresponding vector bundle, and let $f : X \rightarrow S$ be its structural morphism. Then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is an isomorphism.

Corollary 1.7. If $g : S \rightarrow X$ is the zero section of the vector bundle above, then

$$g^* : K_\bullet(X) \longrightarrow K_\bullet(S)$$

is the inverse of f^* . The zero section is a regular immersion, hence has finite Tor-dimension.

2. Projective bundles and torsors

Let S be noetherian, let E be a locally free $\mathcal{O}_S$ -module of rank $r + 1$, and let $X = \mathbb{V}(E)$. Suppose that a torus T with fibres isomorphic to $\mathbb{G}_m^r$ acts freely on X , that the quotient $P = X/T$ exists, and that X is a T -torsor over P . The aim is to compute $K_\bullet(P)$ using Proposition 1.2.

When $T = \mathbb{G}_m$ and X is the complement of the zero section in a line bundle L , the homotopy exact sequence gives

$$K_\bullet(S) \xrightarrow{1-L} K_\bullet(S) \longrightarrow K_\bullet(X) \longrightarrow 0.$$

Thus $K_\bullet(X)$ is the quotient of $K_\bullet(S)$ by the subgroup generated by $(L - 1)K_\bullet(S)$.

More generally, let M be the character group of the torus T , and let $\eta(m)$ denote the invertible sheaf on S associated with a character $m \in M$. Then

$$K_\bullet(P) \simeq K_\bullet(S) \Big/ \sum_{m \in M} (\eta(m) - 1) K_\bullet(S).$$

The proof is by induction on the rank of the torus. Choose a subtorus $T' \subset T$ of rank $r - 1$ with quotient $T/T' \simeq \mathbb{G}_m$. Put $P' = X/T'$. Applying the induction hypothesis to $X \rightarrow P'$ and then the rank-one case to $P' \rightarrow P$ gives the formula.

Remark 2.3. If $T = \mathbb{G}_m^r$, the preceding result may be expressed as

$$K_\bullet(P) \simeq K_\bullet(S) \otimes_{\mathbb{Z}[M]} \mathbb{Z},$$

where $\mathbb{Z}[M]$ acts on $K_\bullet(S)$ through the classes $\eta(m)$ and on $\mathbb{Z}$ through the augmentation.

3. Projective line bundles

Proposition 3.1. Let S be noetherian, let E be locally free of rank 2, put $P = \mathbb{P}(E)$, and let $f : P \rightarrow S$ be the structural morphism. Let L be the canonical invertible sheaf on P . Then:

i) The homomorphism

$$K_{\bullet}(S) \oplus K_{\bullet}(S) \longrightarrow K_{\bullet}(P), \quad (x, y) \longmapsto f^*x + Lf^*y,$$

is an isomorphism.

ii) If E' is a locally free rank-one submodule of E , and $i : S \rightarrow P$ is the corresponding section, then

$$i^*L = (E')^{-1} \otimes (E/E').$$

For a locally free E of rank $r+1$, set $P = \mathbb{P}(E)$ and let $\xi = \text{cl}(\mathcal{O}_P(1))$. One obtains a homomorphism

$$\mathbb{Z}[T]/(T^{r+1}) \longrightarrow K^*(P), \quad T \longmapsto \xi^{-1}.$$

When E is trivial, this gives the familiar formula

$$K^*(\mathbb{P}_S^r) = K^*(S)[\xi]/(\xi - 1)^{r+1}.$$

Proposition 3.3. Let $f : X \rightarrow S$ be flat, and let E be locally free of rank $r+1$ on X . Suppose that for every $s \in S$

$$K^*(\mathbb{P}(E_s)) \simeq K^*(X_s)[\xi_s]/(\xi_s^{r+1}).$$

Then

$$K^*(\mathbb{P}(E)) \simeq K^*(X)[\xi]/(\xi^{r+1}).$$

The proof applies Proposition 1.2 to the elements $1, \xi, \dots, \xi^r$ and then uses noetherian induction to reduce injectivity to the case of a field.

Corollary 3.4. Under the hypotheses of Proposition 3.3, if f has a section, then

$$K^*(S)[T]/(T^{r+1}) \longrightarrow K^*(P), \quad T \longmapsto \xi^{-1},$$

is an isomorphism.

Corollary 3.5. For a locally free $\mathcal{O}_S$ -module E of rank $r+1$,

$$K^*(\mathbb{P}(E)) = K^*(S)[\xi] / \left(\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) \xi^{r+1-i} \right).$$

If $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ is exact with E' of rank 1, and if $s : S \rightarrow \mathbb{P}(E)$ is the corresponding section, then

$$s^*(\xi) = \lambda^1((E')^*) = (E')^{-1}.$$

Corollary 3.7. For $P = \mathbb{P}(E)$ with E of rank $r + 1$,

$$K_{\bullet}(S)^{r+1} \longrightarrow K_{\bullet}(P), \quad (x_0, \dots, x_r) \longmapsto \sum_{i=0}^r \xi^i f^*(x_i),$$

is an isomorphism. The proof is by induction on r , using the section associated with a rank-one subbundle and the open complement, which is a vector bundle over the smaller projective bundle.

4. Flag schemes

Let E be locally free of rank n on a noetherian scheme S . The flag scheme $\text{Flag}(E)$ represents the functor sending $S' \rightarrow S$ to complete flags

$$0 = E_0 \subset E_1 \subset \dots \subset E_n = E_{S'}, \quad \text{rank } E_i = i.$$

Proposition 4.2. Let $D = \text{Flag}(E)$ and $f : D \rightarrow S$. Then:

- i) f is proper and flat;
- ii) $f^* : K^*(S) \rightarrow K^*(D)$ is injective;
- iii) $K^*(D)$ is a free $K^*(S)$ -module of rank $n!$, with a basis given by monomials in the Chern classes of the successive universal quotient line bundles.

The proof is the usual induction on n : the passage from a flag of rank $n - 1$ to a flag of rank n is governed by a projective bundle, and the complement of the closed stratum is a vector bundle.

Proposition 4.3. Let $L_i = E_i/E_{i-1}$ on D , and put $\xi_i = c^1(L_i)$. Then

$$K^*(D) = K^*(S)[\xi_1, \dots, \xi_n]/I,$$

where I is generated by

$$\sigma_k(\xi_1, \dots, \xi_n) = c^k(E), \quad 1 \leq k \leq n,$$

with σ_k the elementary symmetric function. These relations come from the exact sequences

$$0 \rightarrow E_{i-1} \rightarrow E_i \rightarrow L_i \rightarrow 0$$

and the resulting factorization of the total Chern class of E .

5. Torsors under a linear group

Let G be a smooth group scheme over S with connected fibres, and let P be a G -torsor over S . The objective is again to compute $K^*(P)$. If $G = \mathrm{GL}(n)$, then P is the scheme of isomorphisms $\mathrm{Isom}(\mathcal{O}_S^n, E)$ for a locally free module E of rank n . Let $B \subset \mathrm{GL}(n)$ be the Borel subgroup of upper triangular matrices. Then P/B is the flag scheme of E , and $P \rightarrow P/B$ is a B -torsor. Since B is obtained by successive extensions by copies of $\mathbb{G}_m$ and affine spaces, the preceding computations give

$$K^*(P) = K^*(\mathrm{Flag}(E)).$$

More explicitly,

$$K^*(P) = K^*(S)[\xi_1^{\pm 1}, \dots, \xi_n^{\pm 1}] / (\sigma_k(\xi_1, \dots, \xi_n) = c^k(E)).$$

Proposition 5.2. Let G be a split reductive group scheme over S , let T be a split maximal torus, let B be a Borel subgroup containing T , and let W be the Weyl group. For a G -torsor P , set $D = P/B$. Then:

- i) $K^*(S) \rightarrow K^*(D)$ is injective;
- ii) $K^*(D)$ is a free $K^*(S)$ -module of rank $|W|$, with basis represented by the Bruhat cells.

The assertion follows fibrewise from the cellular decomposition and then globally from Proposition 1.2.

Corollary 5.3. Under the same hypotheses, if R is the weight lattice of T , R_+ a system of positive roots defined by B , and $\rho = \frac{1}{2} \sum_{\alpha \in R_+} \alpha$, then

$$K^*(P) = K^*(D)^W = (K^*(S) \otimes \mathbb{Z}[R])^W.$$

6. Appendix: complements on lambda-rings

A lambda-ring is a commutative ring K with unit, equipped with operations $\lambda^n : K \rightarrow K$ satisfying the standard identities. One defines

$$\gamma^n(x) = \lambda^n(x + n - 1)$$

and the gamma-filtration by declaring $F_\gamma^n K$ to be generated by products

$$\gamma^{n_1}(x_1) \cdots \gamma^{n_k}(x_k), \quad \sum n_i \geq n.$$

Proposition 6.2. If K is a lambda-ring and $x \in F_\gamma^1 K$, then

$$\gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n = \exp \left(\sum_{m \geq 1} (-1)^{m-1} \psi^m(x) \frac{t^m}{m} \right),$$

where ψ^m are the Adams operations. The identity is checked on elements of the form $\xi - 1$, with ξ invertible, and then follows for special lambda-rings by the splitting principle.

Corollary 6.3. For $x, y \in F_\gamma^1 K$,

$$\gamma^n(x + y) = \sum_{i=0}^n \gamma^i(x) \gamma^{n-i}(y),$$

and, modulo $F_\gamma^{m+1} K$,

$$\psi^m(x) = \sum_{n \geq m} (-1)^{n-m} m \binom{n}{m} \gamma^n(x).$$

One also considers the codimension filtration

$$F_c^n K = \{x \in K \mid c^i(x) = 0 \text{ for } i < n\},$$

where $c^i(x) = \gamma^i(x - \varepsilon(x))$ and ε is the augmentation.

Expose X – Complements on projection and Riemann–Roch formulae

Complements on the projection and Riemann–Roch formulae

P. Berthelot

1. Filtrations on K -groups

Let X be a noetherian scheme. The group $K_\bullet(X)$ is the Grothendieck group of coherent $\mathcal{O}_X$ -modules, and $K^*(X)$ is the Grothendieck group of locally free $\mathcal{O}_X$ -modules of finite type. There is a canonical map

$$K^*(X) \longrightarrow K_\bullet(X),$$

which is an isomorphism when X is regular.

On $K_\bullet(X)$ one has the filtration by dimension of support:

$$\mathrm{Fil}_n K_\bullet(X) = \langle \mathrm{cl}(F) \mid F \text{ coherent and } \dim \mathrm{Supp}(F) \leq n \rangle.$$

On $K^*(X)$ one has the filtration by codimension:

$$\mathrm{Fil}^n K^*(X) = \{x \in K^*(X) \mid c^i(x) = 0 \text{ for } i < n\}.$$

The gamma-filtration satisfies

$$F_\gamma^n K^*(X) \subset \mathrm{Fil}^n K^*(X),$$

though equality is not known in general.

Proposition 1.3. Let X be noetherian.

- i) The codimension filtration on $K^*(X)$ is compatible with products.
- ii) If X is regular of dimension at most d , then $\text{Fil}^{d+1} K^*(X) = 0$.
- iii) If $f : X \rightarrow Y$ is proper, then

$$f_*(\text{Fil}_n K_\bullet(X)) \subset \text{Fil}_n K_\bullet(Y).$$

The first assertion follows from the formula for Chern classes of tensor products, the second from dimension, and the third from the behaviour of supports under proper direct image.

1.4. Dimension functions

Let S be noetherian. We want, for each proper S -scheme Y and each point $y \in Y$, an integer $D_Y(y)$ satisfying:

- i) $D_Y(y) \geq 0$;
- ii) $D_Y(y) = 0$ precisely at the generic points of irreducible components;
- iii) for an S -morphism $Y_1 \rightarrow Y_2$, one has

$$D_{Y_1}(y_1) \geq D_{Y_2}(f(y_1));$$

- iv) $D_{Y_{\text{red}}}(y) = D_Y(y)$.

If S is locally noetherian and universally catenary, choose an integer n and put

$$D_S(x) = n - \text{codim}(\overline{\{x\}}, S) = n - \dim \mathcal{O}_{S,x}.$$

For $s_Y : Y \rightarrow S$ proper and $y \in Y$ set

$$D_Y(y) = D_S(s_Y(y)) + \text{trdeg}_{k(s_Y(y))} k(y).$$

For an S -morphism $Y_1 \rightarrow Y_2$ this gives

$$D_{Y_1}(y) = D_{Y_2}(f(y)) + \text{trdeg}_{k(f(y))} k(y).$$

The only substantive point is the characterization of generic points. By replacing Y by its reduction and then by an irreducible closed subscheme, and by factoring its structural morphism through its scheme-theoretic image, the assertion follows from the catenarity dimension formula.

2. Polynomial maps

Let G be an abelian group and A a commutative ring. A map $f : G \rightarrow A$ is called polynomial if its restriction to every finitely generated subgroup of G is polynomial in the usual sense. Thus on each finitely generated subgroup one may write it as a polynomial in finitely many additive characters.

Lemma 2.1.1. For a map $f : G \rightarrow A$, the following are equivalent.

- i) f is polynomial.
- ii) Locally on every finitely generated subgroup it is a finite linear combination of functions

$$x \mapsto \binom{\chi(x)}{m},$$

where $\chi : G \rightarrow A$ is a homomorphism and $m \in \mathbb{N}$.

The reason is that the polynomial functions on a finitely generated free abelian group are generated by the binomial polynomials in additive coordinates.

Let X be noetherian and let L be invertible. The map

$$\varphi_L : \text{Pic}(X) \longrightarrow K_{\bullet}(X), \quad M \mapsto \text{cl}(L \otimes M) - \text{cl}(L),$$

is polynomial. Indeed, in $K^*(X)$ one has

$$\text{cl}(L \otimes M) - \text{cl}(L) = L(M - 1),$$

and $M - 1$ lies in the first step of the gamma-filtration.

Proposition 2.3. Let E be locally free of rank $r + 1$ on a noetherian S , and put $P = \mathbb{P}(E)$. The map

$$\text{Pic}(P) \longrightarrow K_{\bullet}(P), \quad L \mapsto \text{cl}(L),$$

is polynomial on each coset coming from $\text{Pic}(S)$. Since

$$K_{\bullet}(P) = \bigoplus_{i=0}^r K_{\bullet}(S) \xi^i,$$

every line bundle on P is of the form $f^*M \otimes \mathcal{O}_P(n)$, and its class is polynomial in the integer n and in the class of M .

2.4. Snapper polynomials

Let X be proper and smooth over a field k , let L be an ample invertible sheaf, and let F be coherent. Snapper's theorem says that

$$n \longmapsto \chi(X, L^{\otimes n} \otimes F)$$

is a polynomial in n . More generally, for invertible sheaves $L_1, \dots, L_d$, the function

$$(n_1, \dots, n_d) \longmapsto \chi(X, L_1^{\otimes n_1} \otimes \dots \otimes L_d^{\otimes n_d} \otimes F)$$

is a polynomial, the Snapper polynomial. Intersection numbers are extracted as its top multi-degree coefficients.

5. Intersection numbers

Let X be proper over a field k , let $L_1, \dots, L_d$ be invertible sheaves, and let F be a coherent sheaf with $\dim \text{Supp}(F) \leq d$. The intersection number

$$(L_1 \cdots L_d; F)$$

is the coefficient of $n_1 \cdots n_d$ in the Snapper polynomial

$$\chi(L_1^{\otimes n_1} \otimes \dots \otimes L_d^{\otimes n_d} \otimes F).$$

Equivalently it may be described by iterated first Chern classes acting on $\text{cl}(F)$ and then applying the Euler characteristic.

This integer is symmetric and d -linear in the variables $L_i \in \text{Pic}(X)$, is zero if $\dim \text{Supp}(F) < d$, and is additive in the class of F in $K_\bullet(X)$. If Y is a closed subscheme of X , $i : Y \rightarrow X$ is the immersion, and $F = i_*G$, then

$$(i^*L_1, \dots, i^*L_d; G) = (L_1, \dots, L_d; F).$$

More generally, the projection formula gives

$$\chi(f^*x \cdot y) = \chi(x \cdot f_*y)$$

for every morphism $f : Y \rightarrow X$ of proper k -schemes.

If $f : Y \rightarrow X$ is a morphism of irreducible proper k -schemes of dimension at most d and of degree n , then

$$(f^*L_1, \dots, f^*L_d; \mathcal{O}_Y) = n(L_1, \dots, L_d; \mathcal{O}_X).$$

The definition of the degree is the usual one: it is zero unless the generic point of Y maps to the generic point of X and the dimensions agree; in that case it is the degree of the corresponding

extension of function fields, expressed in terms of the lengths of the local rings at the generic points.

4.5. Numerical equivalence

Two elements of $\mathrm{Gr}^i(X)$ are numerically equivalent if they define the same homomorphism on $\mathrm{Gr}_i(X)$ by the intersection pairing. The analogous definition is used for $\mathrm{Gr}_i(X)$. Two invertible sheaves are numerically equivalent when their first Chern classes in $\mathrm{Gr}^1(X)$ are numerically equivalent.

An invertible sheaf L is numerically equivalent to zero if and only if

$$\chi(L^{\otimes m} \otimes \mathcal{O}_Z) = \chi(\mathcal{O}_Z)$$

for all integers m and all integral closed curves $Z \subset X$. This is an immediate consequence of the expression of the intersection product in terms of Euler characteristics. Write $\mathrm{Pic}^{\mathrm{num}}(X)$ for the subgroup of $\mathrm{Pic}(X)$ of classes numerically equivalent to zero. One has

$$\mathrm{Pic}^c(X) \subset \mathrm{Pic}^{\mathrm{num}}(X),$$

and a later expose proves the stronger equality $\mathrm{Pic}^{\tau}(X) = \mathrm{Pic}^{\mathrm{num}}(X)$.

5. The isomorphism $\mathrm{Pic}(X) \simeq \mathrm{Gr}^1(X)$

Let X be any ringed topos. The ring $K^{\bullet}(X)$ and its filtration are defined as for schemes. The group $\mathrm{Pic}(X)$ is the group of isomorphism classes of invertible $\mathcal{O}_X$ -modules. The map

$$i : \mathrm{Pic}(X) \longrightarrow K^{\bullet}(X)$$

sends an invertible module L to its class. It is a homomorphism into the group of units of $K^{\bullet}(X)$. The determinant gives a left inverse. If E is locally free of rank n , set

$$\det(E) = \bigwedge^n E.$$

For a short exact sequence of locally free modules

$$0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$$

there is a canonical isomorphism

$$\det(E) \simeq \det(E') \otimes \det(E'').$$

Thus one obtains a group homomorphism

$$\det : K^\bullet(X) \longrightarrow \text{Pic}(X)$$

which is a left inverse to i . Hence $\text{Pic}(X)$ is identified with a subgroup of $K^\bullet(X)^*$.

Define

$$\theta : \text{Pic}(X) \longrightarrow \text{Fil}^1(X), \quad \theta(L) = L - 1.$$

The composition with $\text{Fil}^1(X) \rightarrow \text{Gr}^1(X)$ is the first Chern class

$$c_1 : \text{Pic}(X) \longrightarrow \text{Gr}^1(X).$$

Theorem 5.3.2. The map c_1 is injective. Moreover, $\det : \text{Fil}^1(X) \rightarrow \text{Pic}(X)$ induces a left inverse

$$\alpha : \text{Gr}^1(X) \longrightarrow \text{Pic}(X).$$

If every section of the constant sheaf $\mathbb{Z}_X$ is bounded, for example if X is quasi-compact or if the final object is a finite sum of connected objects, then c_1 is an isomorphism and α is its inverse.

Proof. The identity

$$c_1(L \otimes L') = c_1(L) + c_1(L') + c_1(L)c_1(L')$$

shows that c_1 is a homomorphism after passage to Gr^1 . The determinant kills $\text{Fil}^2(X)$, so it passes to Gr^1 and defines α . Then

$$\alpha c_1(L) = L.$$

If the rank function is bounded, the group $\text{Fil}^1(X)$ is generated by the elements $1 - [E]$ for locally free rank-one modules E , and on these generators one checks directly that $c_1 \alpha$ is the identity.

Lemma 5.3.4. For every ringed topos X and every $x \in \text{Fil}^2(X)$,

$$\det(x) = 1.$$

The determinant formula

$$\det(xy) = \det(x)^{\varepsilon(y)} \det(y)^{\varepsilon(x)}$$

with the augmentation $\varepsilon : K^\bullet(X) \rightarrow \mathbb{Z}$ implies this after applying it to $x \in \text{Fil}^2(X)$, for which $\varepsilon(x) = 0$.

5.3.5. Complement

The boundedness assumption on the rank function is not needed to show that α is well defined, namely that $\det(\text{Fil}^2(X)) = 1$. It enters only in the proof that c_1 is surjective. If E has constant

rank n_0 , then

$$\det(E) = \bigwedge^{n_0} E = \gamma^{n_0}((E - n_0) + 1) = \sum_{0 \leq i \leq n_0} \gamma^i(E - n_0),$$

and modulo $\text{Fil}^2(X)$ this is congruent to

$$1 + n_0(E - n_0),$$

which gives the required congruence.

6. Devissage of $\text{GL}(n)$

Let S be a scheme. Denote by $\text{GL}(n)$ the group scheme of automorphisms of $\mathcal{O}_S^n$, by $\text{SL}(n)$ the subgroup of determinant 1, and by $\mathbb{G}_m$ the multiplicative group scheme. There is an exact sequence

$$1 \longrightarrow \text{SL}(n) \longrightarrow \text{GL}(n) \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1.$$

Proposition 6.2.1. For $n \geq 2$, the group $\text{SL}(n)$ is equal to its commutator subgroup; this commutator group is simply connected.

Proposition 6.2.2. For every positive section n of $\mathbb{Z}$, the étale sheaf of commutators $\text{Der}(\text{GL}(n))$ is represented by

$$\text{SL}(n) = \text{Ker}(\det : \text{GL}(n) \rightarrow \mathbb{G}_m).$$

Corollary 6.2.3. Every representation

$$\sigma : G = \text{GL}(n_1) \times \cdots \times \text{GL}(n_r) \longrightarrow \mathbb{G}_m$$

is of the form

$$\sigma(s_1, \dots, s_r) = \det(s_1)^{m_1} \cdots \det(s_r)^{m_r},$$

with uniquely determined sections $m_1, \dots, m_r$ of $\mathbb{Z}$.

Corollary 6.2.4. With the same notation, for locally free modules $E_1, \dots, E_r$ one has

$$\det(\pi_!(E_1, \dots, E_r)) = \det(E_1)^{m_1} \otimes \cdots \otimes \det(E_r)^{m_r}$$

for uniquely determined exponents m_i .

The exponents are computed by substituting $s_i = \lambda \text{id}$ in the i -th factor and the identity in the others. In particular

$$\det(E_1 \otimes E_2) = \det(E_1)^{n_2} \otimes \det(E_2)^{n_1},$$

$$\det(E_1 \oplus E_2) = \det(E_1) \otimes \det(E_2),$$

and

$$\det\left(\bigwedge^m E\right) = \det(E)^{\binom{n-1}{m-1}}.$$

The proof of Corollary 6.2.3 reduces to the case of a single factor. By Proposition 6.2.2 a character of $\mathrm{GL}(n)$ factors through $\mathrm{GL}(n)/\mathrm{SL}(n)$; the determinant identifies this quotient with $\mathbb{G}_m$, and the characters of $\mathbb{G}_m$ are the powers $\lambda \mapsto \lambda^m$.

For Proposition 6.2.2 one reduces locally on the base to the usual elementary statement for matrices: the commutator subgroup of $\mathrm{GL}(n)$ is $\mathrm{SL}(n)$, generated by elementary matrices. Over an algebraically closed field this follows from Jordan normal form and the elementary row and column operations; descent then gives the sheaf statement.

7. Appendix: specialization in intersection theory

A. Grothendieck

Lemma 7.1. Let $i : Y \rightarrow X$ be a regular closed immersion of bounded codimension, hence a perfect morphism. Let N be its conormal sheaf, locally free of finite type on Y . Then for the direct image

$$i_* : K_\bullet(Y) \longrightarrow K_\bullet(X)$$

and the extraordinary inverse image

$$i^! : K_\bullet(X) \longrightarrow K_\bullet(Y)$$

one has

$$i^! i_*(x) = x \cdot \lambda_{-1}(N), \quad x \in K_\bullet(Y),$$

where multiplication is through the natural $K^\bullet(Y)$ -module structure on $K_\bullet(Y)$. The proof is the same as the corresponding self-intersection formula in $K^\bullet$.

Corollary 7.2. Let $f : X \rightarrow S$ be flat, with $S = \mathrm{Spec}(A)$ the spectrum of a discrete valuation ring. Let s and t be the closed and generic points, and let $i : X_s \rightarrow X$ be the inclusion. Then i has finite Tor-dimension and therefore gives a pullback

$$i^* : K_\bullet(X) \longrightarrow K_\bullet(X_s).$$

Moreover the composite $i^* i_*$ on $K_\bullet(X_s)$ is zero, because the conormal bundle is trivial of rank one and $\lambda_{-1}(\mathcal{O}) = 1 - 1 = 0$.

Proposition 7.3. Let $j : X_t \rightarrow X$ denote the inclusion of the generic fibre. There is a unique homomorphism

$$\sigma : K_\bullet(X_t) \longrightarrow K_\bullet(X_s),$$

called the specialization homomorphism, making the diagram

$$\begin{array}{ccc} K_{\bullet}(X) & \xrightarrow{j^*} & K_{\bullet}(X_t) \\ i^* \downarrow & & \downarrow \sigma \\ K_{\bullet}(X_s) & = & K_{\bullet}(X_s) \end{array}$$

commute. It exists because j^* is the quotient map given by the homotopy exact sequence, and because $i^*i_* = 0$.

When $K_{\bullet}(X_t)$ and $K_{\bullet}(X_s)$ are regarded as $K^{\bullet}(X)$ -modules through j^* and i^* , the map σ is $K^{\bullet}(X)$ -linear.

7.4. Compatibility with filtrations

Put on $K_{\bullet}$ the filtration by dimension of support. If f is of finite type, or more generally if X is catenary, the specialization homomorphism is compatible with these filtrations and induces

$$\sigma : \mathrm{Gr}_{\bullet}(X_t) \longrightarrow \mathrm{Gr}_{\bullet}(X_s).$$

It is enough to check this on the class of an integral closed subscheme $Y_t \subset X_t$. Let Y be its schematic closure in X . Then Y is flat over S , and

$$\sigma(\mathrm{cl}(Y_t)) = \mathrm{cl}(Y_s).$$

Catenarity gives $\dim Y_s \leq \dim Y_t$, so the filtration bound follows.

7.5. Conservation of number

Assume f is proper. Then the fibres X_t and X_s are proper over their residue fields, and one has Euler characteristic homomorphisms

$$\chi_t : K_{\bullet}(X_t) \longrightarrow \mathbb{Z}, \quad \chi_s : K_{\bullet}(X_s) \longrightarrow \mathbb{Z}.$$

They satisfy

$$\chi_s(\sigma(x)) = \chi_t(x), \quad x \in K_{\bullet}(X_t).$$

This is the conservation-of-number theorem. Reducing to $x = \mathrm{cl}(Y_t)$ and taking the schematic closure Y of Y_t in X , it becomes

$$\chi(Y_s) = \chi(Y_t),$$

which is the standard constancy of Euler characteristic in a proper flat family.

7.6. Compatibility with the unit class

It is immediate from the definition that

$$\sigma(\text{cl}(\mathcal{O}_{X_t})) = \text{cl}(\mathcal{O}_{X_s}).$$

Since σ is $K^\bullet(X)$ -linear, the resulting diagram is commutative:

$$\begin{array}{ccc} & K^\bullet(X) & \\ \swarrow j^* & & \searrow i^* \\ K^\bullet(X_t) & & K^\bullet(X_s) \\ \downarrow & & \downarrow \\ K_\bullet(X_t) & \xrightarrow{\sigma} & K_\bullet(X_s), \end{array}$$

where the vertical arrows are the canonical maps from $K^\bullet$ to $K_\bullet$.

7.7. Specialization for $K^\bullet$

Assume now that X is separated and that the fibres X_t and X_s are regular. The canonical maps

$$K^\bullet(X_t) \longrightarrow K_\bullet(X_t), \quad K^\bullet(X_s) \longrightarrow K_\bullet(X_s)$$

are then isomorphisms. We define the specialization homomorphism for $K^\bullet$ as the unique homomorphism making the square

$$\begin{array}{ccc} K^\bullet(X_t) & \xrightarrow{\sigma} & K^\bullet(X_s) \\ \downarrow \sim & & \downarrow \sim \\ K_\bullet(X_t) & \xrightarrow{\sigma} & K_\bullet(X_s) \end{array}$$

commutative.

Since X is regular, hence normal, its connected components coincide with its irreducible components, and they dominate S . Therefore the map

$$H^0(X, \mathbb{Z}) \longrightarrow H^0(X_t, \mathbb{Z})$$

induced by the inclusion of the generic fibre is bijective. The analogous map to $H^0(X_s, \mathbb{Z})$ may therefore be interpreted as a specialization map

$$H^0(X_t, \mathbb{Z}) \longrightarrow H^0(X_s, \mathbb{Z}).$$

With respect to this map, the specialization on $K^\bullet$ is compatible with the augmented algebra structures. In particular, it induces a homomorphism of graded rings

$$\sigma^* : \text{Gr}^\bullet(X_t) \longrightarrow \text{Gr}^\bullet(X_s).$$

The Chern character is compatible with this specialization: the square

$$\begin{array}{ccc} K^\bullet(X_t) & \xrightarrow{c_{X_t}} & \text{Ch Gr}^\bullet(X_t) \\ \downarrow \sigma & & \downarrow \\ K^\bullet(X_s) & \xrightarrow{c_{X_s}} & \text{Ch Gr}^\bullet(X_s) \end{array}$$

commutes.

7.8. Specialization for the Picard group

Keep the hypotheses of 7.7. By the isomorphism

$$\text{Gr}^1 \simeq \text{Pic}$$

constructed in 5.3.2, the specialization map on Gr^1 gives a canonical homomorphism

$$\text{Pic}(X_t) \longrightarrow \text{Pic}(X_s).$$

It may be described more concretely as follows. Since X is regular, its local rings are factorial. Hence every invertible sheaf L_t on X_t extends to an invertible sheaf L on X . The image of L_t is the class of the restriction

$$L_s = L|_{X_s}.$$

This description is independent of the chosen extension. It is enough to check the case $L_t \simeq \mathcal{O}_{X_t}$. Then $L \simeq \mathcal{O}_X(D)$ for a divisor D supported on the special fibre. If X is integral, $D = nX_s$, and this divisor is principal, defined by a power of a uniformizer of the discrete valuation ring. Thus $L_s \simeq \mathcal{O}_{X_s}$.

7.9. Specialization and numerical equivalence

Assume f is proper and that X_t and X_s are regular. We have specialization maps

$$\sigma : \text{Gr}_\bullet(X_t) \longrightarrow \text{Gr}_\bullet(X_s), \quad \sigma^* : \text{Gr}^\bullet(X_t) \longrightarrow \text{Gr}^\bullet(X_s).$$

They are compatible with the intersection pairings. If $x_t \in \text{Gr}_\bullet(X_t)$ and $y_t \in \text{Gr}^\bullet(X_t)$, then after specialization one has the equality of intersection numbers

$$\langle \sigma(x_t), \sigma^*(y_t) \rangle = \langle x_t, y_t \rangle.$$

Equivalently, the specialization homomorphism on $K_\bullet$ respects multiplication by the specialized classes in $K^\bullet$ and respects Euler characteristic by the conservation-of-number theorem.

It follows formally that if the specialization of a class is numerically equivalent to zero, then the original class is numerically equivalent to zero. The converse would follow from surjectivity, modulo torsion, of the specialization map on the appropriate graded group, but this occurs only

in special situations.

7.9.4. It is nevertheless plausible, at least when f is proper and smooth, that the converse should hold: numerical triviality should be preserved by specialization. If true, the specialization maps would give monomorphisms on the quotients by numerical equivalence. At least the preceding argument shows that the numerical group of the generic fibre is a quotient of a subgroup of the numerical group of the special fibre; in particular its rank is no larger.

7.10. Compatibility with direct and inverse images

Let

$$g : X \longrightarrow Y$$

be a morphism of noetherian schemes flat over S . Write σ_X and σ_Y for the specialization maps of X and Y , and write

$$g_t : X_t \rightarrow Y_t, \quad g_s : X_s \rightarrow Y_s$$

for the morphisms on fibres.

If g is proper, then

$$\begin{array}{ccc} K_{\bullet}(X_t) & \xrightarrow{g_{t*}} & K_{\bullet}(Y_t) \\ \downarrow \sigma_X & & \downarrow \sigma_Y \\ K_{\bullet}(X_s) & \xrightarrow{g_{s*}} & K_{\bullet}(Y_s) \end{array}$$

commutes. Indeed, using the defining diagram of specialization, this is reduced to the standard compatibility of proper direct image with restriction to the open generic fibre and with pullback to the special fibre.

If g has finite Tor-dimension, so that inverse images in K -theory are defined, then the analogous diagram

$$\begin{array}{ccc} K_{\bullet}(Y_t) & \xrightarrow{g_t^*} & K_{\bullet}(X_t) \\ \downarrow \sigma_Y & & \downarrow \sigma_X \\ K_{\bullet}(Y_s) & \xrightarrow{g_s^*} & K_{\bullet}(X_s) \end{array}$$

commutes as well. This follows from the functoriality of finite-Tor pullback and from the same quotient description used to define σ .

Consequently, whenever the hypotheses allow passage to associated graded objects, the specialization maps commute with the corresponding Gysin morphisms, proper direct images, and pullbacks.

7.11. Algebraic equivalence of invertible sheaves

Let T be a proper scheme over a field k . An invertible sheaf $L \in \text{Pic}(T)$ is algebraically equivalent to zero if there exist a connected nonsingular curve C over k , two rational points $t_0, t_1 \in C(k)$,

and an invertible sheaf M on $T \times C$ such that

$$L = M_{t_1} \otimes M_{t_0}^{-1}.$$

The algebraically trivial classes form a subgroup $\text{Pic}^a(T)$, and one has

$$\text{Pic}^c(T) \subset \text{Pic}^a(T) \subset \text{Pic}^{\text{num}}(T).$$

If k has characteristic zero, or if T is proper over k , one can characterize $\text{Pic}^a(T)$ as the subgroup of classes some power of which belongs to the connected component subgroup $\text{Pic}^c(T)$.

This recovers the classical geometric notion. A divisor $t_1 - t_0$ on C pulls back to a divisor on $T \times C$, and specialization from one point of the curve to another gives the displayed relation in the Picard group of T .

7.12. Curves and the Picard group

Let C be a proper smooth connected curve over k , and let $t_0, t_1 \in C(k)$. Then there is a homomorphism

$$\text{Pic}(T \times_k C) \xrightarrow{i_1^* - i_0^*} \text{Pic}(T),$$

where $i_\nu : T \rightarrow T \times_k C$ is defined by t_ν for $\nu = 0, 1$. Its image is contained in $\text{Pic}^a(T)$. Conversely, if k is algebraically closed, every algebraically trivial class is obtained in this way for a suitable curve C .

If k is finite and $\bar{k}$ is an algebraic closure, then the Galois group acts on $\text{Pic}^a(T_{\bar{k}})$, and one has the usual exact sequence beginning

$$0 \longrightarrow \text{Pic}^a(T) \longrightarrow \text{Pic}^a(T_{\bar{k}})^{\text{Gal}(\bar{k}/k)} \longrightarrow H^1(k, \text{Pic}^c(T_{\bar{k}})) \longrightarrow \cdots.$$

7.13. Relation with ℓ -adic cohomology

For a locally free sheaf E of finite type on a scheme T , and for a prime ℓ invertible on T , one has ℓ -adic Chern classes

$$c^i(E) \in H^{2i}(T, \mathbb{Z}_\ell(i)).$$

They satisfy the usual multiplicativity for short exact sequences. Hence the total Chern class defines

$$c : K^\bullet(T) \longrightarrow 1 + \prod_{i \geq 1} H^{2i}(T, \mathbb{Z}_\ell(i)).$$

Now let X be proper and flat over the trait S , with regular fibres, and assume S henselian and ℓ different from the residue characteristic. Proper base change gives a specialization isomorphism in

cohomology. The diagram

$$\begin{array}{ccc} K^\bullet(X_t) & \xrightarrow{c^i} & H^{2i}(X_t, \mathbb{Z}_\ell(i)) \\ \downarrow \sigma & & \downarrow \\ K^\bullet(X_s) & \xrightarrow{c^i} & H^{2i}(X_s, \mathbb{Z}_\ell(i)) \end{array}$$

commutes. This is proved by the same argument as for compatibility with pullback: Chern classes commute with inverse image for the immersions of the two fibres.

Passing to finite extensions of the fraction field and then to the limit gives the corresponding statement for geometric fibres

$$\begin{array}{ccc} K^\bullet(X_{\bar{t}}) & \longrightarrow & H^{2i}(X_{\bar{t}}, \mathbb{Q}_\ell(i)) \\ \downarrow & & \downarrow \\ K^\bullet(X_{\bar{s}}) & \longrightarrow & H^{2i}(X_{\bar{s}}, \mathbb{Q}_\ell(i)). \end{array}$$

Equivalently, if a class in $K^\bullet(X_t)$ specializes to a class in $K^\bullet(X_s)$, then their i -th Chern classes are the values at the two geometric points of a single section of $R^{2i}f_*\mathbb{Z}_\ell(i)$.

After passage to associated graded groups, one obtains maps

$$c^i : \mathrm{Gr}^i(T) \longrightarrow H^{2i}(T, \mathbb{Q}_\ell(i)),$$

with the convention that if $\tilde{x} \in \mathrm{Fil}^i K^\bullet(T)$ lifts $x \in \mathrm{Gr}^i(T)$, then

$$c^i(x) = c^i(\tilde{x})/i!.$$

The specialization diagrams remain commutative for these graded Chern class maps.

7.14. Passage to limits on the base

Let the base trait be replaced by a filtered inverse limit of traits, or by the normalization in finite extensions of the generic point. The preceding constructions are compatible with such base change. The key inputs are the compatibility of K -theory with filtered limits under affine transition morphisms, the proper base-change theorem in cohomology, and the preservation of regularity under the smooth base changes considered here.

Thus the specialization maps on K -groups, on associated graded groups, on Picard groups, and on ℓ -adic Chern classes may all be transported to geometric fibres without ambiguity.

7.15. Specialization and geometric fibres

For a smooth proper morphism $X \rightarrow S$, the cohomological specialization map on geometric fibres is an isomorphism. The K -theoretic specialization need not be an isomorphism, but it is compatible with all constructions used above. In particular Chern classes of specialized K -classes correspond under the cohomological specialization isomorphism.

This observation is often used in the following form: a cycle or vector bundle class on the generic fibre whose specialization is numerically trivial has cohomologically trivial Chern classes after specialization, and hence the same Chern classes on the generic fibre.

7.16. Geometric specialization diagrams

Choose an algebraic closure $\bar{K}$ of the fraction field of the trait and an induced algebraic closure $\bar{k}$ of the residue field. For the geometric fibres $X_{\bar{t}}$ and $X_{\bar{s}}$, one has compatible specialization maps

$$K^\bullet(X_{\bar{t}}) \longrightarrow K^\bullet(X_{\bar{s}}),$$

$$\mathrm{Gr}^i(X_{\bar{t}}) \longrightarrow \mathrm{Gr}^i(X_{\bar{s}}),$$

and comparison maps to ℓ -adic cohomology. The squares

$$\begin{array}{ccc} \mathrm{Gr}^i(X_{\bar{t}}) & \xrightarrow{c^i} & H^{2i}(X_{\bar{t}}, \mathbb{Q}_\ell(i)) \\ \downarrow & & \downarrow \sim \\ \mathrm{Gr}^i(X_{\bar{s}}) & \xrightarrow{c^i} & H^{2i}(X_{\bar{s}}, \mathbb{Q}_\ell(i)) \end{array}$$

commute.

7.17. Variants and complements

The same formalism applies to Chow groups when they are identified with the associated graded pieces of K -theory in the cases covered by Riemann–Roch. It also applies to Chern classes with torsion coefficients before passage to $\mathbb{Q}_\ell$, by working modulo ℓ^n and taking limits.

If the morphism is not smooth, one must distinguish the specialization of coherent sheaf classes from that of locally free sheaf classes. The construction of σ for $K_\bullet$ remains available under flatness, while the specialization for $K^\bullet$ requires regularity of the fibres or another hypothesis giving the comparison $K^\bullet \simeq K_\bullet$.

7.18. Questions

These considerations are tied to the standard conjectures. For a smooth projective variety over a field, one may form categories of correspondences modulo rational, algebraic, numerical, or homological equivalence. There are functors

$$\mathcal{C}_{\mathrm{rat}} \longrightarrow \mathcal{C}_{\mathrm{num}} \longrightarrow \mathcal{C}_{\mathrm{hom}}.$$

Grothendieck’s standard conjectures assert, in particular, that the comparison with homological equivalence has the expected equivalence properties. Over finite fields some of these statements are known for special classes such as abelian varieties and certain surfaces, but the numerical and homological comparison remains open in general.

One may also use algebraic equivalence and obtain functors

$$\mathcal{C}_{\text{rat}} \longrightarrow \mathcal{C}_{\text{alg}} \longrightarrow \mathcal{C}_{\text{num}}.$$

The assertion that the first of these is an equivalence is a form of Bloch's conjecture; it remains open in general, although many special cases are known.

Expose XII – A relative representability theorem for the Picard functor

A relative representability theorem for the Picard functor

M. Raynaud, written up by S. Kleiman

Warning

The purpose of Exposes XII and XIII is to prove the nine finiteness theorems for the Picard functor stated by Grothendieck in the comments to the *Fondements de la geometrie algebrique*. In this first expose one proves the first finiteness theorem in the following more precise form. Let X and Y be proper schemes over an integral noetherian base S , and let

$$f : X \rightarrow Y$$

be a surjective S -morphism. Then there exists a non-empty open subset $V \subset S$ over which the induced morphism

$$f^* : \text{Pic}_{Y/S}|_V \longrightarrow \text{Pic}_{X/S}|_V$$

is representable by a quasi-affine morphism of finite presentation. The second expose proves the remaining finiteness results.

The two exposes use rather different methods. In the first, the proof uses devissage of GL_n , non-flat descent, and representability of unramified functors. In the second, one first reduces to cases where the schemes over the base are projective and flat with geometrically integral fibres, sometimes normal or smooth; one then uses the theory of bounded families of sheaves developed by Castelnuovo, Matsusaka, Mumford, Nakai, and Kleiman.

1. Statement of the main theorem and applications

Theorem 1.1. Let S be a noetherian scheme, let X and Y be proper S -schemes, and let $f : X \rightarrow Y$ be a morphism. Assume S integral and f surjective. Then there is a non-empty open subset $V \subset S$ such that

$$f^* : \underline{\text{Pic}}_{Y/S}|_V \longrightarrow \underline{\text{Pic}}_{X/S}|_V$$

between the fppf Picard functors is representable by a quasi-affine morphism of finite presentation.

Corollary 1.2. If X is proper over an integral noetherian scheme S , then after replacing S by a non-empty open subset V , the Picard functor $\underline{\mathrm{Pic}}_{X/S}|_V$ is representable by a group scheme over V which is a union of quasi-projective open subschemes of finite type.

Corollary 1.3. For a surjective morphism $f : X \rightarrow Y$ of proper schemes over a field k , the kernel of

$$f^* : \mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k}$$

is an algebraic group scheme of finite type over k .

1.4. Reductions of the main theorem

The proof of Theorem 1.1 is reduced step by step to more special cases.

First, after shrinking the integral noetherian base S , one may assume suitable normality or regularity properties. Secondly, by flat descent one reduces to the case where Y is integral and f is generically finite. Thirdly, using flattening, one may suppose that f is flat. Finally, devissage reduces the problem to the case where X and Y are projective and flat over S with geometrically integral fibres.

1.5. Plan of proof

The proof is organized as follows. One first treats the theorem when f is a regular immersion of codimension one, and more generally when f is finite and flat. Next one proves stability under composition: if the result is known for $X \rightarrow Y$ and for $Y \rightarrow Z$, it is known for $X \rightarrow Z$. Then one reduces to the generically finite flat case. The final step uses a devissage argument in which the structure of $\mathrm{GL}(n)$, established above, is essential.

Corollary 1.5. Let k be a field.

- a) For a proper k -scheme X , the functor $\underline{\mathrm{Pic}}_{X/k}$ is representable.
- b) If $f : X \rightarrow Y$ is a surjective morphism of proper k -schemes, then

$$f^* : \underline{\mathrm{Pic}}_{Y/k} \longrightarrow \underline{\mathrm{Pic}}_{X/k}$$

is affine.

Corollary 1.6. Let S be integral and noetherian, and let $f : X \rightarrow Y$ be a surjective S -morphism between proper S -schemes.

- a) If Y is normal over S , meaning flat with geometrically normal fibres, then after shrinking S the map f^* on Picard functors is quasi-finite.

- b) If $\underline{\text{Pic}}_{Y/S}$ is essentially proper over S —for instance if Y is smooth over S —then, after shrinking S , the morphism f^* is finite, hence affine.

Indeed, fibrewise the kernel of f_s^* is an affine algebraic group whose identity component is proper; hence it is finite. In the second case f^* is both proper and quasi-affine, therefore finite.

Remark 1.7. It is not known whether Theorem 1.1 remains true with “affine” in place of “quasi-affine”, although the preceding corollaries suggest that this might be the case.

2. First reductions

Applying the preliminary reductions to X/S and Y/S , one finds, after replacing S by a finite flat cover of a non-empty open subset, that each irreducible component of X and Y , endowed with its reduced structure, is integral over S . By fppf descent, this replacement is harmless for the representability and quasi-affineness statements.

The diagram obtained by replacing schemes by their reductions shows that it is enough to prove two assertions:

- I. Theorem 1.1 when f is an immersion, necessarily nilpotent in the reduction step.
- II. Theorem 1.1 when X and Y are reduced and every irreducible component of each, with the induced reduced structure, is integral over S .

For the second assertion, group the irreducible components X_i according to the component Y_j that contains their image. One obtains diagrams comparing the Picard functor of a scheme with the product of the Picard functors of its components. After shrinking S , it is enough to know that the relevant maps to the products are quasi-affine and to treat the case where both X and Y are integral over S .

For the maps from a scheme to the disjoint union of its irreducible components, one reduces to the case of two closed subschemes $Y_1, Y_2 \subset Y$ whose union is Y .

Proposition 2.3. Let $g : Y \rightarrow S$ be proper, and let Y_1, Y_2 be closed subschemes defined by ideals I_1, I_2 . Assume:

- a) Y, Y_1, Y_2 , and $Y_1 \cap Y_2$ are faithfully flat and of finite presentation over S ;
- b) $g_* \mathcal{O}_Y = \mathcal{O}_S$ and $(g_i)_* \mathcal{O}_{Y_i} = \mathcal{O}_S$ universally;
- c) $I_1 \cap I_2 = 0$.

Then

$$\underline{\text{Pic}}_{Y/S} \longrightarrow \underline{\text{Pic}}_{(Y_1 \amalg Y_2)/S}$$

is representable by an affine morphism of finite presentation.

Proof. The assertion is fppf-local on the base. After a base change one may suppose that $Y_1 \cap Y_2$ has a section over S , giving compatible rigidifications. A point of the Picard functor of $Y_1 \amalg Y_2$ is a pair of rigidified invertible sheaves L_1, L_2 on Y_1, Y_2 . To glue them to an invertible sheaf on Y , it is necessary and sufficient to give an isomorphism

$$L_1|_{Y_1 \cap Y_2} \xrightarrow{\sim} L_2|_{Y_1 \cap Y_2}$$

compatible with the rigidifications. Conversely, such an isomorphism defines the glued line bundle as the kernel of

$$(\varepsilon_1)_* L_1 \oplus (\varepsilon_2)_* L_2 \longrightarrow (\varepsilon_{12})_*(L_1|_{Y_1 \cap Y_2}).$$

Thus the fibre of the Picard map over (L_1, L_2) is an isomorphism functor.

Lemma 2.4. Let X be flat, proper, and of finite presentation over S , and let L, M be invertible sheaves on X . The functor

$$T \longmapsto \text{Isom}(L_T, M_T)$$

is representable by an affine S -scheme of finite presentation. If L and M are rigidified along a section, the subfunctor of isomorphisms compatible with the rigidifications is represented by a closed subscheme.

The proof uses the representability of the Hom functors by vector bundles and cuts out the invertible homomorphisms by the equations $uv = 1$ and $vu = 1$.

2.5. Reduction to an effective-epimorphism condition

It remains to reduce the integral case to the following assertion. Assume X and Y are integral over S and that the sequence

$$\mathcal{O}_Y \longrightarrow f_* \mathcal{O}_X \rightrightarrows g_* \mathcal{O}_{X \times_Y X}$$

is exact, where $g : X \times_Y X \rightarrow Y$ is the structural morphism. Then Theorem 1.1 holds.

It is enough to factor f as a finite sequence

$$X = X_r \rightarrow X_{r-1} \rightarrow \cdots \rightarrow X_1 \rightarrow X_0 = Y$$

where each map satisfies this exactness condition and, after shrinking the base, each X_i is integral over S .

Lemma 2.6. Let Y be noetherian and let $f : X \rightarrow Y$ be a finite epimorphism, meaning that $\mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ is injective. Then f factors into a finite sequence of finite morphisms

$$X = X_r \rightarrow X_{r-1} \rightarrow \cdots \rightarrow X_1 \rightarrow X_0 = Y$$

which are effective epimorphisms, i.e. for each step the sequence

$$0 \rightarrow \mathcal{O}_{X_i} \rightarrow (f_i)_* \mathcal{O}_{X_{i+1}} \rightrightarrows (f_i \times f_i)_* \mathcal{O}_{X_{i+1} \times_{X_i} X_{i+1}}$$

is exact.

3. Proof of I: Oort's devissage

Proposition 3.1 (Oort). Let S be locally noetherian, let $f : X \rightarrow S$ be proper, and let $X' \subset X$ be a closed subscheme. Assume:

- i) X' is defined by an ideal I contained in the nilradical N of X , and $NI = 0$;
- ii) X' is flat over S and cohomologically flat over S in dimension 0;
- iii) I is flat over S and cohomologically flat in dimension 0, in such a way that there is a coherent $\mathcal{O}_{X'}$ -module Q with

$$R^q f'_*(I \otimes M) = \mathrm{Hom}_{\mathcal{O}_{X'}}(Q, M)$$

functorially in coherent M ;

- iv) the cokernel E of $f'_* \mathcal{O}_{X'} \rightarrow R^q f'_*(I)$ is locally free;
- v) for every point $s \in S$, the map

$$(f'_* \mathcal{O}_{X'}) \otimes k(s) \longrightarrow (f_* \mathcal{O}_X) \otimes k(s)$$

is surjective.

Then the canonical map

$$\underline{\mathrm{Pic}}_{X/S} \longrightarrow \underline{\mathrm{Pic}}_{X'/S}$$

is representable by an affine morphism of finite presentation. More precisely, there is an exact sequence of fppf sheaves

$$0 \rightarrow \mathbb{V}(E') \rightarrow \underline{\mathrm{Pic}}_{X/S} \xrightarrow{u} \underline{\mathrm{Pic}}_{X'/S} \rightarrow \mathbb{V}(Q),$$

and, if $P = \ker(u)$, then $P \rightarrow \underline{\mathrm{Pic}}_{X/S}$ is represented by a closed immersion and $\underline{\mathrm{Pic}}_{X/S}$ is a torsor over P under the vector group $\mathbb{V}(E')$.

Proof. For an S -scheme T , apply cohomology to the exact sequence

$$0 \rightarrow I_T \rightarrow \mathcal{O}_{X_T} \rightarrow \mathcal{O}_{X'_T} \rightarrow 0.$$

The resulting long exact sequences compare units and Picard groups on X_T and X'_T . The only small point is that the left square obtained from units is not literally commutative. Nevertheless,

one has the inclusion of images

$$\mathrm{Im} \delta_1 \subset \mathrm{Im} \delta_2$$

for the two connecting homomorphisms. To see this, work after making S and T affine. Choose an affine covering (V_λ) of X_T , and lift a unit g on X'_T to units g_λ on the V_λ . The two boundary classes are represented by cocycles which differ by multiplication by the image of g acting on I_T . The hypotheses on the nilradical and the fibrewise surjectivity in (v) allow this multiplicative discrepancy to be absorbed into the image of the other boundary map.

Using (iii) and (iv), the cohomology sequence becomes the functorial exact sequence

$$0 \rightarrow H^0(T, E' \otimes \mathcal{O}_T) \rightarrow H^1(X_T, \mathcal{O}_{X_T}^*) \rightarrow H^1(X'_T, \mathcal{O}_{X'_T}^*) \rightarrow H^0(T, Q \otimes \mathcal{O}_T).$$

After sheafification for the fppf topology this is precisely

$$0 \rightarrow \mathbb{V}(E') \rightarrow \underline{\mathrm{Pic}}_{X/S} \rightarrow \underline{\mathrm{Pic}}_{X'/S} \rightarrow \mathbb{V}(Q).$$

The kernel P is the pullback of the zero section of the separated vector group $\mathbb{V}(Q)$, hence is represented by a closed immersion. The remaining statement says that the first part is a vector-group torsor, and it follows by fppf descent after trivializing the torsor.

Corollary 3.3. Under the hypotheses of Proposition 3.1:

- i) the corresponding connected-component Picard sheaves for X and X' are identified whenever one of the two maps is already known to be an isomorphism;
- ii) if T is affine and $\mathrm{Pic}(X_T) \rightarrow \mathrm{Pic}(X'_T)$ is surjective in the relevant component, then the analogous surjectivity holds for X .

Both assertions are immediate from the exact sequence above and the fact that vector groups have no higher obstruction on affine bases.

Corollary 3.4. Let k be a field and let X be a proper k -scheme. Suppose that $\underline{\mathrm{Pic}}_{X/k}$ possesses a universal invertible sheaf. Then over every affine open subset V of $\underline{\mathrm{Pic}}_{X/k}$ there is a universal invertible sheaf on $X \times V$. This applies, by descent, when each connected component of X has a rational point, for instance over an algebraically closed field.

Proposition 3.5. Let S be integral and noetherian, and let $f : X \hookrightarrow Y$ be a nilpotent immersion of proper S -schemes. Then, after replacing S by a non-empty open subset V , the morphism

$$f^* : \underline{\mathrm{Pic}}_{Y/S}|_V \rightarrow \underline{\mathrm{Pic}}_{X/S}|_V$$

is representable by an affine morphism of finite presentation.

Proof. Filter the nilpotent ideal defining the immersion by successive square-zero pieces satisfying the hypotheses of Proposition 3.1 after shrinking the base. Cohomological flatness in dimension

0 and local freeness of the necessary cohomology modules are open conditions in a proper flat family. Applying Proposition 3.1 successively to the successive thickenings gives an affine morphism of finite presentation at each step; their composite is affine of finite presentation. This proves assertion I.

4. Proof of II: the descent part

The remaining case is the most delicate. Assume X and Y are integral over S , and that

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \rightrightarrows g_*\mathcal{O}_{X \times_Y X} \quad (*)$$

is exact. After shrinking S , one may also assume the auxiliary flatness and finite-presentation hypotheses needed below.

The proof is reduced to the representability of the fibre of f^* over a given line bundle on X . After an fppf base change one may suppose that $X \rightarrow S$ has a section α ; then Y has the section $\beta = f\alpha$. A point of $\underline{\text{Pic}}_{X/S}$ is represented by an invertible sheaf L on X rigidified along α . We must show that the functor of rigidified invertible sheaves M on Y together with an isomorphism

$$f^*M \xrightarrow{\sim} L$$

is quasi-affine and of finite presentation over S .

Lemma 4.2. Under the exactness condition $(*)$, the morphism f is a descent morphism for locally free sheaves. Equivalently, for locally free sheaves L, M on Y , the sequence

$$\text{Hom}_Y(L, M) \rightarrow \text{Hom}_X(f^*L, f^*M) \rightrightarrows \text{Hom}_{X \times_Y X}(f_1^*f^*L, f_2^*f^*M)$$

is exact.

Proof. Put $N = \mathcal{H}om_Y(L, M)$. The assertion is equivalent to exactness of

$$\Gamma(Y, N) \rightarrow \Gamma(Y, f_*\mathcal{O}_X \otimes N) \rightrightarrows \Gamma(Y, g_*\mathcal{O}_{X \times_Y X} \otimes N).$$

This follows from $(*)$ after tensoring with the locally free sheaf N and taking global sections.

Thus invertible sheaves on Y pulling back to L are the same as effective descent data on L relative to f . Forgetting effectivity gives a monomorphism into the functor of gluing data

$$\underline{\text{Isom}}(p_1^*L, p_2^*L),$$

where $p_1, p_2 : X \times_Y X \rightarrow X$ are the projections. By Lemma 2.4 this is represented by an affine scheme of finite presentation over S .

It remains to represent the subfunctor consisting of effective descent data. After the base change to the representing affine scheme of all gluing data, this becomes the following problem: given an

invertible sheaf L on X equipped with a gluing isomorphism

$$\varphi : p_1^*L \xrightarrow{\sim} p_2^*L,$$

determine when this datum descends to Y .

Lemma 4.3. The gluing datum on L is effective if and only if the following conditions hold:

- 1) for all $i \geq 0$, the sheaves $R^i f_* L$ are flat over S ;
- 2) for all $j \geq 0$, the sheaves $R^j g_* p_1^* L$ are flat over S ;
- 3) the cokernel of

$$\pi_1 - \pi_2 : f_* L \rightarrow g_* p_1^* L \simeq g_* p_2^* L$$

is flat over S ;

- 4) the kernel N of the same homomorphism is invertible on Y ;
- 5) the natural composite

$$f^* N \rightarrow f^* f_* L \rightarrow L$$

is an isomorphism.

Proof. The last two conditions are visibly sufficient: the kernel N is then the desired descended invertible sheaf, and its pullback is L . Conversely, if $L \simeq f^* M$ comes from an invertible sheaf M on Y , the assertion may be checked locally on Y after trivializing M . It then reduces to the hypotheses imposed on f and to the exactness of $(*)$.

Each of the five conditions is represented by a monomorphism of finite presentation. For the first two one uses the flattening stratification for higher direct images. For the third, once the first two hold, the cokernel commutes with base change and one applies the same flattening criterion. For the fourth and fifth one uses openness of the locus where a finitely presented module is invertible and where a morphism of finitely presented modules is an isomorphism.

Lemma 4.4. Let $f : X \rightarrow S$ be proper of finite presentation and let F be an S -flat sheaf of finite presentation. For a fixed integer n , the subfunctor of S on which $R^i f_* F$ is flat for all $i \geq n$ is represented by a monomorphism of finite presentation, and over it the sheaves $R^i f_* F$ commute with arbitrary base change for $i \geq n$.

The proof is by descending induction on n , using the theorem on cohomology and base change and the representability of the flatness functor.

Lemma 4.5. For a proper morphism of finite presentation $X \rightarrow S$ and a sheaf F of finite presentation on X , the subfunctor of S on which F is flat over the base is represented by a monomorphism of finite presentation.

Lemma 4.6. Let $u : F \rightarrow G$ be a homomorphism of finitely presented modules on a scheme proper over S , with G flat over S . The subfunctor of S on which u is an isomorphism is represented by an open subscheme of finite presentation. Indeed, first cut out the open on which the cokernel vanishes; over it, the kernel commutes with base change, and one cuts out the open on which the kernel vanishes.

Combining these lemmas, the subfunctor of effective descent data is represented by a monomorphism of finite presentation into the affine scheme of all gluing data. Hence it is quasi-affine of finite presentation. This proves assertion II and therefore Theorem 1.1.

Expose XIII – Finiteness theorems for the Picard functor

Finiteness theorems for the Picard functor

S. Kleiman

1. Bounded sheaves

Let k be an algebraically closed field, and let X be a projective k -scheme endowed with an ample invertible sheaf $\mathcal{O}_X(1)$.

Definition 1.1 (Mumford). Let $m \in \mathbb{Z}$. A coherent sheaf F on X is called *m -regular* if, at every point of the support of F , the sheaf $\mathcal{O}_X(1)$ is generated by $H^0(\mathcal{O}_X(1))$, and if, for every $q > 0$, one has

$$H^q(F(m - q)) = 0.$$

Lemma 1.2. Let

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

be an exact sequence of coherent sheaves on X , where the first map is induced by tensoring with a section of $\mathcal{O}_X(1)$. If F is m -regular, then G is m -regular.

Indeed, the corresponding cohomology sequence contains

$$H^q(F(m - q)) \rightarrow H^q(G(m - q)) \rightarrow H^{q+1}(F(m - q - 1)),$$

and the two outside terms vanish by the definition of m -regularity.

Proposition 1.3. If F is m -regular, then for every $n \geq m$:

- i) F is n -regular;
- ii) the multiplication map

$$H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \longrightarrow H^0(F(n + 1))$$

is surjective;

iii) $F(n)$ is generated by its global sections.

The proof is by induction on the dimension s of the support of F . The case $s = -1$ is empty. Choose a section $\sigma \in H^0(\mathcal{O}_X(1))$ which vanishes at no point of the finite set $\text{Ass}(F)$. It gives an exact sequence

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0,$$

and by Lemma 1.2 the sheaf G is m -regular, with support of dimension $s - 1$.

From the exact sequence

$$H^q(F(n - q - 1)) \rightarrow H^q(F(n - q)) \rightarrow H^q(G(n - q)),$$

the induction hypothesis gives $H^q(G(n - q)) = 0$ for $n \geq m$; hence vanishing for $F(n - q - 1)$ implies vanishing for $F(n - q)$. Starting from the defining vanishing at $n = m$, this proves (i).

For (ii), one uses the commutative diagram obtained by multiplying sections by σ and comparing the exact rows

$$0 \rightarrow H^0(F(n)) \rightarrow H^0(F(n + 1)) \rightarrow H^0(G(n + 1))$$

with the corresponding multiplication maps for F and G . The first row is exact by (i), and the required surjectivity for G is known by the induction hypothesis. This gives the surjectivity for F .

For (iii), write $M_X = M \otimes_k \mathcal{O}_X$ for a vector space M . The maps

$$H^0(F(n))_X \longrightarrow F(n)$$

are compatible with multiplication by $H^0(\mathcal{O}_X(1))$. Since the multiplication maps in (ii) are surjective, once $H^0(F(n))_X \rightarrow F(n)$ is surjective for one n , it is surjective for all larger n . Serre's theorem gives surjectivity for $n \gg 0$, and hence for every $n \geq m$.

Proposition 1.4. Let

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

be an exact sequence on X . If G is m -regular, then:

i) $H^q(F(n)) = 0$ for $q \geq 2$ and $n \geq m - (q - 1)$;

ii) the map $H^1(F(n - 1)) \rightarrow H^1(F(n))$ is an isomorphism for

$$n \geq (m - 1) + h^1(F(m - 1));$$

iii) $H^1(F(n)) = 0$ for

$$n \geq (m - 1) + h^1(F(m - 1)).$$

In particular, F is $[(m - 1) + h^1(F(m - 1))]$ -regular.

Indeed, from

$$H^{q-1}(G(n)) \rightarrow H^q(F(n-1)) \rightarrow H^q(F(n)) \rightarrow H^q(G(n))$$

the two outside terms vanish for $q \geq 2$ and $n \geq m - (q - 1)$ by Proposition 1.3. Thus

$$H^q(F(m-q)) \simeq H^q(F(m-q+1)) \simeq H^q(F(m-q+2)) \simeq \cdots,$$

and the common value is zero by Serre's theorem. This proves (i).

For $n \geq m - 1$ one has the exact sequence

$$0 \rightarrow H^0(F(n-1)) \rightarrow H^0(F(n)) \rightarrow H^0(G(n)) \rightarrow H^1(F(n-1)) \rightarrow H^1(F(n)) \rightarrow 0.$$

If the map $H^0(F(n)) \rightarrow H^0(G(n))$ is surjective, then by the multiplication diagram of Proposition 1.3 the analogous map in degree $n+1$ is also surjective. Hence, once surjectivity has occurred, all subsequent H^1 groups are equal and finally vanish by Serre's theorem. If surjectivity has not yet occurred, the integer $h^1(F(n-1))$ strictly decreases when passing to $h^1(F(n))$. There can be at most $h^1(F(m-1))$ such failures. This proves (ii) and (iii).

Definition 1.5. Let F be a coherent sheaf on X , let $r \geq \dim \text{Supp}(F)$, and let $(b) = (b_0, \dots, b_r)$ be a sequence of $r+1$ integers. We say that F is a (b) -sheaf if it satisfies the following equivalent condition. First, $\mathcal{O}_X(1)$ is generated by $H^0(\mathcal{O}_X(1))$ at every point of the support of F ; secondly $h^0(F(-1)) \leq b_0$; and, if $r \geq 1$, there is a section $\sigma \in H^0(\mathcal{O}_X(1))$ giving an exact sequence

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

where G is a $(b_1, \dots, b_r)$ -sheaf.

Equivalently, there is an F -regular sequence of r sections $\sigma_1, \dots, \sigma_r \in H^0(\mathcal{O}_X(1))$ such that, for $i = 0, \dots, r$, if F_{σ_i} denotes the restriction of F to the common zero scheme of $\sigma_1, \dots, \sigma_i$, then

$$h^0(F_{\sigma_i}(-1)) \leq b_i.$$

Proposition 1.6. Let F be a (b) -sheaf on X . Then:

- i) for every sufficiently general sequence $\sigma_1, \dots, \sigma_r$ of sections of $H^0(\mathcal{O}_X(1))$, the restrictions F_{σ_i} satisfy

$$h^0(F_{\sigma_i}(-1)) \leq b_i \quad (0 \leq i \leq r);$$

- ii) every subsheaf $G \subset F$ is a (b) -sheaf.

Let T be the open subset of the affine space of r -tuples of sections corresponding to F -regular sequences. For fixed i , the sheaves $F_{\sigma_i}(-1)$ for $\sigma \in T(k)$ belong to a flat family over T . Since T is non-empty by hypothesis, the upper semicontinuity theorem for h^0 proves (i). If $G \subset F$, then a sufficiently general sequence is also G -regular and G_{σ_i} is a subsheaf of F_{σ_i} for all i ; (ii) follows.

Lemma 1.7. Let

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

be exact. If the Hilbert polynomial of F is written

$$\chi(F(n)) = \sum_{i=0}^r a_i \binom{n+i}{i},$$

then

$$\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Indeed,

$$\chi(G(n)) = \chi(F(n)) - \chi(F(n-1)) = \sum_i a_i \left(\binom{n+i}{i} - \binom{n-1+i}{i} \right).$$

Proposition 1.8. Let F be a (b) -sheaf on X , put $s = \dim \text{Supp}(F)$, and write

$$\chi(F(n)) = \sum_{i=0}^s a_i \binom{n+i}{i}.$$

Then:

i) for $n \geq -1$,

$$h^0(F(n)) \leq \sum_{i=0}^s b_i \binom{n+i}{i};$$

ii) $a_i \leq b_i$ for every i , and F is also a $(b_0, \dots, b_{s-1}, a_s)$ -sheaf.

The proof is by induction on s . When $s = 0$, the assertion is immediate, since

$$a_0 = h^0(F) = h^0(F(-1)) \leq b_0.$$

For $s \geq 1$, choose

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

with G a $(b_1, \dots, b_s)$ -sheaf and $\dim \text{Supp}(G) = s - 1$. Then

$$h^0(F(n)) - h^0(F(n-1)) = h^0(G(n)),$$

and by induction

$$h^0(G(n)) \leq \sum_{i=0}^{s-1} b_{i+1} \binom{n+i}{i}.$$

Together with $h^0(F(-1)) \leq b_0$, this proves (i) by induction on n . Lemma 1.7 and the induction hypothesis give (ii).

Definition 1.9. The (b) -polynomials are universal polynomials with non-negative rational coefficients, denoted

$$P_r(b_0, \dots, b_{r-1}; x),$$

defined recursively by the preceding estimates for regularity: $P_0 = 0$, and P_r is obtained by applying Proposition 1.4 to a hyperplane section controlled by P_{r-1} . They are increasing in each argument.

Remark 1.10. For these polynomials one has, immediately by induction on $r \geq 1$,

$$P_r(b_0, \dots, b_{r-1}; P_{r-1}(b_1, \dots, b_{r-1}; 0)) = P_r(b_0, \dots, b_{r-1}; 0).$$

Theorem 1.11. Let F be a (b) -sheaf on X , and write

$$\chi(F(n)) = \sum_{i=0}^s a_i \binom{n+i}{i}.$$

Let $(c) = (c_1, \dots, c_s)$ be a sequence of integers with $(c) \geq (b) - (a)$, and put $m = P_s(c)$. Then $m \geq 0$ and F is m -regular. In particular, if $s = \dim \text{Supp}(F)$, then F is

$$P_s(a_0, \dots, a_{s-1}; a_s)$$

regular.

The proof is by induction on s . The case $s = 0$ is clear: $m = 0$, and a zero-dimensional sheaf is n -regular for every $n \geq 0$. For $s \geq 1$, choose

$$0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$$

with G a $(b_1, \dots, b_s)$ -sheaf. By Lemma 1.7 and the induction hypothesis, G is n -regular for an integer

$$n = P_{s-1}(c_2, \dots, c_s)$$

controlled by the same universal polynomials. Proposition 1.4 then shows that F is

$$n + h^1(F(n-1))$$

regular, while the estimates of Proposition 1.8 bound $h^1(F(n-1))$ by a universal expression in the same data. This is precisely the recursive construction of P_s . The final assertion follows from Proposition 1.8(ii) and Remark 1.10.

1.12. Families of coherent sheaves on fibres. Let S be noetherian and let X be of finite type over S . A *family of classes of coherent sheaves on the fibres of X/S* means the following data. For every point $s \in S$ and every field extension $K/k(s)$, one specifies coherent sheaves on X_K . Two such sheaves, defined over extensions K and K' , are in the same class if, after extension

to some common field K'' over $k(s)$, they become isomorphic on $X_{K''}$.

Such a family $\mathcal{F}$ is said to be *bounded by* a coherent sheaf F on $X_T = X \times_S T$, where T is of finite type over S , if $\mathcal{F}$ is contained in the family of classes of the sheaves $F_{k(t)}$ for $t \in T$. It is called *bounded* if such T and F exist.

Assume in addition that X/S is projective and endowed with a relatively ample invertible sheaf $\mathcal{O}_X(1)$. Then $\mathcal{F}$ is called a *(b)-family*, for $(b) = (b_0, \dots, b_r)$, if every class in $\mathcal{F}$ is represented, after extending the ground field to an algebraic closure, by a *(b)-sheaf*.

Theorem 1.13. Let S be noetherian and let X be projective over S , endowed with an ample invertible sheaf $\mathcal{O}_X(1)$ whose restrictions $\mathcal{O}_{X_s}(1)$ are generated by their global sections. Let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibres of X/S . The following conditions are equivalent:

- i) $\mathcal{F}$ is bounded. Moreover, if all sheaves in $\mathcal{F}$ are locally free of rank p , then $\mathcal{F}$ may be bounded by a sheaf on some X_T which is locally free of rank p .
- ii) The set of Hilbert polynomials $\chi(F_s(n))$ for $F \in \mathcal{F}$ is finite, and there is a sequence (b) such that $\mathcal{F}$ is a *(b)-family*.
- iii) The set of Hilbert polynomials $\chi(F_s(n))$ is finite, and there is an integer m such that all sheaves in $\mathcal{F}$ are m -regular.
- iv) The set of Hilbert polynomials is finite, and $\mathcal{F}$ is contained in the family of quotient classes of sheaves E_T , where E is coherent on some X_T . In fact, one may take $T = S$ and $E = \mathcal{O}_X(-m)^{\oplus M}$.
- v) $\mathcal{F}$ is contained in the family of cokernel classes of homomorphisms $E'_T \rightarrow E_T$, where E' and E are coherent on some X_T . In fact, one may take $T = S$ and

$$E' = \mathcal{O}_X(-m')^{\oplus M'}, \quad E = \mathcal{O}_X(-m)^{\oplus M}.$$

Suppose first that $\mathcal{F}$ is bounded by a coherent sheaf F on X_T . Cutting T into finitely many locally closed pieces and using generic flatness, we may assume that F is flat over T . Then the number of Hilbert polynomials $\chi(F_t(n))$ is at most the number of connected components of T . If a fibre F_t is locally free of rank p , then F is locally free of rank p near t , after possibly replacing T by an open neighbourhood.

By the hypotheses on $\mathcal{O}_X(1)$ and the semicontinuity theorem, after another finite stratification of T there is a sequence of global sections $\sigma_1, \dots, \sigma_r$ which is regular for the members of the family, and the dimensions

$$h^0(F_{\sigma_i, t}(-1))$$

are bounded. Thus (i) implies (ii).

The implication (ii) $\Rightarrow$ (iii) is exactly Theorem 1.11. If (iii) holds, Proposition 1.3(iii) gives a surjection

$$\mathcal{O}_X(-m)^{\oplus M} \longrightarrow F$$

for every member of the family, where M may be chosen as the maximum of the finitely many values $\chi(F(m))$. This proves (iii) $\Rightarrow$ (iv).

Assume (iv). For a quotient sequence

$$0 \rightarrow G_T \rightarrow E_T \rightarrow F_T \rightarrow 0,$$

the boundedness of the family of the E_T and the finiteness of the Hilbert polynomials of the F_T imply the corresponding finiteness for the Hilbert polynomials of the kernels G_T . By Proposition 1.6, the G_T are again controlled by the same type of (b)-condition. Applying the already proved implication (ii) $\Rightarrow$ (iv) to the family of kernels gives (v).

It remains to see that (v) implies boundedness. After replacing T by a finite stratification, one may suppose that E and E' are quotients of sums of twists $\mathcal{O}_X(-m)$ and that the relevant kernels commute with passage to fibres. Take presentations

$$L_1 \longrightarrow E', \quad L_2 \longrightarrow E$$

with L_i finite direct sums of twists of $\mathcal{O}_X$. Choosing the twists sufficiently negative kills the relevant Ext^1 obstructions. Hence every morphism $E'_T \rightarrow E_T$ may be lifted, after replacing the presentation, to a morphism between fixed direct sums of twists, and the cokernel is unchanged. Thus the family is contained in the set of cokernels of morphisms

$$\mathcal{O}_{X_T}(-m')^{\oplus M'} \longrightarrow \mathcal{O}_{X_T}(-m)^{\oplus M}.$$

After one more stratification, the sheaves

$$R^q(f_T)_* \mathcal{H}om(E', E)$$

are flat and commute with base change. The affine space

$$\mathbb{V}((f_T)_* \mathcal{H}om(E', E)^*)$$

therefore carries a universal homomorphism $E'_R \rightarrow E_R$ on X_R , and its universal cokernel bounds the family. This proves (v) $\Rightarrow$ (i), and hence the theorem.

2. Several technical lemmas

2.1. Let k be a field and let X be proper over k . If $L_1, \dots, L_s$ are invertible sheaves on X , and if Y is a closed subscheme of dimension at most s , the intersection number

$$\langle c_1(L_1) \cdots c_1(L_s) \cdot Y \rangle$$

is the coefficient of the monomial $n_1 \cdots n_s$ in Snapper's polynomial

$$\chi(L_1^{\otimes n_1} \otimes \cdots \otimes L_s^{\otimes n_s} \otimes \mathcal{O}_Y).$$

2.2. From now on in this section, k is algebraically closed and X is integral, projective, of dimension r , and endowed with a very ample invertible sheaf $H = \mathcal{O}_X(1)$. For an invertible sheaf L on X , the integer

$$\langle c_1(L) c_1(H)^{r-1} \rangle$$

is called the degree of L , relative to H , and is denoted by $d(L)$.

Write

$$\chi(L(n)) = \sum_{i=0}^r a_i \binom{n+i}{i}, \quad \chi(\mathcal{O}_X(n)) = \sum_{i=0}^r b_i \binom{n+i}{i}.$$

Lemma 2.3. One has $a_r = b_r = d(H)$. If $r \geq 1$, the next coefficient is obtained from the corresponding intersection product with $c_1(L)$ and $c_1(H)^{r-2}$; equivalently, it is read off by comparing the Hilbert polynomials of L and of $L \otimes H^{\otimes m}$ at $m = 0, 1$.

Lemma 2.4. If $d(L) \leq 0$ and $H^0(L) \neq 0$, then $L \simeq \mathcal{O}_X$. Hence $d(L) = 0$ and $h^0(L) = 1$.

Indeed, a nonzero section identifies L with $\mathcal{O}_X(Y)$ for an effective divisor Y . If $Y > 0$, then

$$0 < \langle c_1(Y) c_1(H)^{r-1} \rangle = d(L),$$

contrary to the hypothesis.

Lemma 2.5. If $d(L) < 0$, then L is a $(0, \dots, 0, a_r)$ -sheaf, with $r - 1$ zeros.

For $r = 0$ this is immediate. If $r \geq 1$, then $h^0(L(-1)) = 0$ by Lemma 2.4. Choose an integral hyperplane section Y when $r \geq 2$, and use the exact sequence

$$0 \rightarrow L(-1) \rightarrow L \rightarrow L_Y \rightarrow 0.$$

For $r = 1$, Lemma 1.7 gives $h^0(L_Y(-1)) = a_1$; for larger r the assertion follows by induction on the dimension.

Lemma 2.7. Assume $r = 2$ and $d(L) \leq 0$. Then:

i) for every integer n ,

$$h^1(L(n)) \leq a_1(a_1 + 1)a_2 - a_0;$$

ii) $a_0 \leq a_1(a_1 + 1)$.

Let Y be an integral hyperplane section of X . By Lemma 1.7 and the preceding lemma, L_Y is a $(0, a_2)$ -sheaf, with Hilbert polynomial

$$\chi(L_Y(n)) = a_1 + a_2 \binom{n+1}{1}.$$

The bounded-regularity theorem gives that L_Y is $[-a_1]$ -regular. Applying cohomology to

$$0 \rightarrow L(n-1) \rightarrow L(n) \rightarrow L_Y(n) \rightarrow 0$$

shows that $h^1(L(n))$ is controlled by the single middle value $h^1(L(-a_1-2))$ together with the finite sum of the dimensions of the $H^0(L_Y(n))$. Since L is a $(0, 0, a_2)$ -sheaf by Lemma 2.5, Proposition 1.8 bounds these dimensions. Expanding the resulting inequality gives the two asserted estimates.

Lemma 2.8. Assume $r = 2$ and $d(L) < 0$. Then the preceding estimate admits an explicit uniform polynomial refinement in terms of the degree $\delta = d(L)$ and the first Hilbert coefficients of L . In particular, the numbers $h^1(L(n))$ are bounded above by a universal expression depending only on the Hilbert polynomial of L and on the fixed polarization H .

Indeed, write

$$\chi(L^{\otimes m}(n-1)) = a_0^{(m)} + a_1^{(m)}n + a_2 \binom{n+1}{2}.$$

By comparison of Snapper polynomials, $a_0^{(m)}$ and $a_1^{(m)}$ are expressed by the intersection numbers of $c_1(L)^m$ and $c_1(L)^m c_1(H)$, while $a_2 = d(H)$. Applying Lemma 2.7 to the twists $L^{\otimes m}(n)$ gives the announced polynomial bound.

Lemma 2.9. Assume $r \geq 2$, $d(L) < 0$, and X normal. Then:

i) $H^0(L(-n)) = 0$ for

$$n \geq 2 + h^1(\omega_X \otimes L^{-1});$$

ii) $h^1(L(n))$ is bounded, for all n , by a universal polynomial depending only on the last Hilbert coefficients of L and on the fixed polarization;

iii) there are universal polynomials β_i with rational coefficients, independent of k and of X , such that

$$|a_i| \leq \beta_i(a_0, \dots, a_r) \quad (0 \leq i \leq r).$$

For $r = 2$ this is Lemma 2.8, together with Serre duality. In higher dimension the proof proceeds

by induction on r . Choose a normal hyperplane section Y . The exact cohomology sequence

$$H^0(L_Y(-n)) \rightarrow H^1(L(-n-1)) \rightarrow H^1(L(-n)) \rightarrow H^1(L_Y(-n))$$

reduces the assertion to the estimates already known for Y . Since

$$\chi(L_Y(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i},$$

the induction hypothesis bounds the terms occurring on the right. The eventual vanishing for $n \gg 0$ then propagates the bound back to all n . Finally, once L_Y is known to be regular with a universally bounded regularity, Proposition 1.4 and the formula for the Hilbert polynomial express the remaining coefficients of L through bounded quantities.

Remark 2.10. For a fixed integer m , compare the two expansions

$$\sum_{i=0}^r a_i^{(m)} \binom{n+i}{i} = \sum_{i=0}^r a_i \binom{n+m+i}{i}.$$

An elementary induction on r , evaluating at $n = -1$, gives

$$a_i^{(m)} = \sum_{j=i}^r a_j \binom{m-1+j-i}{j-i}.$$

It follows in particular that, for fixed m , the coefficients after twisting are bounded by universal rational polynomials in the original coefficients.

Lemma 2.11. Let S be noetherian and let X be projective over S , with very ample sheaf $H = \mathcal{O}_X(1)$. Assume that the fibres of X/S are geometrically integral and of common dimension r . Let $\mathcal{L}$ be a family of classes of invertible sheaves L_s on the fibres of X/S . Then $\mathcal{L}$ is bounded if and only if, in the Hilbert polynomials

$$\chi(L_s(n)) = \sum_{i=0}^r a_i(s) \binom{n+i}{i},$$

all coefficients $a_i(s)$ remain bounded. If, moreover, the fibres are geometrically normal, it is enough that the three coefficients a_{r-1} , a_1 , and a_0 remain bounded.

Let $\delta = \sup_s d(L_s)$. If $\delta > 0$, replace $\mathcal{L}$ by the family of the $L_s(-\delta)$; the formulae comparing the Hilbert coefficients after a twist show that this changes only the boundedness of the coefficients in a controlled way. Thus one may reduce to the case $\delta < 0$. In that case Lemma 2.5 shows that the sheaves form a (b) -family once the Hilbert coefficients are bounded, and boundedness follows from Theorem 1.13. In the geometrically normal case the deeper estimates of this section show that boundedness of the three indicated coefficients already bounds all the other coefficients; Theorem 1.13 then applies again.

SGA 6, Exposé XIII.3 through Exposé XIV

Exposé XIII – Quotients of Picard schemes by subschemes of Pic, continuation

3. General finiteness theorems

Lemma 3.1. Let S be a scheme and let X be a quasi-compact and quasi-separated S -scheme. Let $\underline{\text{Pic}}_{X/S}$ be the Picard functor localized for the f.p.p.f. topology. Then, for every filtering projective system (Z_α) of quasi-compact and quasi-separated S -schemes whose transition morphisms are affine, the canonical map

$$\varinjlim_\alpha \underline{\text{Pic}}_{X/S}(Z_\alpha) \longrightarrow \underline{\text{Pic}}_{X/S}(\varprojlim_\alpha Z_\alpha) \quad (3.1.1)$$

is bijective. Here one uses the definition of $\underline{\text{Pic}}_{X/S}$ of SGA 3, IV, 6.3 and FGA, no. 232, with the f.p.p.f. topology in place of the fpqc topology.

Consequently, by EGA IV, 8.14.2, if the Picard functor is represented by an S -scheme $\underline{\text{Pic}}_{X/S}$, then this representing scheme is locally of finite presentation over S .

Indeed, an element λ_α of $\underline{\text{Pic}}_{X/S}(Z_\alpha)$ is represented by an invertible sheaf L'_α on $X_{Z'_\alpha}$, where $Z'_\alpha \rightarrow Z_\alpha$ is f.p.p.f. Put $Z = \varprojlim_\alpha Z_\alpha$. The image λ of λ_α in $\underline{\text{Pic}}_{X/S}(Z)$ is represented by the invertible sheaf L' on $X_{Z'}$ obtained by pullback from L'_α , where $Z' = Z'_\alpha \times_{Z_\alpha} Z$.

If λ is the neutral element of $\underline{\text{Pic}}_{X/S}(Z)$, then there exists an f.p.p.f. morphism $Z'' \rightarrow Z'$ such that the pullback of L' to $X_{Z''}$ is isomorphic to $\mathcal{O}_{X_{Z''}}$. By the standard limit theorems of EGA IV, there exist $\beta \geq \alpha$ and an f.p.p.f. morphism

$$Z''_\beta \longrightarrow Z'_\beta = Z'_\alpha \times_{Z_\alpha} Z_\beta$$

such that the pullback of L'_α to $X_{Z''_\beta}$ is $\mathcal{O}_{X_{Z''_\beta}}$. Thus the image of λ_α is already neutral in $\underline{\text{Pic}}_{X/S}(Z_\beta)$, and (3.1.1) is injective. Surjectivity is proved in the same way.

Proposition 3.2. Let S be noetherian and let X be proper over S . Assume that the Picard functor is represented by an S -scheme $\underline{\text{Pic}}_{X/S}$. Then:

- i) $\underline{\text{Pic}}_{X/S}$ is locally of finite type over S .
- ii) Subsets Λ of $\underline{\text{Pic}}_{X/S}$ correspond bijectively to families $\mathcal{L}$ of classes of invertible sheaves on the fibres of X/S .
- iii) Such a subset Λ is quasi-compact if and only if the corresponding family $\mathcal{L}$ is bounded.

Here (i) follows from (3.1), and (ii) from the fact that if $s \in S$ and K is an algebraically closed extension of $k(s)$, then the K -points of $\underline{\text{Pic}}_{X/S}$ are in bijection with isomorphism classes of invertible sheaves on X_K .

For (iii), suppose first that Λ is quasi-compact. Then Λ is contained in an open subscheme U of finite type, and the inclusion $U \rightarrow \underline{\text{Pic}}_{X/S}$ corresponds to an invertible sheaf L on some X_T , where $T \rightarrow U$ is a surjective flat morphism of finite type. Since $T \rightarrow S$ is of finite type, L bounds $\mathcal{L}$.

Conversely, suppose that $\mathcal{L}$ is bounded by a coherent sheaf L on X_T , with T of finite type over S . By 1.13 (i), we may assume L invertible, so that it defines a morphism

$$f : T \longrightarrow \underline{\text{Pic}}_{X/S}.$$

The image of f contains Λ , and is quasi-compact; it is therefore noetherian by (i). Hence Λ is quasi-compact.

Definition 3.3. Let X and Y be proper over a noetherian scheme S . A morphism of Picard functors

$$\underline{\text{Pic}}_{Y/S} \longrightarrow \underline{\text{Pic}}_{X/S}$$

is said to be *of finite type* if the inverse image of every bounded family of classes of invertible sheaves on the fibres of X/S is a bounded family of classes of sheaves on the fibres of Y/S . This extends the usual definition for morphisms between Picard schemes.

Lemma 3.4. Let X, Y, X', Y' be proper over a noetherian scheme S , and let $T \rightarrow S$ be surjective of finite type. Then:

i) A morphism $\Phi : \underline{\text{Pic}}_{Y/S} \rightarrow \underline{\text{Pic}}_{X/S}$ is of finite type if and only if the base changed morphism

$$\Phi_T : \underline{\text{Pic}}_{Y/T} \rightarrow \underline{\text{Pic}}_{X/T}$$

is of finite type.

ii) In a commutative square

$$\begin{array}{ccc} \underline{\text{Pic}}_{Y'/S} & \longrightarrow & \underline{\text{Pic}}_{Y/S} \\ \downarrow & & \downarrow \Phi \\ \underline{\text{Pic}}_{X'/S} & \longrightarrow & \underline{\text{Pic}}_{X/S} \end{array}$$

if the corresponding upper and lower auxiliary morphisms are of finite type, then so is Φ .

If Φ_T is of finite type, let $\mathcal{L}$ be a family on the fibres of X/S , bounded by a sheaf L on X_P . Then the base changed family $\mathcal{L}_T$ on X_T/T is bounded by the pullback of L to $X_{P \times_S T}$. If $\Phi_T(\mathcal{L}_T)$ is bounded by a sheaf M on some Y_Q , then M also bounds $\Phi(\mathcal{L})$. The converse and assertion (ii) are similar.

Theorem 3.5. Let S be noetherian, let X and Y be proper S -schemes, and let $f : X \rightarrow Y$ be a morphism. If f is surjective, then the induced morphism

$$f^* : \underline{\text{Pic}}_{Y/S} \longrightarrow \underline{\text{Pic}}_{X/S}$$

is of finite type.

It is enough, by (3.4) (i), to assume S reduced and irreducible. Then there is a non-empty open $U \subset S$ over which the Picard functors of Y/S and X/S are representable and the restriction of f^* is quasi-affine, by XII, 1.1; hence it is of finite type by 3.2 (i). Giving $T = S - U$ the induced reduced structure, noetherian induction permits one to assume that the corresponding morphism over T is of finite type. The assertion then follows from 3.4 (i), applied to $T \hookrightarrow S$.

Theorem 3.6. Let S be noetherian, let X be proper over S , and let $\underline{\text{Pic}}_{X/S}$ be the Picard functor. For every non-zero integer n , the n th power homomorphism

$$n : \underline{\text{Pic}}_{X/S} \longrightarrow \underline{\text{Pic}}_{X/S}$$

is a morphism of finite type.

Let $f : X' \rightarrow X$ be a Chow presentation: X' is projective over S and f is surjective. If the theorem is known for X' , then it follows for X from 3.5 and 3.4 (ii). Thus we may suppose X/S projective. There exists a non-empty open $U \subset S$, an integral scheme S' , and a finite surjective morphism $S' \rightarrow U$ such that every irreducible component X'_j of $X_{S'}$, with its induced reduced structure, is integral over S' . Giving $T = S - U$ the induced reduced structure, noetherian induction allows the theorem for X_T/T . If it is known for all X'_j/S' , it follows for $\coprod_j X'_j/S'$ from the following lemma, hence for $X_{S'}/S'$ by 3.5 and 3.4 (ii), and finally for X/S by 3.4 (i).

Lemma 3.7. Let $X = \coprod_i X_i$ be of finite type over a noetherian scheme S , and let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibres of X/S . Then $\mathcal{F}$ is bounded if and only if, for every i , the induced family $\mathcal{F}_i$ on the fibres of X_i/S is bounded.

Returning to the proof of 3.6, we have reduced to the case where X is projective and integral over S . Equip X/S with a very ample sheaf $\mathcal{O}_X(1)$. Let $\mathcal{L}$ be a bounded family of classes of invertible sheaves on the fibres of X/S , and let L be an invertible sheaf on X_T , with T of finite type over S , bounding $\mathcal{L}$. After cutting T into pieces, we may assume L flat over T , and after replacing T by its connected components, we may assume T connected. Then the polynomial in q and m

$$P(q, m) = \chi(L_t^{\otimes q}(m))$$

is independent of $t \in T$ by EGA III, 7.9.4.

Let $\mathcal{L}_n$ be the family of those invertible sheaves M_t for which the class of $M_t^{\otimes n}$ lies in $\mathcal{L}$. If

$$\chi(M_t^{\otimes q}(m)) = \sum_i a_i(q) \binom{m+i}{i},$$

then $a_i(q)$ is a polynomial in q of degree at most $r - i$. Since, as a polynomial in q, m , one has $\chi(M_t^{\otimes q}(m)) = P(q, m)$, the polynomials $a_i(q)$ do not depend on the chosen M_t . Consequently the family $\mathcal{L}_n$ has a single Hilbert polynomial $\chi(M_t(m))$, and is bounded by 2.11.

Theorem 3.8. Let S be noetherian. Let Y be projective over S , with relatively ample invertible sheaf $\mathcal{O}_Y(1)$, and let X be the zero-scheme of a section σ of $\mathcal{O}_Y(1)$. Let $f : X \hookrightarrow Y$ be the inclusion. Suppose that, for every $s \in S$ and every irreducible component Y_s^j of Y_s , all irreducible components of

$$X_s \cap Y_s^j = V(\sigma|_{Y_s^j})$$

have dimension at least 2; this is automatic if the irreducible components of the fibres of Y have dimension at least 3. Then

$$f^* : \underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is of finite type.

Moreover, if S is integral, there is a non-empty open $U \subset S$ such that the restricted Picard functors over U are representable and the induced morphism

$$\underline{\mathrm{Pic}}_{Y/S}|_U \longrightarrow \underline{\mathrm{Pic}}_{X/S}|_U$$

is quasi-affine, and hence of finite type.

For the first assertion, by noetherian induction as in the proof of 3.6 one may replace S by any non-empty open. One reduces to the case where S is integral and Y is integral over S ; when Y is replaced by Y' , X is replaced by $X \times_Y Y'$. One then reduces similarly to the case where Y is normal over S , using the following lemma in place of XII, 1.3.

Lemma 3.9. Let S be an integral noetherian scheme and let X be an S -scheme, integral over S . Then there exist a non-empty open $U \subset S$, an integral scheme S' finite and faithfully flat over U , an S' -scheme X' normal and integral over S' , and a finite surjective S' -morphism

$$X' \longrightarrow X_{S'}.$$

Let η be the generic point of S and let K be an algebraic closure of $k(\eta)$. The scheme $\mathrm{Spec}(K)$ is the limit of the filtering projective system of affine integral schemes S_α equipped with finite faithfully flat morphisms $S_\alpha \rightarrow U_\alpha$, where U_α ranges over non-empty open subsets of S . The normalization of X_η therefore descends to an S_α -scheme X' for some α , and the finite surjective morphism $X'_K \rightarrow X_\eta$ descends to a finite surjective S_α -morphism $X' \rightarrow X_{S_\alpha}$. Since X'_K is normal and integral, after replacing S_α by an open S' , the scheme $X'_{S'}$ is normal and integral over S' .

Remark 3.10. Under the notation and hypotheses of 3.9, suppose that there exists a resolving morphism $X'_K \rightarrow X_\eta$ in the sense of EGA IV, 7.9.1; this holds, for example, when $\dim(X_\eta) \leq 2$, by Abhyankar. Then the same argument gives a finite faithfully flat morphism $S' \rightarrow U$, a scheme X' smooth over S' , and a proper surjective birational morphism $X' \rightarrow X_{S'}$.

In the proof of the first part of 3.8, we have reduced to the case where S is integral and Y is normal and integral over S , hence Y is integral. If $\sigma = 0$, then $X = Y$ and there is nothing to prove. Otherwise, after shrinking S , we may assume that $\sigma_s \neq 0$ for all $s \in S$; then σ_s is $\mathcal{O}_{Y_s}$ -regular and

the fibre dimension r of Y_s is at least 3.

Let $\mathcal{L}$ be a bounded family of classes of invertible sheaves on the fibres of X/S . Let M_s be an invertible sheaf on a geometric fibre Y_s of Y/S , and write its Hilbert polynomial

$$\chi(M_s(n)) = \sum_i a_i \binom{n+i}{i}.$$

The Hilbert polynomial of $M_s|_{X_s}$ is

$$\sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

When the class of M_s ranges over $(f^*)^{-1}(\mathcal{L})$, the three coefficients $a_{r-1}, a_{r-2}, a_{r-3}$ remain bounded, by 1.13. If $H = \mathcal{O}_Y(m)$ is a very ample multiple of $\mathcal{O}_Y(1)$, then in the Hilbert polynomials, relative to H , of the M_s in $(f^*)^{-1}(\mathcal{L})$,

$$\chi(M_s^{\otimes q}(n)) = \sum_i a_i^{(m)} \binom{n+i}{i},$$

the coefficients $a_{r-1}^{(m)}, a_{r-2}^{(m)}, a_{r-3}^{(m)}$ remain bounded by 2.10. Thus $\mathcal{L}$ is bounded by 2.11, proving the first part of 3.8.

For the second part, the generic representability and separation of the Picard functors were established in XII, 1.2. By an argument parallel to that of XII, 2, using 3.9 and XII, 1.1, one reduces to the case where Y is normal over S . For the morphism $\text{Pic}_{Y/S} \rightarrow \text{Pic}_{X/S}$ to be quasi-affine, it is enough that it be separated and quasi-finite; this can be checked on geometric fibres and follows from the next lemma.

Lemma 3.11. Let k be algebraically closed. Let Y be a normal projective k -scheme with ample sheaf $\mathcal{O}_Y(1)$, and let X be the zero-scheme of a section σ of $\mathcal{O}_Y(1)$. If $\dim Y \geq 3$, then $\text{Pic}_Y \rightarrow \text{Pic}_X$ is of finite type and has finite kernel. If $\dim Y \geq 2$, the same conclusion holds for

$$\text{Pic}_Y^0 \longrightarrow \text{Pic}_X^0.$$

The finite kernel need not be reduced.

We may assume $\sigma \neq 0$; since Y is integral, σ is regular. The morphism considered is of finite type by the first part of 3.8, respectively by 4.7 (ii). Its kernel N is therefore an algebraic group. It is contained in Pic_Y^0 , and the same kernel occurs in both cases. Since Pic_Y^0 is proper, Y being normal, and since a subgroup is necessarily closed, it is enough, in order to prove N finite, to show that it contains no subgroup of type $\mu_\ell = \mathbb{Z}/\ell\mathbb{Z}$ with $(\ell, \text{char}(k)) = 1$.

Consider Kummer theory, namely the exact sequence of étale sheaves on Y and on X

$$0 \longrightarrow \mu_\ell \longrightarrow \mathbb{G}_m \xrightarrow{\ell} \mathbb{G}_m \longrightarrow 0,$$

and the corresponding commutative diagram

$$\begin{array}{ccccc} H^1(Y, \mu_\ell) & \longrightarrow & H^1(Y, \mathbb{G}_m) & \xrightarrow{\ell} & H^1(Y, \mathbb{G}_m) \\ \downarrow & & \downarrow & & \downarrow \\ H^1(X, \mu_\ell) & \longrightarrow & H^1(X, \mathbb{G}_m) & \xrightarrow{\ell} & H^1(X, \mathbb{G}_m). \end{array}$$

By Hilbert's Theorem 90, $H^1(Y, \mathbb{G}_m) = \text{Pic}_Y(k)$ and similarly for X . Since X is proper and integral, $H^0(X, \mathbb{G}_m) = k^*$, and the map $H^1(Y, \mu_\ell) \rightarrow H^1(Y, \mathbb{G}_m)$ is injective.

It remains to see that

$$H^1(Y, \mathbb{Z}/\ell\mathbb{Z}) \longrightarrow H^1(X, \mathbb{Z}/\ell\mathbb{Z})$$

is injective. This follows from the surjectivity of $\pi_1(X) \rightarrow \pi_1(Y)$, or equivalently, by SGA 1, V, 6.9, from the fact that every connected étale cover $\tilde{Y}$ of Y with group $\mathbb{Z}/\ell\mathbb{Z}$ has connected restriction

$$\tilde{X} = X \times_Y \tilde{Y},$$

a well-known consequence of Bertini's theorem and Zariski's connectedness theorem.

Remark 3.12. With the hypotheses of 3.11:

- i) The kernel N is a finite unipotent group. Indeed, it contains no subgroup of multiplicative type $\mathbb{G}_m$ for any n ; the proof is the same, using in place of Kummer theory the following fact due to Grothendieck: if X is proper over a field k and $H^0(X, \mathcal{O}_X) = k$, then there is an isomorphism of k -group schemes

$$\text{Hom}_k(\mathbb{Z}/n\mathbb{Z}, \text{Pic}_{X/k}^0) \simeq H^1(X, \mathbb{Z}/n\mathbb{Z}).$$

Thus, if k is algebraically closed,

$$\text{Hom}_k(\mathbb{Z}/n\mathbb{Z}, \text{Pic}_{Y/k}^0) \simeq H^1(Y, \mathbb{Z}/n\mathbb{Z}).$$

- ii) If k has characteristic zero, then $N = 0$, since every finite unipotent group is then zero. In this form one recovers, essentially, Kodaira's vanishing theorem in Mumford's formulation.

Theorem 3.13. Let S be noetherian, and let X be projective over S , with relatively ample invertible sheaf $H = \mathcal{O}_X(1)$. Assume that the geometric fibres of X/S are integral and of dimension r . For a family $\mathcal{L}$ of classes of invertible sheaves on the fibres of X/S , the following conditions are equivalent:

- i) $\mathcal{L}$ is bounded.

ii) In the Hilbert polynomials

$$\chi(L_s(n)) = \sum_{i=0}^r a_i(s) \binom{n+i}{i}, \quad L_s \in \mathcal{L},$$

the coefficient a_{r-1} remains bounded and the coefficient a_{r-2} remains bounded below.

iii) As L_s runs through $\mathcal{L}$, the degree

$$d(L_s) = \langle c_1(L_s) c_1(H_s)^{r-1} \rangle$$

remains bounded, and the intersection number

$$\langle c_1(L_s)^2 c_1(H_s)^{r-2} \rangle$$

remains bounded below.

iv) There is an integer N such that

$$\mathcal{L}^{\otimes N} < \mathcal{L} < \mathcal{L}^{\otimes -N}$$

in the order relation defined by ampleness; that is, for every $L_s \in \mathcal{L}$, the two sheaves

$$L_s^{\otimes N} \otimes H_s^{-1}, \quad L_s^{\otimes -N} \otimes H_s$$

are ample.

If $\mathcal{L}$ is bounded by an invertible sheaf over X_T , then EGA II, 4.6.12 gives (iv). If that sheaf is flat over T , the polynomial $\chi(L_t^{\otimes m}(n))$ in m, n depends only on the connected component of T , by EGA III, 7.9.4. This gives (ii), and, using 2.1, (iii).

The family consisting of the classes of H_K is bounded, as is the family consisting of $\mathcal{O}_{X_K}$; hence these satisfy (iii) and (ii), respectively. If (iv) holds, then

$$\langle Nc_1(H_K) + c_1(L_K) \rangle c_1(H_K)^{r-2} > 0,$$

and so $d(L_K) > -N d(H_K)$; similarly $d(L_K) < N d(H_K)$. Also

$$\langle (Nc_1(H_K) + c_1(L_K))^2 c_1(H_K)^{r-2} \rangle > 0,$$

which gives

$$\langle c_1(L_K)^2 c_1(H_K)^{r-2} \rangle > -2N d(L_K) - N^2 d(H_K).$$

Thus (iv) implies (iii).

It remains to prove that (ii), respectively (iii), implies boundedness. By 2.10 and 2.3 one may replace H by a multiple; assume H very ample. After cutting S into pieces, one may suppose that there is a linear section Y of X , integral over S , whose fibres have dimension 2. By 3.8, one may

replace $X, \mathcal{L}$ by $Y, \mathcal{L}|_Y$, using 1.7, respectively the projection formula, to see that (ii), respectively (iii), is preserved. Thus one reduces to $r = 2$. If

$$\delta = \sup\{d(L_K)\} > 0,$$

replace $\mathcal{L}$ by the family of the classes of $L_K(-\delta)$, using 2.3 and 2.10; hence assume $\delta \leq 0$. If the Hilbert polynomials

$$\sum_{i=0}^2 a_i \binom{n+i}{i}$$

have a_1 bounded and a_0 bounded below, then by 2.3 and 2.7 all three coefficients are bounded; hence $\mathcal{L}$ is bounded by 2.11. Thus (ii) implies (i). Finally (iii) implies (ii) by 2.3 and 2.8.

4. Finiteness theorems for $\text{Pic}_{X/S}^\tau$

4.1. Let S be a scheme and let G be a functor in commutative groups on S . We denote by G^0 , respectively G^τ , the subgroup functor whose T -points $f : T \rightarrow G$ are those T -points of G such that for every algebraically closed field K and every K -point t of T , there exist a connected K -scheme Z , two K -points z, z_0 of Z , and a morphism $g : Z \rightarrow G$ with

$$g(z) = f(t), \quad g(z_0) = 0$$

in $G(K)$; respectively, there exists a non-zero integer n such that $nf(t) \in G^0(K)$.

Formation of G^0 and G^τ plainly commutes with arbitrary base change. A morphism $G \rightarrow G'$ of group functors induces $G^0 \rightarrow G'^0$, respectively $G^\tau \rightarrow G'^\tau$. If S is a field and G is represented by a locally finite type group scheme, then G^0 is represented by the connected component of the identity, an open and closed subgroup scheme of finite type, geometrically irreducible.

Lemma 4.2. Let $\Phi : G \rightarrow G'$ be a morphism of commutative group functors on a scheme S . Suppose that for every algebraically closed field K and every morphism $\text{Spec}(K) \rightarrow S$, the functors G_K and G'_K are represented by locally finite type group schemes, and that Φ_K is representable by a morphism of finite type, equivalently quasi-compact. Then

$$\Phi(G^\tau) \subset G'^\tau.$$

The inclusion $\Phi(G^0) \subset G'^0$ has already been noted. Moreover, for $g \in G(K)$, membership of g in $G^\tau(K)$ is equivalent to quasi-compactness of the sequence $(ng)_{n \in \mathbb{N}}$. Since Φ is quasi-compact, the same property for the sequence $\Phi(ng) = n\Phi(g)$ follows; this proves the assertion.

Lemma 4.3. Let X and Y be proper over a scheme S , and let $f : X \rightarrow Y$ be either a surjective morphism or the inclusion of the zero-scheme of a section of an ample sheaf $\mathcal{O}_Y(1)$. Then

$$(f^*)^{-1}(\text{Pic}_{X/S}^\tau) = \text{Pic}_{Y/S}^\tau.$$

This is a formal consequence of 4.2, XII, 1.2, 3.1, and 3.5, respectively 3.8.

4.4. Let k be algebraically closed, let X be proper over k , and let L be an invertible sheaf on X . We say that L is *algebraically equivalent to zero*, respectively *τ -equivalent to zero*, if there exist a connected finite type k -scheme T , an invertible sheaf M on $X_T = X \times_k T$, and two k -points t, t_0 of T such that

$$L \simeq M_t, \quad \mathcal{O}_X \simeq M_{t_0},$$

respectively if some non-zero tensor power $L^{\otimes n}$ is algebraically equivalent to zero. Equivalently, the k -point λ of $\text{Pic}_{X/k}$ corresponding to L lies in $\text{Pic}_{X/k}^0(k)$, respectively in $\text{Pic}_{X/k}^\tau(k)$.

We also recall that L is *numerically equivalent to zero* if, for every closed integral curve Y in X , the intersection number

$$\langle c_1(L) \cdot Y \rangle$$

is zero.

Proposition 4.5 (Functorial behaviour). Let $f : X' \rightarrow X$ be a morphism between proper k -schemes, and let L be an invertible sheaf on X . Then:

- i) If L is τ -equivalent to zero, respectively numerically equivalent to zero, then f^*L is so.
- ii) Conversely, if f is surjective and f^*L is τ -equivalent to zero, respectively numerically equivalent to zero, then L is so.

For τ -equivalence, (i) and (ii) follow immediately from 4.1 and 4.3. For numerical equivalence, let Y' be a closed integral curve in X' , and let $Y = f(Y')$, with the induced reduced structure. If $\dim Y = 1$, the projection formula gives

$$\langle c_1(f^*L) \cdot Y' \rangle = [k(Y') : k(Y)] \langle c_1(L) \cdot Y \rangle. \quad (4.5.1)$$

If $\dim Y = 0$, then $(f^*L)|_{Y'} \simeq \mathcal{O}_{Y'}$. Hence numerical triviality of L implies that of f^*L .

Conversely, when f is surjective and Y is a closed integral curve in X , choose a point y' in the fibre over the generic point y of Y , algebraic over $k(y)$, and let Y' be the closure of y' . Then $f(Y') = Y$, and (4.5.1) shows that

$$\langle c_1(f^*L) \cdot Y' \rangle = 0$$

implies $\langle c_1(L) \cdot Y \rangle = 0$.

Theorem 4.6 (Numerical characterization of $\text{Pic}_{X/k}^\tau$). Let k be algebraically closed, let X be proper over k , and let L be an invertible sheaf on X . The following conditions are equivalent:

- i) L is τ -equivalent to zero.
- ii) For every coherent sheaf F on X , one has $\chi(F \otimes L) = \chi(F)$.

iii) L is numerically equivalent to zero.

If X is projective over k , with ample invertible sheaf $H = \mathcal{O}_X(1)$, these are also equivalent to:

iv) For every integer m , positive or negative, $L^{\otimes m} \otimes H$ is ample.

v) If X is irreducible, then for all integers m, n ,

$$\chi(L^{\otimes m}(n)) = \chi(\mathcal{O}_X(n)).$$

vi) If X is irreducible of dimension $r \geq 2$, then

$$\langle c_1(L)H^{r-1} \rangle = 0, \quad \langle c_1(L)^2H^{r-2} \rangle = 0.$$

To prove (i) $\Rightarrow$ (ii), dévissage reduces, using 4.5 (i), to the case where X is integral and $F = \mathcal{O}_X$. It is enough to show that $\chi(L^{\otimes m})$ is constant. Replacing L by a non-zero tensor power, one may suppose L algebraically equivalent to zero. Then L is deformed to $\mathcal{O}_X$ by an algebraic family M on $X \times T$, with T connected of finite type; the polynomial $\chi(M_t)$ is independent of t by EGA III, 7.9.5.

If Y is a closed integral curve in X , apply (ii) to $F = \mathcal{O}_Y$. Then

$$\chi(\mathcal{O}_Y) = \chi(L|_Y),$$

and Riemann's theorem on curves gives

$$\chi(L|_Y) = \langle c_1(L) \cdot Y \rangle + \chi(\mathcal{O}_Y).$$

Thus (ii) implies (iii).

Let $f : X' \rightarrow X$ be a Chow presentation, with X' projective and f surjective. If (iii) $\Rightarrow$ (i) is known for X' , it follows for X by 4.5. We may therefore suppose X projective with ample sheaf H . Passing to the disjoint union of the reduced irreducible components reduces, by 4.5 and EGA III, 2.6.2, to the case where X is integral.

If X is irreducible, then (ii) plainly implies (v), (iii) implies (vi), and by 2.1, (v) implies (vi); moreover (vi) is unchanged by replacing X by X_{red} . It remains to prove that (vi) implies (i) and (iv), and that (iv) implies (i), for X integral. Under either (vi) or (iv), the family of classes

$$\{L^{\otimes m}\}_{m \in \mathbb{N}}$$

satisfies 3.13 (iii), respectively 3.13 (iv), and is therefore bounded. Hence there is an invertible sheaf on some X_T , with T of finite type over k , which bounds this family. Choose two distinct integers p, q such that $L^{\otimes p}$ and $L^{\otimes q}$ correspond to the same connected component of T ; then $L^{\otimes(p-q)}$ is algebraically equivalent to zero. This gives (i). Similarly, (vi) implies that for some N , and every m , the sheaf $L^{\otimes m} \otimes H^{\otimes N}$ is ample; hence $(L^{\otimes m} \otimes H)^{\otimes N}$ is ample for every m , proving (iv).

Theorem 4.7. Let $f : X \rightarrow S$ be proper and flat, with S noetherian. Then:

- i) The subfunctor $\text{Pic}_{X/S}^\tau$ of $\text{Pic}_{X/S}$ consisting of classes of invertible sheaves numerically equivalent to zero on the fibres is represented by an open subscheme of $\text{Pic}_{X/S}$.
- ii) If, in addition, f is Cohen–Macaulay and its geometric fibres are integral, then the subfunctor $\text{Pic}_{X/S}^{\text{num}}$ of $\text{Pic}_{X/S}$ consisting of classes of invertible sheaves algebraically equivalent to zero on the fibres is also represented by a closed subscheme.
- iii) $\text{Pic}_{X/S}^\tau$ is of finite type over S ; equivalently, the family of classes of invertible sheaves τ -equivalent to zero on the fibres of X/S is bounded.

For (i), respectively (ii), it is enough to treat a morphism $T \rightarrow \text{Pic}_{X/S}$, where T is an arbitrary S -scheme, and show that the inverse image of $\text{Pic}_{X/S}^\tau$, respectively $\text{Pic}_{X/S}^{\text{num}}$, is represented by an open, respectively by an open and closed, subscheme of T . Taking $T = \text{Spec}(B)$ over an affine open $\text{Spec}(A) \subset S$, write T as the limit of the filtering projective system of affine S -schemes

$$T_i = \text{Spec}(B_i),$$

where B_i runs through the finite type A -subalgebras of B . By 3.1, the morphism in question descends to some $T_i \rightarrow \text{Pic}_{X/S}$. Thus one may assume T of finite type over S .

The morphism $T \rightarrow \text{Pic}_{X/S}$ corresponds to an f.p.p.f. morphism $R \rightarrow T$ and an invertible sheaf L on X_R . By f.p.p.f. descent, T may be replaced by R . After the base change $R \rightarrow S$, and using compatibility of formation of $\text{Pic}_{X/S}^\tau$ with base change, (i) and (ii) reduce to the case of a section $S \rightarrow \text{Pic}_{X/S}$ defined by an invertible sheaf L on X .

Let S_0 be the set of $s \in S$ such that L_s is τ -equivalent to zero. One must prove that S_0 is open, and closed under the supplementary assumptions of (ii). Since S is noetherian, this amounts to:

- (a) every generalization of a point of S_0 remains in S_0 ;
- (b) every specialization of a point of S_0 , sufficiently close to it, remains in S_0 ;
- (c) under the hypotheses of (ii), every specialization of a point of S_0 remains in S_0 .

Let $f : X' \rightarrow X$ be a Chow presentation: X' is projective over S and f is surjective. If (a), (b), (c), and (iii) are known for X' , then they follow for X by 4.3 and 3.5. Thus one may suppose X projective over S .

Let H be ample for X/S . If, for some $s \in S$, the sheaf $(L \otimes H)_s$ is ample, then $(L \otimes H)_t$ is ample for every generalization t of s . Using the equivalence in 4.6 between (i) and (iv), this proves (a).

For (b), (c), and 4.7 (iii), one may assume S integral, with generic point s in (b) and (c). There exists an integral scheme S' and an f.p.p.f. morphism $S' \rightarrow S$ such that every irreducible component X_j of $X_{S'}$, with the induced reduced structure, is integral over S' . It is enough to prove (b) and (iii) after this base change: for (b) by descent, and for (iii) by noetherian induction and 3.4 (i).

Considering the canonical surjective morphism

$$\coprod X'_j \longrightarrow X_{S'},$$

one reduces, by 4.3, 3.5, and 3.7, to the case where S is integral with generic point s , and X is projective and integral over S . Let $\mathcal{O}_X(1)$ be very ample for X/S . Then for all m, n , the polynomials

$$\chi(L^m(n)), \quad \chi(\mathcal{O}_X(n))$$

are independent of $t \in S$. Thus (c) follows from the equivalence 4.6 (i) $\Leftrightarrow$ (v), and the same condition shows that the family of classes of invertible sheaves τ -equivalent to zero on the fibres of X/S has a single Hilbert polynomial. Hence (iii) follows from 2.11.

Remark 4.8. Let X be proper and integral over a noetherian scheme S . Without assuming X/S projective, it is not known whether $\mathrm{Pic}_{X/S}^\tau \hookrightarrow \mathrm{Pic}_{X/S}$ is still a closed immersion, except when X is normal over S and $\mathrm{Pic}_{X/S}$ is representable.

5. Finiteness theorems of Néron–Severi type

Theorem 5.1. Let X be proper over a noetherian scheme S . Then the Néron–Severi groups

$$\mathrm{NS}(X_{\overline{K}}) = \mathrm{Pic}_{X_{\overline{K}}/\overline{K}}(\overline{K}) / \mathrm{Pic}_{X_{\overline{K}}/\overline{K}}^\tau(\overline{K})$$

of the geometric fibres $X_{\overline{K}}$ of X/S are finitely generated, and their ranks and the orders of their torsion subgroups remain bounded.

By noetherian induction, it is enough to prove the theorem after replacing S by an arbitrary non-empty open of S_{red} . Thus one may suppose $\mathrm{Pic}_{X/S}^\tau$ represented by an S -scheme of finite type. Then the order of $\mathrm{Tors}(\mathrm{NS}(X_{\overline{K}}))$ is the number of connected components of $(\mathrm{Pic}_{X/S}^\tau)_{\overline{K}}$, and this number is constant over a non-empty open of S .

It remains to bound the rank of $\mathrm{NS}(X_{\overline{K}})$. As in the proof of 4.7 (iii), one reduces to S integral and X projective and integral over S . Let r be the common dimension of the fibres. For $r = 0$, respectively $r = 1$, the rank is plainly 0, respectively 1. If $r \geq 2$, one reduces to the case $r = 2$ and X smooth over S : after shrinking S , choose a linear section Y of X which is integral over S and has fibres of dimension 2, and replace X by Y using 4.3. Then, after a flat cover of a non-empty open of S , 4.3 and 3.10, using resolution of singularities for surfaces, allow one to replace X by a smooth S -scheme.

Now the theorem follows from 4.6 (i) $\Rightarrow$ (iii), the finiteness and specialization theorem for étale cohomology groups, and the following proposition.

Proposition 5.2. Let k be algebraically closed and let X be proper and smooth over k . Then the group $C^p(X)$ of algebraic cycles of codimension p , modulo numerical equivalence, is free of rank at most the Betti number b^{2p} .

Ignoring the action of roots of unity, étale cohomology

$$H_{\text{et}}^i(X, \mathbb{Q}_\ell) = \left[\varprojlim H_{\text{et}}^i(X, \mathbb{Z}/\ell^n \mathbb{Z}) \right] \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \quad (\ell \neq \text{char}(k))$$

is a Weil cohomology: it satisfies Poincaré duality and the Künneth formula, and receives a non-zero functorial homomorphism from the ring of algebraic cycles modulo rational equivalence,

$$C^p(X) \longrightarrow H_{\text{et}}^{2p}(X, \mathbb{Q}_\ell(p)).$$

Let $a_1, \dots, a_m \in C^{r-p}(X)$, where $r = \dim X$, be classes whose images form a basis of the $\mathbb{Q}_\ell$ -vector subspace of

$$H_{\text{et}}^{2(r-p)}(X, \mathbb{Q}_\ell(r-p))$$

spanned by algebraic cycles. Consider

$$\alpha : C^p(X) \longrightarrow \mathbb{Q}_\ell^m, \quad \alpha(b) = (\langle b \cdot a_1 \rangle, \dots, \langle b \cdot a_m \rangle).$$

Then α is injective on numerical classes, and the bound on the rank follows.

Remark 5.3. If k is algebraically closed and X is a proper smooth surface over k , the second Betti number can be written

$$b^2(X) = c_2(X) - 2 + 2b^1(X),$$

where $b^1(X) = 2 \dim(\text{Pic}_{X/k}^0)$ and $c_2(X)$ is the second Chern class, equivalently the Euler–Poincaré characteristic or the self-intersection number of the diagonal in $X \times X$. Thus $b^2(X)$ has an interpretation independent of cohomology. In characteristic > 0 , the inequality

$$\text{rank } C^1(X) \leq b^2(X)$$

in this form is due to Igusa, before étale cohomology, using deep results on the structure of the tame fundamental group of an algebraic curve minus finitely many points.

Theorem 5.4. Let k be algebraically closed and let X be proper over k . Then there exist finitely many closed integral curves $Y_1, \dots, Y_p$ in X such that a subset A of the scheme $\text{Pic}_{X/k}^\tau$ is quasi-compact if and only if the intersection numbers

$$\langle L, Y_i \rangle, \quad L \in A,$$

remain bounded. Moreover p may be taken to be the rank of the Néron–Severi group

$$\text{Pic}_{X/k}(k) / \text{Pic}_{X/k}^0(k).$$

Since $\text{Pic}_{X/k}^0$ is an open quasi-compact subgroup of $\text{Pic}_{X/k}^\tau$, a subset A is quasi-compact if and

only if the corresponding set of points in

$$\left(\text{Pic}_{X/k}^\tau(k) / \text{Pic}_{X/k}^0(k) \right) \otimes \mathbb{Q}$$

is finite. But τ -equivalence is the same as numerical equivalence, by 4.6, so one may choose curves in X whose numerical classes form a basis of the dual vector space.

6. Appendix: (b) -sheaves on $P = \mathbb{P}_k^r$

Let k be algebraically closed and let $P = \mathbb{P}_k^r$. Let $\mathcal{O}_P(1)$ be the canonical invertible sheaf on P , and set

$$S = k[x_0, \dots, x_r].$$

Recall that a coherent sheaf F on P is called (b) -regular, or simply b -regular, if

$$H^i(P, F(b-i)) = 0 \quad (i > 0).$$

Theorem 6.2. Let F be a non-zero coherent sheaf on P , b -regular. Let

$$s \in H^0(P, F(b))$$

be a section whose zero subscheme $Z(s)$ has codimension > 0 , i.e. contains no irreducible component of $\text{Supp}(F)$. Then

$$F' = \text{Ker}(F \xrightarrow{s} F(b))$$

is $(b+1)$ -regular.

The proof uses the exact sequence

$$0 \longrightarrow F' \longrightarrow F \xrightarrow{s} F(b) \longrightarrow G \longrightarrow 0,$$

where G has finite length, together with the vanishing implied by b -regularity. If $m = P_F(b)$, then

$$h^0(G) = m, \quad h^i(G) = 0 \quad (i > 0).$$

Since $s \neq 0$, the quotient G is non-zero; in particular F' is not contained in $F(-1)$, i.e. the section is not divisible by the equation of a hyperplane. The regularity assertion follows by the standard Castelnuovo–Mumford argument.

6.3. The converse of 6.2 is also useful. Suppose there is an exact sequence

$$0 \longrightarrow F' \longrightarrow F \xrightarrow{s} F(b) \longrightarrow G \longrightarrow 0$$

with F' $(b+1)$ -regular and G of finite length. Then F is b -regular. In the case $q = 1$, the support of the relevant module N_* has dimension 0, and one has

$$h^1(N_*(b-1)) = 0, \quad h^0(N_*(b-1)) \leq h^0(N_*(b)).$$

Moreover, if $G = 0$ then

$$h^0(F'(b)) = h^0(F(b)) - m.$$

Corollary 6.8 (Grothendieck). Let X be projective over a noetherian scheme S , let $\mathcal{O}_X(1)$ be very ample for X/S , and let $\mathcal{F}$ be a bounded family of classes of coherent sheaves on the fibres of X/S . Then there exists an integer m such that, for every class $\xi \in \mathcal{F}$ and every coherent sheaf F representing ξ , the sheaf F is m -regular on the corresponding fibre.

After replacing the base field by the algebraic closure of a purely transcendental extension, one may assume the ground field infinite. Then the assertion follows from the previous regularity estimates, from 6.10 and 1.8 (i), and from 6.2 applied to $F(n)$ for $n \geq m$.

Corollary 6.14 (L. Hermann). For every $N \geq 0$ there is a polynomial $R_N(x)$ such that for every algebraically closed field k , every projective k -scheme X with very ample sheaf $\mathcal{O}_X(1)$, and every coherent sheaf F on X satisfying $\chi(F(n)) = N$ for all n , one has

$$h^0(F) \leq R_N(\deg X).$$

7. Appendix: The Hodge index theorem

We recover and complement here some earlier results by a direct argument.

Theorem 7.1. Let X be an irreducible surface, proper over an algebraically closed field. Suppose that there exists an invertible sheaf H on X such that

$$\langle c_1(H)^2 \rangle > 0$$

(for example H ample). Then an invertible sheaf L on X satisfying

$$\langle c_1(L), c_1(H) \rangle = 0, \quad \langle c_1(L)^2 \rangle \geq 0$$

is numerically equivalent to zero. Consequently, by 4.6, the intersection form on

$$\mathrm{NS}(X) \otimes \mathbb{Q} = [\mathrm{Pic}(X)/\mathrm{Pic}^0(X)] \otimes \mathbb{Q}$$

is non-degenerate of type $(1, \rho - 1)$.

First suppose that X is reduced, hence integral, that H is ample, that

$$\langle c_1(L), c_1(H) \rangle = 0, \quad \langle c_1(L)^2 \rangle > 0.$$

Arguing by contradiction, choose $m > 0$ such that $H_1 = L \otimes H^{\otimes m}$ is ample, and choose n so that, with

$$N = L^{\otimes n} \otimes H^{-1},$$

one has

$$\langle c_1(N), c_1(H_1) \rangle = n \langle c_1(L)^2 \rangle - m \langle c_1(H)^2 \rangle > 0.$$

At the same time

$$\langle c_1(N), c_1(H) \rangle = -\langle c_1(H)^2 \rangle < 0, \quad \langle c_1(N)^2 \rangle = n^2 \langle c_1(L)^2 \rangle + \langle c_1(H)^2 \rangle > 0,$$

contradicting the next lemma.

Lemma 7.1.2. Let X be an integral surface, projective over an algebraically closed field. For an invertible sheaf N on X with $\langle c_1(N)^2 \rangle > 0$, the following conditions are equivalent:

- i) For every ample sheaf H , one has $\langle c_1(N), c_1(H) \rangle > 0$.
- i') There exists an ample sheaf H with $\langle c_1(N), c_1(H) \rangle > 0$.
- ii) There exists an integer $n > 0$ such that $H^0(X, N^{\otimes n}) \neq 0$.

If (ii) holds, there is an effective divisor D such that $N \simeq \mathcal{O}_X(D)$, and $\langle c_1(N)^2 \rangle > 0$ forces $D > 0$, giving (i). Conversely, to see that (i') implies (ii), assume $H = \mathcal{O}_X(1)$ very ample. There is a closed integral locally principal curve Y with $H = \mathcal{O}_X(Y)$. Riemann's theorem gives vanishing of the relevant cohomology for $N^{\otimes n}$ for n large. The exact sequence

$$H^0(N^{\otimes n}(m)) \rightarrow H^0(N^{\otimes n}(m)|_Y) \rightarrow H^1(N^{\otimes n}(m-1)) \rightarrow H^1(N^{\otimes n}(m)) \rightarrow 0$$

then implies $H^1(N^{\otimes n}) = 0$ for all sufficiently large n . Hence

$$h^0(N^{\otimes n}) = \chi(N^{\otimes n}) = \frac{1}{2} \langle c_1(N)^2 \rangle n^2 + \dots > 0$$

for $n \gg 0$.

Now suppose in 7.1 that there is an invertible sheaf M on X with

$$\langle c_1(L), c_1(M) \rangle \neq 0.$$

Arguing by contradiction, replace X by the underlying reduced scheme of a Chow presentation; this is allowed by the projection formula. Let ξ be ample on X . Choose $m > 0$ such that $H_1 = \xi^{\otimes m}$ is ample and

$$\langle c_1(L), c_1(H_1) \rangle \neq 0.$$

Then choose integers p, q with $q \neq 0$ and set

$$L_1 = L^{\otimes p} \otimes H_1^{\otimes q}$$

so that

$$\langle c_1(L_1), c_1(H) \rangle > 0, \quad \langle c_1(L_1)^2 \rangle > 0.$$

This reduces to the preceding case.

Finally, in the general case, assume again by contradiction that L is not numerically equivalent to zero. Replace X by the normalization $\tilde{X}$ of X_{red} ; then $\tilde{X}$ has only finitely many singular points, and the replacement is justified by the projection formula and 4.5 (ii). There is a closed integral curve Y on $\tilde{X}$ with

$$\langle c_1(L), Y \rangle \neq 0.$$

The curve Y is locally principal outside a finite set Z . Let $f : X' \rightarrow \tilde{X}$ be the blow-up of the ideal I defining the defect. The ideal

$$M = I\mathcal{O}_{X'}$$

is invertible and defines a subscheme of X' which is the union of two subschemes Y' and Y'' with $f(Y') = Y$ and $f(Y'') \subset Z$. By the projection formula,

$$\langle c_1(f^*L), c_1(M) \rangle = -\langle c_1(f^*L), Y' \rangle - \langle c_1(f^*L), Y'' \rangle = -\langle c_1(L), Y \rangle \neq 0.$$

Replacing X, L, H by X', f^*L, f^*H brings us back to the preceding case and completes the proof of 7.1.

Remark 7.2. It is not known whether every irreducible proper surface over an algebraically closed field which carries an invertible sheaf H with $\langle c_1(H)^2 \rangle > 0$ must be projective. Nor is it known whether, for every irreducible proper surface, the intersection form on $\text{NS}(X) \otimes \mathbb{Q}$ is non-degenerate. It follows from 7.1 that, when it is degenerate, it must be negative, not indefinite.

Corollary 7.3. Under the hypotheses of 7.1, let M be an invertible sheaf on X not numerically equivalent to zero, and set

$$a = \langle c_1(M), c_1(H) \rangle, \quad h = \langle c_1(H)^2 \rangle.$$

Then

$$\langle c_1(M)^2 \rangle < a^2/h.$$

Indeed, take $L = M^{\otimes h} \otimes H^{-a}$. Since $\langle c_1(L), c_1(H) \rangle = 0$, Theorem 7.1 gives $\langle c_1(L)^2 \rangle < 0$.

Corollary 7.4. Let X be an irreducible projective scheme of dimension $r \geq 2$ over an algebraically closed field, and let H be ample on X . Then:

- i) An invertible sheaf L on X is numerically equivalent to zero if and only if

$$\langle c_1(L)c_1(H)^{r-1} \rangle = 0, \quad \langle c_1(L)^2c_1(H)^{r-2} \rangle \geq 0.$$

ii) For every invertible sheaf M on X not numerically equivalent to zero,

$$\langle c_1(M)^2 c_1(H)^{r-2} \rangle \langle c_1(H)^r \rangle < \langle c_1(M) c_1(H)^{r-1} \rangle^2.$$

Assertion (ii) follows immediately from (i) by the argument of 7.3. To prove (i), let Y be a closed integral curve in X ; it is enough to show $\langle c_1(L), Y \rangle = 0$. Embed X in projective space by $H^{\otimes m}$, $m \gg 0$. Let $I = \bigoplus I_\nu$ be the homogeneous ideal of Y , and choose ν_0 so that I_{ν_0} generates I for $\nu \geq \nu_0$. Hypersurfaces Z of degree ν containing Y have no common point outside Y . By induction on r and Bertini's theorem, choose $r - 2$ such hypersurfaces $Z_1, \dots, Z_{r-2}$ so that

$$X' = Z_1 \cap \dots \cap Z_{r-2} \cap X$$

is irreducible. Since $\langle c_1(L), Y \rangle = \langle c_1(L|_{X'}), Y \rangle$, it remains to prove that $L|_{X'}$ is numerically equivalent to zero on the surface X' . This follows from 7.1 by the projection formula.

Corollary 7.5. Let k be algebraically closed. Let Y be projective over k , with ample sheaf H , and let X be the zero-scheme of a section of H . Suppose that, for every irreducible component Y^* of Y , all irreducible components of

$$X \cap Y^* = V(\omega|_{Y^*})$$

have dimension > 2 . Then

$$\text{NS}(Y) \longrightarrow \text{NS}(X)$$

is injective. It is enough to assume Y integral, and the assertion follows from 7.4 (i) and the projection formula.

Bibliography for Exposé XIII.

- [1] Abhyankar, S. *Resolution of singularities of arithmetical surfaces*, in *Arithmetical Algebraic Geometry*, Purdue University, Harper & Row, New York, 1965.
- [2] Mumford, D. *Pathologies III*, Amer. J. Math. 1967, pp. 94–103.
- [3] Grothendieck, A. and Dieudonné, J. *EGA, Éléments de Géométrie Algébrique*, Publ. Math. IHÉS, 1960.
- [4] Grothendieck, A. *FGA, Fondements de la géométrie algébrique*, extracts from the Séminaire Bourbaki, 1957–1962.
- [5] Grothendieck, A. et al. *SGA, Séminaire de géométrie algébrique*, 1960–66, IHÉS.
- [6] Jussila, O. Exposé X, *Formalism of intersections on proper algebraic schemes*, in this seminar.
- [7] Raynaud, M. Exposé XII, *A relative representability theorem for the Picard functor*, in this seminar.

Exposé XIV – Open problems in intersection theory

OPEN PROBLEMS IN INTERSECTION THEORY

by A. Grothendieck

In this exposé we review several problems connected with questions encountered in this seminar and deserving further study. We leave aside the problems connected with the local intersection theory of Serre, which were not used in the seminar, and refer to Serre's book for the two fundamental conjectures still open in that direction. We also omit the very important problems, especially for arithmetic applications, concerning the structure of groups of cycle classes on projective smooth algebraic schemes; for these questions see S. Kleiman, *Algebraic cycles and the Weil conjectures*, in *Dix exposés sur la cohomologie des schémas*. In no. 4 below we also give complements to the seminar, mainly concerning the Chow ring, which had not found a place in the preceding exposés.

1. λ -operations in the derived category $D(X)$

Let X be a locally ringed topos, and let $K^\bullet(X)$ be the ring of classes of perfect complexes on X (Exposé IV). One would like to define on $K^\bullet(X)$ a λ -ring structure, functorial in X , such that the natural homomorphism from the ring of classes of locally free modules on X , endowed with the λ -structure of Exposé V, to $K^\bullet(X)$ is a λ -homomorphism. For this it would be necessary to define functors, generally non-additive, of exterior powers

$$\Lambda^i : D(X) \longrightarrow D(X),$$

sending pseudo-coherent, respectively perfect, complexes to pseudo-coherent, respectively perfect, complexes, and having the formal properties with respect to exact triangles needed to define

$$\lambda^i(\mathrm{cl}(L_\bullet)) = \mathrm{cl}(\Lambda^i(L_\bullet)). \quad (1.1)$$

When X has characteristic zero, i.e. when its structure sheaf is a sheaf of $\mathbb{Q}$ -algebras, the problem has a simple solution:

$$\Lambda^i(L_\bullet) = (L_\bullet^{\otimes i})^{\mathrm{alt}}. \quad (1.2)$$

Here the i th tensor power is taken in the derived category and regarded as an object of $D(\mathcal{O}_X[\mathfrak{S}_i])$, and “alt” denotes passage to the alternating part for the action of the symmetric group. This is meaningful because of the characteristic-zero hypothesis. A formula of this type was already used in RRR, in the first proof of the Riemann–Roch theorem.

Without a characteristic hypothesis, if $L_\bullet$ is zero in positive degrees, then by Dold–Puppe it is canonically homotopic to the complex associated to a semi-simplicial module $L'_\bullet$ canonically associated with it. For flat $L_\bullet$, say, a natural candidate for $\Lambda^i(L_\bullet)$ is the complex associated with the semi-simplicial module $\Lambda^i L'_\bullet$. To extend this to complexes not assumed zero in positive degrees, one would have to study the behaviour of this operation under translation $L_\bullet \mapsto L_\bullet[1]$, or

suspension. Unfortunately there seems to be no isomorphism

$$\Lambda^i(L_\bullet[1]) \simeq \Lambda^i(L_\bullet)[i]$$

that would make the desired extension formal.

2. The Riemann–Roch formula without projectivity assumptions

2.1. Once a natural λ -structure on $K^\bullet(X)$ is available, the Riemann–Roch theorem in the form considered in Exposé VIII can be formulated for any proper complete-intersection morphism

$$f : X \rightarrow Y.$$

More generally, it should apply to a relative scheme X over a locally ringed topos Y , subject to the corresponding relative conditions. The Todd class of f is defined from the class in $K^\bullet(X)$ of the relative cotangent complex $\mathbb{L}_\bullet$ of f , constructed in Exposé 0, 4.4; this complex is perfect by the local hypothesis on f . A proof of Riemann–Roch in this generality will probably require essentially new ideas. These same ideas may also give the key to the analogous formula in the complex analytic, or rigid analytic, case. In complex analytic geometry, only the case of a locally free sheaf F on a compact nonsingular variety X , over a point Y , is presently known, by Atiyah–Singer’s index theorem; the case of a merely coherent F still seems beyond the known methods.

2.2. As another essential ingredient in Riemann–Roch, recall the finiteness problem raised in Exposé III: if $f : X \rightarrow Y$ is proper and $L_\bullet$ is a complex on X pseudo-coherent relative to Y , is $\mathbb{R}f_*(L_\bullet)$ pseudo-coherent? A general proof should probably use algebraic ideas also capable of proving Grauert’s finiteness theorem under hypotheses substantially more general than the classical analytic ones, with base spaces or topoi topologically ringed in $\mathbb{C}$ -algebras: locally compact spaces with the sheaf of continuous complex-valued functions, C^∞ manifolds with the sheaf of complex C^∞ functions, real analytic spaces, and so on.

2.3. Finally, in these questions the properness assumption on f should be replaced by the hypothesis that the cohomology sheaves $H^i(L_\bullet)$ of the complexes considered on X have support proper over Y .

3. A “denominator-free” Riemann–Roch formula for an immersion

3.1. Let $f : Y \rightarrow X$ be a regular closed immersion, with conormal sheaf N . For simplicity suppose that X admits an ample invertible sheaf. The Riemann–Roch formula for $y \in K^\bullet(Y)$,

$$\mathrm{ch}(f_*(y)) = f_*(\mathrm{ch}(y) \mathrm{Td}(-N)),$$

together with the general formula

$$f_*(y)\xi = f_*(yf^*\xi) \quad (\xi \in K^\bullet(X)),$$

allows one to express the Chern classes of $f_*(y)$ in $\mathrm{Gr}^i(X)$ in the form

$$c_i(f_*(y)) = P_i(c_j(y), c_k(N)), \quad (3.1)$$

where P_i is a universal polynomial, a priori with rational coefficients. In fact, RRR shows that these polynomials have integral coefficients. This raises the question of the validity of a denominator-free Riemann–Roch formula.

Such a formula cannot in general be sought in $\mathrm{Gr}^\bullet(X)$ itself, since $f_! : K^\bullet(Y) \rightarrow K^\bullet(X)$ is not compatible with the γ -filtrations, shifted by d , except up to torsion. Thus there is generally no homomorphism

$$f_* : \mathrm{Gr}^\bullet(Y) \rightarrow \mathrm{Gr}^{\bullet+d}(X)$$

realizing the rational map $f_!$ and making (3.1) a formula in $\mathrm{Gr}^\bullet(X)$ rather than merely in $\mathrm{Gr}^\bullet(X)_\mathbb{Q}$.

If X is regular, so that $K^\bullet(X) \simeq K_\bullet(X)$, one may endow $K^\bullet(X)$ with the topological filtration by codimension of supports. It is plausible that this filtration is compatible with the ring structure; this can be proved at least when X is smooth over a field, by using the Chow ring. Then the associated graded

$$\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$$

is a graded ring, and, if Y is regular, there is a natural homomorphism of graded groups

$$f_* : \mathrm{Gr}_{\mathrm{top}}^\bullet(Y) \longrightarrow \mathrm{Gr}_{\mathrm{top}}^{\bullet+d}(X).$$

Since the γ -filtration is finer than the topological filtration, there is a canonical homomorphism of graded groups, indeed of graded rings,

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}^\bullet(X),$$

which is an isomorphism modulo torsion. This gives a theory of Chern classes with values in $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$, and hence a meaning to (3.1) as a formula in $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$. In this form the formula is proved in RRR when X has characteristic zero, at least when X is smooth over a field. The characteristic-zero hypothesis is used essentially in order to have formulas of the type (1.1), (1.2); essentially the same proof should work without a characteristic hypothesis once the question raised in no. 1 is answered positively.

3.2. When X is smooth over a field, there is also the Chow ring $A(X)$ and a theory of Chern classes with values in $A(X)$, finer than the theory with values in $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$. The latter Chern classes are images of the Chow Chern classes under the canonical surjection

$$\phi : A(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}^\bullet(X).$$

This map is an isomorphism modulo torsion, but not an isomorphism in general. If X and Y are smooth, the inclusion $f : Y \hookrightarrow X$ still defines a degree d homomorphism of graded groups

$$f_* : A^\bullet(Y) \longrightarrow A^{\bullet+d}(X),$$

and (3.1) can be interpreted as a formula in $A(X)$. The formula is trivial for $i < d$ and interesting only for $i > d$. Even for $i = d = 2$, neither this formula nor the weaker one obtained by replacing rational equivalence by algebraic equivalence is presently proved. Unlike the formula in $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$, there is no compelling reason to expect the denominator-free formula to hold in $A(X)$ itself. One practical way to test it, given our ignorance about the groups $A^i(X)$ for $i > 2$, is to push the two sides of (3.1) forward to groups $A^2(Z) = \mathrm{Pic}(Z)$ for suitable proper morphisms $g : X \rightarrow Z$.

3.3. One may also interpret (3.1) cohomologically whenever Gysin homomorphisms

$$f_* : H^{\bullet-2d}(Y, \mathbb{Z}_\ell(\bullet - d)) \longrightarrow H^\bullet(X, \mathbb{Z}_\ell(\bullet))$$

are available. This is the case at least when both X and Y are smooth over a scheme S , by relative cohomological purity, and conjecturally after absolute purity in the expected generality. When X and Y are smooth over $\mathbb{C}$, one can also use the integral Chern class on the locally compact space of complex points. In that case, Atiyah and Hirzebruch proved the formula for analytic embeddings, using naive K -theory. By Lefschetz's principle and SGA 4, the ℓ -adic formula follows for X and Y smooth over an algebraically closed field of characteristic zero. Its validity is therefore very plausible, but it is not currently proved even over a non-algebraically-closed field of characteristic zero.

3.4. One may likewise use De Rham or Hodge cohomological Chern classes. The problem of proving (3.1) in this setting only arises outside characteristic zero, since the formula is known in any case modulo torsion. For smooth quasi-projective schemes over a field, the De Rham or Hodge version would follow from the formula for cycle classes modulo algebraic equivalence, and so it gives another accessible way of testing the cycle-class version.

3.5. The proof of the transcendental cohomological version of (3.1) is essentially transcendental and uses tubular neighbourhoods of Y in X . This suggests that a proof of a formula of type (3.1) may be tied to a systematic use of K -groups and cohomology groups with support in Y , a technical means of localizing around Y .

4. Relations between $K^\bullet(X)$ and the Chow ring $A(X)$

4.1. For simplicity, suppose that X is smooth and quasi-projective over a field k . Then one has the Chow ring $A(X)$ of cycle classes modulo rational equivalence. There are also the graded rings

$$\mathrm{Gr}^\bullet(X) \quad \text{and} \quad \mathrm{Gr}_{\mathrm{top}}^\bullet(X),$$

the first associated with the γ -filtration on $K^\bullet(X)$, the second with the filtration on $K^\bullet(X) = K_\bullet(X)$ by codimension of supports. There are Chern-class theories with values in each of these rings, and also an ℓ -adic cohomological theory

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

for $\ell \neq \text{char}(k)$.

These theories are related by the canonical homomorphisms

$$\begin{array}{ccc} & A^\bullet(X) & \\ & \downarrow \phi & \searrow \psi \\ \text{Gr}^\bullet(X) & \searrow j & \\ & \text{Gr}_{\text{top}}^\bullet(X) & \rightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)). \end{array} \quad (4.1)$$

Here

$$H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)) = \bigoplus_i H^{2i}(X, \mathbb{Z}_\ell(i)).$$

The map j comes from the fact that the γ -filtration is finer than the topological filtration, and is an isomorphism modulo torsion. The map ϕ sends the class of a codimension i closed subscheme $Y \subset X$ to the class of $\mathcal{O}_Y$ in

$$\text{Fil}_{\text{top}}^i K_\bullet(X) / \text{Fil}_{\text{top}}^{i+1} K_\bullet(X).$$

It is surjective, and, as in 4.2, is an isomorphism modulo torsion. Hence, modulo torsion, the three rings

$$A(X), \quad \text{Gr}_{\text{top}}^\bullet(X), \quad \text{Gr}^\bullet(X)$$

are canonically isomorphic, and Riemann–Roch formulas expressed in any one of them after tensoring with $\mathbb{Q}$ are equivalent. The ring $\text{Gr}^\bullet(X)$ has the advantage of being defined for every locally ringed topos, hence for every scheme, without regularity or ampleness assumptions, and of acquiring covariance with respect to proper morphisms. Its disadvantage is that this covariance only appears after tensoring with $\mathbb{Q}$.

The Chow ring has another advantage: one has a homomorphism

$$A(X) \rightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet))$$

which interprets cohomological Chern classes as images of Chow-valued Chern classes. In general this homomorphism does not factor through $\text{Gr}_{\text{top}}^\bullet(X)$, and it is at least doubtful that there is in general a homomorphism

$$\text{Gr}^\bullet(X) \rightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet))$$

carrying Chern classes to Chern classes. All these difficulties disappear after tensoring with $\mathbb{Q}$:

$$\mathrm{Gr}^\bullet(X)_{\mathbb{Q}} \simeq \mathrm{Gr}_{\mathrm{top}}^\bullet(X)_{\mathbb{Q}} \simeq A^\bullet(X)_{\mathbb{Q}} \longrightarrow H^{2\bullet}(X, \mathbb{Q}_\ell(\bullet)). \quad (4.2)$$

4.2. To prove that the surjection ϕ of (4.1) is an isomorphism modulo torsion, one uses the Chern homomorphism

$$K_\bullet(X) \longrightarrow \mathrm{Ch}(A(X)),$$

where the target is the Chern ring of $A(X)$. This is compatible with the topological filtration of $K^\bullet(X)$ and the natural filtration of $\mathrm{Ch}(A(X))$, hence gives a homomorphism of graded groups

$$\psi : \mathrm{Gr}_{\mathrm{top}}^\bullet(X) \longrightarrow A(X). \quad (4.3)$$

One proves

$$\psi \circ \phi(x) = (-1)^{i-1}(i-1)!x \pmod{\text{torsion}}, \quad x \in A^i(X), \quad (4.4)$$

and similarly

$$\phi \circ \psi(x) = (-1)^{i-1}(i-1)!x \pmod{\text{torsion}}, \quad x \in \mathrm{Gr}_{\mathrm{top}}^i(X).$$

The first formula implies the second, since ϕ is surjective. Thus ϕ and ψ are isomorphisms modulo torsion, inverse to each other up to the factor $(-1)^{i-1}(i-1)!$ in degree i .

To prove (4.4), reduce to x represented by an irreducible closed subscheme Y of codimension i . The formula is equivalent to

$$\mathrm{cl}(\mathcal{O}_Y) = (-1)^{i-1}(i-1)! \mathrm{cl}(Y) \pmod{\text{torsion}}. \quad (4.5)$$

Removing a closed subset of codimension $> i$, one reduces to the case where Y is regular, hence smooth over k if k is perfect. In that case (4.5) is precisely the Riemann–Roch formula for the inclusion $Y \hookrightarrow X$ and the element $1 \in K^\bullet(Y)$, written with Chern classes in $A(X)$. Borel–Serre prove the needed formula in the circumstances considered here. This proves, at least for k perfect, that ϕ is an isomorphism modulo torsion. The general case reduces to the algebraically closed one using the trace map for a finite extension k'/k .

4.3. It is plausible that (4.4), equivalently (4.5), holds exactly and not merely modulo torsion. This is a special case of the denominator-free Riemann–Roch formula in $A(X)$ for an immersion. In characteristic zero, the denominator-free formula is known at least in $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$, which implies that the two sides of (4.5) have the same image in $\mathrm{Gr}_{\mathrm{top}}^i(X)$.

There is a simple support argument for (4.5) in $A(X)$ itself. Since the restriction of the left side to $X - Y$ is zero, and Y is irreducible, the left side must be an integral multiple of $\mathrm{cl}(Y)$:

$$\mathrm{cl}(\mathcal{O}_Y) = a(Y) \mathrm{cl}(Y).$$

This proves the exact formula whenever $\mathrm{cl}(Y)$ is not torsion. For example, if X is projective over

k and $\text{cl}(Y) \cdot \xi^{r-i} > 0$ for a polarization ξ , the formula follows. Thus the formula holds whenever X is an open subscheme of a smooth projective scheme over k , as happens whenever an adequate resolution of singularities theorem is available; in particular it holds in characteristic zero by Hironaka.

If exact (4.5) holds, then the kernel of

$$\phi : A^i(X) \rightarrow \text{Gr}_{\text{top}}^i(X)$$

is killed by $(i-1)!$, and in particular $\phi : A^2(X) \rightarrow \text{Gr}_{\text{top}}^2(X)$ is an isomorphism. Apart from ϕ^1 , which is already known to be an isomorphism, this does not seem to have been proved in general, even when $\dim X = 3$. The earlier assertion of the validity of (4.4) and (4.5) in the cited notes was therefore premature.

4.4. The main present advantage of $\text{Gr}_{\text{top}}^\bullet(X)$, or $\text{Gr}^\bullet(X)$, over the Chow ring $A(X)$ is that it is better suited to explicit computations of certain intersection formulas in cases of excess intersection. The moving-lemma definition of intersection classes in $A(X)$ is poorly adapted to computation. A striking example is the fundamental formula, for a closed immersion $f : Y \rightarrow X$ of smooth quasi-projective schemes over k ,

$$f_*(y) = y \cdot c_d(N), \quad (4.6)$$

valid in $\text{Gr}_{\text{top}}^\bullet(X)$, where $y \in \text{Gr}^\bullet(Y)$ and N is the conormal sheaf of Y in X , assumed everywhere of rank d . In Exposé VII this formula follows from a simple computation of Tor sheaves. By 4.2, it implies the same formula in $A(Y)$ only modulo torsion. It is remarkable that (4.6) in the Chow ring itself, and even the particular self-intersection formula

$$f_*(1) = c_d(N), \quad (4.7)$$

has not at this point been proved in general, even over the complex numbers. The same problem appears in the computation of the Chow ring of the blow-up X' of X along Y , where the key formula is

$$c_i(\xi) = j_*(\pi^* c_{i-d}(F)). \quad (4.8)$$

It is known in $\text{Gr}_{\text{top}}^\bullet(X')$ by a Tor computation, but its validity in $A(X')$ remains to be established. Even the ℓ -adic analogues of (4.6), (4.7), (4.8) are not presently proved in full generality; when y is the cohomology class of an algebraic cycle, the formulas are at least true modulo torsion.

4.5. We next give examples where, in diagram (4.1), either j or ϕ is not an isomorphism. The image of

$$j : \text{Gr}^\bullet(X) \rightarrow \text{Gr}_{\text{top}}^\bullet(X)$$

is the subring of $\text{Gr}_{\text{top}}^\bullet(X)$ generated by Chern classes of locally free sheaves. If $K^\bullet(X)$ is generated as a $\mathbb{Z}$ -algebra by classes of invertible sheaves, this is the subring generated by $\text{Gr}_{\text{top}}^1(X)$.

Take X to be a flag variety

$$X = G/B,$$

where G is a simply connected semisimple algebraic group over an algebraically closed field k , and B is a Borel subgroup. Over $k = \mathbb{C}$, the Bruhat cell decomposition implies that the canonical map

$$K^\bullet(X) \longrightarrow K^\bullet(X(\mathbb{C}))$$

is an isomorphism, and the same is true for the corresponding Chow/cohomology comparison. Hodgkin proved that $K^\bullet(X(\mathbb{C}))$ is generated by classes of line bundles; hence the same is true for $K^\bullet(X)$, and by specialization also over any algebraically closed field. The cell decomposition also implies that $A(X)$ is torsion-free, hence $\phi : A(X) \rightarrow \mathrm{Gr}_{\mathrm{top}}^\bullet(X)$ is an isomorphism. Therefore j is an isomorphism if and only if $A(X)$ is generated by its degree-one elements, or, over $\mathbb{C}$, if and only if the integral cohomology of $X(\mathbb{C})$ is generated in degree 2, equivalently if BG is torsion-free. This occurs exactly when G is a product of groups $SL(n)$ and $Sp(m)$. Thus $G = SO(n)$ with $n \geq 7$, or a simple exceptional group, gives examples where j is not an isomorphism.

4.6. In this example the good ring is plainly the torsion-free ring

$$A(X) = \mathrm{Gr}_{\mathrm{top}}^\bullet(X),$$

with basis given by Bruhat cells and, over $\mathbb{C}$, identified with integral cohomology. By contrast, $\mathrm{Gr}^\bullet(X)$ has torsion of a rather pathological kind. This confirms the view that $\mathrm{Gr}^\bullet(X)$ is mainly useful modulo torsion, i.e. after tensoring with $\mathbb{Q}$.

The same example shows that only after tensoring with $\mathbb{Q}$ does $\mathrm{Gr}^\bullet(X)$ have a covariant character for proper morphisms, even for closed immersions of smooth quasi-projective schemes. If j is not an isomorphism, there is an integral closed subscheme $Y \subset X$ such that

$$c_i^\gamma(Y) \in K^\bullet(X)$$

does not lie in $\mathrm{Fil}_{\mathrm{top}}^i(X)$, where $i = \mathrm{codim}(Y, X)$. Choosing Y of minimal dimension and removing its singular locus, one obtains a closed immersion

$$f : Y' \hookrightarrow X'$$

of codimension i between smooth quasi-projective schemes such that

$$f_* : K^\bullet(Y') \rightarrow K^\bullet(X')$$

is not compatible with the γ -filtrations shifted by i , since $f_*(1)$ is not in $\mathrm{Fil}_{\mathrm{top}}^i(X')$.

4.7. With the notation of 4.5, Exposé IX shows that Hodgkin's theorem is equivalent to

$$K^\bullet(G) = \mathbb{Z}.$$

Thus $\mathrm{Gr}^\bullet(G) = \mathbb{Z}$. The map $K^\bullet(G/B) \rightarrow K^\bullet(G)$ is a priori surjective, with kernel generated by the $c_1(L) - 1$, where L runs through the line bundles on G/B associated with characters $B \rightarrow \mathbb{G}_m$. Similarly,

$$A(G/B) \rightarrow A(G)$$

is surjective with kernel generated by $A^1(G/B)$. It follows that $A(G) = \mathbb{Z}$ if and only if G is torsion-free, i.e. a product of groups $SL(n)$ and $Sp(m)$. Thus, when G has torsion, the underlying scheme of G gives an example where

$$\phi : A(G) \rightarrow \mathrm{Gr}_{\mathrm{top}}^\bullet(G)$$

is not bijective.

4.8. Similarly, one gets examples where

$$\psi : A(X) \rightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)) \quad (4.9)$$

does not factor through $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$. It is enough, after specialization, to consider $k = \mathbb{C}$, and hence the homomorphism

$$A(G) \rightarrow H^{2\bullet}(G(\mathbb{C}), \mathbb{Z}), \quad (4.10)$$

induced from

$$A(G/B) = H^{2\bullet}(G/B(\mathbb{C}), \mathbb{Z}) \rightarrow H^{2\bullet}(G(\mathbb{C}), \mathbb{Z}).$$

It is tempting to conjecture that (4.10) is injective. If so, whenever G has torsion the map (4.10) would not factor through ϕ , and the same would hold for (4.9) for a suitable ℓ . At least for the group G_2 over $\mathbb{C}$, the map (4.10) is non-zero in positive degree: there is an element of $A^3(G)$ whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ has order 2. This shows that (4.9) does not factor through ϕ for $\ell = 2$. The fact about G_2 follows from the corresponding statement in the topology of compact Lie groups, communicated by J.-P. Serre.

5. Complements on Chern homomorphisms

5.1. The example in 4.5 does not rule out the possible existence, for an arbitrary scheme X , of a ring homomorphism

$$g^* : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet))$$

which sends each Chern class c_i of a locally free sheaf E to the cohomological Chern class of SGA 5, VII. Since $\mathrm{Gr}^\bullet(X)$ is generated by the $c_i(E)$, such a g^* would be uniquely determined by compatibility with Chern classes.

5.2. Using the Chern homomorphism

$$K^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)),$$

and passing to associated graded for the filtration on the source and the degree filtration on the target, one obtains a homomorphism of graded groups, but not of rings,

$$\gamma^* : \mathrm{Gr}_{\mathrm{top}}^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet))$$

satisfying

$$\gamma^*(\mathrm{cl}(E)) = c_i^{(\ell)}(E) \pmod{\text{torsion}}.$$

After the same divisions as before, this gives a ring homomorphism

$$g^* : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)) \otimes \mathbb{Z}[1/(i-1)!],$$

carrying Chern classes to Chern classes. Thus the existence question has an affirmative answer after replacing cohomology by its quotient modulo torsion. The first doubtful case for defining the components g^i is g^2 , where a priori one reaches only

$$2c_2^{(\ell)} = \gamma^*(\mathrm{cl}(\Lambda^2 E - \mathrm{Sym}^2 E)).$$

A negative answer for g^2 would follow, for example, from the existence of three invertible sheaves L_1, L_2, L_3 on X such that

$$(L_1 - 1)(L_2 - 1)(L_3 - 1) = 0 \quad \text{in } K^\bullet(X),$$

but

$$c_1^{(\ell)}(L_1)c_1^{(\ell)}(L_2)c_1^{(\ell)}(L_3) \neq 0 \quad \text{in } H^6(X, \mathbb{Z}_\ell(3)).$$

5.3. When X is smooth and quasi-projective over a field k , one can similarly ask whether there exists a ring homomorphism

$$g : \mathrm{Gr}^\bullet(X) \longrightarrow A(X)$$

with values in the Chow ring and carrying Chern classes to Chern classes. The preceding discussion applies, and in particular shows how to construct directly

$$(-1)^{i-1}(i-1)!g^i$$

when g^i itself is not available.

5.4. The same question also arises for a compact topological space X , since one may form the graded ring $\mathrm{Gr}^\bullet(X)$ of $K(X)$ with its λ -filtration and ask for a ring homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z})$$

carrying Chern classes to Chern classes.

6. Cohomological Riemann–Roch and the Gysin homomorphism

6.1. For simplicity, restrict to schemes X admitting an ample invertible sheaf, so that naive and non-naive $K^\bullet(X)$ need not be distinguished. Consider first Riemann–Roch with denominators for a proper complete-intersection morphism $f : X \rightarrow Y$. For a prime ℓ invertible on Y , one has ℓ -adic Chern classes

$$c_i^{(\ell)}(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

and hence a homomorphism

$$K^\bullet(X) \longrightarrow H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)). \quad (6.1)$$

The missing ingredient for a cohomological Riemann–Roch formula is a Gysin homomorphism

$$f_* : H^{\bullet-2d}(X, \mathbb{Z}_\ell(\bullet-d)) \longrightarrow H^\bullet(Y, \mathbb{Z}_\ell(\bullet)), \quad (6.2)$$

where d is the virtual relative dimension of f . Assuming such a homomorphism is defined, Riemann–Roch for f in the form of Exposé VIII, with values in $\mathrm{Gr}^\bullet(Y)$, is equivalent to commutativity of

$$\begin{array}{ccc} \mathrm{Gr}^\bullet(X) & \longrightarrow & H^{2\bullet}(X, \mathbb{Q}_\ell(\bullet)) \\ f_* \downarrow & & \downarrow f_* \\ \mathrm{Gr}^{\bullet-d}(Y) & \longrightarrow & H^{2(\bullet-d)}(Y, \mathbb{Q}_\ell(\bullet-d)). \end{array} \quad (6.3)$$

The horizontal arrows are obtained from (6.1) by passing to associated graded and dividing so that Chern classes go to Chern classes.

6.2. The Gysin homomorphism is defined at least when Y is smooth over a field, using relative cohomological purity. The same construction applies when Y is excellent, regular, and of characteristic zero, by the absolute purity theorem of SGA 4, XIX, and it will apply for every excellent regular Y once absolute purity is known in full generality. It also applies when X and Y are both smooth over a common scheme S and f is an S -morphism, by relative purity. In each case one should conjecture that (6.3) is commutative. The only case presently verified is the case in which X and Y are both smooth over a field; it follows from commutativity of the analogous diagram with Chow rings:

$$\begin{array}{ccc} A^\bullet(X) & \longrightarrow & H^{2\bullet}(X, \mathbb{Z}_\ell(\bullet)) \\ f_* \downarrow & & \downarrow f_* \\ A^{\bullet-d}(Y) & \longrightarrow & H^{2(\bullet-d)}(Y, \mathbb{Z}_\ell(\bullet-d)). \end{array} \quad (6.4)$$

6.3. In each case where a Gysin homomorphism can be constructed, the construction is integral, with coefficients in $\mathbb{Z}_\ell(i)$, not only with coefficients in $\mathbb{Q}_\ell(i)$. When f is an immersion, this permits the denominator-free Riemann–Roch question of no. 3. If X and Y are both smooth over a field k , then commutativity of (6.4) shows that it would be enough to prove the denominator-free formula in the Chow ring.

6.4. Without regularity on Y , even when X and Y are of finite type over $\mathbb{C}$, there is no evident definition of a cohomological Gysin morphism (6.2), even for a regular immersion of codimension one. Can one nevertheless define one with reasonable properties, especially making (6.3) commute? This is closely related to defining, for X a complete intersection of codimension i in Y , its cohomology class

$$\mathrm{cl}(X) \in H^{2i}(Y, \mathbb{Z}_\ell(i))$$

so that proper intersections correspond to cup-products. This latter question has an affirmative answer at least modulo torsion, by setting

$$\mathrm{cl}(X) = \frac{c_i(\mathcal{O}_X)}{(-1)^{i-1}(i-1)!} \in H^{2i}(Y, \mathbb{Q}_\ell(i)).$$

It is not known whether $c_i(\mathcal{O}_X)$ is divisible by $(i-1)!$ in $H^{2i}(Y, \mathbb{Z}_\ell(i))$. For instance, if Y is a projective threefold over $\mathbb{C}$ and X is a singular point of Y , is $c_3(\mathcal{O}_X)$ divisible by 2 in $H^6(Y, \mathbb{Z}_\ell(3))$? Equivalently, is the mod-2 Chern class zero? Similarly, if Y is a projective fourfold and X a one-dimensional closed subscheme that is globally the intersection of three positive divisors D_1, D_2, D_3 , is

$$c_1(D_1)c_1(D_2)c_1(D_3) \in H^6(Y, \mathbb{Z}/2\mathbb{Z})$$

independent of the chosen D_i cutting out X ?

7. Chern classes of perfect complexes

7.1. To develop a cohomological Riemann–Roch formula, with or without denominators, outside the artificial assumption that the schemes admit ample line bundles, one is forced to define, for a perfect complex $L_\bullet$ on a scheme X , Chern classes

$$c_i(L_\bullet) \in H^{2i}(X, \mathbb{Z}_\ell(i))$$

generalizing those of locally free modules. The same question arises for general locally ringed topoi, and for perfect complexes on paracompact spaces ringed by continuous complex-valued functions, using ordinary integral cohomology. For compact spaces, the $K^\bullet(X)$ built from perfect complexes agrees with naive K -theory built from locally free modules, so the problem becomes classical. Already for locally compact spaces, however, the problem is non-trivial, since integral cohomology is not always the inverse limit of the cohomologies of compact subsets.

7.2. More precisely, if Y is a closed subset of X such that $L_\bullet|_{X-Y}$ is zero in the derived category, i.e. all cohomology sheaves of $L_\bullet$ are supported on Y , one would like to define Chern classes with support in Y ,

$$c_i^Y(L_\bullet) \in H_Y^{2i}(X, \mathbb{Z}_\ell(i)),$$

whose image under

$$H_Y^{2i}(X, \mathbb{Z}_\ell(i)) \longrightarrow H^{2i}(X, \mathbb{Z}_\ell(i))$$

is the ordinary class of 7.1. Even when ordinary Chern classes reduce to those of locally free modules, such supported Chern classes have not yet been constructed. They will probably have to be constructed together with the general Chern classes postulated in 7.1. These supported classes should play a role in proving the denominator-free cohomological Riemann–Roch formula and should also yield an improved Riemann–Roch formula involving families of supports.

8. The Chow ring of regular schemes

The problem is to define rational equivalence of cycles, and to study the resulting ring of classes and its functorial properties, for a noetherian regular scheme X admitting an ample invertible sheaf, generalizing the theory known when X is smooth over a field. The key technical result should again be a moving lemma, ensuring that a cycle can be varied within its class so as to meet a given cycle in the proper dimension.

This problem is natural, especially because 4.5–4.8 show that the substitutes $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$ are unsatisfactory when one wants more than a theory modulo torsion. If X is obtained from a projective smooth algebraic scheme Y over a field k by taking the local scheme at the vertex of the projective cone over Y and deleting the vertex, then, assuming the expected limiting property for Chow rings, one finds that $A(X)$ is canonically isomorphic to

$$A(Y)/\xi A(Y),$$

where $\xi \in A^1(Y)$ is induced by the chosen projective embedding. This quotient may be called the primitive part of $A(Y)$, in a sense suggested by Lefschetz–Hodge theory. Thus local Chow rings of local rings with isolated singularity are non-trivial geometric invariants and should play a role in the arithmetic study of such local rings.

Once a reasonable notion of Chow ring is available for a noetherian regular scheme with an ample line bundle, the relations established and conjectured in no. 4 between that ring and $\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$ should be generalized, and the denominator-free Riemann–Roch question of no. 3 should be formulated in terms of that ring.

Bibliography for Exposé XIV.

- [1] M. F. Atiyah and F. Hirzebruch, *The Riemann–Roch theorem for analytic embeddings*, Topology, vol. 1, pp. 151–166 (1962).
- [2] A. Borel and J.-P. Serre, *Le théorème de Riemann–Roch*, Bull. Soc. Math. France, vol. 86, pp. 97–136 (1958).
- [3] R. Bott and H. Samelson, *Applications of the theory of Morse to symmetric spaces*, Amer. J. Math., vol. 80, pp. 964–1029 (1958).
- [4] C. Chevalley, *Les classes d’équivalence rationnelle*, I and II, Séminaire at the ÉNS, 1958, exposés 2 and 3.

- [5] C. Chevalley, *Classification des groupes de Lie algébriques*, Séminaire at the ÉNS, 1956/58.
- [6] A. Dold and D. Puppe, *Homologie nicht-additiver Funktoren. Anwendungen*, Annales de l'Institut Fourier, t. XI, pp. 201–312 (1961).
- [7] H. Grauert, *Ein Theorem der analytischen Garbentheorie*, Publ. Math. IHÉS no. 5, pp. 5–64 (1960).
- [8] A. Grothendieck, *Classes de faisceaux et théorème de Riemann–Roch*, multigraphed notes from the Institute for Advanced Study, 1957, cited here as RRR and reproduced in this seminar as an appendix to Chapter 0.
- [9] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. Math. France, vol. 86, pp. 137–154 (1958).
- [10] A. Grothendieck, *Classes de Chern et représentations linéaires de groupes discrets*, in *Dix exposés sur la cohomologie des schémas*.
- [11] H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. 79, pp. 109–326 (1964).
- [12] L. Hodgkin, to appear.
- [13] L. Illusie, work in preparation.
- [14] J.-P. Serre, *Algèbre locale, Multiplicités*, Lecture Notes in Math. no. 11, Springer, 1965.
- [15] A. Borel, *Sous-groupes commutatifs et torsion des groupes de Lie compacts connexes*, Tohoku Math. Journal, vol. 13, pp. 216–240 (1961).
- [16] P. Deligne, Exposé XVIII in SGA 4.
- [17] Yu. Manin, *Correspondences, motives and monoidal transformations* (Russian), Mat. Sbornik, t. 77 (119), no. 4, 1968.

Note to page 1. The theory of λ -operations in $D(X)$, and of the operations σ_i in $K^\bullet(X)$, was written by P. Deligne in 1968. He proves that $K^\bullet(X)$ thereby becomes a pre- λ -ring and, when $\mathcal{O}_X$ is a sheaf of algebras over a field, a λ -ring. It is plausible that a refinement of Deligne's method would remove the restrictive hypothesis on $\mathcal{O}_X$.

Notes to page 11. In April 1969, D. Mumford communicated a proof of (4.6) in the Chow ring, without neglecting torsion, obtained in a Harvard seminar in 1959. The proof of this result appears in SGA 5, VII, 9. The corresponding cohomological result is proved by J.-P. Jouanolou under more general hypotheses, without quasi-projective hyperplane assumptions and in the case smooth relative to an arbitrary base.

In January 1969, J.-P. Jouanolou proved these formulas for all y , without neglecting torsion; see SGA 5, VII, 4.9. Deligne and Manin had previously observed independently, by rather different

elementary arguments, that it suffices to establish the formulas for algebraic y , indeed for $y = 1$ in (4.6), in order to deduce the general relation.

Terminological index

Term	Exposé
additive map	I, III
augmented λ -algebra	V
algebraically equivalent to zero in $K^\bullet$	XIII
algebraically equivalent to zero, invertible sheaf	XIII
$\mathfrak{C}$ -perfect amplitude contained in $[a, b]$, complex	I, III
finite $\mathfrak{C}$ -perfect amplitude, complex	I
finite perfect amplitude, complex	I
flat amplitude contained in $[a, b]$, complex	I
finite flat amplitude, complex	I
perfect amplitude relative to f or Y , complex	III
perfect amplitude contained in $[-n, 0]$, morphism	III
finite perfect amplitude relative to f , complex	III
flat amplitude relative to f or Y , complex	III
flat amplitude contained in $[-n, 0]$, morphism	III
finite flat amplitude relative to f , complex	III
projective amplitude	II App. II
λ -ring	V
free λ -ring generated by a family of generators	V
Chern ring	V
polynomial map	X
binomial ring	V
λ -ring generated by generators and relations	V
Chern character	V
Todd character	V
S -abelian category	I
flat S -abelian category	I
S -additive category	I
S -triangulated category	I
Chern class	V
completed Chern class	V
total Chern class	V
codimension of a regular immersion	VII
coherent complex	I
coherer	II
relative cotangent complex	VIII
elliptic complex	II App. II
cup-product	IV
special positive k -cycle	XIII

Term	Exposé
degree of a polynomial map	X
degree of an invertible sheaf	XIII
degree of a morphism of schemes	X
dense subset of an abelian or triangulated category	IV
perfect dimension	I
virtual relative dimension	VIII
divisorial	II
dual	I
perfect duality	I
numerical equivalence in $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_\bullet(X)$	XIII
τ -equivalence	XIII
complex pseudo-analytic space	II App. II
Stein complex pseudo-analytic space	II App. II
exact subset of an abelian or triangulated category	IV
(b) -sheaf	XIII
conormal sheaf	VII
relative cotangent sheaf	II App. II
relative tangent sheaf	II App. II
(b) -family	XIII
ample family	XIII
family of classes of coherent sheaves	XIII
bounded family of classes of coherent sheaves	XIII
family of classes of coherent sheaves bounded by a coherent sheaf	XIII
relative cotangent bundle	II App. II
relative tangent bundle	II App. II
faithfully flat morphism of ringed topoi	I
filtration of an augmented λ -algebra	V
S -additive functor	I
S -exact functor	I
projection formula	IV
Grothendieck group	IV
λ -homomorphism	V
specialization homomorphism	I
λ -ideal	V
analytic index	II App. II
complete-intersection morphism	VIII
n -isomorphism	I
homogeneous free graded module	I
locally stable under kernels of epimorphisms	I

Term	Exposé
locally quasi-liftable	I
locally quasi-stable under kernels of epimorphisms	I
locally liftable	I
Chern number	V
intersection number	XIII
intersection number for a family of invertible sheaves	XIII
numerically equivalent to zero, invertible sheaf	XIII
universal differential operator of order n	II App. II
Adams operations	V
$\mathfrak{C}$ -perfect complex	I
perfect complex	I
perfect morphism	III
perfect complex relative to f	III
precoherent	I
of finite n - $\mathfrak{C}$ -presentation	I
of finite presentation	I
of finite éc- $\mathfrak{C}$ -presentation	I
n - $\mathfrak{C}$ -pseudo-coherent complex	I
$\mathfrak{C}$ -pseudo-coherent complex	I
n -pseudo-coherent complex	I
n -pseudo-coherent morphism	III
pseudo-coherent complex	I
pseudo-coherent morphism	III
n -pseudo-coherent relative to f , complex	III
pseudo-coherent relative to f , complex	III
(b) -polynomial	XIII
Snapper polynomial	X
universal polynomial	V
pre- λ -ring	V
n -quasi-isomorphism	I
rank of a perfect complex	X
m -regular sheaf	XIII
regular homomorphism	VII
regular ideal	VII
regular immersion	VII
triangulated S -subcategory	I
special λ -ring	V
special morphism	III
Stein locally ringed topos	II App. II
strictly $\mathfrak{C}$ -perfect	I

Term	Exposé
strictly $\mathfrak{C}$ - n -pseudo-coherent	I
strictly $\mathfrak{C}$ -pseudo-coherent	I
system of generators	VII

Index of notation

Notation	Exposé
ch	V
$\mathrm{Ch}(A), \mathrm{Ch}_\bullet(A)$	I
$\mathrm{Coh}(S)$	I
$\mathfrak{C}_n\text{-parf-amp}(F)$	II App. II
$c^{(\ell)}(E), c_i^{(\ell)}(E)$	II App. II
$c_i(E), c_i(x)$	V
$D^{\mathfrak{C}}(S), D^+(S), D^-(S), D^b(S)$	I
$D_{\mathrm{qc}}^{\mathfrak{C}}(S), D_{\mathrm{qc}}(S), D_{\mathrm{qc},n\text{-cons}}(S)$	III
$D_{\mathrm{parf}}^{\mathfrak{C}}(S), D_{\mathrm{parf},\mathrm{cons}}^{\mathfrak{C}}(S), D_{\mathrm{parf}}(S)$	III
$D_{n\text{-pseudo-coh}}^{\mathfrak{C}}(S), D_{\mathrm{pseudo-coh}}^{\mathfrak{C}}(S),$	III
$D_{n\text{-pseudo-coh}}(S)$	
$D_{\mathrm{parf}}^{\mathfrak{C}}(X), D_{\mathrm{parf}}^{\mathfrak{C},f\text{-parf}}(X), D^{\mathfrak{C}}(X), D(X)$	I
$D_{\mathrm{parf}}^{\mathfrak{C},f\text{-torf}}(X), D_{\mathrm{parf}}^{\mathfrak{C}}(X)$	III
$D_{n\text{-cons}}(X), D_{\mathrm{cons}}(X), D_{\mathrm{parf}}(X), D_{\mathrm{torf}}(X)$	III
$D_{\mathrm{coh}}^{\mathfrak{C}}(X), D_{\mathrm{coh}}(X)$	I
$E^\vee$	I
$E \otimes F, E \otimes^{\mathbf{L}} F$	I
f^* between $K_\bullet$ groups	IV
f^* between $K^\bullet$ groups	IV
f_* between $K_\bullet$ groups	IV
$f_!$ between $K^\bullet$ groups	IV
f_* for a projective bundle	VI
$f_!$	IV
$\mathrm{Filt}_{\mathrm{top}}(X)$	X
$\mathrm{Gr}^\bullet(X), \mathrm{Gr}_\bullet(X)$	V, X
$\mathrm{Gr}_{\mathrm{top}}^\bullet(X)$	XIV
$\mathrm{Gr}^i(X)$	V
$\mathrm{Hom}^\bullet(E, F)$	I
$\mathrm{Hom}(E, F)$	I
$K^\bullet(X), K_\bullet(X)$	IV
$K^\bullet(X)_{\mathrm{naive}}, K_\bullet(X)_{\mathrm{naive}}$	IV
$K^\bullet(f), K_\bullet(f)$	IV
$K_{\mathrm{num}}^\bullet(X), K_{\mathrm{alg}}^\bullet(X)$	XIII
$K^\bullet(X/Y), K_\bullet(X/Y)$	IV
$\mathrm{Lib}(U), \mathrm{Lib}_{\mathrm{hom}}(U)$	I
$\Lambda^i(E), \Lambda^i(L_\bullet)$	V, XIV
$\lambda^i(x), \lambda_t(x)$	V
$\lambda_t^{-1}(M)$	V

Notation	Exposé
$\lambda_T^i(x)$	V
$\mathcal{M}\mathrm{od}(U)$	I
$\mathrm{ob}\, \mathcal{C}$	I

Notation	Exposé
$\mathrm{par}\text{-}\mathrm{amp}_{\bullet}(E), \mathrm{par}\text{-}\mathrm{amp}^{\bullet}(E)$	XIII
$\mathrm{Parf}(\mathfrak{C}, S), \mathrm{Parf}(\mathcal{C}, \mathcal{C}'), \mathrm{Parf}(S)$	I, III
$\mathrm{Pic}^{\tau}(X), \mathrm{Pic}_{X/S}^{\tau}$	XIII
$\mathrm{Pic}^0(X), \mathrm{Pic}_{X/k}^0$	XIII
$\mathrm{Pic}^{\mathrm{num}}(X), \mathrm{Pic}_{X/S}^{\mathrm{num}}$	XIII
$\mathrm{Pic}^{[n]}(X)$	XIII
$\mathrm{Pic}(X), \mathrm{Pic}_{X/S}$	XIII
$P_{X/S}$	XIII
$P_{X/S}^{(n)}$	XIII
$P_{X/S}(E)$	XIII
Proj	VI
$\mathcal{P}_{\gamma, \infty}(X), \mathcal{P}_{\lambda, \infty}(X)$	V
$\mathrm{Qcoh}(S)$	I
$T^{\bullet}(E), T_{\bullet}(E)$	VIII
$\mathrm{Todd}(E), \mathrm{Todd}(f), \mathrm{Td}(E)$	VIII
$\mathrm{tor}\text{-}\mathrm{amp}(F), \mathrm{tor}\text{-}\mathrm{amp}_{\bullet}(E), \mathrm{tor}\text{-}\mathrm{amp}^{\bullet}(E)$	II App. II
Trip	I
X/S	III
$X^{(n)}$	V
$1 + A[[t]]^*$	V
$Z(x)$	V

End of the Séminaire de Géométrie Algébrique du Bois Marie 1966–67